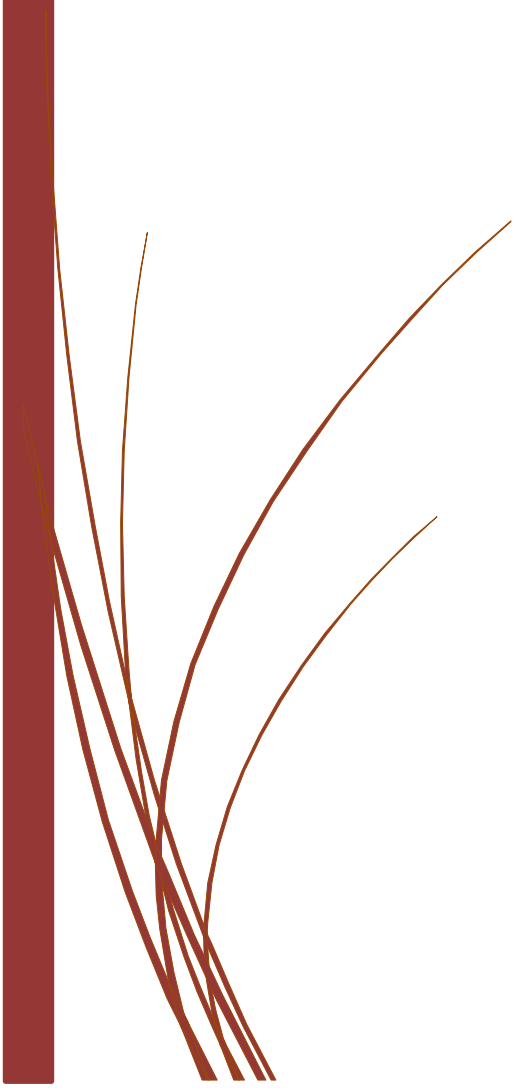
CHAPTER-06

DETERMINATION OF NATIONAL INCOME

Unit 1 – National Income Accounting

Unit 2 – The Keynesian theory for determination of National Income



UNIT – I
NATIONAL INCOME ACCOUNTING

Topic References:

1.1 Introduction to National Income
1.2 Usefulness and Significance of National Income Estimates
1.3 Concepts of National Income
1.4 Measurement of National Income in India
1.5 System of Regional Accounts in India
1.6 Limitation and Challenges in computation of National Income

1.1 Introduction to National Income

- National Income Accounting, pioneered by the Nobel prize-winning economists Simon Kuznets and Richard Stone, is the system of macro-economic accounts from the stage of production of goods and services to the stage of their final disposal.
- The Central Statistical Organisation (CSO) in the Ministry of Statistics and Programme Implementation (MoSP&I) is responsible for the compilation of National accounts statistics.
- At the State level, State Directorates of Economics and Statistics (DESs) have the responsibility of compiling their State Domestic Product and other aggregates.

1.2 Usefulness and Significance of National Income Estimates

National income accounts are fundamental aggregate statistics in macroeconomic analysis and are extremely useful, especially for the emerging and transition economies.

- Businesses to forecast the future demand for their products.
- The estimates of national income show the composition and structure of national income in terms of different sectors of the economy, the periodical variations in them and the broad sectoral shifts in an economy over time.
- Sectoral contribution to National Income information is used by the government to decide various sector-specific development policies to increase growth rates.
- National income statistics also provide a quantitative basis for macroeconomic modelling and analysis, for assessing and choosing economic policies and for objective statements as well as evaluation of governments' economic policies.
- National income estimates throw light on income distribution and the possible inequality in the distribution among different income categories. It facilitates the process of comparisons of structural statistics, such as ratios of investment to growth, taxes proceeds and fiscal deficit, or government expenditures to GDP.
- International comparisons in respect of incomes and living standards assist in determining eligibility for loans, and/or other funds or conditions under which such loans, and/ or funds are made available.
- Combined with financial and monetary data, national income data provides a guide to make policies for growth and inflation.

1.3 Different Concepts of National Income

- The production side of the economy transforms inputs, such as labour and capital, into output, GDP.

1.3.1 Gross Domestic Product

Nominal GDP or GDP $_{MP}$
Gross domestic product at Market Price (GDP) $_{MP}$ is a measure of
- the market value,
- of all final economic goods and services,
- gross of depreciation,
- produced within the domestic territory of a country,
- during a given time period.

The term 'gross' implies that GDP is measured 'gross' of depreciation.
It includes the value of goods produced, such as houses and mobiles, and the value of services, such as telecom, health, insurance.
The output of each of these is valued at its market price, and the values are added together to get GDP $_{MP.}$

Real GDP

Nominal GDP increases over time for two reasons:
1. The production of most of goods increases over time
2. The prices of most goods also increase over time.

If our goal is to measure production and its change over time, we need to eliminate the effect of increasing prices on our measure of GDP.
That's why real GDP is constructed as the sum of the quantities of final goods times constant (rather than current prices)

GDP Deflator

The calculation of real GDP gives us a useful measure of inflation known as GDP deflator.
The GDP deflator is the ratio of nominal GDP in a given year to real GDP of that year.

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

The GDP deflator, as the name implies, can be used to 'deflate' or take inflation out of GDP.
In other words, the GDP deflator is a price index used to convert nominal GDP to real GDP

$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{GDP Deflator}} \times 100$$

The deflator measures the current level of prices relative to the level of prices in the base year. Since nominal GDP and real GDP must be the same in the base year, the deflator for the base year is always 100.

Inflation Rate

Using the GDP deflator, the inflation rate between two consecutive years can be compute using the following procedure:

$$\text{Inflation rate in year 2} = \frac{\text{GDP deflator in year 2 - GDP deflator in year 1}}{\text{GDP deflator in year 1}} \times 100$$

GDP Deflator in India is expected to reach 154.87 points by the end of 2022, according to Trading Economics global macro models and analysts expectations. In the long-term, the India GDP Deflator is projected to trend around 167.94 points in 2023 and 175.67 points in 2024, according to econometric models.
Inflation Rate in 2023 = (167.94 - 154.87) / 154.87 * 100
Inflation Rate in 2023 as compared to 2022 will be 8.439 %

ILLUSTRATION 1 **[SM]**

Find out GDP Deflator? Interpret It

(In Billion Rs.)			
Years	**Nominal GDP**	**Real GDP**	**GDP Deflator**
2014	500	500	
2015	800	650	
2016	1150	800	
2017	1300	950	
2018	1550	1190	
2019	1700	1240	

[**ANS.** 2014-100, 2015-123.08, 2016-143.75, 2017-136.84, 2018-130.25, 2019-137.10]

ILLUSTRATION 2 **[SM]**

The nominal and real GDP respectively of a country in a particular year are Rs. 3000 Crores and Rs. 4700 Crores respectively. Calculate GDP deflator and comment on the level of prices of the year in comparison with the base year.

[**ANS.** 63.83]

ILLUSTRATION 3 **[SM]**

Find nominal GDP if real GDP = 450 and price index = 120

[**ANS.** 540]

ILLUSTRATION 4 **[SM]**

Suppose nominal GNP of a country in 2010 is given at Rs. 600 Crores and price index is given as base year 2010 is 100. Now let the nominal GDP increases to Rs. 1200 Crores in 2018 and the price index rises to 110, find out real GDP?

[**ANS.** 1090.9 Crores]

1.3.2 Net Domestic Product

NDP thus comes closer to measuring the net amount of goods and services produced in the country in a given period of time.

It is the total value of production minus the value of capital used up in producing that output.

Other considerations such as asset obsolescence and complete destruction are also taken into account by the NDP.

$NDP_{MP} = GDP_{MP} - Depreciation$

The basis of distinction between 'gross' and 'net' is depreciation or consumption of fixed capital.

Gross = Net + Depreciation
or
Net = Gross – Depreciation

1.3.3 Gross National Product (GNP)

Gross National Product at Market Price $(GNP)_{MP}$ is a measure of

- the market value,
- of all final economic goods and services,
- gross of depreciation,
- produced within the domestic territory of a country,
- by normal residents,
- during an accounting year,
- including net factor incomes from abroad.

It is the total income earned by a nation's permanent residents (called nationals).

It differs from GDP by including income that our citizens earn abroad and excluding income that foreigners earn here.

Profits earned by Apple from its Indian manufacturing operations is part of India's GDP but these profits are part of USA's GNP.

Gross National Product (GNP) is evaluated at market prices and therefore it is in fact Gross National Product at market prices (GNP_{MP}).

GNP_{MP} = GDP_{MP} +Factor income earned by the domestic factors of production employed in the rest of the world - Factor income earned by the factors of production of the rest of the world employed in the domestic territory.

$GNP_{MP} = GDP_{MP}$ + Net Factor Income from Abroad (NFIA)
$GDP_{MP} = GNP_{MP}$ – Net Factor Income from Abroad (NFIA)

NFIA is the difference between the aggregate amount that a country's citizens and companies earn abroad, and the aggregate amount that foreign citizens and overseas companies earn in that country.

NFIA = Net compensation of employees + Net income from property and entrepreneurship + Net retained earnings

If Net Factor Income from Abroad is positive, then GNP_{MP} would be greater than GDP_{MP}.

The distinction between 'national' and 'domestic' is net factor income from abroad.

National = Domestic + Net Factor Income from Abroad

1.3.4 Net National Product at Market Prices (NNP_{MP})

Net National Product at Market Prices (NNP_{MP}) is a measure of

- the market value
- of all final economic goods and services,
- produced by normal residents
- within the domestic territory of a country
- including Net Factor Income from Abroad
- during an accounting year
- excluding depreciation.

NNP_{MP} = GNP_{MP} – Depreciation
NNP_{MP} = NDP_{MP} + Net Factor Income from Abroad
NNP_{MP} = GDP_{MP} + Net Factor Income from Abroad – Depreciation

1.3.5 Gross Domestic Product at Factor Cost (GDP_{FC})

Gross domestic product (GDP) at factor cost is GDP at market prices minus net indirect taxes. The money value of output produced within a country's domestic limits in a year, as received by the factors of production, is measured by GDP at factor cost.

GDP_{FC} = GDP_{MP} – Indirect Taxes + Subsidies
GDP_{FC} = Compensation of employees
+ Operating Surplus (rent + interest+ profit)
+ Mixed Income of Self- employed
+ Depreciation

Thus, we find that the basis of distinction between market price and factor cost is net indirect taxes (i.e., Indirect taxes - Subsidies).

Market Price = Factor Cost + Net Indirect Taxes
= Factor Cost + Indirect Taxes – Subsidies

Factor Cost = Market Price - Net Indirect Taxes
= Market Price - Indirect Taxes + Subsidies

Factor Cost vs Basic Price vs Market Price

GDP at Basic Price excludes any taxes on products the producer receives from the purchaser and passes on to the government (Eg: GST or Sales Tax or Services Tax) but includes any subsidies the producer receives from the government and uses to lower the prices charged to purchasers. In simple terms, the basic price is the subsidised price without tax.

Relationship between Factor Cost and Basic Price:

Factor cost + production tax – production subsidies = Basic prices.

Relationship between Basic Price and Market Price:

Basic Price + Product tax – Product Subsidy = Market Price.

Note: Thus, market price includes both product tax as well as production tax while excluding both product and production subsidies.

ILLUSTRATION 5

Compute the amount of subsidies from the following data : *[PEQ-Nov-19- 3 Marks]*

	(₹ In crores)
GDP at Market Price	*7,79,567*
Indirect Taxes	*4,54,367*
GDP at Factor Cost	*3,60,815*

[**ANS.** ₹ 35,615 Crore]

1.3.6 Net Domestic Product at Factor Cost (NDP $_{FC}$)

Net Domestic Product at Factor Cost (NDP $_{FC}$) is defined as the total factor incomes earned by the factors of production.

In other words, it is sum of domestic factor incomes or domestic income net of depreciation.

As mentioned above, market price includes indirect taxes imposed by government. We have to deduct indirect taxes and add the subsidies in order to calculate that part of domestic product which actually accrues to the factors of production.

The measure that we obtain so is called Net Domestic Product at factor cost.

NDP $_{FC}$ = NDP $_{MP}$ – Net Indirect Taxes
NDP $_{FC}$ = Compensation of employees
+ Operating Surplus (rent + interest+ profit)
+ Mixed Income of Self- employed

1.3.7 Net National Product at Factor Cost (NNP $_{FC}$) or National Income

National Income is defined as the factor income accruing to the normal residents of the country during a year.

It is the sum of domestic factor income and net factor income from abroad.

In other words, national income is the value of factor income generated within the country plus factor income from abroad in an accounting year.

If NFIA is positive, then national income will be greater than domestic factor incomes.

NNP $_{FC}$ = National Income = FID (factor income earned in domestic territory) + NFIA.

1.3.8 Per Capita Income

The GDP per capita is a measure of a country's economic output per person.

It is obtained by dividing the country's gross domestic product, adjusted by inflation, by the total population.

It serves as an indicator of the standard of living of a country.

1.3.9 Personal Income (PI)

While national income is income earned by factors of production, Personal Income is the income received by the household sector including Non-Profit Institutions Serving Households.

Thus, national income is a measure of income earned and personal income is a measure of actual current income receipts of persons from all sources which may or may not be earned from productive activities during a given period of time.

Examples of this include transfer payments such as social security benefits, unemployment compensation, welfare payments etc.

Individuals also contribute income which they do not actually receive; for example, undistributed corporate profits and the contribution of employers to social security.

Personal income excludes retained earnings, indirect business taxes, corporate income taxes and contributions towards social security.

Households receive interest payments from the firms and governments; they also make interest payments to firms and governments. As such, the net interest paid by households to firms and government is also deducted from national income.

Personal income forms the basis for consumption expenditures and is derived from national income as follows:

PI = NI + income received but not earned – income earned but not received.

PI = NI - Undistributed profits – Net interest payments made by households – Corporate Tax +Transfer Payments to the households from firms and government.

PI= National income – Income from property and entrepreneurship accruing to government administrative departments – Saving of non-departmental enterprises + National debt interest + Current transfers from government + Current transfers from rest of the world – Saving of private corporate sector – Corporate profit tax

An important point to remember is that national income is not the sum of personal incomes because personal income includes transfer payments (eg. pension) which are excluded from national income.

Further, not all national income accrues to individuals as their personal income.

1.3.10 Disposable Personal Income (DI)

Disposable personal income is a measure of the amount of the money in the hands of the individuals that is available for their consumption or savings.

Disposable personal income is derived from personal income by subtracting the direct taxes paid by individuals and other compulsory payments made to the government.

DI = PI - Personal Income Taxes – Non tax payments

Two more concepts need to be understood, namely:

1. **Net National Disposable Income (NNDI)**

Net National Disposable Income (NNDI) = Net National Income + other net current transfers from the rest of the world (Receipts less payments)

Net National Disposable Income (NNDI) = NNI + net taxes on income and wealth receivable from abroad + net social contributions and benefits receivable from abroad.

2. **Gross National Disposable Income (GNDI)**

NNDI + CFC = GNI + other net current transfers from the rest of the world (Receipts less payments)

Domestic Income may be categorized into:

1. Income from domestic product accruing to the public sector which includes income from property and entrepreneurship accruing to government administrative departments and savings of non-departmental enterprises.
2. Income from domestic product accruing to private sector = NDP_{FC} - Income from property and entrepreneurship accruing to government administrative departments - Savings of non-departmental enterprises.

1.3.11 Private Income

Private income is a measure of the income (both factor income and transfer income) which accrues to private sector from all sources within and outside the country.

Private Income = Factor income from net domestic product accruing to the private sector + Net factor income from abroad + National debt interest + Current transfers from government + Other net transfers from the rest of the world.

ILLUSTRATION 6 **[SM]**

From the following data, estimate National Income and Personal Income.

Items	*₹ in Crores*
Net national product at market price	*1,891*
Income from property and entrepreneurship accruing to government administrative departments	*45*
Indirect taxes	*175*
Subsidies	*30*
Saving of non-departmental enterprises	*10*
Interest on National debt	*15*
Current transfers from government	*35*
Current transfers from rest of the world	*20*
Saving of private corporate sector	*25*
Corporate profit tax	*25*

[**ANS.** Personal Income = 1711 Crores, National Income = 1746 Crores]

ILLUSTRATION 7 **[SM]**

Calculate the aggregate value of depreciation when the GDP at market price of a country in a particular year was ₹ 1,100 Crores. Net Factor Income from Abroad was ₹ 100 Crores. The value of Indirect taxes – Subsidies was ₹ 150 Crores and National Income was ₹ 850 Crores.

[ANS. Depreciation = ₹ 200 Crores]

ILLUSTRATION 8 **[SM]**

On basis of following information, calculate NNP at market price and Disposable personal income

Items	*₹ in Crores*
NDP at factor cost	*14900*
Income from domestic product accruing to government	*150*
Interest on National debt	*170*
Transfer payment by government	*60*
Net private donation from abroad	*30*
Net factor income from abroad	*80*
Indirect taxes	*335*
Direct taxes	*100*
Subsidies	*262*
Taxes on corporate profits	*222*
Undistributed profits of corporations	*105*

[ANS. DI = 14663 Crores, NNP_{MP} = 15053 Crores]

1.4 Measurement of National Income in India

- National Accounts Statistics (NAS) in India are compiled by National Accounts Division in the Central Statistics Office, Ministry of Statistics and Programme Implementation.
- Annual as well as quarterly estimates are published.
- This publication is the key source-material for all macroeconomic data of the country.
- As per the mandate of the Fiscal Responsibility and Budget Management Act 2003, the Ministry of Finance uses the GDP (at current prices) to determine the fiscal targets.
- The Ministry of Statistics and Programme Implementation has released the new series of national accounts, revising the base year from 2004-05 to 2011-12.
- In the revision of National Accounts statistics done by Central Statistical Organization (CSO) in January 2015, it was decided that sector-wise estimates of Gross Value Added (GVA) will now be given at basic prices instead of at factor cost. In simple terms, for any commodity the 'basic price' is the amount receivable by the producer from the purchaser for a unit of a product minus any tax on the product plus any subsidy on the product.

1.4.1 The Circular Flow of Income

Q. Explain the circular flow of Income in an Economy. [PEQ-NOV 19- 3 Marks]

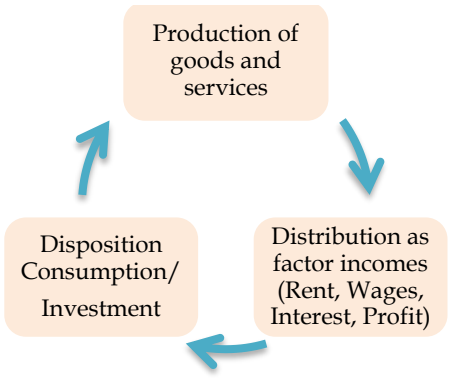
There are three different interlinked phases in a circular flow of income, namely: production, distribution and disposition as can be seen from the figure

(i) In the production phase, firms produce goods and services with the help of factor services.

(ii) In the income or distribution phase, the flow of factor incomes in the form of rent, wages, interest and profits from firms to the households occurs

(iii) In the expenditure or disposition phase, the income received by different factors of production is spent on consumption goods and services and investment goods. This expenditure leads to further production of goods and services and sustains the circular flow.

Figure - Circular Flow of Income



These processes of production, distribution and disposition keep going on simultaneously and enable us to look at national income from three different angles namely: as a flow of production or value added, as a flow of income and as a flow of expenditure.

Corresponding to the 3 phases, there are 3 methods of measuring national income:

- Value Added Method
- Income Method, and
- Expenditure Method

1.4.2 Value Added Method / Product Method / Industrial Origin Method / Net Output Method

- National income by value added method is the sum total of net value added at factor cost across all producing units of the economy.
- The value added method measures the contribution of each producing enterprise in the domestic territory of the country in an accounting year and entails consolidation of production of each industry less intermediate purchases from all other industries.
- This method of measurement shows the unduplicated contribution by each industry to the total output.
- This method involves the following steps:

Step 1. Identifying the producing enterprises and classifying them into different sectors according to the nature of their activities

- Primary sector,
- Secondary sector, and
- Tertiary sector or service sector

These sectors are further divided into sub-sectors and each sub-sector is further divided into commodity group or service-group.

Step 2. Estimating the gross value added (GVA $_{MP}$) by each producing enterprise

Gross value added	**= Value of output – Intermediate consumption**
(GVA $_{MP}$)	**= (Sales + change in stock) – Intermediate consumption**

Step 3. Estimation of National income

- For each individual unit, Net value added is found out.
- Adding the net value-added by all the units in one sub-sector, we get the net value- added by the sub-sector.
- By adding net value-added or net products of all the sub-sectors of a sector, we get the value-added or net product of that sector.
- For the economy as a whole, we add the net products contributed by each sector to get Net Domestic Product.
- We subtract net indirect taxes and add net factor income from abroad to get national income.

The values of the following items are also included:

(i) Own account production of fixed assets by government, enterprises and households.
(ii) Production for self- consumption, and
(iii) Imputed rent of owner occupied houses.

ILLUSTRATION 9 **[SM]**

Calculate National Income by Value Added Method with the help of following data-

Particulars	*₹ (in Crores)*
Sales	*700*
Opening stock	*500*
Intermediate Consumption	*350*
Closing Stock	*400*
Net Factor Income from Abroad	*30*
Depreciation	*150*
Excise Tax	*110*
Subsidies	*50*

[**ANS.** 70 Crores]

ILLUSTRATION 10 **[SM]**

Calculate national income by value added method.

Particulars	*(₹ in crores)*
Value of output in primary sector	*2000*
Intermediate consumption of primary sector	*200*
Value of output of secondary sector	*2800*
Intermediate consumption of secondary sector	*800*
Value of output of tertiary sector	*1600*
Intermediate consumption of tertiary sector	*600*
Net factor income from abroad	*(30)*
Net indirect taxes	*300*
Depreciation	*470*

[**ANS.** 4000 Crores]

ILLUSTRATION 11 **[SM]**

Calculate Net Value Added by Factor Cost from the following data

Items	*₹ in Crores*
Purchase of materials	*85*
Sales	*450*
Depreciation	*30*
Opening stock	*40*
Closing stock	*30*
Excise tax	*45*
Intermediate consumption	*200*
Subsidies	*15*

[**ANS.** 180 Crores]

ILLUSTRATION 12 **[SM]**

From, the following data calculate the Gross National Product at Market Price using Value Added method

Particulars	*₹ in Crores*
Value of output in primary sector	*500*
Net factor income from abroad	*(20)*
Value of output in tertiary sector	*700*
Intermediate consumption in secondary sector	*400*
Value of output in secondary sector	*900*
Government Transfer Payments	*600*
Intermediate consumption in tertiary sector	*300*
Intermediate consumption in primary sector	*250*

[**ANS.** 1,130 Crores]

ILLUSTRATION 13 ***[PEQ-May-18 -3 Marks]***

From the following data, compute the Gross National Product at Market Price (GNP_{MP}) using value added method:

	(₹ In crores)
Value of output in Secondary Sector	*1,000*
Intermediate consumption in Primary Sector	*300*
Value of output in Tertiary Sector	*3,000*
Intermediate consumption in Secondary Sector	*400*
Net factor income from abroad	*(100)*
Value of output in Primary Sector	*800*
Intermediate consumption in Tertiary Sector	*900*

[**ANS.** 3,100 Crores]

1.4.3 Income Method / Factor Income Method / Factor Payment Method / Distributed Share Method

- Whatever is produced by a producing unit is distributed among the factors of production for their services.
- National income is calculated by summation of factor incomes paid out by all production units within the domestic territory of a country as wages and salaries, rent, interest, and profit.
- By definition, it includes factor payments to both residents and non-residents.
- Only incomes earned by owners of primary factors of production are included in national income. Transfer incomes are excluded from national income. Thus, while wages of labourers will be included, pensions of retired workers will be excluded from national income.
- Labour income includes, apart from wages and salaries, bonus, commission, employers' contribution to provident fund and compensations in kind.
- Non-labour income includes dividends, undistributed profits of corporations before taxes, interest, rent, royalties and profits of unincorporated enterprises and of government enterprises.
- However, normally, it is difficult to separate labour income from capital income because in many instances people provide both labour and capital services. Such is the case with self-employed people like lawyers, engineers, traders, proprietors etc.
- In economies where subsistence production and small commodity production is dominant, most of the incomes of people would be of mixed type.
- In sectors such as agriculture, trade, transport etc. in underdeveloped countries (including India), it is difficult to differentiate between the labour element and the capital element of incomes of the people. In order to overcome this difficulty a new category of incomes, called 'mixed income' is introduced which includes all those incomes which are difficult to separate..

NDP_{FC} = Sum of factor income paid out by all production units within the domestic territory of a county

NNP_{FC} = Compensation of employees + Operating Surplus (rent + interest+ profit) + Mixed Income of Self- employed + Net Factor Income from Abroad

ILLUSTRATION 14 **[SM]**

From the following data, calculate NNP_{FC}, NNP_{MP}, GNP_{MP} and GDP_{MP}.

Items	*(₹ in Crores)*
Operating surplus	*2000*
Mixed income of self-employed	*1100*
Rent	*550*
Profit	*800*
Net indirect tax	*450*
Consumption of fixed capital	*400*
Net factor income from abroad	*-50*
Compensation of employees	*1000*

[**ANS.** GDP_{MP} = 4950 Crore, GNP_{MP} = 4900 Crore, NNP_{MP} = 4500 Crore, NNP_{FC} = 4050 Crore]

ILLUSTRATION 15 **[SM]**

Calculate the Operating Surplus with the help of following data-

Particulars	*₹ in Crores*
Sales	*4000*
Compensation of employees	*800*
Intermediate consumption	*600*
Rent	*400*
Interest	*300*
Net indirect tax	*500*
Consumption of Fixed Capital	*200*
Mixed Income	*400*

[**ANS.** 1500 Crores]

ILLUSTRATION 16 **[SM]**

From the following data calculate (a) Gross Domestic Product at Factor Cost, and (b) Gross Domestic Product at Market price

Items	*₹ in Crores*
Net exports	*(-) 50*
Compensation of employees	*3000*
Rent	*800*
Interest	*900*
Profit	*1,300*
Net indirect taxes	*300*
Net domestic capital formation	*800*
Gross domestic capital formation	*900*
Factor income to abroad	*80*

[**ANS.** GDP $_{FC}$ = 6100 Crores, GDP $_{MP}$ = 6400 Crores]

ILLUSTRATION 17 ***[PEQ-Nov 19 - 3 Marks]***

Compute NNP at Factor Cost or National Income from the following data using income method

	(₹ In crores)
Compensation of Employees	*3,000*
Mixed Income of Self Employed	*1,050*
Indirect Tax	*480*
Subsidies	*630*
Depreciation	*428*
Rent	*1,020*
Interest	*2,010*
Profit	*980*
Net Factor Income from Abroad	*370*

[**ANS.** 8,430 Crores]

1.4.4 Expenditure Method / Income Disposal Approach

In the expenditure approach to measuring GDP, we add up the value of the goods and services purchased by each type of final user mentioned below.

1. *Final Consumption Expenditure*

(a) Private Final Consumption Expenditure (PFCE)

To measure this, the volume of final sales of goods and services to consumer households and non-profit institutions serving households acquired for consumption (not for use in production) are multiplied by market prices and then summation is done.

It also includes the value of primary products which are produced for own consumption by the households, payments for domestic services which one household renders to another, the net expenditure on foreign financial assets or net foreign investment.

Land and residential buildings purchased or constructed by households are not part of PFCE. They are included in gross capital formation.

Thus, only expenditure on final goods and services produced in the period for which national income is to be measured and net foreign investment are included in the expenditure method of calculating national income.

(b) Government Final Consumption Expenditure (GFCE)

Since the collective services provided by the governments such as defence, education, healthcare etc. are not sold in the market, the only way they can be valued in money terms is by adding up the money spent by the government in the production of these services.
This total expenditure is treated as consumption expenditure of the government. Government expenditure on pensions, scholarships, unemployment allowance etc. should be excluded because these are transfer payments.

2. *Gross Domestic Capital formation*

Gross domestic fixed capital formation includes final expenditure on machinery and equipment and own account production of machinery and equipments, expenditure on construction, expenditure on changes in inventories, and expenditure on the acquisition of valuables such as, jewelry and works of art.

3. *Net Exports*

Net exports are the difference between exports and imports of a country during the accounting year. It can be positive or negative.

How to arrive at national income or NNP $_{FC}$ using expenditure method?

- First find the sum of final consumption expenditure, gross domestic capital formation and net exports. The resulting figure is GDP $_{MP}$
- To this, we add the NFIA and obtain GNP $_{MP}$
- Subtracting indirect taxes from GNP $_{MP}$, we get GNP $_{FC}$
- National income or NNP $_{FC}$ is obtained by subtracting depreciation from GNP $_{FC}$

ILLUSTRATION 18 **[SM]**

Compute National income

Items	*(in Crores of ₹)*
Consumption	*750*
Investment	*250*
Government Purchases	*100*
Exports	*100*
Imports	*200*

[**ANS.** 1,000 Crores]

ILLUSTRATION 19 **[SM]**

Calculate NNP_{FC}. By expenditure method with the help of following information-

Items	*₹ in Crores*
Private final consumption expenditure	*10*
Net Import	*20*
Public final consumption expenditure	*05*
Gross domestic fixed capital formation	*350*
Depreciation	*30*
Subsidy	*100*
Income paid to abroad	*20*
Change in stock	*30*
Net acquisition of valuables	*10*

[**ANS.** NNP_{FC} = 435 Crores]

ILLUSTRATION 20 **[SM]**

Calculate GDP_{MP} and derive national income from the following data

Items	*₹ in Crores*
Inventory Investment	*100*
Exports	*200*
Indirect taxes	*100*
Net factor income from abroad	*(50)*
Personal consumption expenditure	*3,500*
Gross residential construction investment	*300*
Depreciation	*50*
Imports	*100*
Government purchases of goods and services	*1,000*
Gross public investment	*200*
Gross business fixed investment	*300*

[**ANS.** GDP_{MP} = 5,500 Crores, NNP_{FC} = 5,300 Crores]

ILLUSTRATION 21 ***[PEQ-MAY 19-5 Marks]***

Compute GNP at Factor Cost and NDP at market price using expenditure method from the following data :

	(₹ In crores)
Personal Consumption Expenditure	*2,900*
Imports	*300*
Gross Public Investment	*500*
Consumption of Fixed Capital	*60*
Exports	*200*
Inventory Investment	*170*
Government purchases of Goods and Services	*1,100*
Gross Residential Construction Investment	*450*

Net Factor Income from Abroad	*(30)*
Gross Business Fixed Investment	*410*
Subsidies	*80*

[**ANS.** GNP $_{FC}$ = 5,480 Crores, NDP $_{MP}$ = 5370 Crores]

ILLUSTRATION 22 [SM]

Calculate NI with the help of Expenditure method and income method with the help of following data:

Items	*₹ in Crores*
Compensation of employees	*1,200*
Net factor income from Abroad	*20*
Net indirect taxes	*120*
Profit	*800*
Private final consumption expenditure	*2,000*
Net domestic capital formation	*770*
Consumption of fixed capital	*130*
Rent	*400*
Interest	*620*
Mixed income of self-employed	*700*
Net export	*30*
Govt. final consumption expenditure	*1100*
Operating surplus	*1820*
Employer's contribution to social security scheme	*300*

[**ANS.** Expenditure Method = ₹ 3,800 Crores, Income Method = ₹ 3,740 Crores]

ILLUSTRATION 23 [SM]

Find GDP_{MP} and GNP_{MP} from the following data using income method. Show that it is the same as that obtained by expenditure method.

Items	*₹ in Crores*
Personal Consumption	*7,314*
Depreciation	*800*
Wages	*6,508*
Indirect Business Taxes	*1,000*
Interest	*1,060*
Domestic Investment	*1,442*
Government Expenditures	*2,196*
Rental Income	*34*
Corporate Profits	*682*
Exports	*1,346*
Net Factor Income from Abroad	*40*
Mixed Income	*806*
Imports	*1,408*

[**ANS.** GDP $_{MP}$ = 10,890 Crores, GNP $_{MP}$ = 10,930 Crores]

- Ideally, all the three methods of national income computation should arrive at the same figure. When national income of a country is measured separately using these methods, we get a three dimensional view of the economy.
- Each method of measuring GDP is subject to measurement errors and each method provides a check on the accuracy of the other methods.
- By calculating total output in several different ways and then trying to resolve the differences, we will be able to arrive at a more accurate measure than would be possible with one method alone.
- Income method may be most suitable for developed economies where people properly file their income tax returns.
- With the growing facility in the use of the commodity flow method of estimating expenditures, an increasing proportion of the national income is being estimated by expenditure method.
- Countries like India are unable to estimate their national income wholly by one method. Thus, in agricultural sector, net value added is estimated by the production method, in small scale sector net value added is estimated by the income method and in the construction sector net value added is estimated by the expenditure method.

1.5 System of Regional Accounts in India

- Regional accounts provide an integrated database and help decision making at the regional level.
- At present, practically all the states and union territories of India compute state income estimates and district level estimates.
- State Income or Net State Domestic Product (NSDP) is a measure in monetary terms of the volume of all goods and services produced in the state within a given period of time (generally a year) accounted without duplication.
- Per Capita State Income is obtained by dividing the NSDP (State Income) by the midyear projected population of the state.
- The state level estimates are prepared by the State Income Units of the respective State Directorates of Economics and Statistics (DESs). The Central Statistical Organisation assists the States in the preparation of these estimates by rendering advice on conceptual and methodological problems.
- In the preparation of state income estimates, certain activities such as railways, communications, banking and insurance and central government administration, that cut across state boundaries, and thus their economic contribution cannot be assigned to any one state directly are known as the 'Supra-regional sectors' of the economy. The estimates for these supra regional activities are compiled for the economy as a whole and allocated to the states on the basis of relevant indicators.

1.6 Limitation and Challenges in computation of National Income

Can the GDP of a country be taken as an index of the welfare of people in that country?

There are many reasons to dispute the validity of GDP as a perfect measure of well-being. GDP measures exclude the following which are critical for the overall wellbeing of citizens.

(a) Income distributions and, therefore, GDP per capita is a completely inadequate measure of welfare. Countries may have significantly different income distributions and, consequently, different levels of overall well-being for the same level of per capita income.

(b) Quality improvements in systems and processes due to technological as well as managerial innovations which reflect true growth in output from year to year.

(c) Productions hidden from government authorities, either because those engaged in it are evading taxes or because it is illegal (drugs, gambling etc.).

(d) Nonmarket production (with a few exceptions) and Non-economic contributors to well-being for example: health of a country's citizens, education levels, political participation, or other social and political factors that may significantly affect well- being levels.

(e) The disutility of loss of leisure time. We know that, other things remaining the same, a country's GDP rises if the total hours of work increase.

(f) Economic 'bads' for example: crime, pollution, traffic congestion etc which make us worse off.

(g) The volunteer work and services rendered without remuneration undertaken in the economy, even though such work can contribute to social well-being as much as paid work.

(h) Many things that contribute to our economic welfare such as, leisure time, fairness, gender equality, security of community feeling etc,

(i) Both positive and negative externalities which are external effects that do not form part of market transactions

(j) The distinction between production that makes us better off and production that only prevents us from becoming worse off.

For e.g. defense expenditures such as on police protection. Increased expenditure on police due to increase in crimes may increase GDP but these expenses only prevent us from becoming worse off. However, no reflection is made in national income of the negative impacts of higher crime rates.

As another example, automobile accidents result in production of repairs, output of medical services, insurance, and legal services all of which are production included in GDP just as any other production.

Q. What are the conceptual difficulties in measurement of national income. [PEQ -MAY 19 - 2 Marks]

There are many conceptual difficulties related to measurement which are difficult to resolve, such as:

(a) lack of an agreed definition of national income,

(b) accurate distinction between final goods and intermediate goods,

(c) issue of transfer payments,

(d) services of durable goods,

(e) difficulty of incorporating distribution of income,

(f) valuation of a new good at constant prices, and

(g) valuation of government services

Other challenges relate to:

(a) Inadequacy of data and lack of reliability of available data,
(b) presence of non-monetised sector,
(c) production for self-consumption,
(d) absence of recording of incomes due to illiteracy and ignorance,
(e) lack of proper occupational classification, and
(f) Accurate estimation of consumption of fixed capital

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. *The concept of 'resident unit' involved in the definition of GDP denotes*
   (a) *A business enterprise which belongs to a citizen of India with production units solely situated in India*
   (b) *The unit having predominant economic interest in the economic territory of the country for one year or more irrespective of the nationality or legal status*
   (c) *A citizen household which had been living in India during the accounting year and one whose economic interests are solely in India*
   (d) *Households and business enterprises composed of citizens of India alone living in India during the accounting year*

2. *Read the following statements and answer the following question.*
   I. *Intermediate consumption consists of the value of the goods and services consumed as inputs by a process of production,*
   II. *Intermediate consumption excludes fixed assets whose consumption is recorded as consumption of fixed capital.*
   (a) *Only I is true*
   (b) *Both I and II are true*
   (c) *Only II is true*
   (d) *Neither I nor II is true*

3. *Gross Domestic Product (GDP) of any nation*
   (a) *excludes capital consumption and intermediate consumption*
   (b) *is inclusive of capital consumption or depreciation*
   (c) *is inclusive of indirect taxes but excludes subsidies*
   (d) *None of the above*

4. *Read the following statements*
   I. *'Value added' refers to the difference between value of output and purchase of intermediate goods.*
   II. *'Value added' represents the contribution of labour and capital to the production process.*
   (a) *Statements I and II are incorrect*
   (b) *Statements I and II are correct*
   (c) *Statement I is correct and II is incorrect*
   (d) *Statement II is correct and I is incorrect*

5. *Non-economic activities are*
   (a) *those activities whose value is excluded from national income calculation as it will involve double counting*
   (b) *those which produce goods and services, but since these are not exchanged in a market transaction they do not command any market value*
   (c) *those which do not involve production of goods and services as they are meant to provide hobbies and leisure time activities*
   (d) *those which result in production for self consumption and therefore not included in national income calculation*

6. *Which of the following does not enter into the calculation of national income?*
   (a) *Exchange of previously produced goods*
   (b) *Exchange of second hand goods*
   (c) *Exchange of stocks and bonds*
   (d) *All the above*

7. *Which of the following enters into the calculation of national income?*
   - (a) *The value of the services that accompany the sale*
   - (b) *Additions to inventory stocks of final goods and materials*
   - (c) *Stocks and bonds sold during eth current year*
   - (d) *(a) and (b) above*

8. *Gross National Product at market prices GNP_{MP} is*
   - (a) *GDP_{MP} + Net Factor Income from Abroad*
   - (b) *GDP_{MP} - Net Factor Income from Abroad*
   - (c) *GDP_{MP} – Depreciation*
   - (d) *GDP_{MP} + Net Indirect Taxes*

9. *Choose the correct statement*
   - (a) *GNP includes earnings of Indian corporations overseas and Indian residents working overseas; but GDP does not include these*
   - (b) *NNP_{FC} = National Income = FID (factor income earned in domestic territory) – NFIA.*
   - (c) *Capital goods and inventory investment are excluded from computation of GDP*
   - (d) *NDP_{MP} = GDP_{MP} + Depreciation*

10. *The basis of distinction between market price and factor cost is*
    - (a) *net factor income from abroad*
    - (b) *net indirect taxes (i.e., Indirect taxes - Subsidies)*
    - (c) *net indirect taxes (i.e., Indirect taxes + Subsidies)*
    - (d) *depreciation (consumption of fixed capital)*

11. *If net factor income from abroad is positive, then*
    - (a) *national income will be greater than domestic factor incomes.*
    - (b) *national income will be less than domestic factor incomes.*
    - (c) *net exports will be negative*
    - (d) *domestic factor incomes will be greater than national income*

12. *The GDP per capita is*
    - (a) *a measure of a country's economic output per person*
    - (b) *actual current income receipts of persons*
    - (c) *national income divided by population*
    - (d) *(a)and (c) above*

13. *Which of the following is an example of transfer payment?*
    - (a) *Old age pensions and family pensions*
    - (b) *Scholarships given to deserving diligent students.*
    - (c) *Compensation given for loss of property due to floods*
    - (d) *All the above*

14. *Mixed income of the self -employed means*
    - (a) *net profits received by self -employed people*
    - (b) *outside wages received by self- employed people*
    - (c) *combined factor payments which are not distinguishable,*
    - (d) *wages due to non- economic activities*

15. *Which of the following is added to national income while calculating personal income?*
    - (a) *Transfer payments to individuals*
    - (b) *Undistributed profits of corporate*
    - (c) *Transfer payments made to foreigners*
    - (d) *Mixed income of self employed*

PEQ–JUNE-2024

16. *When NFIA is positive, then:*
    (a) GNPmp < GDPmp (b) GNPmp < GDPmp
    (c) GDPmp < GNPmp (d) GDPmp > GNPmp (1 mark)

17. *GDP at Market price:*
    (a) GDP at factor cost plus indirect taxes less subsidies
    (b) NNP at market prices minus indirect taxes
    (c) GNP at market price plus net factor income from abroad.
    (d) None (1 mark)

18. *Gross value added at market price $GVA_{(MP)}$ =*
    (a) Value of output + Change in stock
    (b) Value of output + Intermediate consumption
    (c) Value of output + Change in stock - Intermediate consumption
    (d) Value of output - Intermediate consumption (1 mark)

19. *Income received by household sector including non-profit institutions serving households is called:*
    (a) Per capita income (b) Personal income
    (c) National income (d) Disposal income (1 mark)

20. *Joseph had a shoe company in US. The profits earned from its manufacturing unit operating in India will come under:*
    (a) GDP of India, GNP of US (b) GDP of US and GNP of India
    (c) GNP of US and GNP of India (d) GDP of US and GDP of India (1 mark)

21. *If Nominal GDP and real GDP for year 2015 are ₹ 5,000 crores and ₹ 6,700 crores respectively, then what will be the impact on change in prices with respect to base year?*
    (a) Prices will fall (b) Prices will rise
    (c) Not defined as base year is not given (d) No effect on prices (1 mark)

PEQ-SEPTEMBER-2024

22. *Which of the following refers to the mixed income of the self-employed?*
    (a) Wages due to non-economic activities
    (b) Combined factor payments which are not distinguishable
    (c) Net profits received by self-employed
    (d) Outside wages received by self-employed (1 mark)

23. *What does Disposable Personal Income (DPI) represent?*
    (a) The income individuals have for consumption or saving
    (b) The total income earned by individuals before taxes
    (c) The total savings accumulated by individuals
    (d) The total income of a country's government (1 mark)

24. *The ratio between nominal GDP and real GDP is known as:*
    (a) Gross Domestic Product deflator (b) Inflation Rate
    (c) Gross National Product (d) Net National Product (1 mark)

25. *If the GDP Deflator of a country in the year 2020 is 140.75 and in the year 2021 is 157.33, then what is the inflation rate in the year 2021?*
(a) 10.97% (b) 9.09% (c) 11.78% (d) 12.34% (1 mark)

26. *If GDP at market price is ₹ 6,700 crore and depreciation is ₹ 530 crore, then what will be the NDP at market price?*
(a) ₹6,170 crore (b) ₹ 6,200 crore (c) ₹ 7,230 crore (d) ₹ 6,690 crore

27. *What are the phases included in the circular flow of income₹*
(a) Production, Income and Taxation (b) Production, Saving and Expenditure
(c) Production, Distribution and Disposition (d) Income, Saving and Investment
(1 mark)

28. *Which of the following is not the limitation in the computation of the national income?*
(a) Services of durable goods
(b) Issue of transfer payments
(c) Difficulty of incorporating the distribution of income
(d) Quality improvements due to technological innovations (1 mark)

29. *Which method measures the contribution of each producing enterprise in the domestic territory of a country over an accounting year by consolidating production of each industry less intermediate purchases from all other industries?*
(a) Value Added Method (b) Income Disposal Method
(c) Income Method (d) Expenditure Method (1 mark)

30. *What will be the GDP at market price (GDP_{MP}) on the basis of following information?*

Particulars	*Amount (₹) crores*
GNP_{MP}	*1,100*
Net Factor Income from Abroad	*200*
Depreciation	*50*
Net Indirect Tax	*100*

(a) ₹ 1,050 crores (b) ₹ 1,000 crores (c) ₹ 900 crores (d) ₹ 1,300 crores
(1 mark)

31. *Which of the following is not a correct formula to calculate NNPMP?*
(a) $NNP_{MP} = NDP_{MP}$ + Net Factor Income from Abroad
(b) $NNP_{MP} = GNP_{MP}$ - Depreciation
(c) $NNP_{MP} = GDP_{MP}$ + Net Factor Income from Abroad-Depreciation
(d) $NNP_{MP} = NDP_{MP}$ – Depreciation (1 mark)

32. *The state-level estimates are typically prepared by the State Income Units of the respective states. Which organization is primarily responsible for this task?*
(a) Central Statistical Organization (CSO) (b) Directorates of Economics and Statistics (DESs)
(c) Ministry of Finance (d) Reserve Bank of India (1 mark)

PEQ-JANUARY-2025

33. *The value of all final goods and services produced in the country within a given period is called:*
(a) National Income (b) Gross Domestic Product (GDP)
(c) Net National Product (NNP) (d) Gross National Product (GNP) (1 mark)

34. *If GDP_{MP} is more than GNP_{MP} it means:*
(a) The aggregate amount that a country's citizens and companies earn abroad is greater than the aggregate amount that foreign citizens and overseas companies earn in that country.
(b) NFIA is negative.
(c) The aggregate amount that a country's citizens and companies earn abroad is equal to the aggregate amount that foreign citizens and overseas companies earn in that country.
(d) NFIA is positive. (1 mark)

35. *Find the real GDP if nominal GDP = 720 and price index = 120.*
(a) 864 (b) 500 (c) 600 (d) 700 (1 mark)

36. *Which of the following is not true for personal income?*
(a) It is income received by household sector.
(b) It includes Non-profit Institutions serving households.
(c) It is a measure of actual current income receipts of persons only from productive activities.
(d) It excludes retained earnings. (1 mark)

37. *Product Method or Value-added method for calculation of National Income is also called:*
(a) Industrial Origin Method (b) Income Disposal Method
(c) Factor Payment Method (d) Distributed Share Method (1 mark)

38. *While using the income method, which of the following income is included while calculating national income?*
(a) Capital gains (b) Windfall profits
(c) Income from sale of second-hand goods (d) Commissions and brokerages (1 mark)

39. *Calculate Gross value added at market price if sales = 750, opening stock = 300, closing stock = 200 and intermediate consumption is 250.*
(a) 400 (b) 450 (c) 600 (d) 650 (1 mark)

40. *Nominal GDP is:*
(a) Same as real GDP (b) Real GDP less depreciation
(c) GDP at current prices (d) GDP at constant prices (1 mark)

PEQ-MAY-2025

41. *Which of the following organisation is responsible for the compilation of National accounts statistics?*
(a) Central Statistical Organisation (CSO)
(b) Directorate of Economics and Statistics (DES)
(c) Ministry of Finance of Central Government
(d) Reserve Bank of India (RBI)

42. *Which of the following is not a significance and usefulness of the National Income estimates?*
   (a) *Businesses use it to forecast the future demand of their products*
   (b) *To identify the sector wise composition of national income*
   (c) *To assist in making policies related to inflation*
   (d) *To assist in making policies related to population growth*

43. *Considering the data given in the table below calculate the Inflation rate of year 3:*

Year	Nominal GDP	Real GDP	GDP Deflator
0	900	900	100
1	1200	1000	120
2	1500	1200	125
3	1800	1250	144
4	2000	1600	125

   (a) *13.19%*
   (b) *15.20%*
   (c) *19%*
   (d) *-19%*

44. *Which of the following defines the Gross National Product (GNP) of a country under the concept of National Income?*
   (a) *It is market value of all final economic goods & services produced within the domestic territory of a country including Net Factor Income from abroad.*
   (b) *It is market value of all final economic goods & services produced within the domestic territory of a country excluding Net Factor Income from abroad.*
   (c) *It is production cost of all final economic goods & services produced within the domestic territory including depreciation.*
   (d) *It is production cost of all final economic goods & services produced within the domestic territory excluding depreciation.*

45. *The Gross Domestic Product at Factor Cost (GDPFC) is calculated by which of the following formula (Here GDPMP is Gross Domestic Product at market prices):*
   (a) $GDP_{FC} = GDP_{MP}$ + Net Factor Income from Abroad
   (b) $GDP_{FC} = GDP_{MP}$ + Net Factor Income from Abroad - Depreciation
   (c) $GDP_{FC} = GDP_{MP}$ - Indirect Taxes + Subsidies
   (d) $GDP_{FC} = GDP_{MP}$ + Indirect Taxes - Subsidies

46. *Which of the following is true about the basic price in the determination of National Income?*
   (a) *Basic Price = Factor Cost Subsidy Manufacturing Taxes + Manufacturing*
   (b) *Basic Price = Factor Cost - Depreciation*
   (c) *Basic Price = Factor Cost + Net Factor Income from Abroad*
   (d) *Basic Price = Factor Cost + Production Tax + Production Subsidy*

47. *Which of the following is an indicator of the standard of living of a country under the concept of National Income?*
   (a) *Personal Income*
   (b) *Disposable Personal Income*
   (c) *Per Capita Income*
   (d) *GDP at factor cost*

48. *The amount of money in the hands of the individual that is available for their consumption or savings is known as*
   (a) *Private Income*
   (b) *Per Capita Income*
   (c) *Disposable Personal Income*
   (d) *Personal Income*

49. *Which of the following is included in the calculation of Personal Income under the concept of National Income?*
   (a) *Unemployment Compensation*
   (b) *Retained Earnings*
   (c) *Indirect Business Taxes*
   (d) *Contribution towards Social Security*

50. *Which of the following is true in relation with Private Income under the concept of National Income?*
   (a) *It is personal income adjusted by inflation rate.*
   (b) *It is personal income less personal income tax.*
   (c) *It is the income (both factor & transfer income) accrued to private sector from all sources within the country only.*
   (d) *It is income (both factor and transfer income) accrued to the private sector from all sources within and outside the country.*

51. *In the preparation of state income estimates, certain activities like railways, banking, insurance etc. that cut across state boundaries, and thus their economic contribution assigned to more than one state are known as_____.*
   (a) *Central Sectors of economy*
   (b) *Supra Regional Sectors of economy*
   (c) *Tertiary Sectors of economy*
   (d) *Secondary Sectors of economy*

PEQ-SEPTEMBER-2025

52. *What will be the inflation rate in FY 2022-23 as compared to FY 2021-22?*
   (a) 9.54% *(b) 8.76%* *(c) 9.17%* *(d) 7.44%*

53. *In which FY the nominal GDP and real GDP are the same?*
   (a) FY 2021-22 *(b) FY 2022-23* *(c) FY 2020-21* *(d) FY 2023-24*

54. *In which FY the price level has fallen?*
   (a) FY 2020-21 *(b) FY 2023-24* *(c) FY 2022-23* *(d) FY 2021-22*

55. *Calculate the consumption of fixed capital (depreciation) from the following data:*

Particulars	Amount (₹) in crores
GDP_{MP}	8,000
NNP_{MP}	5,600
Indirect tax	400
Subsidy	300
Net factor income from abroad	200

(a) ₹ 1,700 crores *(b) ₹ 5,200 crores* *(c) ₹ 7,600 crores* *(d) ₹ 2,600 crores*

56. *Calculate the net factor income from abroad from the following data:*

Particulars	Amount (₹) in crores
NDP_{MP}	10,000
NNP_{MP}	12,000
Net indirect tax	400
Depreciation	200

(a) ₹ 2,400 crores *(b) ₹ 2,000 crores* *(c) ₹ 3,600 crores* *(d) ₹ 1,800 crores*

57. *If the GDP deflator is greater than 100, it indicates that:*
   (a) *Real GDP is greater than nominal GDP.*
   (b) *Nominal GDP is greater than real GDP.*
   (c) *Price levels are lower as compared to the base year.*
   (d) *Real GDP is equal to nominal GDP.*

58. *Gamma Ltd., a South Korean company, earns profits from its manufacturing operations in India. How these profits will be reflected in the economic measures of both countries?*
   (a) *They will be included in India's GDP and South Korea's GDP.*
   (b) *They will be included in India's GDP but not in South Korea's GNP.*
   (c) *They will be included in India's GDP and South Korea's GNP.*
   (d) *They will be included only in India's GNP.*

59. *Who are the economists credited with pioneering National Income Accounting?*
   (a) *John Maynard Keynes and Milton Friedman*
   (b) *Simon Kuznets and Richard Stone*
   (c) *Adam Smith and David Ricardo*
   (d) *Paul Samuelson and Joseph Schumpeter*

60. *If a country has the following economic data: GDPmp ₹ 2,000 billion, indirect taxes = ₹ 250 billion and subsidies = ₹ 100 billion, what is its GDP at Factor Cost?*
   (a) *₹ 1,850 billion*
   (b) *₹ 2,100 billion*
   (c) *₹ 2,150 billion*
   (d) *₹ 2,000 billion*

PEQ-JANUARY-2026

61. *Refer to the following data:*
Sales: 1,000 Crores
Depreciation: 70 Crores
Opening stock:100 Crores
Closing stock: 90 Crores
GST: 100 Crores
Intermediate consumption: ₹ 440 Crores
Based on the above data, compute the Gross Value Added at Market Price (GVAmp).
   (a) *380 crores*
   (b) *620 crores*
   (c) *480 crores*
   (d) *550 crores*

62. *Which item is not included when calculating national income using the income method?*
   (a) *Compensation of employees*
   (b) *Mixed income of self-employed*
   (c) *Interest paid on consumption loans*
   (d) *Rent*

63. *Calculate real GDP if nominal GDP is 450 and price index is 120.*
   (a) *540*
   (b) *350*
   (c) *375*
   (d) *475*

Use the following data to answer question 64 and 65:

Particular	₹ Crores
Operating surplus	2,500
Mixed income of Self Employed	1,500
Rent	680
Profit	720
Interest	1,100
Net indirect tax	400
Consumption of fixed capital	300
Compensation of employees	1,000

64. *Calculate GDPMP*
   (a) *₹ 5,700 crores*
   (b) *₹ 8,200 crores*
   (c) *₹ 5,000 crores*
   (d) *₹ 7,500 crores*

65. *Calculate GDPFC*
   (a) *₹ 5,000 crores*
   (b) *₹ 7,100 crores*
   (c) *₹ 7,500 crores*
   (d) *₹ 5,300 crores*

66. *Operating surplus includes:*
   (a) *Rent, Interest and Compensation of employees*
   (b) *Rent, Interest and Profit*
   (c) *Rent, Profit and Compensation of employees*
   (d) *Rent, Interest and Taxes*

67. *Which of the following factor of production is not included in calculating the National Income?*
   (a) *Pensions of retired workers*
   (b) *Wages of labourers*
   (c) *Employers' contribution to provident fund*
   (d) *Interest on loans availed for productive services*

68. *If Net Factor Income from Abroad is negative:*
   (a) *GNPMP Would be smaller than* GDP_{MP}
   (b) *Gross Income earned by country's citizen and companies abroad is negative*
   (c) *NNPMP would be greater than* GNP_{MP}
   (d) *Huge depreciation component in* NNP_{MP}

69. *In National Accounts, which of the following is considered an 'economic bad' excluded from GDP?*
   (a) *Traffic Congestion*
   (b) *Volunteer services*
   (c) *Rent from self-owned houses*
   (d) *Consumption of fixed capital*

ANSWERS / HINTS

I. Multiple Choice Type Questions

1.	(b)	2.	(b)	3.	(b)	4.	(b)	5.	(b)	6.	(d)	7.	(d)
8.	(a)	9.	(a)	10.	(b)	11.	(a)	12.	(d)	13.	(d)	14.	(c)
15.	(a)	16.	(c)	17.	(a)	18.	(d)	19.	(b)	20.	(a)	21	(a)
22.	(b)	23	(a)	24.	(c)	25.	(c)	26.	(a)	27.	(c)	28.	(d)
29.	(a)	30.	(c)	31.	(d)	32.	(b)	33.	(b)	34.	(b)	35.	(c)
36.	(c)	37.	(a)	38.	(d)	39.	(a)	40.	(c)	41.	(a)	42.	(d)
43.	(b)	44.	(a)	45.	(c)	46.	NA	47.	(c)	48.	(c)	49.	(a)
50.	(d)	51.	(b)	52.	(c)	53.	(c)	54.	(b)	55.	(d)	56.	(b)
57.	(b)	58.	(c)	59.	(b)	60.	(a)	61.	(d)	62.	(c)	63.	(c)
64.	(a)	65.	(b)	66.	(a)	67.	(a)	68.	(a)	69.	(a)		

II. Illustration

1. *(In Billion Rs.)*

Years	Nominal GDP	Real GDP	GDP Deflator
2014	500	500	100
2015	800	650	123.08
2016	1150	800	143.75
2017	1300	950	136.84
2018	1550	1190	130.25
2019	1700	1240	137.10

A deflator above 100 is an indication of price levels being higher as compared to the base year. From years 2015 through 2019, we find that price levels are higher than that of the base year, the highest being in the year 2016.If the GDP deflator is greater than 100, then nominal GDP is greater than real GDP. If the GDP deflator next year is less than the GDP deflator this year, then the price level has fallen; if it is greater, price levels have increased.

2. *Nominal GDP* = ₹ *3000 Crores*

Real GDP = ₹ *4700 Crores*

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

$$= \frac{3000}{4700} \times 100 = 63.83$$

The price level has fallen since GDP deflator is less than 100 at 63.83.

3. *Nominal GDP = Real GDP x Price Index / 100*

Nominal GDP = 450 X 120/100

= 540

---

4. *Real GDP = Nominal GDP X 100/Price Index*

= 1200 X 100/110 = 1090.9 Crores

---

5. GDP_{MP} = GDP_{FC} + Indirect Tax – Subsidy

∴ Subsidy = GDP_{FC} + Indirect Tax – GDP_{MP}

= 3,60,815 + 4,54,367 – 7,79,567

---

6. *National Income = Net national product at market price – Indirect taxes + Subsidies*

= 1,891 – 175 + 30 = 1746 crores

Personal Income = National income – Income from property and entrepreneurship accruing to government administrative departments – Saving of non-departmental enterprises + National debt interest + Current transfers from government + Current transfers from rest of the world – Saving of private corporate sector – Corporate profit tax

= 1746 – 45 –10+ 15 + 35 + 20 – 25 – 25

= 1711 Crores

---

7. *Given*

GDP_{MP} *= 1100 Crores, NFIA = 100 Crores, NIT =150 Crores,* NNP_{FC} *= 850 Crores*

∴ GDP_{FC} = GDP_{MP} - NIT

= 1100 – 150 = 950

GNP_{FC} = GDP_{FC} + NFIA

= 950 + 100 = 1050

NNP_{FC} = GNP_{FC} - Depreciation

850 = 1050 - Depreciation

Depreciation = 1050 – 850 = 200 Crores.

---

8. *NNP at Market price = NNP at factor cost +indirect tax-subsidies*

Where NNP at factor cost = NDP_{FC} + NFIA

= 14900 + 80=14980

Therefore, NNP_{MP} = 14980 + 335 – 262

= 15053

Disposable personal income (DI) = PI- Personal income tax

PI = NI + income received but not earned – income earned but not received

= 14980+ 170+60 +30 -150 -222- 105 = 14763

Therefore, DI= 14763- 100 = 14663 Crores

---

9. NVA_{FC} = GDP_{MP} –Depreciation +NFIA- Net Indirect Tax
Where GVA_{MP} = Value of output- intermediate consumption
Value of Output = Sales+ change in stock
= 700+ (400-500) = 600
GVA ($_{MP}$) = 600 – 350= 250
Therefore NI = 250-150 +30-(110-50)
= 70 Crores

---

10. GDP_{MP} =
(Value of output in primary sector - intermediate consumption of primary sector) +
(value of output in secondary sector - intermediate consumption of secondary sector) +
(value of output in tertiary sector - intermediate consumption of tertiary sector)

Value of output in primary sector	=	2000
- Intermediate consumption of primary sector	=	(200)
+ Value of output in secondary sector	=	2800
- Intermediate consumption in secondary sector	=	(800)
+ Value of output in tertiary sector	=	1600
- Intermediate consumption of tertiary sector	=	(600)

GDP $_{MP}$ = ₹ 4800 Crores
NNP_{FC} = GDP_{MP} + NFIA -NIT-Depreciation
NNP_{FC} = National income= 4800 + (-30)-300-470 = 4000 Crores

---

11. GVA $_{MP}$ = Sales+ change in stock - Intermediate consumption
= 450+ (30-40) -200
= 240Crores
NVA_{MP} = GVA_{MP} – Depreciation
NVA_{MP} = 240-30 = 210 Crores
NVA_{FC} = NVA_{MP} – (indirect tax - subsidies)
= 210 – (45 -15) = 180Crores

---

12. Value of Output in Primary Sector = ₹500 Crores
Intermediate Consumption in Primary Sector = ₹250 Crores
Value Added in Primary Sector = 500−250=250 Crores

Value of Output in Secondary Sector = ₹900 Crores
Intermediate Consumption in Secondary Sector = ₹400 Crores
Value Added in Secondary Sector = 900−400=500 Crores

Value of Output in Tertiary Sector = ₹700 Crores
Intermediate Consumption in Tertiary Sector = ₹300 Crores
Value Added in Tertiary Sector = 700−300=400 Crores

GNP $_{MP}$ =
Value Added in Primary Sector+Value Added in Secondary Sector+Value Added in Tertiary Sector+Net Factor Income from Abroad

GNP $_{MP}$ =250+500+400+ (−20)

GNP $_{MP}$ = ₹ 1,130 crore

---

13. Value of Output in Primary Sector = ₹800 Crores

Intermediate Consumption in Primary Sector = ₹300 Crores

Value Added in Primary Sector = 800−300=500 Crores

Value of Output in Secondary Sector = ₹1,000 Crores

Intermediate Consumption in Secondary Sector = ₹400 Crores

Value Added in Secondary Sector = 1,000−400=600 Crores

Value of Output in Tertiary Sector = ₹3,000 Crores

Intermediate Consumption in Tertiary Sector = ₹900 Crores

Value Added in Tertiary Sector = 3,000−900=2,100 Crores

GNP $_{MP}$ =

Value Added in Primary Sector+Value Added in Secondary Sector+Value Added in Tertiary Sector+Net Factor Income from Abroad

GNP $_{MP}$ = 500+600+2,100+(−100)

GNP $_{MP}$ = ₹ 3,100 Crore

---

14. *GDP $_{MP}$ = Compensation of employees + mixed income of self-employed + operating surplus + depreciation + net indirect taxes*

(Note: operating surplus = rent+ profit + interest)

=1000 + 1100 + 2000 + 400 + 450 =4950

GNP_{MP} = GDP_{MP} + NFIA

= 4950 + (-50) = 4900

NNP_{MP} = GNP_{MP} - consumption of fixed capital

= 4900 – 400 = 4500

NNP_{FC} or NI = NNP_{MP} – NIT

= 4500 – 450 = = 4050 Crores

---

15. *GVA_{MP} = Gross Value Output $_{MP}$ – Intermediate consumption*

= (Sales + change in stock) – Intermediate consumption

= 4000-600 = 3400

GDP_{MP} = GVA_{MP} = 3400 Crores

NDP_{MP} = GDP_{MP} – consumption of fixed capital

= 3400 – 200

= 3200 Crores

NDP $_{FC}$ = NDP_{MP} – NIT

= 3200 – 500 = 2700 Crores

NDP_{FC} = Compensation of employees + Operating surplus + Mixed income

2700 = 800 + Operating Surplus + 400

Operating surplus = 1500 Crores

16.

(a) *GDP at factor cost = NDP at factor cost + Depreciation*

= Compensation of employees+ Rent+ Interest + Profit + Mixed income+ (Gross domestic capital formation - Net domestic capital formation)

= ₹ 3,000 + ₹ 800 + ₹ 900 + ₹ 1,300 + (₹ 900 - ₹ 800)

= ₹ 6100 Crores

(b) *Gross Domestic Product at Market Price*

= GDP at factor cost + Net Indirect taxes

= ₹ 6100 + ₹ 300

= 6400 Crores

---

17. NNP_{FC} = *NNP $_{FC}$ = Compensation of Employees + Rent + Interest + Profit + Mixed Income of Self - Employed + Net Factor Income from Abroad*

NNP_{FC} = *3,000+1,020+2,010+980+1,050+370*

= *8,430 crores*

---

18. *Net Exports = Exports−Imports*

Net Exports = 100 −200=−100Crores

National Income = Consumption + Investment + Government Purchases + Net Exports

National Income = 750+250+100+(−100)

National Income = ₹ 1,000 crore

---

19. GDP_{MP} *= Government final consumption expenditure (Public final consumption expenditure) + Private final consumption expenditure + Gross domestic capital formation (Gross domestic fixed capital formation + change stock + Net acquisition of valuables) + Net export*

(Note: As net import is20, hence, net export is -20)

= 5 +10 + [350 + 30 +10]+(- 20)

= 5+10+390-20 = 385 Crores

NNP_{FC} *= GDP_{MP} – Depreciation + Net factor income from abroad (Income from abroad – Income paid to abroad) – Net Indirect tax (Indirect tax – subsidies)*

= 385 – 30 + [0 – 20] – [0-100]

= 385 – 30 – 20 + 100 = 435 Crores.

---

20. Investment=

Gross Residential Construction Investment+Gross Public Investment+Gross Business Fixed Investment+Inventory Investment

Investment=300+200+300+100=900Crores

GDP $_{MP}$=Consumption + Investment + Government Purchases + Net Exports

GDP $_{MP}$=3,500+900+1,000+100=5,500Crores

NI = GDP $_{MP}$−Depreciation−Indirect Taxes+Subsidies+Net Factor Income from Abroad
NI=5,500−50−100+(−50)
NI=5,500−50−100−50=5,300Crores

---

21. ***GDCF*** ***= Gross Residential Construction + Gross Public Investment+ Gross Business Fixed Investment + Inventory Investment***
= *450 + 500 + 410 + 170 =1,530 = 450 + 500 + 410 + 170 = 1,530*

GNP $_{MP}$ ***= C+I+G+(X−M)+NFIA***
= *2,900 + 1,530 + 1,100 + (200 − 300) + (−30)*
= *2,900 + 1,530 + 1,100 + (200 − 300) + (−30)*
= *2,900 + 1,530 + 1,100 – 100 – 30 = 5,400*
= *2,900 + 1,530 + 1,100 − 100 – 30 = 5,400*

GNP $_{FC}$ ***= GNP*** $_{MP}$ ***+ Subsidies*** ***= 5,400 + 80 = ₹ 5,480 crores***

GDP $_{MP}$ ***= GNP*** $_{MP}$ ***− NFIA*** *= 5,400 − (−30)= 5,430*

NDP $_{MP}$ ***= GDP*** $_{MP}$ ***– Depreciation*** **=** *5,430 − 60 = ₹ 5,370 crores*

NDP $_{MP}$ ***= GDP*** $_{MP}$ ***− Depreciation*** *= 5,430 – 60 = ₹ 5,370 crores*

---

22. *GDP*$_{MP}$ *= Private final consumption expenditure + Government final consumption expenditure + Gross domestic capital formation (Net domestic capital formation+ depreciation) + Net export*
= *2000 + 1100 + (770+ 130) + 30= 4030Crores*

NNP$_{FC}$ *or NI* = *GDP*$_{MP}$*- depreciation + NFIA – NIT*
= *4030 – 130 + 20 – 120= 3800 Crores*

By Income method

NNP$_{FC}$ *or NI* = *compensation of employees+ operating surplus+ Mixed income of self- employed + NFIA*
= *1200+ 1820+ 700+ 20= 3740Crores*

---

23. ***GDP*** $_{MP}$ ***= C + I + G + (X − M)***
= *7,314 + 1,442 + 2,196 + (1,346 − 1,408)*
= *7,314 + 1,442 + 2,196 + (1,346 − 1,408)*
= *7,314 + 1,442 + 2,196 – 62 = 10,890*
= *7,314 + 1,442 + 2,196 − 62 = 10,890*

GNP $_{MP}$ ***= GDP*** $_{MP}$ ***+ NFIA***
= *10,890 + 40 = 10,930*

For Your Notes

For Your Notes

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For Your Notes

UNIT – II

THE KEYNESIAN THEORY FOR DETERMINATION OF NATIONAL INCOME

Topic References:

2.1 Introduction
2.2 Circular Flow in a Simple Two Sector Model
2.3 Basic Concepts and Functions
2.4 The Two Sector Model
2.5 The Investment Multiplier
2.6 The Three Sector Model
2.7 The Four Sector Model

2.1 Introduction

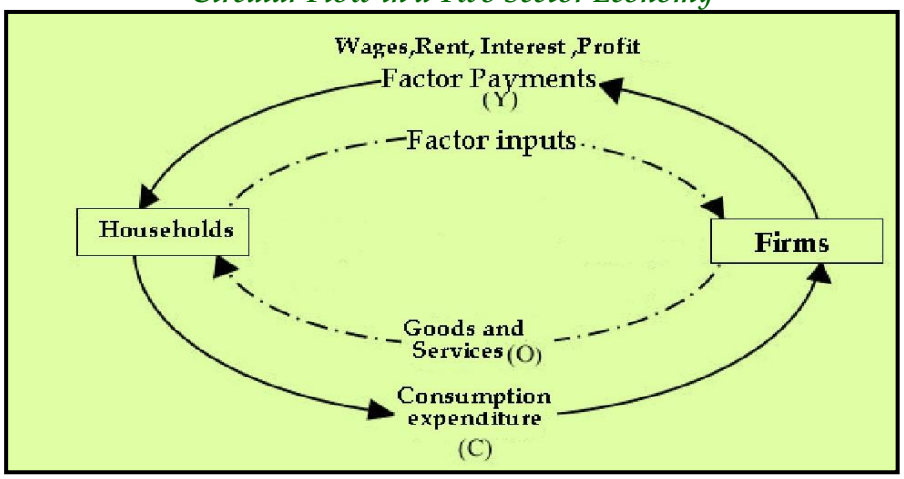
- This unit shall focus on two issues namely,
  - the factors that determine the level of national income and
  - the determination of equilibrium aggregate income and output in an economy.
- A comprehensive theory to explain these phenomena was first put forward by the British economist John Maynard Keynes in his masterpiece 'The General Theory of Employment Interest and Money' published in 1936.
- The Keynesian theory of income determination is presented in three models:
  - The two-sector model consisting of the household and the business sectors,
  - The three-sector model consisting of household, business and government sectors, and
  - The four-sector model consisting of household, business, government and foreign sectors

2.2 Circular Flow in a Simple Two Sector Model

- Even though an economy of this kind does not exist in reality, it provides a simple and convenient basis for understanding the Keynesian theory of income determination.
- The simple two sector economy model assumes that there are only two sectors in the economy viz., households and firms, with only consumption and investment outlays.
- Households own all factors production and they sell their factor services to earn factor incomes which are entirely spent to consume all final goods and services produced by business firms.
- The business firms are assumed to hire factors of production from the households; they produce and sell goods and services to the households and they do not save.
- The total income produced, Y, accrues to the households and equals their disposable personal income Y_d i.e. $Y = Y_d$.
  - There are no corporations, corporate savings or retained earnings.
  - All prices (including factor prices), supply of capital and technology remain constant.
  - The government sector does not exist and therefore, there are no taxes, government expenditure or transfer payments.
  - The economy is a closed economy i.e. foreign trade does not exist; there are no exports and imports and external inflows and outflows.
  - All investment outlay is autonomous (not determined either by the level of income or the rate of interest);

all investment is net and, therefore, national income equals the net national product.
In the figure, the circular flow of income and expenditu re which presents the working of the two- sector economy is illustrated in a simple manner.

Figure
Circular Flow in a Two Sector Economy



The circular broken lines with arrows show factor and product flows and present '**real flows**' and the continuous line with arrows show '**money flows**' which are generated by real flows.
These two circular flows - real flows and money flows are in opposite directions and the value of real flows equal the money flows because the factor payments are equal to household incomes.
There are no injections into or leakages from the system. Since the whole of household income is spent on goods and services produced by firms, household expenditures equal the total receipts of firms which equal value of output.

Factor Payments = Household Income = Household Expenditure = Total Receipts of Firms = Value of Output.

Equilibrium output occur when the desired amount of output demanded by all the agents in the economy exactly equals the amount produced in a given time period.
'Equilibrium' (defined as a state in which there is no tendency to change; or a position of rest).
Logically, an economy can be said to be in equilibrium when the production plans of the firms and the expenditure plans of the households match.

2.3 Basic Concepts and Functions

2.3.1 Aggregate Demand Function : Two Sector Model

In a simple two-sector economy aggregate demand (AD) or aggregate expenditure consists of only two components:

(i) aggregate demand for consumer goods (C), and

(ii) aggregate demand for investment goods (I)

$$AD = C + I$$

Of the two components, consumption expenditure accounts for the highest proportion of the GDP. In a simple economy, the variable 'I' is assumed to be determined exogenously and constant in the short run. Therefore, the short-run aggregate demand function can be written as:

$$AD = C + \bar{I}$$

Where $\bar{I}$ = constant investment.

From the equation (2.2), we can infer that, in the short run, AD depends largely on the aggregate consumption expenditure.

2.3.2 The Consumption Function

Q. Explain the consumption function using a suitable table and diagram. **[PEQ -Nov19-3 Marks]**

The positive relationship between consumption spending and disposable income is described by the consumption function.

Consumption function expresses the functional relationship between aggregate consumption expenditure and aggregate disposable income, expressed as:

$$C = f(Y)$$

The private demand for goods and services accounts for the largest proportion of the aggregate demand in an economy and plays a crucial role in the determination of national income.

According to Keynes, the total volume of private expenditure in an economy depends on the total current disposable income of the people and the proportion of income which they decide to spend on consumer goods and services. The specific form of consumption–income relationship termed the consumption function, proposed by Keynes is as follows:

$$C = a + bY$$

Where:

C = Aggregate consumption expenditure;

Y = Total disposable income;

a = Constant term which denotes the (positive) value of consumption at zero level of disposable income; and

b = The slope of the function, ($\Delta C / \Delta Y$) is the marginal propensity to consume (MPC) i.e the increase in consumption per unit increase in disposable income.

The consumption function shows the level of consumption (C) corresponding to each level of disposable income (Y) and is expressed through a linear consumption function,

When income is low, consumption expenditures of households will exceed their disposable income and households dis-save i.e. they either borrow money or draw from their past savings to purchase consumption goods.

The Keynesian assumption is that consumption increases with an increase in disposable income, but that the increase in consumption will be less than the increase in disposable income ($b < 1$). i.e. $0 < b < 1$.

2.3.3 Marginal Propensity to Consume (MPC)

The concept of MPC describes the relationship between change in consumption (ΔC) and the change in income (ΔY).

The value of the increment to consumer expenditure per unit of increment to income is termed the Marginal Propensity to Consume (MPC).

$$MPC = \Delta C / \Delta Y = b$$

Although the MPC is not necessarily constant for all changes in income (in fact, the MPC tends to decline at higher income levels), analysis of consumption generally works with a constant MPC.

2.3.4 Average Propensity to Consume (APC)

The average propensity to consume is a ratio of consumption defining income consumption relationship.

The ratio of total consumption to total income is known as the average propensity to consume (APC).

APC = Total Consumption / Total Income

The table below shows the relationship between income consumption and saving.

Table

Relationship between Income and Consumption

Income (Y)	Consumption (C)	APC (C / Y)	MPC ($\Delta C / \Delta Y$)	MPS (1-MPC)
0	500	$500/0 = \infty$	-	-
1,000	1,250			
2,000	2,000			
3,000	2,750			
6,000	5,000			
10,000	8,000			

2.3.5 The Saving Function

The saving function shows the level of saving (S) at each level of disposable income (Y). The intercept for the saving function, (-a) is the (negative) level of saving at zero level of disposable income at consumption equal to 'a'.

By definition, national income Y = C + S which shows that disposable income is, by definition, consumption plus saving. Therefore, S = Y – C.

2.3.6 The Marginal Propensity to Save (MPS)

This increment to saving per unit increase in disposable income (1 - b) is called the marginal propensity to save (MPS).

MPS = ΔS / ΔY = 1 - b

In other words, the MPS is the increase in saving per unit increase in disposable income. MPC is always less than unity, but greater than zero, i.e., 0 < b < 1

Also, MPC + MPS = 1; we have MPS 0 < b < 1.

Thus, saving is an increasing function of the level of income because the marginal propensity to save (MPS) = 1- b is positive, i.e. saving increase as income increases.

2.3.7 Average Propensity to Save (APS)

The ratio of total saving to total income is called average propensity to save (APS). Alternatively, it is that part of total income which is saved.

APS = Total Saving / Total Income

Disposable Income (Y_d) (₹ Crores)	Consumption (C) (₹ Crores)	Saving (S) (₹ Crores)
0	20	-20
60	70	-10
120	120	0
180	170	10
240	220	20

2.3.8 Aggregate Supply:

Ex ante or planned aggregate supply is the total supply of goods and services which firms in a national economy plan on selling during a specific time period. It is equal to the national income of the economy, which is either consumed or saved.

AS = C + S

2.4 The Two Sector Model

According to Keynesian theory of income determination, the equilibrium level of national income is a situation in which aggregate demand (C+ I) is equal to aggregate supply (C + S) i.e.

In a two sector economy, the aggregate demand (C+ I) refers to the total spending in the economy i.e. it is the sum of demand for the consumer goods (C) and investment goods (I) by households and firms respectively.

According to Keynes, aggregate demand will not always be equal to aggregate supply.

Aggregate demand depends on households plan to consume and to save. Aggregate supply depends on the producers' plan to produce goods and services.

For the aggregate demand and the aggregate supply to be equal so that equilibrium is established, the house holds' plan must coincide with producers' plan.

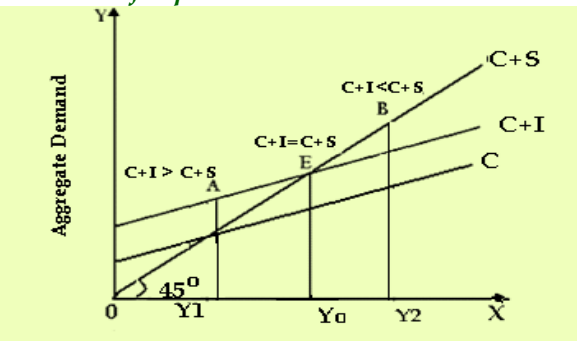
At equilibrium, expected value equals realized value. However, Keynes held the view that that there is no reason to believe that:

(i) consumers' consumption plan always coincides with producers' production plan, and

(ii) that producers' plan to invest matches always with households plan to save Putting it differently, there is no reason for C + I and C + S to be always equal.

In figure , the aggregate demand curve is linear and positively sloped indicating that as the level of national income rises, the aggregate demand (or aggregate spending) in the economy also rises. The aggregate expenditure line is flatter than the 45-degree line because, as income rises, consumption also increases, but by less than the increase in income.

Figure
Determination of Equilibrium Income: Two Sector Model



The figure depicts the determination of equilibrium income. Income is measured along the horizontal axis and the components of aggregate demand, C and I, are measured along the vertical axis.

Since the autonomous expenditure component (I) does not depend directly on income, the (C+I) schedule lies above the consumption function by a constant amount. Equilibrium level of income is such that aggregate demand equals output (which in turn equals income).

Only at point E and at the corresponding equilibrium levels of income and output (Y0), does aggregate demand exactly equal output. At that level of output and income, planned spending precisely matches production. Once national income is determined, it will remain stable in the short run.

Why the other points on the graph are not points of equilibrium?

Let us consider a level of income below Y_0, say Y_1, generates consumption as shown along the consumption function. When this level of consumption is added to the autonomous investment expenditure (I), the aggregate demand exceeds income; i.e the (C + I) schedule is above the 45° line. The aggregate expenditures exceed aggregate output.

Conversely, at levels of income above Y0, for example at Y2, output exceed demand (the 45° line is above the C + I schedule).

Since C + S = Y, the national income equilibrium can be written as

$$Y = C + I$$

The equality between saving and investment can be seen directly from the identities in national income accounting. Since income is either spent or saved, Y = C + S. Without government and foreign trade, aggregate demand equals consumption plus investment, Y = C + I. Putting the two together, we have C + S = C + I, or S = I.

2.4.1 Equilibrium with Unemployment or Inflation

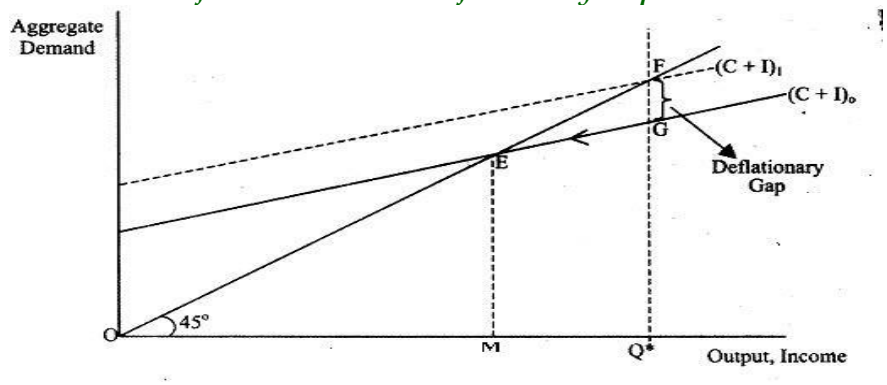
Keynesian equilibrium with equality of planned aggregate expenditures and output need not take place at full employment.

(i) Deflationary Gap

If the aggregate demand is for an amount of output less than the full employment level of output, then we say there is deficient demand.

Deficient demand gives rise to a 'deflationary gap' or 'recessionary gap'. Recessionary gap also known as 'contractionary gap' arises in the Keynesian model of the macro economy when the equilibrium level of aggregate production achieved in the short-run falls short of what could be produced at full employment. Recessionary gap occurs when the economy is in a business-cycle contraction or recession.

Figure
Deficient Demand - Deflationary Gap



In figure, OQ* is the full employment level of output. For the economy to be at full employment equilibrium, aggregate demand should be Q*F. If the aggregate demand is Q*G, it represents a situation of deficient demand. The resulting deflationary gap is FG. Firms will experience unplanned build-up of inventories of unsold goods and they will respond by cutting production and employment leading to decrease in output and income until the under-employment equilibrium is reached at E.

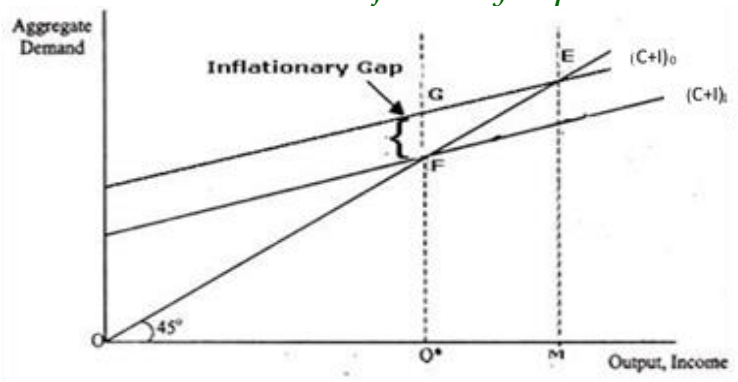
(ii) Inflationary Gap

If the aggregate demand is for an amount of output greater than the full employment level of output, then we say there is excess demand.

Excess demand gives rise to 'inflationary gap' which is the amount by which actual aggregate demand exceeds the level of aggregate demand required to establish the full employment equilibrium.

This is the sort of gap that tends to occur during a business-cycle expansion and sets in motion forces that will cause demand pull inflation.

Figure
Excess Demand - Inflationary Gap



The economy will be at full employment equilibrium at F with OQ* full employment level of output and income. Suppose the aggregate demand is for Q*G, there is excess demand and the resulting inflationary gap FG.

The real output will be constant, but the rise in the price level will cause an increase in the nominal output until the new equilibrium is reached at point E. Point E is an equilibrium point because the aggregate demand ME is equal to output OM. At the new equilibrium, real output, real income and employment will be the same; nominal output and income has increased due to inflation.

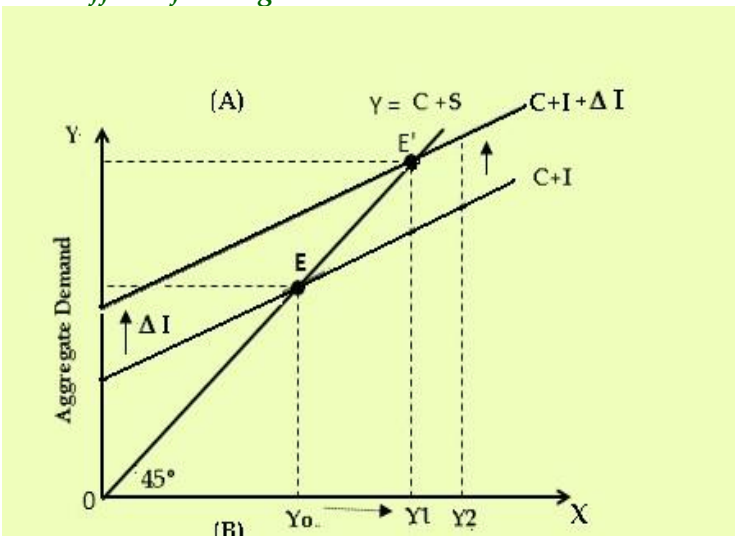
Therefore, output will remain at less than the full employment rate as long as there is insufficient spending in the economy.

2.5 The Investment Multiplier

In our two-sector model, a change in aggregate demand may be caused by change in consumption expenditure or in business investment or in both.

We shall now examine the effect of an increase in investment (upward shift in the investment schedule) causing an upward shift in the aggregate demand function.

Figure
Effect of Changes in Autonomous Investment



In the figure, an increase in autonomous investment by Δ I shifts the aggregate demand schedule from C+I to C+I+ΔI.
Correspondingly, the equilibrium shifts from E to E¹ and the equilibrium income increases more than proportionately from Y0 to Y1.

Why and how does this happen?
This occurs due to the operation of the investment multiplier.
The multiplier refers to the phenomenon whereby a change in an injection of expenditure will lead to a proportionately larger change (or multiple change) in the level of national income.

Multiplier explains how many times the aggregate income increases as a result of an increase in investment. When the level of investment increases by an amount say ΔI, the equilibrium level of income will increase by some multiple amounts, Δ Y. The ratio of ΔY to ΔI is called the investment multiplier, k.
The size of the multiplier effect is given by Δ Y = k ΔI.

$$\mathbf{k = \Delta Y / \Delta I}$$

For example - If a change in investment of ₹2,000 million causes a change in national income of ₹6,000 million, then the multiplier is 6,000/2,000 = 3.

Thus multiplier indicates the change in national income for each rupee change in the desired investment. The value 3 in the above example tells us that for every ₹ 1 increase in desired investment expenditure, there will be ₹3 increase in equilibrium national income.

Multiplier, therefore, expresses the relationship between an initial increment in investment and the resulting increase in aggregate income. Since the increase in national income (ΔY) is the result of increase in investment (ΔI), the multiplier is called 'investment multiplier.'

The process behind the multiplier can be compared to the 'ripple effect' of water.

We find at the end that the increase in income per rupee increase in investment is:

$$\frac{\Delta Y}{\Delta I} = \frac{1}{1-MPC} = \frac{1}{MPS}$$

From the above, we find that the MPC is the determinant of the value of the multiplier and that there exists a direct relationship between MPC and the value of multiplier.
Higher the MPC more will be the value of the multiplier, and vice-versa.
On the contrary, higher the MPS, lower will be the value of multiplier and vice-versa.
The maximum value of multiplier is infinity when the value of MPC is 1 i.e. the economy decides to consume the whole of its additional income.
We conclude that the value of the multiplier is the reciprocal of MPS.
For example, if the value of MPC is 0.75, then the value of the multiplier is multiplier is: $\frac{1}{0.25} = 4$
Causes responsible for the decline in income are called leakages.
The more powerful these leakages are the smaller will be the value of multiplier.

Q. Explain the leakages and injections in the circular flow of income? [PEQ-May 18 - 2 Marks]

The leakages are caused due to:

1. progressive rates of taxation which result in no appreciable increase in consumption despite increase in income
2. high liquidity preference and idle saving or holding of cash balances and an equivalent fall in marginal propensity to consume
3. increased demand for consumer goods being met out of the existing stocks or through imports
4. additional income spent on purchasing existing wealth or purchase of government securities and shares from shareholders or bond holders
5. undistributed profits of corporations
6. part of increment in income used for payment of debts
7. case of full employment additional investment will only lead to inflation, and
8. scarcity of goods and services despite having high MPC

2.6 The Three Sector Model

Aggregate demand in the three sector model of closed economy (neglecting foreign trade) consists of three components namely, household consumption(C), desired business investment demand(I) and the government sector's demand for goods and services(G).

Thus in equilibrium, we have

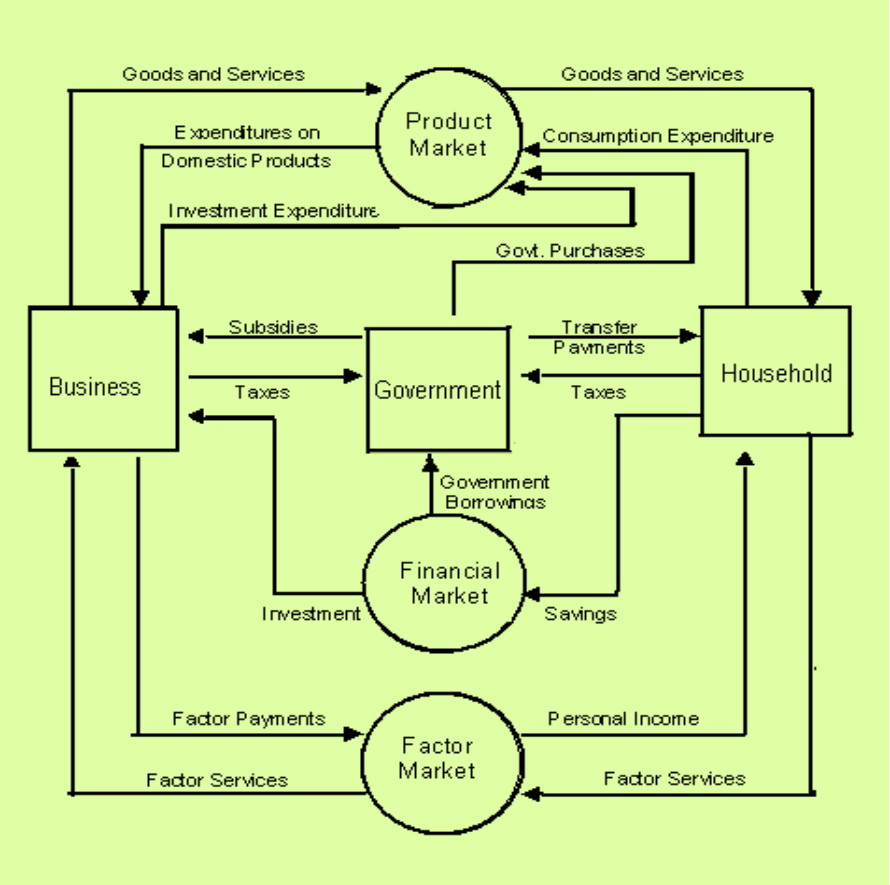
$$Y = C+I+G$$

Since there is no foreign sector, GDP and national income are equal. As prices are assumed to be fixed, all variables are real variables and all changes are in real terms.

Through the below flow chart, we can find that the government sector adds the following key flows to the model:

i) Taxes on households and business sector to fund government purchases
ii) Transfer payments to household sector and subsidy payments to the business sector
iii) Government purchases goods and services from business sector and factors of production from household sector, and
iv) Government borrowing in financial markets to finance the deficits occurring when taxes fall short of government purchases

Figure
Circular Flow in a Three Sector Economy



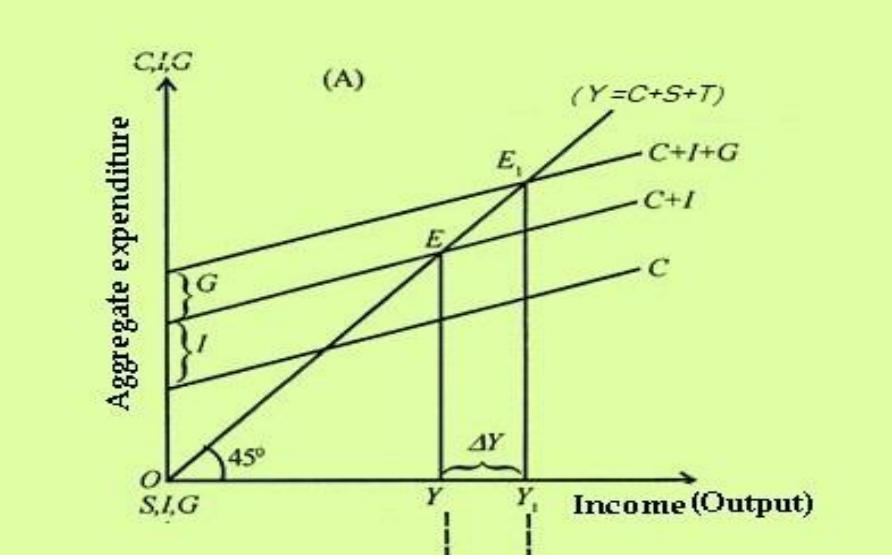
There are two flows out of the household sector in addition to consumption expenditure namely, saving flow and the flow of tax payments to the government. These are actually leakages.

The saving leakage flows into financial markets, which means that the part of that is saved is held in the form of some financial asset (currency, bank deposits, bonds, equities, etc.). The tax flow goes to the government sector.

The leakages which occur in household sector do not necessarily mean that the total demand must fall short of output. There are additional demands for output on the part of the business sector itself for investment and from the government sector. In terms of the circular flow, these are injections. The investment injection is shown as a flow from financial markets to the business sector.

The three-sector Keynesian model is commonly constructed assuming that government purchases are autonomous.

Figure
Determination of Equilibrium Income: Three Sector Model



The variables measured on the vertical axis are C, I and G. The autonomous expenditure components namely, investment and government spending do not directly depend on income and are exogenous variables determined by factors outside the model. Therefore, C + I + G schedule lies above the consumption function by a constant amount.

The line S + T in the graph plots the value of savings plus taxes. This schedule slopes upwards because saving varies positively with income. Just as government spending, level of tax receipts (T) is decided by policy makers.

The equilibrium level of income is shown at the point E_1 where the (C + I + G) schedule crosses the 45° line, and aggregate demand is therefore equal to income (Y). In equilibrium, it is also true that the (S + T) schedule intersects the (I + G) horizontal schedule.

Why the other points on the graph are not points of equilibrium?

Consider a level of income below Y. We find that it generates consumption as shown along the consumption function. When this level of consumption is added to the autonomous expenditures (I + G), aggregate demand exceeds income; the (C + I + G) schedule is above the 45° line. With demand outstripping production, desired investments will exceed actual investment and there will be an unintended inventory shortfall and therefore a tendency for output to rise.

Conversely, at levels of income above Y_1, output will exceed demand;

It is only at Y that output is equal to aggregate demand;

An important thing to note is that the change in total spending, followed by changes in output and employment, is what will restore equilibrium in the Keynesian model, not changes in prices.

2.6.1 The Government Sector and Income Determination

We have seen above that the government influences the level of income through taxes, transfer payments, government purchases and government borrowing.

(i) Income Determination with Lump Sum Tax

The consumption function is defined as –

$C = a + b Y_d$

Where $Y_d = Y - T$ (disposable income), T = lump sum tax

$Y = a + b (Y - T) + I + G$

$Y = a + b Y - bT + I + G$

$Y - bY = a - bT + I + G$

$Y(1-b) = a - bT + I + G$

$Y = \frac{1}{1-b} (a-bT+I+G)$

(ii) Income Determination with Lump Sum Tax and Transfer payments

The consumption function is defined as –

$C = a + b Y_d$

Where $Y_d = Y - T + TR$

where T is a lump sum tax and TR is autonomous transfer payments

$C = a + b (Y - T + TR)$

$Y = C + I + G$

$Y = a + b (Y - T + TR) + I + G$

$Y = a + bY - bT + bTR + I + G$

$Y - bY = a - bT + bTR + I + G$

$Y(1-b) = a - bT + bTR + I + G$

$Y = \frac{1}{1-b} (a-bT + bTR + I+G)$

(iii) Income Determination with tax as a function of Income

In (i) and (ii) above, we have analysed the effect of balanced budget with an autonomous lump sum tax. In reality, the tax system consists of both lump sum tax and proportional taxes. The tax function is defined as;

Tax function $T = \bar{T} + t Y$

Where

$\bar{T}$ = autonomous constant tax

t = income tax rate

T = total tax

The consumption function is

$C = a + b Y_d$

Where $Y_d = Y - T$ or $Y - \bar{T} - t Y$

$C = a + b(Y - \bar{T} - t Y)$

Therefore, the equilibrium level of national income can be measured as-

$Y = C + I + G$

$Y = a + bY_d + I + G$

$Y = a + b(Y - \bar{T} - tY) + I + G$

$Y = a + bY - b\bar{T} - b\,t\,Y + I + G$

$Y - bY + btY = a - b\bar{T} + I + G$

$Y(1 - b + bt) = a - b\bar{T} + I + G$

$$Y = \frac{1}{1-b\,(1-t)}(a - b\bar{T} + I + G)$$

Where $\frac{1}{1-b\,(1-t)}$ (represent the tax multiplier)

(iv) Income Determination with Tax (as a Function of Income), Government Expenditure and Transfer Payments

Here consumption function is written as

$C = a + b(Y - \bar{T} - tY + TR)$

$Y = a + b(Y - \bar{T} - tY + TR) + I + G$

$$Y = \frac{1}{1-b\,(1-t)}(a - b\bar{T} + bTR + I + G)$$

2.7 The Four Sector Model

The four sector model includes all four macroeconomic sectors, the household sector, the business sector, the government sector, and the foreign sector. The foreign sector includes households, businesses, and governments that reside in other countries.

In the four sector model, there are three additional flows namely: exports, imports and net capital inflow which is the difference between capital outflow and capital inflow.

$$\mathbf{Y = C + I + G + (X-M)}$$

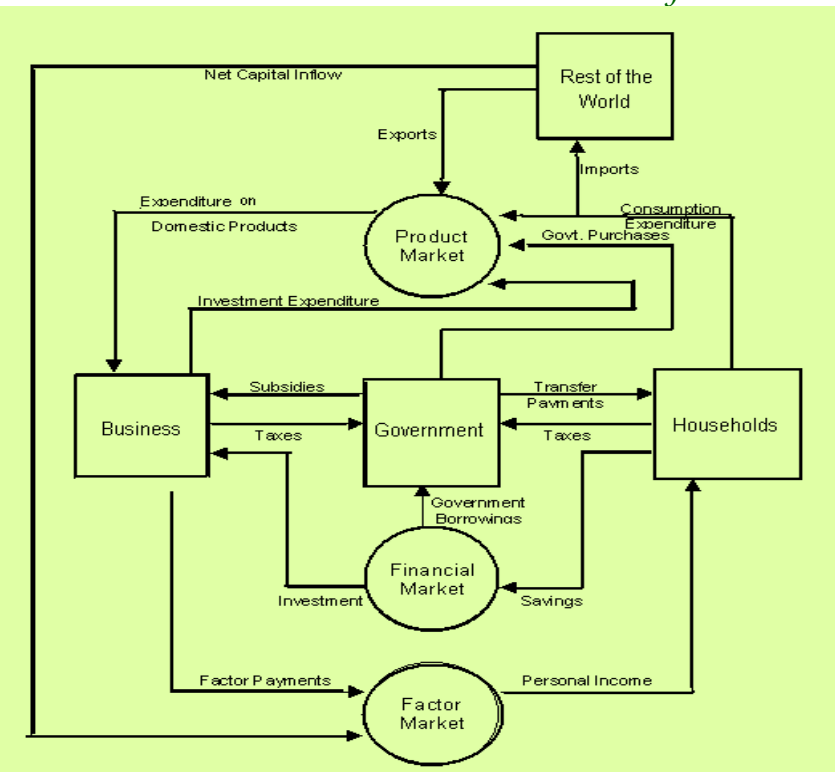
The domestic economy trades goods with the foreign sector through exports and imports. Exports are the injections in the national income, while imports act as leakages or outflows of national income.

Exports represent foreign demand for domestic output and therefore, are part of aggregate demand. Since imports are not demands for domestic goods, we must subtract them from aggregate demand.

The demand for imports has an autonomous component and is assumed to depend on income. Imports depend upon marginal propensity to import which is the increase in import demand per unit increase in GDP.

The demand for exports depends on foreign income and is therefore exogenously determined. Imports are subtracted from exports to derive net exports, which is the foreign sector's contribution to aggregate expenditures.

Figure
Circular Flow in a Four Sector Economy



Since import has an autonomous component ($\bar{M}$) and is assumed to depend on income(Y) and marginal propensity to import (m), the import function is expressed as $M = \bar{M} + mY$.
Marginal propensity to import $m = \Delta M / \Delta Y$ is assumed to be constant.
As the aggregate demand in the four sector model is given in equation , the equilibrium condition is expressed as follows-

$Y = C + I + G + (X - M)$
Where $C = a + b(Y-T)$
$M = \bar{M} + mY$

The equilibrium level of National Income can now be expressed by –

$Y = C + I + G + (X - M)$
$Y = a + b(Y - T) + I + G + X - \bar{M} - mY$
$Y - bY + mY = a - bT + I + G + X - \bar{M}$
$Y = \frac{1}{1-b+m}(a - b\,T + I + G + X - \bar{M})$

The economy being in equilibrium, suppose export of country increases by ΔX autonomously, all other factors remaining constant. By incorporating the increase in exports by ΔX, the equilibrium equation of the country can be expressed as

$Y + \Delta Y = \frac{1}{1-b+m}(a - b\,T + I + G + X - \bar{M} + \Delta X)$ or
$Y + \Delta Y = \frac{1}{1-b+m}(a - b\,T + I + G + X - \bar{M}) + \frac{1}{1-b+m}\Delta X$
As, $Y = \frac{1}{1-b+m}(a - b\,T + I + G + X - \bar{M})$

We get, $Y + \Delta Y = Y + \frac{1}{1-b+m} \Delta X$

Subtracting Y from both sides, we get

$\Delta Y = \frac{1}{1-b+m} \Delta X$

By rearranging

$\Delta Y = \frac{1}{1-b+m} \Delta X$, we get

$\frac{\Delta Y}{\Delta X} = \frac{1}{1-b+m}$

Or alternatively written as

$\frac{\Delta Y}{\Delta X} = \frac{1}{1-(b-m)}$

The term $\frac{1}{1\text{-}b+m}$ is known as foreign trade multiplier whose value is determined by marginal propensity to consume (b) and marginal propensity to import (m).

If in the model proportional income tax and government transfer payments are incorporated, then only the denominator of multiplier will change. If income tax is of form $T = \bar{T} + t Y$ where $\bar{T}$ is constant lump-sum, t is the proportion of income tax and TR > 0 and autonomous, then the four sector model can be expressed as: –

$Y = C + I + G + (X - M)$

Where

$C = a + b(Y - \bar{T} - t Y + TR)$

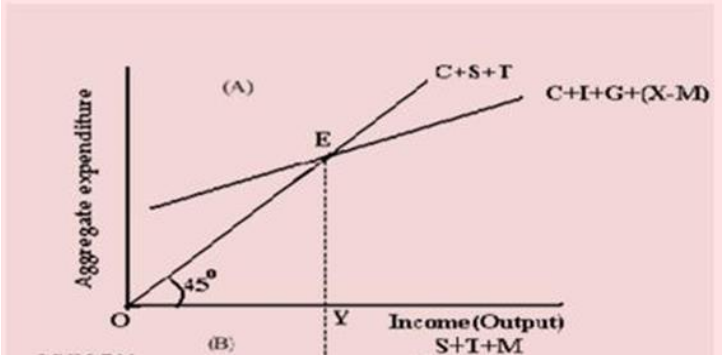
$M = \bar{M} + mY.$

The equilibrium level of National Income can now be expressed as:

$Y = \frac{1}{1-b(1-t)+m} (a - b\bar{T} + b\,TR + I + G + X - \bar{M})$

Figure

Determination of Equilibrium Income: Four Sector Model

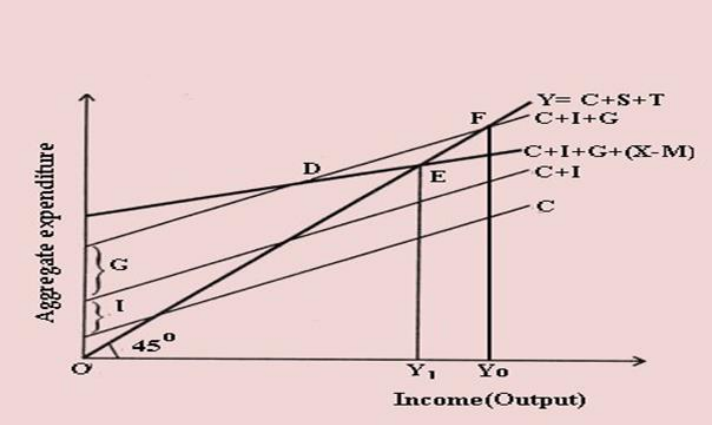


Equilibrium is identified as the intersection between the C + I + G + (X - M) line and the 45-degree line. The equilibrium income is Y. From panel B, we find that the leakages(S+T+M) are equal to injections (I+G+X) only at equilibrium level of income.

We have seen above that only net exports(X-M) are incorporated into the four sector model of income determination.

We know that injections increase the level of income and leakages decrease it. Therefore, if net exports are positive (X > M), there is net injection and national income increases. Conversely, if X<M, there is net withdrawal and national income decreases.

Figure
Effects on Income When Imports are Greater than Exports



We have seen above that equilibrium income is expressed as a product of two terms: $\Delta Y = k\ \Delta I$; i.e. the level of autonomous investment expenditure and the investment multiplier.

The autonomous expenditure multiplier in a four sector model includes the effects of foreign transactions and is stated as $\frac{1}{1-(b-m)}$ where 'm' is the propensity to import which is greater than zero. You may recall that the multiplier in a closed economy is $\frac{1}{1-b}$

The greater the value of 'm', the lower will be the autonomous expenditure multiplier. The more open an economy is to foreign trade, (the higher mis) the smaller will be the response of income to aggregate demand shocks, such as changes in government spending or autonomous changes in investment demand.

The higher the value of 'm', the larger the proportion of this induced effect on demand for foreign, not domestic, consumer goods. The increase in imports per unit of income constitutes an additional leakage from the circular flow of (domestic) income at each round of the multiplier process and reduces the value of the autonomous expenditure multiplier.

An increase in demand for exports of a country is an increase in aggregate demand for domestically produced output and will increase equilibrium income just as an increase in government spending or an autonomous increase in investment.

In summary, an increase in the demand for a country's exports has an expansionary effect on equilibrium income, whereas an autonomous increase in imports has a contractionary effect on equilibrium income.

2.8 Conclusion

According to the Keynesian theory of income and employment, national income depends upon the aggregate effective demand. If the aggregate effective demand falls short of that, it will result in unemployment in the economy.

On the contrary, if the aggregate effective demand exceeds the economy's full employment output (production capacity), it will result in inflation.

Appropriate changes in fiscal policy by adjusting government expenditure and taxes could keep the autonomous expenditure constant even in the face of undesirable changes in the investment.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. *In the Keynesian model, equilibrium aggregate output is determined by*
   (a) aggregate demand (b)consumption function
   (c) the national demand for labour (d) the price level

2. *Keynes believed that an economy may attain equilibrium level of output*
   (a) only at the full-employment level of output (b) below the full-employment level of output
   (c) only if prices were inflexible (d) (a) and (c) above

3. *According to Keynes, consumption expenditure is determined by*
   (a) The level of interest rates (b)Extent of government taxes and subsidies
   (c) disposable income (d) autonomous investment expenditure

4. *The marginal propensity to consume (MPC) can be defined as*
   (a) a change in spending due to a change in income
   (b) a change in income that is saved after consumption
   (c) part of income that is spent on consumption.
   (d) part of income that is not saved

5. *If the consumption function is expressed as C = a + bY then b represents*
   (a) autonomous consumer expenditure when income is zero
   (b) the marginal propensity to consume
   (c) the expenditure multiplier when consumption is increased
   (d) part of disposable income

6. *If the consumption function is expressed as C = a + bY then a represents*
   (a) autonomous consumer expenditure (b) the marginal propensity to consume
   (c) the consumption income relationship (d) non-linear consumption function

7. *If the consumption function is $C = 20 + 0.5Y_d$, then an increase in disposable income by ₹ 100 will result in an increase in consumer expenditure by ____*
   (a) 25 (b) 70 (c) 50 (d) 100

8. *If the autonomous consumption equals ₹ 2,000 and the marginal propensity to consume equals 0.8. If disposable income equals ₹ 10,000, then total consumption will be ____*
   (a) 8,000 (b) 6,000 (c) 10,000 (d) None of the above

9. *In the Keynesian cross diagram, the point at which the aggregate demand function crosses the 45 degree line indicates the*
   (a) level of full employment income.
   (b) less than full employment level of income.
   (c) equilibrium level of income which may or may not be full employment level of income
   (d) autonomous level of income which may not be full employment level of income

10. *In a closed economy, aggregate demand is the sum of*
    (a) Consumer expenditure, demand for exports and government spending
    (b) Consumer expenditure, planned investment spending and government spending
    (c) Consumer expenditure, actual investment spending, government spending and net exports
    (d) Consumer expenditure, planned investment spending, government spending and net exports

11. *Under equation C= a+by, b=0.8, what is the value of 2 sector expenditure multiplier?*
(a) 4 (b) 2 (c) 5 (d) 1

12. *In an open economy, aggregate demand is the sum of*
(a) C+I+G (b) C+I (c) C+I+G+(X-M) (d) None of the above

13. *Aggregate output and autonomous consumer expenditure are related*
(a) negatively (b) positively (c) proportionately (d) Collectively

14. *The slope of the consumption function is called*
(a) autonomous consumption when income is zero (b)average propensity to consume.
(c) marginal propensity to consume (d) None of the above

15. *The saving function shows*
(a) the incremental saving caused by incremental disposable income
(b) the level of saving at each level of disposable income
(c) position where saving is equal to investment
(d) change in saving due to autonomous investment

16. *Which of the following statement is true?*
(a) Saving is an increasing function of level of income and MPS is positive
(b) MPC is always greater than Zero but less than unity i.e. $0 < b < 1$
(c) MPC + MPS = 1 and MPS $0 < b < 1$
(d) All the Above

PEQ – JUNE – 2024

17. *Disposable Personal Income is:*
(a) Total personal income left after tax deduction at source
(b) Personal income-Direct tax paid by individual-indirect tax paid by individual
(c) Personal income-Personal income taxes-non-tax payments
(d) Personal income-indirect taxes paid by individual (1 mark)

18. *Calculate GDP deflator for year 2022 and 2023:*

Year	Nominal GDP (₹ billion)	Real GDP (₹ billion)
2022	1550	1054
2023	1850	1158

(a) 147.06 and 159.67 (b) 100 and 161.54
(c) 68 and 62.67 (d) 400 and 560 (1 mark)

19. *The GDP deflator for 2022 is 159.67, and for 2023 it is 146.17. The inflation rate between year 2022-23 is:*
(a) -7.84 (b) -8.45 (c) 8.51 (d) 7.84 (1 mark)

20. *Which of the following is not a reason of leakage?*
(a) Part of increment in income used for payment of debts
(b) Distributed profits of corporation
(c) High liquidity preference
(d) Progressive rate of taxation (1 mark)

21. *Find aggregate demand, when consumption is ₹ 2,000 crores, investment is ₹ 700 crores, government spending is ₹ 750 crores, total exports is ₹ 150 crores and total expenditure on imports are ₹ 50 crores.*
(a) ₹ 3,550 crores (b) ₹ 3,300 crores
(c) ₹ 3,600 crores (d) ₹ 3,350 crores (1 mark)

22. *$C = 200 + 0.8Y_d$, $G = T = 150$, $I = 200$, $TR = 75$. Find equilibrium income:*
(a) ₹ 625 (b) ₹ 2,450 (c) ₹ 2,250 (d) ₹ 800 (1 mark)

23. *Which of following is true for MPC?*
(a) It is always greater than one.
(b) It is always less than unity but greater than zero.
(c) It can be even less than zero.
(d) It can attain any value depending upon the disposable income. (1 mark)

24. *Investment multiplier is:*
(a) Rate of change in investment due to increase in income.
(b) Rate of increase in national income due to increase in investment.
(c) Rate of change in investment due to increase in savings.
(d) Rate of change in savings due to increase in investment. (1 mark)

25. *if aggregate demand for an amount is less than the full employment level of output in economy then it gives rise to:*
(a) Inflationary gap (b) Deflationary gap
(c) Potential gap (d) Excess demand (1 mark)

26. *Increase in investment by ₹ 500 crores will increase the national income by ₹ 1,500 crores. Find MPC?*
(a) 3 (b) 0.75 (c) 0.70 (d) 0.66 (1 mark)

PEQ-SEPTEMBER-2024

27. *Suppose the consumption of an economy is given by $C = 30 + 0.5Y$ and investment $I = 12 + 0.15Y$. What will be the equilibrium level of National Income?*
(a) 115 (b) 120 (c) 105 (d) 112 (1 mark)

28. *In the three-sector model of a closed economy (neglecting foreign trade), what are the three components of aggregate demand?*
(a) Consumption, Exports, and Government Spending
(b) Consumption, Savings, and Investment
(c) Consumption, Investment, and Government Spending
(d) Consumption, Imports, and Exports (1 mark)

29. *According to the Keynesian theory of income and employment, national income depends upon:*
(a) Aggregate Effective Savings (b) Aggregate Effective Supply
(c) Aggregate Effective Demand (d) Aggregate Effective Production (1 mark)

30. *What will be the Total Income if the value of Average Propensity to Consume (APC) is 0.45 and Total Consumption = ₹ 900 crores?*
(a) ₹ 2,000 crores (b) ₹ 2,500 crores
(c) ₹ 2,050 crores (d) ₹ 2,100 crores (1 mark)

PEQ-JANUARY-2025

31. *The consumption function is a functional relationship between aggregate consumption and:*
(a) aggregate disposable income *(b) aggregate demand*
(c) aggregate supply *(d) savings* *(1 mark)*

32. *Calculate average propensity to save when C = 300 and Y = 1200.*
(a) 0.25 *(b) 0.50* *(c) 0.75* *(d) 0.80* *(1 mark)*

33. *Marginal propensity to consume is:*
(a) zero when the income is zero.
(b) always less than unity but greater than zero.
(c) greater than one when income rises.
(d) it does not depend on income. *(1 mark)*

34. *In a 3-sector model, suppose C = 8 + 0.6 Y_d, I = 90, G = T = 10, where C is consumption, I is investment, Y_d is disposable income, G is government expenditure and T is tax.*
Find out the equilibrium level of national income
(a) 120 *(b) 150* *(c) 180* *(d) 200* *(1 mark)*

35. *In an economy investment expenditure is increased by ₹ 300 crores and marginal propensity to consume is 0.6. Calculate the total increase in income.*
(a) ₹ 300 crores *(b) ₹ 100 crores*
(c) ₹ 650 crores *(d) ₹ 750 crores* *(1 mark)*

36. *The investment multiplier is defined as the ratio of:*
(a) change in investment due to change in saving
(b) change in demand due to change in investment
(c) change in consumption due to change in investment
(d) change in national income due to change in investment *(1 mark)*

37. *The ratio of total consumption to total income is known as:*
(a) Average Propensity to Consume (APC) *(b) Marginal Propensity to Consume (MPC)*
(c) Saving function *(d) Income function* *(1 mark)*

PEQ-MAY-2025

38. *Which of the following is not a sector to be included in the three sector model of Keynesian theory of Income determination?*
(a) Household Sector
(b) Business Sector
(c) Foreign Sector
(d) Government Sector

39. *Under the Keynesian theory of National Income determination, which of the following term is given to the demand for money?*
(a) Investment multiplier
(b) Liquidity preference
(c) Aggregate demand
(d) Marginal Propensity

40. *Under the Keynesian theory of determination of national income, the assumption is that the consumption increases with an increase in disposable income but the increase in consumption will be_____ the increase in disposable income.*
   (a) Equal to
   (b) opposite to
   (c) Greater than
   (d) Less than

41. *Which of the following is true in respect of relation of Marginal Propensity to Consume (MPC) and Marginal Propensity to Save (MPS) as per the Keynesian theory of determination of National Income?*
   (a) MPC = MPS
   (b) MPC+ MPS = 1
   (c) MPC+ MPS = 0
   (d) No relation exists between MPC and MPS

PEQ-SEPTEMBER-2025

42. *If the consumption function C=10+0.5Y and Investment is ₹ 200, then what will be the equilibrium level of Income?*
   (a) ₹ 510
   (b) ₹ 325
   (c) ₹ 420
   (d) ₹ 350

43. *The maximum value of investment multiplier will be the infinity when the value of:*
   (a) MPC is 1.
   (b) MPC is -1.
   (c) APC is 1.
   (d) MPS is-1.

44. *There are three different interlinked phases in a circular flow of income, namely:*
   (a) Production, consumption and disposition
   (b) Production, consumption and investment
   (c) Production, disposition and distribution
   (d) Production, expenditure and disposition

45. *If national income increases from ₹ 500 crores to ₹ 2,500 crores due to an increase in investment of ₹ 200 crores, what will be the value of the Marginal Propensity to Consume (MPC)?*
   (a) 0.90
   (b) 0.85
   (c) 0.75
   (d) 0.95

46. *The slope of the consumption function represents _____.*
   (a) Average Propensity to Save (APS).
   (b) Marginal Propensity to Consume (MPC).
   (c) Marginal Propensity to Save (MPS).
   (d) Average Propensity to Consume (APC).

47. *If the Average Propensity to Save (APS) of a household is 0.4, and their total income is ₹ 25,000, what is the total saving?*
   (a) ₹15,000
   (b) ₹10,000
   (c) ₹6,250
   (d) ₹12,500

PEQ-JANUARY-2026

48. *National income of the economy which is either consumed or saved is equal to the:*
    (a) *Aggregate supply*
    (b) *Marginal propensity to consume*
    (c) *Average supply*
    (d) *Marginal propensity to save*

49. *Given the saving function S-20-0.5Y and autonomous investment I = 100 crores, what is the consumption?*
    (a) *₹ 240 crores*
    (b) *₹ 140 crores*
    (c) *₹ 340 crores*
    (d) *₹ 440 crores*

50. *If marginal propensity to consume is 0.6 and marginal propensity to import is 0.04, what is the value of the foreign trade multiplier?*
    (a) *3.33*
    (b) *2.27*
    (c) *3.66*
    (d) *2.54*

51. *All of the following are included in government final consumption expenditure except:*
    (a) *Government expenditure on defence*
    (b) *Government expenditure on education*
    (c) *Government expenditure on healthcare*
    (d) *Government expenditure on pensions*

52. *An autonomous increase in imports has _____ effect on equilibrium income.*
    (a) *Expansionary*
    (b) *Contractionary*
    (c) *Balanced*
    (d) *No*

PEQ-SEPTEMBER-2026

53. *In the four-sector Keynesian model, the equilibrium income condition is:*
    (a) $Y = C + I$
    (b) $Y = C + I + G$
    (c) $Y = C + I + G + (X - M)$
    (d) $Y = C + S + T$

II. Illustration

1. *What will be the value of average propensity to save when -*
(i) *C = 200 at Y = 1,000*
(ii) *S = 450 at Y = 1,200*

2. *Calculate marginal propensity to consume and marginal propensity to save from the following data about an economy which is in equilibrium:*
National income = 2500, Autonomous consumption expenditure = 300,
Investment expenditure= 100

3. *An economy is in equilibrium. Calculate national income from the following-*
Autonomous consumption = 100; Marginal propensity to save= 0.2; Investment expenditure= 200

4. *Suppose the consumption of an economy is given by C = 20+ 0.6 Y and investment I= 10+ 0.2Y. What will be the equilibrium level of National Income?*

5. *Suppose the consumption function C= 7+ 0.5Y, Investment is ₹ 100, Find out equilibrium level of Income, consumption and saving?*

6. *If the consumption function is C= 250 + 0.80 Y and I = 300. Find out equilibrium level of Y, C and S?*

7. *If saving function S = -10 + 0.2Y and autonomous investment I = 50 Crores. Find out the equilibrium level of income, consumption and if investment increases permanently by ₹ 5 Crores, what will be the new level of income and consumption?*

8. *Given the empirical consumption function C= 100+0.75Y and I = 1000, calculate equilibrium level of national income. What would be the consumption expenditure at equilibrium level national income?*

9. *In an economy investment expenditure is increased by ₹ 400 Crores and marginal propensity to consume is 0.8. Calculate the total increase in income and saving.*

10. *An increase in investment by 400 Crores leads to increase in national income by 1,600 Crores. Calculate marginal propensity to consume.*

11. *In an economy, investment is increased by Rs 600 Crores. If the marginal propensity to consume is 0.6, calculate the total increase in income and consumption expenditure.*

12. *Suppose in a country investment increases by ₹ 100 Crores and consumption is given by C = 10 + 0.6Y (where C = consumption and Y = income). How much increases will there take place in income?*

13 *Suppose we have the following data about a simple economy:*
C = 10 + 0.75Yd, I = 50, G = T = 20where C is consumption, I is investment, Yd is disposable income, G is government expenditure and T is tax.
(a) Find out the equilibrium level of national income.
(b) What is the size of the multiplier?

---

14. *Suppose the structural model of an economy is given –*
C = 100+ 0.75 Yd; I = 200, G = T = 100; TR= 50, find the equilibrium level of income?

---

15. *For a closed economy, the following data is given –*
Consumption C = 75 + 0.5 (Y-T); Investment I = 80; Total tax T = 25 + 0.1Y; Government expenditure G = 100.
(a) Find out equilibrium income?
(b) What is the value of multiplier?

---

16. *Suppose C= 100 + 0.80 (Y- T + TR); I = 200; T= 25+0.1Y; TR= 50; G = 100*
Find out equilibrium level of Income?

---

17. *The consumption function is C = 40 + 0.8Y_d,T= 0.1Y, I = 60 Crores G = 40 Crores, X = 58 and M = 0.05 Y. Find out equilibrium level of income, Net Export, net export if export were to increase by 6.25.*

---

18. *An economy is characterised by the following equation-*

Consumption	*C = 60+0.9Y_d*
Investment	*I = 10*
Government expenditure	*G = 10*
Tax	*T = 0*
Exports	*X = 20*
Imports	*M = 10 +0.05 Y*

What is the equilibrium income?
Calculate trade balance and foreign trade multiplier.

III. Application Oriented Questions

1. *Assume that an economy's consumption function is specified by the equation C = 500 + 0.80Y.*
(a) What will be the consumption when disposable income (Y) is ₹ 4,000, ₹ 5,000 and ₹6,000?
(b) Find saving when disposable income is ₹ 4,000, ₹ 5,000 and ₹6,000?
(c) What amount of consumption for consumption function C is autonomous?
(d) What amount is induced when disposable income is ₹ 4,000, ₹ 5,000 and ₹6,000?

2. *Find the value of multiplier when*
(a) MPC is 0.2 (b) MPC is 0.5 (c) MPC is 0.8

3. *For the linear consumption function C = 700 + 0.8Y; I is ₹ 1200 and Net exports X-M = ₹ 100. Find equilibrium output?*

IV. Past Exam Questions

1. *Calculate the Marginal Propensity to Consume (MPC) and Marginal Propensity to Save (MPS) from the following data:* ***[PEQ-May 18 -2 Marks]***

Income (Y)	*Consumption (C)*	*Level*
₹ 8,000	*₹ 6,000*	*Initial level*
₹ 12,000	*₹ 9,000*	*Changed level*

[ANS. MPC = 0.75 ; MPS = 0.25]

2. *Suppose in an economy:*

Consumption Function	$C = 150 + 0.75\ Y_d$
Investment spending	$I = 100$
Government spending	$G = 115$
Tax	$Tx = 20 + 0.20\ Y$
Transfer Payments	$Tr = 40$
Exports	$X = 35$
Imports	$M = 15 + 0.1\ Y$

Where, Y and Y_d are National Income and Personal Disposable Income respectively. All figures are in rupees. Find:

(a) *The equilibrium level of National Income*

(b) *Consumption at equilibrium level*

(c) *Net Exports at equilibrium level* ***[PEQ-May 18- 5 Marks]***

[ANS. (a) ₹ 800 ; (b) ₹ 645 ; (c) ₹ (60) : adverse balance of trade]

3. *Calculate the Average Propensity to Consume (APC) and Average Propensity to Save (APS) from the following data:* ***[PEQ-Nov 18 - 2 Marks]***

Income (Y)	*Consumption (C)*
₹ 4,000	*₹ 3,000*

[ANS. APC = 0.75 ; APS = 0.25]

4. *Given Consumption Function C = 300 + 0.75Y ; Investment = ₹ 800; Net Imports = ₹ 100. Calculate equilibrium level of output?* ***[PEQ-May 19 - 3 Marks]***

[ANS. Equilibrium Output = ₹ 4,000]

5. *When investment in an economy increased from ₹ 10,000 crores to ₹ 14,000 crores and as a result of this national income rises from ₹ 80,000 crores to ₹ 92,000 crores. Compute Investment Multiplier?* ***[PEQ-May 19 - 2 Marks]***

[ANS. Investment Multiplier = 3]

ANSWERS / HINTS

I. Multiple Choice Type Questions

1.	(a)	2.	(b)	3.	(c)	4.	(a)	5.	(b)	6.	(a)	7.	(c)	8.	(c)
9.	(c)	10.	(b)	11.	(c)	12.	(c)	13.	(b)	14.	(c)	15.	(b)	16.	(d)
17.	(c)	18.	(a)	19.	(b)	20.	(d)	21.	(a)	22.	(b)	23.	(b)	24.	(b)
25.	(b)	26.	(d)	27.	(b)	28.	(c)	29.	(c)	30.	(a)	31.	(a)	32.	(c)
33.	(b)	34.	(b)	35.	(d)	36.	(d)	37.	(a)	48.	(c)	39.	(b)	40.	(d)
41.	(b)	42.	(c)	43.	(a)	44.	(c)	45.	(a)	46.	(b)	47.	(b)	48.	(a)
49.	(b)	50.	(b)	51.	(d)	52.	(b)	53.	(c)						

II. Illustration

1.

(i) $APS = S/Y$;

$S = Y - C$

$= 1,000 - 200 = 800.$

Therefore,

$APS = S/Y$

$= 800/1000 = 0.8$

(ii) When $S = 450$ and $Y = 1,200$;

$APS = S = 450/1200 = 0.375$

2. $Y = C + I$

By putting the value we get,

$2500 = C + 100$

$C = 2500 - 100 = 2400$

$C = a + bY$

$2400 = 300 + 2500\ b$

$2400 - 300 = 2500b$

$b = 0.84;$

$MPS = 1 - MPC$

$= 1 - 0.84 = 0.16$

3. $Y = C + I$

$Y = (a + bY) + I$

where

$MPC = 1 - MPS$

$Y = 100 + 0.8Y + 200$

$= 300 + 0.8Y$

$Y - 0.8Y = 300$

$0.2Y = 300,$

$Y = 1500$

4. $Y = C + I$
$= 20 + 0.6\,Y + 10 + 0.2\,Y$
$Y = 30 + 0.8\,Y$
$Y - 0.8\,Y = 30$
$Y = 150$

---

5. *Equilibrium Condition–*
$Y = C + I,$
Given $C = 7 + 0.5Y$ and $I = 100$
Therefore
$Y = 7 + 0.5Y + 100$
$Y - 0.5Y = 107$
$Y = 107 * 2 = 214$
$Y = C + I,$
$214 = C + 100$
$C = 114$
$S = Y - C = 100$

---

6. $Y = \frac{1}{1-b}(a + \bar{I})$ *or* $Y = C + I$
$Y = \frac{1}{1-.80}(250 + 300) = 2750$
$C = \frac{a + b}{1-b}(a + \bar{I})$ *or* $C = 250 + 0.80\,Y$
$C = 250 + 0.8(2750)$
$C = 2450$
$S = Y - C$ *where* $C = a + bY$
$S = Y - (a + bY)$
$S = -a + (1 - b)\,Y$
$= -250 + (1 - 0.80)2750$
$= 300$
Or directly, $S = Y - C$
$S = 2750 - 2450 = 300.$

---

7. $S = I$
$-10 + 0.2Y = 50$
$0.2Y = 50 + 10$
$Y = 300$ Crores
$C = Y - S$
Where $S = -10 + 0.2\,(300) = 50$
$C = 300 - 50$
$= 250$ Crores
With the increase in investment by ₹ 5 Crores, the new investment will become equal to ₹ 55 Crores.
$S = I$
$-10 + 0.2Y = 55$
$Y = 325$ Crores
$C = 270$ Crores

8. $C = 100+0.75Y$ *and*
$I = 1000$,
$Y = C+I$ *in equilibrium*
$Y = 100+0.75\ Y + 1000$
$Y = \frac{1}{1-0.75}(100+1000)$
$Y = \frac{1}{1-0.75}(1100)$
$= 1/0.25\ (1100)$
$= 4400.$
$Y = C + I;$
$C = 4400 - 1000$
$= 3400$

9. $MPC = 0.8;$
$\Delta I = 400$ *Crores*
Multiplier
$(K) = 1/1- MPC$
$= 1/1- 0.8$
$= 1/0.2$
$= 5$
$MPS = 1 - MPC$
$= 1 - 0.8$
$= 0.2$
Increase in income
$(\Delta Y) = K \times \Delta I$
$= 5 \times 400 = 2{,}000$ *Crores*
Increase in saving $= \Delta Y \times MPS$
$= 2{,}000 \times 0.2$
$= 400$ *Crores*

10. *Increase in investment* $(\Delta I) = 400$ *Crores*
Increase in national income $(\Delta Y) = 1{,}600$ *Crores*
Multiplier $(K) = \Delta Y/\Delta I$
$K = 1{,}600/400 = 4$
We know, $K = 1/1 - MPC$
$4 = 1/1 - MPC$
$MPC = 0.75$

11. $MPC = 0.6;\ \Delta I = ₹\ 600$ *Crores*
Multiplier $(K) = 1/1- MPC$
$= 1/1 - 0.6 = 1/0.4 = 25.$
Increase in income $(\Delta Y) = K \times \Delta I$
$= 2.5 \times$ *Rs 600 Crores* $= ₹\ 1{,}500$ *Crores*
Increase in consumption $(\Delta C) = \Delta Y \times MPC = Rs1,$
$500 Crores \times 0.6 = ₹\ 900$ *Crores.*

12. *Multiplier = k = $\frac{1}{1 - MPC}$* $k = \frac{1}{-.6} = 2.5$

Substituting the value of k and ΔI value in ΔY= kΔI

ΔY= 2.5 X 100= ₹ 250Crores

Thus, increase in investment by Rs 100 Crores will cause equilibrium income to rise by ₹ 250 Crores.

13.

(a) Since G = T, budget of the government is balanced

Substituting the values of C, I and G in Y we have

$Y = C+I+G$

$Y = a + bY_d + I + G$

$Y = 10 + 0.75 (Y - 20) + 50 + 20$

$Y = 10 + 0.75 Y - 15 + 50 + 20$

$or, Y - 0.75 Y = 65$

$or, Y (1 - 0.75) = 65$

$or, 0.25 Y = 65$

$or, Y = 65 / .25 = 260$

The equilibrium value of

$Y = 260$

(b) The value of the multiplier is $= 1/ (1 - MPC)$

$= 1/ (1 - b)$

$= 1/ (1 - 0.75)$

$= 1/0.25 = 4$

14. $Y = C+ I+ G$

$Y = 100+ 0.75 Y_d + 200 + 100$

$Y = 100 + 0.75(Y - 100 + 50) + 200 + 100$

$Y = 100 + 0.75Y - 75 + 37.5 + 200 + 100$

$Y = 1450$

Or use $Y = \frac{1}{1-b} (a - bT + bTR + I + G)$ *to calculate income.*

15.

a) $Y = C+ I+G$

$Y = 75+ 0.5(Y- 25- 0.1Y) + 80+ 100$

$Y (1-0.5+0.05) = 75-12.5+80+100$

$Y = \frac{1}{1-0.5+0.05}(242.5)$

$Y = 440.91$

b) Multiplier = 1 / [1-b (1-t)]

$= 1 / [1-0.5(1-0.1)] = 1.82$

16. $Y = C + I + G$

$Y = 100 + 0.80 (Y- T + TR) + I+ G$

$Y = 100 + 0.80(Y - 25 - 0.1Y + 50) + 200 + 100$

$Y - 0.80\ Y + 0.08\ Y = 420$
$Y(1-0.8+0.08) = 420$
$Y = 1500$

---

17. $C = 40 + 0.8Yd$
$C = 40 + 0.8\ (Y - 0.1Y)$
$Y = C + I + G + (X - M)$
$Y = 40+0.8(Y-0.1Y)+60+40+(58-0.05Y)$
$Y = 40+0.8(0.9Y)+60+40+58-0.05Y$
$Y - 0.72Y + 0.05Y = 198$
$Y(1-0.72+0.05) = 198$
$Y(0.33) = 198$
$Y = 198/0.33$
$= 600$ Crores
Net Export $= X - M$
$= 58 - 0.05Y$
$58 - 0.05\ (600) = 58 - 30 = 28$
If exports increase by 6.25, then exports $= 64.25$
$Y = 40 + 0.8\ (Y - 0.1Y) + 60 + 40 + (64.25 - 0.05Y)$
$Y\ (1-0.72+0.05) = 204.5$
$Y\ (0.33) = 204.5$
$Y = 204.5/0.33 = 619.697$
Then import $= .05 \times 619.697 = 30.98$
Net Export $= 64.25 - 30.98 = 33.27$ Crores
Thus, there is surplus in balance of trade as Net Exports are positive.

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18. $Y = C + I + G + (X - M)$
$= 60 + 0.9(Y - 0) + 10 + 10 + (20 - 10 - 0.05Y)$
$= 60 + 0.9\ Y + 30 - 0.05\ Y$
$Y = 600$
Trade Balance = $X - M$ $= 20 - 10 - 0.05(600) = -20$
Thus, trade balance in deficit.
Foreign trade multiplier $= 1 / 1 - b + m$
$= 1 / 1 - 0.9 + 0.05$
$= 6.66$

III. Application Oriented Questions

1.

(a) Consumption for each level of disposable income is found by substituting the specified disposable income level into the consumption equation.
Thus, for Y = ₹ 4,000, C = ₹ 500 + 0.80(₹ 4,000) = ₹ 500 + ₹ 3,200 = ₹ 3,700.
Likewise C is ₹ 4,500 when Y = ₹ 5,000, and ₹ 5,300 when Y = ₹ 6,000.

(b) Saving is the difference between disposable income and consumption. It is the difference between the consumption line and the 45 line at each level of disposable income
Using the calculation from part a) above, we find that saving is ₹ 300 when Y is ₹ 4,000; ₹ 500 when Y

is ₹ 5,000 and ₹ 700 when Y is ₹ 6,000.

(c) Autonomous consumption is the amount consumed when disposable income is zero; autonomous consumption is ₹ 500, i.e. the consumption expenditure when the consumption line C intersects the vertical axis and disposable income is 0.

Since autonomous consumption is unrelated to income, autonomous consumption is ₹ 500 for all levels of income.

(d) Induced consumption is the amount of consumption that depends upon the level of income. Consumption is ₹ 3,700 when disposable income is ₹ 4,000. Since ₹ 500 is autonomous (i.e. consumed regardless of the income level), ₹ 3,200 out of the ₹ 3,700 level of consumption is induced by disposable income.

Similarly, Induced consumption is ₹ 4,000 when disposable income is ₹ 5,000, and ₹ 4,800 when disposable income is ₹6,000

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2. The value of the multiplier (k) is found by relating the change in output (ΔY) to the initial change in aggregate spending. The value of the multiplier is directly related to the level of MPC, i.e., the greater the MPC, the larger the value of the multiplier. The value of the multiplier is found from the equation k = 1/ (1− MPC).

(a) When MPC is 0.2, the multiplier is 1.25

(b) When MPC is 0.5, the multiplier is 2

(c) When MPC = 0.80, the multiplier is 5

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3. The Equilibrium level of output can be found by equating output and aggregate spending i.e. by solving

Y	*= C + I + X – M*
Y	*= 700 + 0.8Y + 1200 + 100*
Y – 0.8Y	*= 700 + 1200 +100*
0.2Y	*= 2000*
Y = 2000 / 0.2	*= 10,000*

For Your Notes

For Your Notes

For Your Notes

For Your Notes

CHAPTER - 07

PUBLIC FINANCE

Unit 1 – Fiscal Function: An Overview Centre and State Finance

Unit 2 – Market Failure / Government Intervention to Correct Market Failure

Unit 3 – The Process of Budget Making: Sources of Revenue, Expenditure Management and Management of Public Debt

Unit 4 – Fiscal Policy

UNIT – I

FISCAL FUNCTION: AN OVERVIEW CENTRE AND STATE FINANCE

Topic References:

1.1 Introduction
1.2 The Role of Government in an economic system
1.3 Allocation Function
1.4 Redistribution Function
1.5 Stabilization Function

1.1 Introduction

The governments of all nations have important economic functions even where markets constitute the basic resource allocation mechanism.

The primary goal of the state is to promote the general welfare of the society.

Governments at various levels involve in several operations for running the state.

For example; the government raises money from various sources, incurs expenditures, consumes goods and services, borrows money, employs people, and provides key institutions such as property rights. The governments also establish and administer rules and regulations and puts in place policies concerning all aspects of life of people.

There are three main macroeconomic goals for any nation.

The first is economic growth. If the real gross domestic product grows at a faster rate than population, then people can enjoy higher standard of living.

The second goal is high levels of employment which will ensure higher income and higher output. When unemployment occurs, it harms not only the unemployed, but the society as a whole because there is loss of output that could have been produced.

The third macroeconomic goal is stable price levels. Inflation reduces real incomes and purchasing power of some people, and disproportionately affects lower income families. On the contrary, deflation signals a downturn in economic activity which may cause recession or even depression and large scale unemployment.

The government does not expect the economy to function automatically; rather it intervenes to direct them to function in particular directions.

1.2 The Role of Government in an economic system

Why an economic system should be in place?

The basic economic problem of scarcity arises from the fact that on account of qualitative as well as quantitative constraints, the resources available to any society cannot produce all economic goods and services that its members desire to have.

Therefore, an economic system should exist to answer the basic questions such as what, how and for whom to produce and how much resources should be set apart to ensure growth of productive capacity.

The modern society, in general, offers three alternate economic systems through which the decisions of resource reallocation may be made namely - the market, the government and a mixed system where both markets and governments simultaneously determine resource allocation.

Adam Smith, a bold advocate of free markets and minimal governmental activity, saw an important resource allocation role for government when he underlined the role of government in national defence, maintenance of justice and the rule of law, establishment and maintenance of highly beneficial public institutions and public works which the market may fail to produce on account of lack of sufficient profits.

Since the 1930s, more specifically as a consequence of the great depression, the state's role in the economy has been distinctly gaining in importance and therefore, the traditional functions of the state as described above, have been supplemented with what is referred to as economic functions (also called fiscal functions or public finance function)

Richard Musgrave, in his classic treatise 'The Theory of Public Finance' (1959), introduced the three branch taxonomy of the role of government in a market economy.

- Resource allocation, (efficiency),
- Income redistribution (fairness) and
- Macroeconomic stabilization.

The allocation and distribution functions are primarily microeconomic functions, while stabilization is a macroeconomic function.

1.3 Allocation Function

- ✓ Resource allocation refers to the way in which the available factors of production are allocated among the various uses to which they might be put.
- ✓ It determines how much of the various kinds of goods and services will actually be produced in an economy.
- ✓ One of the most important functions of an economic system is the optimal or efficient allocation of scarce resources so that the available resources are put to their best use and no wastages are there.
- ✓ The private sector resource allocation is characterized by market supply and demand and price mechanism as determined by consumer sovereignty and producer profit motives.
- ✓ The state's allocation is accomplished through the revenue and expenditure activities of governmental budgeting.

- ✓ In the real world, resource allocation is both market determined and government determined.
- ✓ There is also the problem of non-existence of markets in a variety of situations.
- ✓ While private goods will be sufficiently provided by the market, public goods will not be produced in sufficient quantities by the market.

Why do markets fail to give right answers to the question as to what goods should be produced and in what quantities? Or, why do markets generate misallocation of resources?

Efficient allocation of resources is assumed to take place only in perfectly competitive markets. In reality, markets are never perfectly competitive.

Market failures which hold back the efficient allocation of resources occur mainly due to the following reasons:

- ***Market Power -*** Imperfect competition and presence of monopoly power in different degrees leading to under-production and higher prices than would exist under conditions of competition.
- ***Externalities -*** Externalities which arise when the production and consumption of a good or service affects people and they cannot influence through markets the decision about how much of the good or service should be produced e.g. pollution.
- ***Public Goods -*** Markets typically fail to provide collective goods which are, by their very nature, consumed in common by all the people.
- ***Incomplete Information -*** Imperfect information leads to market failure.
- ***Factor Immobility -*** which causes unemployment and inefficiency
- ***Financial Gap -*** Inequalities in the distribution of income and wealth.

In the absence of appropriate government intervention, market failures may occur and the resources are likely to be misallocated by too much production of certain goods or too little production of certain other goods.

In many cases, the government can provide us with goods and services that we cannot produce on our own or buy at a price from the market.

For example, the government establishes property rights and makes the necessary arrangements for enforcing contracts through provision of law enforcement and courts.

Goods which involve externalities that are not met by the market require intervention by the government for corrective measures.

Merit goods which are greatly beneficial to the society also fall under the purview of provision by the government.

These interventions do not imply that markets are replaced by government action.

In its allocation role, the government acts as a complement rather than as a substitute to the market system in an economy.

The resource allocation role of government's fiscal policy focuses on the potential for the government to improve economic performance through its expenditure and tax policies.

The allocative function in budgeting determines

(a) Who and what will be taxed

(b) How and on what the government revenue will be spent.

(c) It is concerned with the provision of public goods

(d) The process by which the total resources of the economy are divided among various uses

(e) An optimum mix of various social goods (both public goods and merit goods).

(f) The reallocation of society's resources from private use to public use.

A variety of allocation instruments are available by which governments can influence resource allocation in the economy like -

- government may directly produce the economic good (For example, electricity and public transportation services)
- government may influence private allocation through incentives and disincentives (For example, tax concessions and subsidies may be given for the production of goods that promote social welfare and higher taxes may be imposed on goods such as cigarettes and alcohol)
- government may influence allocation through its competition and merger policies which will affect the structure of industry and commerce (For example, the Competition Act in India promotes competition and prevents anti-competitive activities)
- governments' regulatory activities such as licensing, controls, minimum wages, and directives on location of industry influence resource allocation
- government sets legal and administrative frameworks, and
- any of a mixture of intermediate techniques may be adopted by governments

Maximizing social welfare is one of the primary and most commonly manifest reasons for government intervention in the market.

However, it is also possible that instead of eliminating market distortions, sometimes governments may contribute to generate them. The possible sources of this type of government failures are inadequate information, conflicting objectives and administrative costs involved in government intervention.

1.4 Redistribution Function

Over the past decades there has been tremendous expansion in economic activities which has generated enormous increase in aggregate output and wealth. However, the outcomes of this growth have not spread evenly across the households.

A major function of present-day governments therefore involves changing the pattern of distribution of income from what the market would offer to a more egalitarian one.

The distribution responsibility of the government arises from the fact that, left to the market, the distribution of income and wealth among individuals in the society is likely to be skewed and therefore the government has to intervene to ensure a more desirable and just distribution.

The distributive function of budget is related to the basic question of for whom should an economy produce goods and services.

As such, it is concerned with the adjustment of the distribution of income and wealth so as to ensure distributive justice namely, equity and fairness.

Effective demand is determined by the level of income of the households and this in turn determines the distribution of real output among the population.

The distribution function of the government aims at:

- redistribution of income to achieve an equitable distribution of societal output among households
- advancing the well-being of those members of the society who suffer from deprivations of different types
- providing equality in income, wealth and opportunities

- providing security for people who have hardships, and
- ensuring that everyone enjoys a minimal standard of living

Examples of the redistribution function performed by governments are:

- taxation policies of the government whereby progressive taxation of the rich is combined with provision of subsidy to the poor households
- proceeds from progressive taxes used for financing public services, especially those that benefit low-income households (example, supply of essential food grains at highly subsidized prices to BPL households)
- employment reservations and preferences to protect certain segments of the population,
- regulation of the manufacture and sale of certain products to ensure the health and well-being of consumers, and
- special schemes for backward regions and for the vulnerable sections of the population

In modern times, most of the egalitarian welfare states provide free or subsidized education and health-care system, unemployment benefits, pensions and such other social security measures.

There is, nevertheless, an argument that in exercising the redistributive function, there exists a conflict between efficiency and equity.

Governments' redistribution policies which interfere with producer choices or consumer choices are likely to have efficiency costs or deadweight losses.

For example, greater equity can be achieved through high rates of taxes on the rich; but high rates of taxes could also act as a disincentive to work, and discourage people from savings and investments and risk taking. This in turn will have negative consequences for productivity and growth of the economy. Consequently, the potential tax revenue may be reduced and the scope for government's welfare activities would get seriously limited.

1.5 Stabilization Function

Macroeconomic stability is said to exist when:

- An economy's output matches its production capacity,
- The economy's total spending matches its total output
- The economy's labour resources are fully employed, and
- Inflation is low and stable.

A market economy does not automatically generate full employment and price stability and therefore the governments should pursue deliberate stabilization policies.

Business cycles are natural phenomena in any economy and they tend to occur periodically.

The market mechanism is limited in its capacity to prevent or to resolve the disruptions caused by the fluctuations in economic activity.

In the absence of appropriate corrective intervention by the government, the instabilities that occur in the economy in the form of recessions, inflation etc. may be prolonged for longer periods causing enormous hardships to people especially the poorer sections of society.

It is also possible that a situation of stagflation (a state of affairs in which inflation and unemployment exist side by side) may set in and make the problem more intricate.

The stabilization issue also becomes more complex as the increased international interdependence causes forces of instability to get easily transmitted from one country to other countries. This is also known as contagion effect.

The stabilization function is one of the key functions of fiscal policy and aims at eliminating macroeconomic fluctuations arising from suboptimal allocation.

The stabilization function is concerned with the performance of the aggregate economy in terms of:

- Labour employment and capital utilization,
- Overall output and income,
- General price levels,
- Balance of international payments, and
- The rate of economic growth.

Government's fiscal policy has two major components which are important in stabilizing the economy:

1. an overall effect generated by the balance between the resources the government puts into the economy through expenditures and the resources it takes out through taxation, charges, borrowing etc.
2. a microeconomic effect generated by the specific policies it adopts.

Government's stabilization intervention may be through monetary policy as well as fiscal policy.

Monetary policy has a singular objective of controlling the size of money supply and interest rate in the economy which in turn would affect consumption, investment and prices.

Fiscal policy for stabilization purposes attempts to direct the actions of individuals and organizations by means of its expenditure and taxation decisions.

On the expenditure side, Government can choose to spend in such a way that it stimulates other economic activities.

For example, government expenditure on building infrastructure may initiate series of productive activities. Production decisions, investments, savings etc. can be influenced by its tax policies.

Government expenditure injects more money into the economy and stimulates demand. On the other hand, taxes reduce the income of people and therefore, reduce effective demand.

- During recession, the government increases its expenditure or cuts down taxes or adopts a combination of both so that aggregate demand is boosted up with more money put into the hands of the people.
- On the other hand, to control high inflation the government cuts down its expenditure or raises taxes or adopts a combination of both.

In other words, expansionary fiscal policy is adopted to alleviate recession and contractionary fiscal policy is resorted to for controlling high inflation.

The nature of the budget (surplus or deficit) also has important implications on a country's economic activity. While deficit budgets are expected to stimulate economic activity, surplus budgets are thought to slow down economic activity.

There is often a conflict between the different goals and functions of budgetary policy. Effective policy design to meet the diverse goals of government is very difficult to conceive and to implement. The challenge before any government is how to design its budgetary policy so that the pursuit of one goal does not jeopardize the other.

Centre and State Finance

Fiscal federalism, a term introduced by Richard Musgrave, deals with the division of governmental functions and financial relations among the different levels of government.

Musgrave argued that the federal or central government should be responsible for performing functions related to economic stabilization and income redistribution, and the allocation of resources should be the responsibility of the state and local governments.

India is a federation of 28 states and 8 union territories.

Fundamentally, federalism is an institutional arrangement to accommodate two sets of government — one at the national level and the other at the regional level.

Each government is autonomous in its own sphere. An independent judiciary is established to resolve disputes between the central government and the states on issues related to division of power.

The constitution of India has provided for the division of powers between the central and the state governments.

Article 246 of the Constitution demarcates the powers of the union and the state by classifying their powers into three lists, namely union list, state list and the concurrent list.

The union list contains items on which the union parliament alone can legislate, the state list has items on which the state legislative assemblies alone can legislate, and the concurrent list, on which both the parliament and the legislative assemblies can legislate.

In the event of conflicting legislation in concurrent list, the law passed by the centre prevails. Taxes are levied by the centre and the states.

The union government can levy taxes such as tax on income, other than agricultural income, customs and export duties, excise duties on certain goods, corporation tax, tax on capital value of assets excluding agricultural land, terminal taxes, security transaction tax, central GST, union excise duty, taxes other than stamp duties etc.

The state governments can levy taxes on agricultural income, lands and buildings, mineral rights, electricity, vehicles, tolls, professions, collect land revenue and impose excise duties on certain items. The property of the union is exempt from state taxation. The property and income of the states are not liable to be taxed by the centre.

A significant element of fiscal federalism is inter-governmental transfers and revenue-sharing to fulfill diverse national objectives.

Since the states have comparatively less sources of income, their revenues may not be sufficient to meet their expenditure responsibilities.
Articles 268 to 281 of the constitution contain specific provisions in respect of distribution of finances among states.

Distribution of revenue between the union and states is based on the constitutional provisions as follows:

Article 268	Duties levied by the union but collected and appropriated by the states.
Article 269	Taxes levied and collected by the union but assigned to the states.
Article 270	Taxes levied and collected by the union and distributed between the union and states as prescribed in clause 2 and the States.
Article 271	Surcharge on certain duties and taxes for purposes of the union
Article 275	Statutory Grants – in–aid from the union to certain states.
Article 282	Grants for any public purpose
Article 293	Loans for any public purpose

Article 280 provides for an institutional mechanism, namely the Finance Commission, to facilitate such transfers.

The Finance Commission is a constitutionally mandated body that is at the centre of fiscal federalism.

It is responsible for evaluating the state of finances of the union and state governments, recommending the sharing of taxes between them and laying down the principles determining the distribution of these taxes among states.

The Finance Commission helps in maintaining fiscal federalism in India by performing following functions:

(a) The distribution between the union and the states of the net proceeds of taxes which are to be divided between them and the allocation between the states of the respective shares of such proceeds.
(b) Determination of principles and quantum of grants-in-aid to states which are in need of such assistance.
(c) To make recommendations to the President on measures needed to augment the consolidated fund of a state to supplement the resources of the panchayats and municipalities in the state on the basis of the recommendations made by the Finance Commission of the state.
(d) Any other matter referred to the Commission by the President in the interests of sound finance.

While recommending transfers, the Finance Commission considers issues related to vertical equity (deciding about the share of all states in the revenue collected by centre) and horizontal equity (allocation among states their share of central revenue).

The Finance Commission broadly assesses the overall gross tax revenues of the union; cesses, surcharges and non-tax revenue are netted out from gross tax revenue to arrive at the net divisible pool (NDP).

The 15th Finance Commission was constituted on 27, November 2017 against the background of the abolition of Planning Commission (as also of the distinction between Plan and non-Plan expenditure) and the introduction of the goods and services tax (GST).

The commission recommended the share of states in the central taxes (vertical devolution) for the 2021-26 to be 41%, which is the same as that for 2020-21.

This is less than the 42% share recommended by the 14th Finance Commission for 2015-20.

The adjustment of 1% is to provide for the newly formed union territories of Jammu and Kashmir, and Ladakh from the resources of the centre.

The criteria for distribution of central taxes among states for 2021-26 period are same as that for 2020-21. They are:

(a) Income Distance i.e the distance of a state's income from the state with the highest income.

(b) Area

(c) Population (2011)

(d) Demographic performance (to reward efforts made by states in controlling their population)

(e) Forest and ecology:

(f) Tax and fiscal efforts:

The introduction of GST, which was rolled out across the country on 1 July 2017, has significantly changed the state of affairs of financial relations between the centre and states.

The GST subsumes the majority of indirect taxes – excise, services tax, sales tax, octroi (entry tax).

The states levy and collect state GST (SGST) and the union levies and collects the central GST (CGST). For any particular good or service or a combination of the two, the SGST and CGST rates are equal.

An integrated GST (IGST) is applied on inter-state movement of goods and services and on imports and exports.

IGST is simply a combination of SGST and CGST administered and collected by the union government, kept in a separate account, and distributed between the union and states, GST accounts for 35 per cent of the gross tax revenue of the union and around 44 per cent of own tax revenue of the states.

The GST system replaced the then prevailing production-based taxation system with a consumption based one.

Since the manufacturing states had apprehension about loss of revenue, it was decided to provide compensation to states for loss of revenue arising on account of implementation of the Goods and Services Tax for a period of 5 years from the date of its implementation.

For providing compensation to states, a cess is levied on some luxury goods and demerit goods and the proceeds are credited to the compensation fund. GST compensation was extended beyond 5 years to enable states to tide over the pandemic induced economic slowdown.

During the five-year transition period, the top five GST compensation-receiving states were Maharashtra, Karnataka, Gujarat, Tamil Nadu, and Punjab.

The total amount of compensation released to the states and union territories during the year 2022 -23 is 1,15,662 crore.

In so far as expenditure decentralization is concerned,

the central government is entrusted with the responsibilities of provision of nationally important areas like defence, foreign affairs, foreign trade and exchange management, money and banking, cross-state transport and communication.

The state governments are entrusted with the responsibility of facilitating agriculture and industry, providing social sector services such as health and education, police protection, state roads and infrastructure.

The **loc**al self -governments such as municipalities and panchayats are entrusted with the responsibility of providing public utility services such as water supply and sanitation, local roads, electricity etc. For items that fall in the concurrent list, both central and state governments are responsible for providing services.

Borrowing by the government of India and borrowing by states are defined under Article 292 and 293 of Constitution of India.

The centre may borrow within the limits fixed by parliament by law upon the security of the Consolidated Fund of India or give guarantees within such limits, if any.

The state governments may borrow within the territory of India upon the security of the Consolidated Fund of the State within such limits, if any, as may from time to time be fixed by the Legislature of such state by law, or give guarantees within such limits.

The centre may give loans to the states within limits fixed under article 292 and give guarantees in respect of loans raised by the states.

States need to obtain the centre's consent in order to borrow in case the state is indebted to the centre over a previous loan.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. Redistribution policies are likely to have efficiency costs because
   (a) *They will reduce the efficiency of governments*
   (b) *They may create disincentives to work and save*
   (c) *Governments have to forego taxes*
   (d) *They are likely to make the poor people dependent on the rich*

2. Macroeconomic stabilization may be achieved through
   (a) *Free market economy* (b) *Fiscal policy*
   (c) *Monetary policy* (d) *(b) and (c) above*

3. Which of the following policies of the government fulfils the redistribution function
   (a) *Parking the army on the northern borders of the country*
   (b) *Supply of food grains at subsidized prices to the poor people*
   (c) *Controlling the supply of money through monetary policy*
   (d) *All of the above*

4. Choose the correct statement
   (a) *Fiscal policy involves the use of changes in taxation and government spending; while monetary policy involves the use of price and profit controls.*
   (b) *Fiscal policy involves the use of price and profit controls; while monetary policy involves the use of taxation and government spending.*
   (c) *Fiscal policy involves the use of changes in taxation and government spending; while monetary policy involves the use of changes in the supply of money and interest rates.*
   (d) *Fiscal policy involves the use of changes in the supply of money and interest rates; while monetary policy involves the use of changes in taxation and government spending.*

5. The justification for government intervention is best described by
   (a) *The need to prevent recession and inflation in the economy*
   (b) *The need to modify the outcomes of private market actions*
   (c) *The need to bring in justice in distribution of income and wealth*
   (d) *All the above*

6. Read the following statements:

I. The market-generated allocation of resources is usually imperfect and leads to inefficient allocation of resources in the economy

II. Market failures can at all times be corrected through government intervention

III. Public goods will not be produced in sufficient quantities in a market economy

Of the three statements above:
   (a) *I, II and III are correct* (b) *I and III are correct*
   (c) *II and III are correct* (d) *III alone is correct*

7. When a government offers unemployment benefits and also resorts to progressive taxation which function does it seem to fulfill?
   (a) *It is trying to establish stability in an economy*
   (b) *It is trying to redistribute income and wealth*

(c) *It is trying to allocate resources to their most efficient use*
(d) *It is creating a source of market failure*

8. Government of Emeline Land decides to provide most modern road infrastructure throughout the nation. This can be classified as
(a) *Distribution function* (b) *Allocation function*
(c) *Stabilization function* (d) *None of the above*

9. Which function does the government perform when it provides transfer payments to offer support to the underprivileged
(a) *Allocation* (b) *Efficiency* (c) *Distribution* (d) *None of the above*

10. Which of the following is true in respect of centre and state government finances?
(a) *The centre can tax agricultural income and mineral rights*
(b) *Finance commission recommends distribution of taxes between the centre and states*
(c) *GST subsumes majority of direct taxes and a few indirect taxes*
(d) *IGST is collected by the state governments*

11. GST compensation is given to
(a) *to the industries which have made losses due to the introduction of GST*
(b) *to compensate for the lower rates of GST on essential items*
(c) *to the states to compensate for the loss of revenue due to the introduction of GST*
(d) *to compensate for the loss of input tax credit in manufacturing*

12. Which of the following is true in respect of the role of Finance Commissions in India?
I. The distribution between the union and the states of the net proceeds of taxes
II. Allocation between the states of the respective shares of such proceeds.
III. Make Recommendations on integrated GST on inter-state movement of goods and services
IV. To recommend expenditure decentralization among different states
(a) *I and II are correct* (b) *II and III are correct*
(c) *I, II and III are correct* (d) *All the above are correct*

13. In a federal set up, the stabilization function can be effectively performed by
(a) *Respective state governments* (b) *Ministry of taxes*
(c) *The government at the centre* (d) *None of the above*

14. Which of the following is concerned with division of economic responsibilities between the central and state Government of India?
(a) *NITI Aayog* (b) *central bank*
(c) *Finance Commission* (d) *Parliament*

15. Fiscal Federalism refers to _____.
(a) *Organizing and implementing development plans*
(b) *Sharing of political power between centers and states*
(c) *The management of fiscal policy by a nation*
(d) *Division of economic functions and resources among different layers of the government*

16. Which one of the following taxes is levied by the state government only?
(a) *Corporation tax* (b) *Wealth tax*
(c) *Income tax* (d) *None of the above*

17. The percentage of share of states in central taxes for the period 2021-26 recommended by the Fifteenth Finance Commission is
(a) *38 percent* (b) *41 percent*
(c) *42 percent* (d) *The commission has not submitted its report*

18. Which of the following is not a criterion for determining distribution of central taxes among states for 2021-26 period
(a) *Demographic performance* (b) *Forest and ecology*
(c) *Infrastructure performance* (d) *Tax and fiscal efforts*

19. As per the supreme court verdict in May 2022
(a) *The union has greater powers than the states for enacting GST laws*
(b) *The union and state legislatures have "equal, simultaneous powers "to make laws on Goods and Services Tax*
(c) *The union legislature's enactments will prevail in case of a conflict between those of union and states*
(d) *The state legislatures can make rules only with the permission of central government*

20. Providing social sector services such as health and education is
(a) *the responsibility of the central government*
(b) *the responsibility of the respective state governments*
(c) *the responsibility of local administrative bodies*
(d) *none of the above*

21. When the government decides to produce fertilizers and supply them to the agriculturists, it aims
(a) *to achieve equity and fairness to the agriculturists*
(b) *to influence the way resources are allocated in the economy*
(c) *to ensure higher profits to agriculturists*
(d) *to make greater profits for the public sector*

22. The allocation and distribution functions are primarily :
(a) *Micro-economic functions*
(b) *Macro-economic functions*
(c) *both micro as well as macro-economic functions*
(d) *aimed at bringing in price stability and economic growth*

PEQ -JUNE - 2024

23. Allocative and distributive functions of a government is a _____ economic function.
(a) micro (b) macro (c) both (d) none (1 mark)

24. Who is responsible for economic stabilization and the income redistribution in an economy?
(a) State Government *(b) Central Government*
(c) Central and State *(d) Central, State, local bodies* *(1 mark)*

25. In a market economy, the allocation of resources is primarily determined by:
(a) *Central planning by the government* (b) *Consumer preferences and demand*
(c) *The availability of natural resources* (d) *The level of government spending* *(1 mark)*

PEQ - SEPTEMBER - 2024

26. Which Articles of the Indian Constitution contain specific provisions regarding the distribution of finances among states?
(a) Articles 248 to 257 (b) Articles 356 to 369
(c) Articles 268 to 281 (d) Articles 292 to 304 (1 mark)

27. Richard Musgrave introduced a three-branch taxonomy describing the role of government in a market economy. What are these branches?
(a) Resource allocation, income redistribution, and macro economic stabilization
(b) Resource allocation, income generation, and price stability
(c) Production efficiency, income equality, and fiscal sustainability
(d) Price regulation, employment generation, and fiscal responsibility (1 mark)

28. What actions can be taken during Inflation?
(a) Implement austerity measures and reduce government spending
(b) Introduce subsidies and reduce tax rates Levy new taxes and raise the rates of existing taxes
(c) Increase government borrowing and stimulate demand (1 mark)

PEQ - JANUARY - 2025

29. Taxes on agriculture income is levied by:
(a) Central government
(b) State government
(c) Both central and state government as they are in concurrent list.
(d) Local self-government (1 mark)

30. Government's direct production of an economic good e.g. electricity and public transportation services are example of:
(a) Allocation function (b) Distribution function
(c) Stabilization function (d) Protection function (1 mark)

31. A progressive direct tax system ensures:
(a) Economic growth with stability because it distributes the burden of taxes unequally.
(b) Those who have greater ability to pay contribute more and the tax burden is distributed fairly among the population.
(c) Uniform taxes for all.
(d) Luxuries are taxed heavily. (1 mark)

32. Which of the following is applied on inter-state movement of goods and services and on imports and exports?
(a) CGST (b) SGST (c) IGST (d) Income tax (1 mark)

33. Which constitutional body maintains fiscal federalism in India?
(a) Central government (b) Parliament
(c) Reserve Bank of India (d) Finance Commission (1 mark)

PEQ - MAY - 2025

34. Which of the following is one of the key functions of fiscal policy and aims at eliminating the macroeconomic fluctuations arising from sub optimal allocation of resources?
    (a) The Allocation Function
    (b) The Redistribution Function
    (c) The Stabilization Function
    (d) The Utilisation Function

35. Which of the following statement is true in respect of Social cost?
    (a) Social Cost = Private Cost + External Cost
    (b) Social Cost = Private Cost + External Cost - Total Negative Externalities
    (c) Social Cost = Private Cost + External Cost-Government Taxes
    (d) Social Cost = Private Cost + Total Negative Externalities - Government Taxes

PEQ – SEPTEMBER - 2025

36. What does Article 275 of the Indian Constitution deal with?
    (a) Surcharge on certain duties and taxes for purposes of the union.
    (b) Statutory grants-in-aid from the union to certain states.
    (c) Grants for any public purpose.
    (d) Loans for any public purpose.

37. The government's use of proceeds from progressive taxes to supply essential food grains at highly subsidized prices to BPL households is an example of:
    (a) Allocation Function
    (b) Stabilization Function
    (c) Redistribution Function
    (d) Production Function

PEQ – JANUARY - 2026

38. Governments redistribution function leads to dead weight losses because:
    (a) It leads to lower taxes on the rich for the government
    (b) It emphasizes on production of merit goods only
    (c) It acts as a disincentive to entrepreneurship and work
    (d) It decreases social security costs

39. Statement 1: During recession, the government increases its expenditure or cuts down taxes.
    Statement 2: Deficit budgets are expected to stimulate economic activity while surplus budget tends to slowdown economic growth.
    (a) Both statements (1) and (2) are correct
    (b) Statement (1) is correct while statement (2) is incorrect
    (c) Statement (1) is incorrect while statement (2) is correct
    (d) Both statements (1) and (2) are incorrect

40. Which of the following is not one of the three main macroeconomic goals for any nation:
    (a) Universal education to all
    (c) High level of employment
    (8) Economic growth
    (d) Price stability

41. Stagflation is:
   (a) Increased taxes leading to reduced disposable income
   (b) Economic situation where inflation and unemployment exist side by side
   (c) Easy transmission of economic instability from one country to other countries
   (d) Improved economic stability

42. GST is a:
   (a) Production based taxation system
   (E) Income based taxation system
   (F) Turnover based taxation system
   (G) Consumption based taxation system

43. What is Net Divisible Pool (NDP)?
   (a) Total funds from which loans can be taken by various states
   (b) Total loans recovered from IMF to be divided among center and states
   (c) Distribution of total revenue of a nation among various sectors of the economy
   (d) Portion of gross tax revenue which is distributed between the center and the states

44. Which of following taxes has not been replaced by the Goods and Service Tax? 91.
   (a) VAT
   (b) Income Tax
   (c) Excise duty
   (d) Service Tax

ANSWERS / HINTS

I. Multiple Choice Type Questions

1	(b)	2	(d)	3	(b)	4	(c)	5	(d)	6	(b)	7	(b)	8	(b)
9	(c)	10	(b)	11	(c)	12	(a)	13	(c)	14	(c)	15	(d)	16	(d)
17	(b)	18	(c)	19	(b)	20	(b)	21	(b)	22	(a)	23	(a)	24	(b)
25	(b)	26	(c)	27	(a)	28	(a)	29	(b)	30	(a)	31	(b)	32	(c)
33	(d)	34	(c)	35	(a)	36	(b)	37	(c)	38	(c)	39	(a)	40	(a)
41	(b)	42	(d)	43	(d)	44	(b)								

For Your Notes

For Your Notes

For Your Notes

For Your Notes

UNIT – II

PART – I : MARKET FAILURE

Topic References: (Part I)

2.1 Introduction
2.2 The Concept of Market Failure
2.3 Why do Market Fail
2.4 Market Power
2.5 Externalities
2.6 Public Goods
2.7 Incomplete Information

2.1 Introduction

In a well functioning market, prices provide the accurate signals to producers and consumers and the right quantity of whatever consumers choose to consume will be produced and supplied at the right price. However, this is not always true.

2.2 The Concept of Market Failure

- Under certain circumstances, 'market failure' occurs, i.e. the market fails to allocate resources efficiently and therefore, market outcomes become inefficient.
- Market failure is a situation in which the free market leads to misallocation of society's scarce resources in the sense that there is either overproduction or underproduction of particular goods and services leading to a less than optimal outcome.
- The reason for market failure lies in the fact that though perfectly competitive markets work efficiently, most often the prerequisites of competition are unlikely to be present in an economy.
- Market failures are situations in which a particular market, left to itself, is inefficient.
- There are 2 types of market failure.
- Complete market failure. This is a case of "missing markets" and occurs when the market does not supply products at all despite the fact that such products and services are wanted by people. E.g. Pure public goods.
- Partial market failure occurs when the market does actually function, but it produces either the wrong quantity of a product or at the wrong price. This results in loss of economic welfare.

2.3 Why Do Markets Fail?

There are four major reasons for market failure. They are:

- Market power,
- Externalities,
- Public goods, and
- Incomplete information

2.4 Market Power

- Market power or monopoly power is the ability of a firm to profitably raise the market price of a good or service over its marginal cost.
- Firms that have market power are price makers and therefore, can charge a price that gives them positive economic profits.
- Excessive market power causes the single producer or a small number of producers to produce and sell less output than would be produced in a competitive market.
- Market power can cause markets to be inefficient because it keeps price higher and output lower than the outcome of equilibrium of supply and demand.
- In the extreme case, there is the problem of non-existence of markets or missing markets resulting in failure to produce various goods and services, despite the fact that such products and services are wanted by people.
  For example, the markets for pure public goods do not exist.

2.5 Externalities

['spillover effects', 'neighborhood effects', 'third-party effects' or 'side-effects']
Anything that one individual does, may have, at the margin, some effect on others.

For example, if individuals decide to switch from consumption of ordinary vegetables to consumption of organic vegetables, they would, other things equal, increase the price of organic vegetables and potentially reduce the welfare of existing consumers of organic vegetables.

All these operate through price mechanism i.e. through changes in prices. The price system works efficiently because market prices convey information to both producers and consumers.

However, sometimes, the actions of either consumers or producers result in costs or benefits that do not reflect as part of the market price. Such costs or benefits which are not accounted for by the market price are called externalities because they are "external" to the market.

Externalities may be uni-directional or reciprocal.
Suppose a workshop creates earsplitting noise and imposes an externality on a baker who produces smoke and disturbs the workers in the workshop, then this is a case of reciprocal externality.

If an accountant who is disturbed by loud music but has not imposed any externality on the singers, then the externality is unidirectional.

Externalities can be positive or negative.
Negative externalities occur when the action of one party imposes costs on another party. Positive externalities occur when the action of one party confers benefits on another party.
The four possible types of externalities are:

- Negative production externalities
- Positive production externalities
- Negative consumption externalities
- Positive consumption externalities

Negative Production Externalities

A negative externality initiated in production which imposes an external cost on others may be received by another in consumption or in production.

For example, a negative production externality occurs when a factory which produces aluminum discharges untreated waste water into a nearby river and pollutes the water causing health hazards for people who use the water for drinking and bathing. Pollution of river also affects fish output as there will be less catch for fishermen due to loss of fish resources.

The former is a case where a negative production externality is received in consumption and the latter presents a case of a negative production externality received in production.

The firm, however, has no incentive to account for the external costs that it imposes on consumers of river water or fishermen when making its production decision. Additionally, there is no market in which these external costs can be reflected in the price of aluminum.

Positive Production Externalities

A positive production externality initiated in production that confers external benefits on others may be received in production or in consumption.
Compared to negative production externalities, positive production externalities are less common.

For example, positive production externality received in production, we can cite the case of a firm which offers training to its employees for increasing their skills. The firm generates positive benefits on other firms when they hire such workers as they change their jobs.

A positive production externality is received in consumption when an individual raises an attractive garden and the persons walking by enjoy the garden. These external effects were not in fact taken into account when the production decisions were made.

Negative Consumption Externalities

Negative consumption externalities are extensively experienced by us in our day to day life.
Such negative consumption externalities initiated in consumption which produce external costs on others may be received in consumption or in production.

For example, Negative Externalities where they affect consumption of others are smoking cigarettes in public place causing passive smoking by others, creating litter and diminishing the aesthetic value of the room, playing the radio loudly obstructing one from enjoying a concert.

The act of undisciplined students talking and creating disturbance in a class preventing teachers from making effective instruction and the case of excessive consumption of alcohol causing impairment in efficiency for work and production are instances of negative consumption externalities affecting production.

Positive Consumption Externalities

A positive consumption externality initiated in consumption that confers external benefits on others may be received in consumption or in production.
For example, if people get immunized against contagious diseases, they would confer a social benefit to others as well by preventing others from getting infected. Consumption of the services of a health club by the employees of a firm would result in an external benefit to the firm in the form of increased efficiency and productivity.

What is meant by Private Cost and Social Cost?

Private cost is the cost faced by the producer or consumer directly involved in a transaction.

Social costs refer to the total costs to the society on account of a production or consumption activity. Social costs are private costs borne by individuals directly involved in a transaction together with the external costs borne by third parties not directly involved in the transaction.

Social Cost = Private Cost + External Cost

When negative production externalities exist, social costs exceed private cost because the true social cost of production would be private cost plus the cost of the damage from externalities.

If producers do not take into account the externalities, there will be over-production and market failure.

Applying the same logic, negative consumption externalities lead to a situation where the social benefit of consumption is less than the private benefit.

Thus, we conclude that when there is negative externality, a competitive market will produce too much output relative to the social optimum. This is a clear case of market failure where prices fail to provide the correct signals.

2.6. Public Goods

Paul A. Samuelson who introduced the concept of 'collective consumption good' in his path-breaking 1954 paper 'The Pure Theory of Public Expenditure' is usually recognized as the first economist to develop the theory of public goods.

A public good (also referred to as collective consumption good or social good) is defined as one which all enjoy in common in the sense that each individual's consumption of such a good leads to no subtraction from any other individuals' consumption of that good.

2.6.1 Characteristics of Public Goods

- Public goods yield utility to people and are products (goods or services) whose consumption is essentially collective in nature. No direct payment by the consumer is involved in the case of pure public goods.
- Public good is non-rival in consumption. It means that consumption of a public good by one individual does not reduce the quality or quantity available for all other individuals. When consumed by one person, it can be consumed in equal amounts by the rest of the persons in the society.
- That is, your consumption of a public good in no way interferes with its consumption by other people. For example, if, you eat your apple, another person cannot eat it. But, if you walk in street light, other persons too can walk without any reduced benefit from the street light.
- Public goods are non-excludable. Consumers cannot (at least at less than prohibitive cost) be excluded from consumption benefits. If the good is provided, one individual cannot deny other individuals' consumption. Provision of a public good at all by government means provision for all. For example, national defence once provided, it is impossible to exclude anyone within the country from consuming and benefiting from it.
- Public goods are characterized by indivisibility. For example, you can buy chocolates or ice cream as separate units, but a lighthouse, a highway, an airport, defence, clean air etc. cannot

be consumed in separate units. In the case of public goods, each individual may consume all of the good i.e. the total amount consumed is the same for each individual.

- Public goods are generally more vulnerable to issues such as externalities, inadequate property rights, and free rider problems.
- Once a public good is provided, the additional resource cost of another person consuming the goods is 'zero'. For example, a Lighthouse near a sea shore to guide the ships. Once the beacon is lit, an additional ship can use it without any additional cost of provision.
- A unique feature of public goods is that they do not conform to the settings of market exchange. The property rights of public goods with extensive indivisibility and non-exclusive properties cannot be determined with certainty. Therefore, the owners of such products cannot exercise sufficient control over their assets. For example, if you maintain a beautiful garden, you cannot exercise full control over it so as to charge your neighbours for the enjoyment which they get from your garden.
- Producers are not motivated to produce a socially-optimal amount of products if they cannot charge a positive price for them or make profits from them. As such, though public goods are extremely valuable for the well-being of the society, left to the market, they will not be produced at all or will be grossly under produced.

2.7 Incomplete Information

Complete information is an important element of competitive market.

Perfect information implies that both buyers and sellers have complete information about anything that may influence their decision making.

However, this assumption is not fully satisfied in real markets due to the following reasons.

- Often, the nature of products and services tends to be highly complex e.g. Cardiac surgery, financial products (such as pension products mutual funds etc.)
- In many cases consumers are unable to quickly / cheaply find sufficient information on the best prices as well as quality for different products. Sometimes they misunderstand the true costs or benefits of a product or are uncertain about the true costs and benefits.
- People are ignorant or not aware of many matters in the market. Generally they have inaccurate or incomplete data and consequently make potentially 'wrong' choices / decisions.

Information failure is widespread in numerous market exchanges. When this happens misallocation of scarce resources takes place and equilibrium price and quantity is not established through price mechanism. This results in market failure.

2.7.1 Asymmetric Information

Asymmetric information occurs when there is an imbalance in information between buyer and seller i.e. when the buyer knows more than the seller or the seller knows more than the buyer.

This can distort choices.

For example, the landlords know more about their properties than tenants, a borrower knows more about their ability to repay a loan than the lender, a used-car seller knows more about vehicle quality than a buyer and some traders may possess insider information in financial markets.

These are situations in which one party to a transaction knows a material fact that the other party does not.

This phenomenon, which is sometimes referred to as the 'lemons problem', is an important source of market failure.

With asymmetric information, low-quality goods can drive high-quality goods out of the market.

2.7.2 Adverse Selection

Adverse selection is a situation in which asymmetric information about quality eliminates high-quality goods from a market.

For example, health insurance. The people who are most likely to purchase health insurance are those who are most likely to use it, i.e. people with unhealthy life styles and those with underlying health issues. The insurance company being aware of this raises the average price of insurance cover. This prices healthy consumers out of the market as healthy people will be unwilling to pay such high premium. The result is that only high risk individuals buy insurance. This is a market failure.

Another example is the used car market i.e. the 'market for lemons'. The owner of a car knows much more about its quality than anyone else. The buyer's willingness to pay for any particular car will be based on the 'average quality' of used cars. Anyone who sells a 'lemon' (an unusually poor car) stands to gain. The market becomes flooded with lemons. Eventually the market may offer nothing but lemons. The good-quality cars disappear because they are kept by their owners or sold only to friends.

Briefly put, buyers expect hidden problems in items offered for sale, leading to low prices and the best items being kept off the market.

2.7.3 Moral Hazard

Moral hazard is opportunism characterized by an informed person's taking advantage of a less - informed person through an unobserved action.

It arises from lack of information about someone's future behavior.

Moral hazard occurs when an individual knows more about his or her own actions than other people do.

This leads to a distortion of incentives to take care or to exert effort when someone else bears the costs of the lack of care or effort.

When someone is protected from paying the full costs of their harmful actions, they tend to act irresponsibly, making the harmful consequences more likely.

Moral hazard occurs when a party whose actions are unobserved can affect the probability or magnitude of a payment associated with an event.

For example, the insured consumers are likely to take greater risks, knowing that a claim will be paid for by the insurance company. The more of one's costs that are covered by the insurance company, the less a person cares whether the doctor charges excessive fees or uses inefficient and costly procedures as part of his health care. This causes insurance premiums to rise for everyone, driving many potential customers out of the market.

This became a big issue in India when the health insurance providers and big private hospitals came in conflict and the issue was resolved by putting in place a 'third party administration' to settle the medical claims.

Asymmetric information, adverse selection and moral hazard affect the ability of markets to efficiently allocate resources and therefore lead to market failure because the party with better information has a competitive advantage.

PART - II: GOVERNMENT INTERVENTION TO CORRECT MARKET FAILURE

Topic References: (Part II)

2.8 Government Intervention to Minimize Market Power
2.9 Government Intervention to Correct Externalities
2.10 Government Intervention in case of Merit Goods
2.11 Government Intervention in case of De-Merit Goods
2.12 Government Intervention in case of Public Goods
2.13 Price Intervention: Non Market Pricing
2.14 Government Intervention for Correcting Market Failure
2.15 Government Intervention for Equitable Distribution

2.8 Government Intervention to Minimize Market Power

Because of the social costs imposed by monopoly, governments intervene by establishing rules and regulations designed to promote competition and prohibit actions that are likely to restrain competition.

These legislations differ from country to country.

For example, in India, we have the Competition Act, 2002 (as amended by the Competition (Amendment) Act, 2007) to promote and sustain competition in markets.

The Antitrust laws in the US and the Competition Act, 1998 of UK etc are designed to promote competitive economy by prohibiting actions that are likely to restrain competition. Such legislations generally aim at prohibiting contracts, combinations and collusions among producers or traders which are in restraint of trade and other anticompetitive actions such as predatory pricing.

Other measures include:

- Market liberalisation by introducing competition in previously monopolistic sectors such as energy, telecommunication etc.
- Controls on mergers and acquisitions if there is possible market domination
- Price capping and price regulation
- Profit or rate of return regulation
- Patronage to consumer associations
- Tough investigations into cartelisation and unfair practices such as collusion and predatory pricing
- Restrictions on monopsony power of firms
- Reduction in import controls and
- Nationalisation

However, sometimes we find that governments protect monopoly positions of firms that have developed unique innovations. For example, patent and copyright laws grant exclusive rights of products or processes to provide incentives for invention and innovation.

Another example is that of permitted natural monopoly. Natural monopolies can produce the entire output of the market at a cost that is lower than what it would be if there were several firms. Examples of such natural monopoly are electricity, gas and water supplies.

In order to control the market power of such natural monopolies, governments usually regulate the price of the goods and services provided by them.

2.9 Government Intervention to Correct Externalities

I. Negative Externality

How government regulation deal with the inefficiencies that arise from negative externalities?

Government initiatives towards negative externalities may be classified as:

1. Direct controls that openly regulate the actions of those involved in generating negative externalities, and
2. Market-based policies that would provide economic incentives so that the self- interest of the market participants would achieve the socially optimal solution.

1. Direct controls:

(a) Direct controls prohibit specific activities that explicitly create negative externalities or require that the negative externality be limited to a certain level
For Example, limiting emissions.

(b) Production, use and sale of many commodities and services are prohibited in our country.
For Example, Smoking is completely banned in many public places. Stringent rules are in place in respect of tobacco advertising, packaging and labeling etc.

(c) Governments may pass laws to alleviate the effects of negative externalities.
For Example, India has enacted the Environment (Protection) Act, 1986.

(d) The government may, through legislation, fix emissions standard which is a legal limit on how much pollutant a firm can emit. The set standard ensures that the firm produces efficiently. If the firm exceeds the limit, it can invite monetary penalties or/and criminal liabilities.
For Example, The firms have to install pollution-abatement mechanisms to ensure adherence to the emission standards. This means additional expenditure to the firm leading to rise in the firm's average cost. New firms will find it profitable to enter the industry only if the price of the product is greater than the average cost of production plus abatement expenditure.

(e) Another method is to charge an emissions fee which is levied on each unit of a firm's emissions. The firms can minimize costs and enhance their profitability by reducing emissions.

(f) Government may also form special bodies/boards to specifically address the problem:
For Example, the Ministry of Environment & Forest, the Pollution Control Board of India and the State Pollution Control Boards.

2. Market Based Poliicies

The market-based approaches – environmental taxes and cap-and-trade – operate through price mechanism to create an incentive for change.
This is achieved by:

(a) Setting the price directly through a pollution tax

(b) Setting the price indirectly through the establishment of a cap-and-trade system.

(a) Setting the price directly through a pollution tax

One method of internalization of negative externalities is imposing pollution taxes.

These taxes are named Pigouvian taxes after A.C. Pigou who argued that an externality cannot be alleviated by contractual negotiation between the affected parties and therefore taxation should be resorted to.

These taxes, by 'making the polluter pay', seek to internalize external costs into the price of a product or activity.

The tax is placed on the externality itself (the amount of pollution emissions) rather than on output (say, amount of steel).

For each unit of pollution, the polluter must choose either to pay the tax or to reduce pollution through any means at its disposal.

The proceeds from the tax, can be specifically earmarked for projects that protect or enhance environment.

Problems in administering an efficient pollution tax?

- ✓ Pollution taxes are difficult to determine and administer because it is difficult to discover the right level of taxation that would ensure that the private cost plus taxes will exactly equate with the social cost.
- ✓ The method of taxing the polluters has many limitations because it involves the use of complex and costly administrative procedures for monitoring the polluters.
- ✓ This method does not provide any genuine solution to the problem. It only establishes an incentive system for use of methods which are less polluting.
- ✓ In the case of goods which have inelastic demand, producers will be able to easily shift the tax burden in the form of higher product prices. This will have an inflationary effect and may reduce consumer welfare.
- ✓ Pollution taxes also have potential negative consequences on employment and investments because high pollution taxes in one country may encourage producers to shift their production facilities to those countries with lower taxes.

(b) Setting the price indirectly through the establishment of a cap-and-trade system.

These are marketable licenses to emit limited quantities of pollutants and can be bought and sold by polluters.

Under this method, each firm has permits specifying the number of units of emissions that the firm is allowed to generate.

A firm that generates emissions above what is allowed by the permit is penalized with substantial monetary sanctions.

These permits are transferable, and therefore different pollution levels are possible across the regulated entities.

Permits are allocated among firms, with the total number of permits so chosen as to achieve the desired maximum level of emissions.

By allocating fewer permits than the free pollution level, the regulatory agency creates a shortage of permits which then leads to a positive price for permits.

This establishes a price for pollution, just as in the tax case.

The high polluters have to buy more permits, which increases their costs, and makes them less competitive and less profitable.

The low polluters receive extra revenue from selling their surplus permits, which makes them

more competitive and more profitable.
Therefore, firms will have an incentive not to pollute.
India is experimenting with cap-and-trade in the form of Perform, Achieve & Trade (PAT) scheme and carbon tax in the form of a cess on coal.

The advantages claimed for tradable permits are:

- ✓ The system allows flexibility and rewards efficiency
- ✓ It is administratively cheap and simple to implement and ensures that pollution is minimised in the most cost-effective way
- ✓ It also provides strong incentives for innovation.
- ✓ Consumers may benefit if the extra profits made by low pollution firms are passed on to them in the form of lower prices.
- ✓ The main argument in opposition to the employment of tradable emission permits is that they do not in reality stop firms from polluting the environment; they only provide an incentive to them to do so.
- ✓ if firms have monopoly power of some degree along with a relatively inelastic demand for its product, the extra cost incurred for procuring additional permits so as to further pollute the atmosphere, could easily be compensated by charging higher prices to consumers.

II. Positive Externality

When positive externalities are present, government may attempt to solve the problem through -

- Corrective subsidies to the producers aimed at increasing the supply of the good
- Corrective subsidies to consumers aimed at increasing the demand for the good.

As we are aware, a corrective production subsidy involves government paying part of the cost to the firms in order to promote the production of goods having positive externalities. This is in fact a market-based policy as subsidies to producers would lower their cost of production.

E.g. fertilizer subsidy. A subsidy on fee for education is an example of consumption subsidy.

In the case of products and services whose externalities are vastly positive, the government enters the market directly as an entrepreneur to produce and provide them. Public education, health care and fundamental research are the obvious examples.

Governments also engage in direct production of environmental quality. Examples are: a forestation, reforestation, protection of water bodies, treatment of sewage and cleaning of toxic waste sites.

2.10 Government Intervention In Case of Merit Goods

Merit goods are goods which are deemed to be socially desirable and therefore the government deems that its consumption should be encouraged.

Substantial positive externalities are involved in the consumption of merit goods.

Left to the market, only private benefits and private costs would be reflected in the price paid by consumers.

This means, compared to what is socially desirable, people would consume inadequate quantities.

Examples of merit goods - education, health care, welfare services, housing, fire protection, waste management, public libraries, museum and public parks.

What are the reasons for government provision of merit goods?

- ✓ Information failure is widely prevalent with merit goods and therefore individuals may not act in their best interest because of imperfect information.
- ✓ Equity consideration demand that merit goods like health and education should be provided free on the basis of need rather than individual's ability to pay
- ✓ There is a lot of uncertainty as to the need for merit goods E.g. health care. Due to uncertainty about the nature and timing of healthcare required in future, individuals may be unable to plan their expenditure and save for their future medical requirements. The market is unlikely to provide the optimal quantity of health care when consumers actually need it, because they may be short of the necessary finances to pay the market price.

What are the government responses to under-provision of merit goods?

The possible government responses to under-provision of merit goods are regulation, subsidies, direct government provision and a combination of government provision and market provision.

Regulation determines how a private activity may be conducted. For example, the way in which education is to be imparted is government regulated.

Governments can prohibit some type of goods and activities, set standards and issue mandates making others oblige. For example, government may make it compulsory to avail insurance protection.

Compulsory immunization may be insisted upon as it helps not only the individual but also the society at large.

Government could also use legislation to enforce the consumption of a good which generates positive externalities. For example use of helmets, seat belts, The Right of Children to Free and Compulsory Education Act, 2009 which mandates free and compulsory education for every child of the age of six to fourteen years.

When governments provide merit goods, it may give rise to large economies of scale and productive efficiency apart from generating substantial positive externalities and overcoming the problems mentioned above.

2.11 Government Intervention In Case Of Demerit Goods

Demerit goods are goods which are believed to be socially undesirable. Examples of demerit goods - cigarettes, alcohol, intoxicating drugs etc.

The consumption of demerit goods imposes significant negative externalities on the society as a whole and therefore the private costs incurred by individual consumers are less than the social costs experienced by the society.

The production and consumption of demerit goods are likely to be more than optimal under free markets.

The price that consumers pay for a packet of cigarettes is market determined and does not account for the social costs that arise due to externalities.

All goods with negative externalities are not essentially demerit goods; e.g. Production of steel causes pollution, but steel is not a socially undesirable good.

How do governments correct market failure resulting from demerit goods?

- ✓ At the extreme, government may enforce complete ban on a demerit good. For Example, Intoxicating drugs. In such cases, the possession, trading or consumption of the good is made illegal.
- ✓ Through persuasion which is mainly intended to be achieved by negative advertising campaigns which emphasize the dangers associated with consumption of demerit goods.
- ✓ Through legislations that prohibit the advertising or promotion of demerit goods in whatsoever manner.
- ✓ Strict regulations of the market for the good may be put in place so as to limit access to the good, especially by vulnerable groups such as children and adolescents.
- ✓ Regulatory controls in the form of spatial restrictions. For Example, smoking in public places, sale of tobacco to be away from schools, and time restrictions under which sale at particular times during the day is banned.
- ✓ Imposing unusually high taxes on producing or purchasing the good making them very costly and unaffordable to many is perhaps the most commonly used method for reducing the consumption of a demerit good. For Example, the GST Council has bracketed four items namely, high end cars, pan masala, aerated drinks and tobacco products into demerit goods category and therefore these would be taxed (with a cess being added on to the basic tax) at much higher rates than the top GST slab of 40 %.
- ✓ The government can fix a minimum price below which the demerit good should not be exchanged.

  The demand for demerit goods such as, cigarettes and alcohol is often highly inelastic, so that any increase in price resulting from additional taxation causes a less than proportionate decrease in demand. Also, sellers can always shift the taxes to consumers without losing customers.

2.12 Government Intervention In Case Of Public Goods

Public goods which are non-excludable is highly prone to free rider problem and therefore markets are unlikely to get established.

The non-rival nature of consumption provides a strong argument for the government rather than the market to provide and pay for public goods.

In the case of such pure public goods where entry fees cannot be charged, direct provision by governments through the use of general government tax revenues is the only option.

Excludable public goods can be provided by government and the same can be financed through entry fees.

A very commonly followed method is to grant licenses to private firms to build a public good facility.

Under this method, the goods are provided to the public on payment of an entry fee.

In such cases, the government regulates the level of the entry fee chargeable from the public and keeps strict watch on the functioning of the licensee to guarantee equitable distribution of welfare.

2.13 Price Intervention : Non Market Pricing

Price intervention takes the form of **price controls** which are legal restrictions on price. Price controls may take the form of either a

- ✓ Price floor - a minimum price buyers are required to pay for a good or service
  For Example, Fixing of minimum wages; or
- ✓ Price ceiling - a maximum price sellers are allowed to charge for a good or service
  For Example, rent controls; price of stents etc.

PRICE FLOOR

Government usually intervenes in many primary markets which are subject to extreme as well as unpredictable fluctuations in price. For example in India, in the case of many crops the government has initiated the Minimum Support Price (MSP) programme as well as procurement by government agencies at the set support prices. The objective is to guarantee steady and assured incomes to farmers.

PRICE CEILINGS

When prices of certain essential commodities rise excessively, government may resort to controls in the form of price ceilings for making a resource or commodity available to all at reasonable prices. For example - maximum prices of food grains and essential items are set by government during times of scarcity.

With the objective of ensuring stability in prices and distribution, governments often intervene in grain markets through building and maintenance of buffer stocks. It involves purchases from the market during good harvest and releasing stocks during periods when production is below average.

2.14 Government Intervention for Correcting Information Failure

Governments actively intervene in the market for combating the problem of market failure due to information problems and considering the importance of information in making rational choices. A few examples are:

- Government makes it mandatory to have accurate labeling and content disclosures by producers. E.g. Labeling on cigarette packets, display of nutritional information on food packages.
- Mandatory disclosure of information, for example: SEBI requires that accurate information be provided to prospective buyers of new stocks.
- Public dissemination of information to improve knowledge
- Regulation of advertising and setting of advertising standards to make advertising more responsible, informative and less persuasive.

2.15 Government Intervention For Equitable Distribution

One of the most important activities of the government is to redistribute incomes so that there is equity and fairness in the society.

Some common policy interventions include: progressive income tax, targeted budgetary allocations, unemployment compensation, transfer payments, subsidies, social security schemes, job reservations, land reforms, gender sensitive budgeting etc.

Government also intervenes to combat black economy and market distortions associated with a parallel black economy.

However, we cannot be sure whether the government interventions would be effective or whether it would make the functioning of the economy less efficient.

Government failures where government intervention in the economy to correct a market failure creates inefficiency and leads to a misallocation of scarce resources occur very often.

Government failure occurs when:

- intervention is ineffective causing wastage of resources expended for the intervention
- intervention produces fresh and more serious problems

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. 'Market failure' is a situation which occurs when
   (a) *private goods are not sufficiently provided by the market*
   (b) *public goods are not sufficiently provided by public sector*
   (c) *The market fail to form or they allocate resources efficiently*
   (d) *(b) and (c) above*

2. Which of the following is an example of market failure?
   (a) *Prices of goods tend to rise because of shortages*
   (b) *Merit goods are not sufficiently produced and supplied*
   (c) *Prices fall leading to fall in profits and closure of firms*
   (d) *None of the above*

3. Which of the following is an outcome of market power?
   (a) *makes price equal to marginal cost and produce a positive external benefit on others*
   (b) *can cause markets to be efficient due to reduction in costs*
   (c) *makes the firms price makers and restrict output so as to make allocation inefficient*
   (d) *(b) and(c) above*

4. Markets do not exist
   (a) *for goods which have positive externalities* (b) *for pure public goods*
   (c) *for goods which have negative externalities* (d) *none of the above*

5. Which of the following is the right argument for provision of public good by government?
   (a) *Governments have huge resources at their disposal*
   (b) *Public goods will never cause any type of externality*
   (c) *Markets are unlikely to produce sufficient quantity of public goods*
   (d) *Provision of public goods are very profitable for any government*

6. Adequate amount of a pure public good will not be provided by the private market because of
   (a) *the possibility of free riding*
   (b) *the existence of very low prices and low profits*
   (c) *governments would any way produce them, so there will be overproduction*
   (d) *there are restrictions as well as taxes on production of public goods*

7. The free rider problem arises because of
   (a) *ability of participants to produce goods at zero marginal cost*
   (b) *marginal benefit cannot be calculated due to externalities present*
   (c) *the good or service is non excludable*
   (d) *general poverty and unemployment of people*

8. A chemical factory has full information regarding the risks of a product, but continues to sell it. This is possible because of
   (a) *asymmetric information* (b) *moral hazard*
   (c) *free riding* (d) *(a) and (c) above*

9. If an individual tends to drive his car in a dangerously high speed because he has a comprehensive insurance cover, it is a case of
   (a) *free riding* (b) *moral hazard*
   (c) *poor upbringing* (d) *Inefficiency*

10. Smoking in public is a case of
   (a) *Negative consumption externality* (b) *Negative production externality*
   (c) *Internalising externality* (d) *None of the above*

11. Read the following statements
I The market-based approaches to control externalities operate through price mechanism
II. When externalities are present, the welfare loss would be eliminated
III. The key is to internalizing an externality is to ensure that those who create the externalities include them while making decisions
Of the above statements
   (a) *II and III are correct* (b) *I only is correct*
   (c) *II only is correct* (d) *I and III are correct*

12. Which of the following statements is false?
   (a) *Tradable permits provide incentive to innovate and reduce negative externalities*
   (b) *A subsidy on a good which has substantial positive externalities would reduce its cost and consequently its price would be lower*
   (c) *Substantial negative externalities are involved in the consumption of merit goods.*
   (d) *Merit goods are likely to be under-produced and under consumed through the market mechanism*

13. Which one of the following would you suggest for reducing negative externality?
   (a) *Production subsidies* (b) *Excise duty*
   (c) *Pigouvian taxes* (d) *All of the above*

14. A Pigouvian subsidy
   (a) *cannot be present when externalities are present*
   (b) *is a good solution for negative externality as prices will increase*
   (c) *is not measurable in terms of money and therefore not practical*
   (d) *may help production to be socially optimal when positive externalities are present*

15. If governments make it compulsory to avail insurance protection, it is because
   (a) *Insurance companies need to be running profitably*
   (b) *Insurance will generate moral hazard and adverse selection*
   (c) *Insurance is a merit good and government wants people to consume it*
   (d) *None of the above*

16. The Competition Act, 2002 aims to -
   (a) *protect monopoly positions of firms that have developed unique innovations*
   (b) *to promote and sustain competition in markets*
   (c) *to determine pricing under natural monopoly.*
   (d) *None of the above*

17. Rules regarding product labelling
    (a) *Seeks to correct market failure due to externalities*
    (b) *Is a method of solving the problem of public good*
    (c) *May help solve market failure due to information failure*
    (d) *Reduce the problem of monopolies in the product market*

18. Identify the incorrect statement
    (a) *A minimum support price for agricultural goods is a Government intervention method to guarantee steady and assured incomes to farmers.*
    (b) *An externality is internalised if the ones that generated the externality incorporate them into their private cost- benefit analysis*
    (c) *The production and consumption of demerit goods are likely to be less than optimal under free markets*
    (d) *Compared to pollution taxes, the cap and trade method is administratively cheap and simple to implement and ensures that pollution is minimised in the most cost-effective way.*

19. The incentive to let other people pay for a good or service, the benefits of which are enjoyed by an individual
    (a) *Is a case of negative externality* (b) *Is a case of market efficiency*
    (c) *Is a case of free riding* (d) *Is inappropriate and warrant action*

20. A government subsidy
    (a) *is a market-based policy*
    (b) *involves the government paying part of the cost to the firms in order to promote the production of goods having positive externalities*
    (c) *is generally provided for merit goods*
    (d) *all the above*

21. The production and consumption of demerit goods are
    (a) *likely to be more than optimal under free markets.*
    (b) *likely to be less than optimal under free markets*
    (c) *likely to be subjected to price intervention by government*
    (d) *(a) and c) above*

22. The argument for education subsidy is based on
    (a) *Education is costly* (b) *the ground that education is merit good*
    (c) *education creates positive externalities* (d) *(b) and c) above*

23. Read the following statements

I. Social costs are the total costs incurred by the society when a good is consumed or produced.
II The external costs are not included in firms' income statements or consumers' decisions
III. Each firm's cost which is considered for determining output would be only private cost or direct cost of production which does not include external costs
IV. Production and consumption decisions are efficient only when private costs are considered

Of the above
    (a) *Statements I and III are correct* (b) *Statements I, II and III are correct*
    (c) *Statement I only is correct* (d) *All the above are correct*

24. Government failure occurs when
   (a) *Government fails to implement its election promises on policies*
   (b) *A government is unable to get reelected*
   (c) *Government intervention is ineffective and produces fresh and more serious problems*
   (d) *None of the above*

25. A thermal power plant uses coal and creates pollution in an otherwise unpolluted area. Which of the following would ensure that a socially optimal output of electricity is produced?
   (a) *Where marginal private cost equals marginal private benefit.*
   (b) *Where marginal private cost equals marginal social benefit.*
   (c) *Where marginal social cost equals marginal private benefit.*
   (d) *Where marginal social cost equals marginal social benefit.*

26. Which of the following statements is true?
I. The market based approaches to control market operate through price mechanism
II. When externalities are present, the welfare loss are eliminated
III. The key to internalize an externality is to ensure that those who create externalities include them while making decision
   (a) *Both II and III* (b) *I only* (c) *II only* (d) *Both I and III*

27. If merit goods are provided free by the government
   (a) *The quantity demanded of merit good will be less than supply*
   (b) *The quantity demanded of merit good will be equal to supply*
   (c) *The quantity demanded of merit good is likely to be more than supply*
   (d) *Any of the above can happen*

28. The government should intervene in the marketplace to discourage the production and consumption of _
   (a) *Goods which are not necessary* (b) *Demerit goods*
   (c) *Goods having no externalities* (d) *Goods which the markets produce less*

29. If government produces and supplies a public good
   (a) *It may correct market failure as there will be no externalities*
   (b) *It may correct market failure caused by free riding*
   (c) *It may correct market failure because people may free ride*
   (d) *It may correct market failure because people may not free ride*

PEQ - JUNE - 2024

30. When smoking is completely banned by the government in public places like bus stand, schools, parks etc. There is an example of:
   (a) *Direct control on negative production externality*
   (b) *Direct control on positive production externality*
   (c) *Command solution*
   (d) *Direct control on positive consumption externality* *(1 mark)*

31. Externalities leads to market failure because:
    (a) *They always result in negative outcomes for markets causing market failure*
    (b) *They represent cost or benefit which are not reflected in price*
    (c) *They are always associated with public goods and services*
    (d) *They prevent the production of private good having mankind.* *(1 mark)*

PEQ - SEPTEMBER - 2024

32. Sarah is considering purchasing a used car from a private seller named John. John claims that the car is in excellent condition and has never been in an accident. However, John is aware that the car had significant damage in the past due to an accident, but he has repaired it without disclosing this information to Sarah. Based on the case described, what concept of economics does this scenario illustrate?
    (a) Lemons Problem *(b) Moral Hazard*
    (c) Market Power *(d) Government Interventions* *(1 mark)*

33. All revenues received, loans raised and all moneys received by the government in repayment of loans are credited to the:
    (a) Contingency fund of India *(b) Consolidated fund of India*
    (c) Public provident fund of India *(d) Public Account* *(1 mark)*

PEQ - JANUARY - 2025

34. The cap and trade method used by government to ensure that pollution is minimized in the most cost effective way is an example of:
    (a) *Government intervention to correct externalities*
    (b) *Government intervention in the case of merit goods*
    (c) *Government intervention in the case of demerit goods*
    (d) *Government intervention for correcting market failure* *(1 mark)*

PEQ - MAY - 2025

35. Which of the following is not a characteristic of Public Goods?
    (a) *Non-rival in consumption*
    (b) *Indivisibility*
    (c) *More vulnerable to externalities*
    (d) *Excludable*

36. What does the term "lemon" means in the model of "Lemons problem" developed by "George Akeil of" in relation to used car market:-
    (a) *Only good quality used cars*
    (b) *Both poor and good quality used cars*
    (c) *Average quality used cars*
    (d) *Scrapped used cars*

37. The statutory disclaimer that "Mutual Fund investments are subject to market risks please read the offer documents carefully before investing" is which of the following type of government intervention?
(a) *Government intervention to correct externalities.*
(b) *Government intervention for equitable distribution.*
(c) *Government intervention to correct information failure.*
(d) *Government intervention in case of public goods.*

PEQ – SEPTEMBER - 2025

38. Who introduced the concept of ₹collective consumption goods, which later came to be known as public goods?
(a) *Milton Friedman*
(b) *John Maynard Keynes*
(c) *Paul A. Samuelson*
(d) *Adam Smith*

39. The Right of Children to Free and Compulsory Education Act, 2009, which mandates free and compulsory education for every child between the ages of six to fourteen years, is an example of:
(a) *Government Intervention in the case of Merit Goods.*
(b) *Government Intervention in the case of Demerit Goods.*
(c) *Government Intervention in the case of Public Goods.*
(d) *Government Intervention to correct Externalities.*

40. Alex has comprehensive car insurance, which covers the cost of repairs if his car is damaged in an accident. As a result, he drives more recklessly, knowing that his insurance will cover any damages, Over time, he gets into more accidents and files more insurance claims due to his careless driving. This scenario illustrates which of the following concepts?
(a) *Moral Hazard*
(b) *Market Power*
(c) *Lemon's Problem*
(d) *Government Interventions*

41. The situation when the market does not supply products at all despite the fact that such products and services are wanted by the people is known as:
(a) *Temporary market failure*
(b) *Complete market failure*
(c) *Partial market failure*
(d) *Relative market failure*

PEQ – SEPTEMBER - 2026

42. Maximum price ceilings imposed on essential commodities aim to:
(a) *Protect producers*
(b) *Increase exports*
(c) *Protect consumers*
(d) *Increase profits*

ANSWERS / HINTS

I. Multiple Choice Type Questions

1.	(c)	2	(b)	3	(c)	4.	(b)	5.	(c)	6.	(a)	7.	(c)	8.	(a)
9.	(b)	10.	(a)	11.	(d)	12.	(c)	13.	(c)	14.	(d)	15.	(c)	16.	(b)
17.	(c)	18.	(c)	19.	(c)	20.	(d)	21.	(d)	22.	(d)	23.	(b)	24.	(c)
25.	(d)	26	(d)	27.	(c)	28.	(d)	29.	(c)	30.	(c)	31.	(b)	32.	(a)
33.	(b)	34.	(a)	35.	(d)	36.	(b)	37.	(c)	38.	(c)	39.	(a)	40.	(a)
41.	(b)	42.	(c)												

For Your Notes

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UNIT – III

THE PROCESS OF BUDGET MAKING: SOURCES OF REVENUE, EXPENDITURE MANAGEMENT AND MANAGEMENT OF PUBLIC DEBT

Topic References:

3.1 Introduction
3.2 The Process of Budget Making
3.3 Sources of Revenue
3.4 Public Expenditure Management
3.5 Public Debt Management

3.1 Introduction

Budget is a powerful policy instrument in the hands of government to regulate and to restructure a country's economic priorities.

The need for budgeting arises from the need to efficiently allocate limited resources to ensure maximum social welfare.

By proper budgeting, the government is able to ensure redistribution of income and wealth.

The other objectives of budgets are reduction/ elimination of economic fluctuations to bring in stability, sustainable increase in real GDP and reduction in regional disparities.

In simple terms, a budget is a statement that presents the details of 'where the money comes from' and 'where the money goes to'.

The government budget is a document presented for approval and legislation by a government and contains estimates of the proposed expenditure for a given period and the proposed means of financing them.

In other words, a government budget is a schedule of the entire revenues and expenditures that the government expects to receive and plans to spend during the following year.

The budget also contains estimates of the government's accounts for the next fiscal year called budgeted estimates.

Being the document which consolidates revenues from all sources and outlays for all activities, the budget is the most comprehensive report of the government's finances.

3.2 The Process of Budget Making

The budgetary process is the means by which the executive and legislative branches together formulate a coherent set of taxing and spending proposals.

The finances of the government of India have traditionally been controlled by the Ministry of Finance.

The budget is prepared by the Ministry of Finance in consultation with NITI Aayog and other relevant ministries.

The budget must be presented and approved by both houses of parliament before the beginning of the fiscal year (April 1 to March 31).

Despite the fact that the term 'budget' has not been used in the Indian Constitution, the process of making it is generally referred to as budgeting.

Article112 of the constitution provides that in respect of every financial year the 'president shall cause to be laid before both the houses of parliament a statement of the estimated receipts and expenditure of the government of India for that year, referred to as the "Annual Financial Statement''.

The budgetary procedures are -

(i) Preparation of the budget
(ii) Presentation and enactment of the budget and
(iii) Execution of the budget.

The budget process mainly consists of two types of activities:

1. The administrative process, wherein the budget along with the accompanying documents are prepared in consultation with various stakeholders;
2. The legislative process wherein the budget is passed by the parliament after discussions.

Despite the fact that the union budget is presented on 1st February (or any other suitable date as decided by the government), the process of budget preparation commences in August-September of the previous year. The Budget Division of the Ministry of Finance prepares a comprehensive schedule for carrying out the budget preparation activities.

The process of budget making is set off with the Budget Division issuing the budget circular containing detailed instructions and formats for preparing the estimates to all ministries, states, union territories and autonomous bodies.

The detailed estimates of expenditure are prepared by ministries and departments according to their assessment of requirements for the subsequent year. Every department prepares estimates for receipts and expenditure separately.

A series of pre-budget consultations are done by the union finance minister with the finance ministers and chief ministers of states, various stakeholders and interest groups including industry associations, representatives from agriculture and social and welfare sectors, labour organisations, experts from NITI Aayog, economists etc. to elicit their suggestions on the proposed budget.

The budget is presented in the Parliament in such form as the Finance Ministry may decide after considering the suggestions (if any) of the Estimates Committee.

Broadly, the budget documents depict information relating to receipts and expenditure for two years. They are:

(i) Budget estimates (BE) of receipts and expenditure in respect of current and ensuing financial year
(ii) For the current year through Revised Estimates (RE); and
(iii) Actuals of the year preceding the current year

The budget speech is mainly a policy document which draws attention to the proposed policies and programmes of the government.

The finance minister makes a detailed budget speech at the time of presenting the budget before the Lok Sabha.

The budget speech of the Finance Minister is usually in two parts.

- Part A of the budget speech gives an outline of the prevailing macro economic situation of the country and the budget estimates for the next financial year. Elaborating the priorities of the government, the minister presents a broad framework of the total funds raised by the government via taxes or borrowings, proposed government expenditure allocations for different sectors and fresh schemes for different sectors.

- Part B of the budget speech details the progress the government has made on various developmental measures, the direction of future policies and the government's tax proposals for the upcoming financial year including variations in the current taxation system.

The Annual Financial Statement shows the receipts and expenditure of government in three separate parts under which government accounts are maintained, namely:

1. Consolidated Fund of India
2. Contingency Fund of India, and the
3. Public Account.

The list of budget documents presented to the parliament, besides the finance minister's budget speech, is given below:

(a) Annual Financial Statement (AFS)
(b) Demands for Grants (DG)
(c) Finance Bill
(d) Statements mandated under FRBM Act:
    i. Macro -Economic Framework Statement
    ii. Medium-Term Fiscal Policy cum Fiscal Policy Strategy Statement

Nine other documents which are in the nature of explanatory statements supporting the mandated documents are also presented along with the documents mentioned above.

The expenditures of certain categories (e.g. the emoluments and allowances of the President of India and his/her office, and emoluments of Judges of supreme courts and high ranking personnel of constitutional bodies across India) are 'charged' on the Consolidated Fund of India and are not subject to the vote of parliament, are also indicated separately in the budget.

By convention in an election year, the budget may be presented twice. The first one is to first to secure a Vote on Account for a few months. This is followed by the Annual financial statement for that year or the full-fledged Budget.

The budget is discussed in two stages in the Lok Sabha.

First, there is the general discussion on the budget as a whole.

After the first stage of general discussion on the union budget is over, the house is adjourned for a fixed period.

During this period, the demands for grants of various ministries/ departments are considered by the standing committees concerned, and once the reports are presented by these committees within the stipulated time, the house proceeds to discussion and conducts ministry-wise voting on demands for grants.

The Lok Sabha has the power to concur or to refuse any demand or even to reduce the amount of grant sought by government.

The budget is laid on the table of the Rajya Sabha soon after the Finance Minister has completed her/his budget speech in the Lok Sabha.

The Rajya Sabha, does not vote on the demands for grants and there is only a general discussion on the budget.

After the general discussion on the budget proposals and voting on demands for grants have been completed, the government introduces the Appropriation Bill.

The Appropriation Bill is intended to give authority to government to incur expenditure from and out of the Consolidated Fund of India.

Motions for reduction to various demands for grants are made in the form of 'cut motions' seeking to reduce the sums sought by government.

The Finance Bill seeking to give effect to the government's taxation proposals is introduced in Lok Sabha immediately after the presentation of the general budget.

It is accompanied by a memorandum explaining the provisions of the bill and their effect on the finances of the country. The motion for leave to introduce a finance bill cannot be opposed.

The finance bill is taken up for consideration and passing after the Appropriation Bill is passed.

The finance bill seeks to give effect to the financial proposals of the government for the next financial year.

The Parliament has to pass the Finance Bill within 75 days of its introduction.

On the last day of the days allotted for discussion on the demands for grants, the speaker puts all the outstanding demands for grants to the vote of the house. This process is known as 'Guillotine'. It is a device for bringing the debate on financial proposals to an end within a specified time.

After the Finance Bill has been passed by the Lok Sabha, it is transmitted to the Rajya Sabha for Its recommendations.

The bill being a money bill, Rajya Sabha has to return it within a period of 14 days, with or without recommendations. The recommendations of Rajya Sabha may be accepted or rejected by the Lok Sabha.

However from 2017-18, the date of presentation of the budget has been advanced to 1st February. An important budgetary reform was the merger of railway budget with the general budget from the budget for financial year 2017-18.

3.3 Sources of Revenue

The Department of Revenue of the Ministry of Finance exercises control in respect of the revenue matters relating to direct and indirect union taxes.

The department is also entrusted with the administration and enforcement of regulatory measures provided in the enactments concerning goods and services tax (GST), central sales tax, stamp duties and other relevant fiscal statutes.

The Department of Revenue exercises control in respect of matters relating to all the direct and indirect union taxes through two statutory boards, namely,

1. the Central Board of Direct Taxes (CBDT) and
2. the Central Board of Indirect Taxes and Customs (CBIC).

Matters relating to the levy and collection of all direct taxes are looked after by the CBDT whereas those relating to levy and collection of goods and service taxes (GST), Customs and central excise duties, service tax and other Indirect taxes fall within the purview of the CBIC.

Government receipts are classified under two categories:

1. Revenue receipts which consists of tax revenue and non tax revenue.
2. Capital receipts which consists of debt receipts and non debt capital receipts

The broad sources of revenue are:

(a) Corporation tax
(b) Taxes on income
(c) Wealth tax
(d) Customs duties
(e) Union excise duties
(f) Goods and services tax including GST compensation cess
(g) Taxes on union territories

Centre's net tax revenue is the total of tax revenue after paying of the states' share and the National Calamity Contingent duty (NCCD) transferred to the National Calamity Contingency

Non-tax revenues comprise the following:

(a) Interest receipts,
(b) Dividends and profits from public sector enterprises and surplus transfers from Reserve Bank of India
(c) Other Non-tax revenues and
(d) Receipts of union territories

Various social services provided by the government such as medical services, public health: broadcasting, education, sports, art and culture, housing: and economic services such as communication, energy, transport, science, technology and environment, railways and general administrative services also yield revenue for the government.

Capital Receipts include:

1. Non debt capital receipts which include
   (a) Recoveries of loans and advances
   (b) Miscellaneous capital receipts (disinvestments and others)

2. Debt capital receipts which include
   (a) Market loans for different purposes
   (b) Short term /Treasury bill borrowings
   (c) Securities issued against small savings,
   (d) State provident fund (Net)
   (e) Net external debts
   (f) Other receipts (Net)

In short, non-debt receipts include recoveries of loans advanced by the government to PSEs, state governments, foreign governments and union territories and sale proceeds of government assets, including those realized from divestment of government equity in public sector undertakings (PSUs).

Debt capital receipts comprise of market loans and short term borrowings by the government, borrowing from the Reserve Bank of India and loans taken from foreign governments/institutions. Examples of 'Other receipts' include Sovereign Gold Bond Scheme, receipts from international financial institutions and saving bonds.

3.4 Public Expenditure Management

Developing economies like India require enormous amount of public spending to initiate and accelerate economic growth and to promote employment opportunities.

Effective reduction in fiscal deficit requires an ingenious mix of revenue and expenditure policies.

Government expenditure affects allocation of resources among various uses and therefore, great care should be taken to channelize the resources to socially desirable areas.

Public expenditure management is the process that allows governments to be fiscally responsible.

Public expenditure programmes or projects should be designed and implemented to provide given levels of outputs or achieve specific objectives at minimum cost.

The economic costs of unproductive public expenditures can be extensive .and may have far reaching effects such as:

- larger deficits
- higher levels of taxation,
- lower economic growth,
- fewer resources available for use elsewhere, and
- greater debt burden in the future.

The Department of Expenditure of the Ministry of Finance is the nodal department for overseeing the public financial management system in the central government and matters connected with state finances.

It is responsible for

- the implementation of the recommendations of the Finance Commission and the Central Pay Commission,
- monitoring of audit comments/observations, and
- preparation of central government accounts.

Additionally, it also assists central ministries/departments in

- controlling the costs and prices of public services,
- reviewing systems and procedures to optimize outputs and outcomes of public expenditure.

The requirements of funds for all categories of expenditure including various programmes and schemes, along with receipts of the departments are discussed during the pre-budget meetings chaired by Secretary (Expenditure).

Expenditure estimates are provisionally finalized and communicated to ministries / departments after the approval of Finance Minister.

One of the explanatory documents of the budget document is the 'Expenditure Profile' (earlier known as expenditure budget) consisting of relevant data across all ministries/departments to outline a profile of the general financial performance of the government of India.

It gives an aggregation of various types of expenditure and certain other items across demands.

The total expenditure through budget (both current and capital) of various ministries and departments is composed of central expenditure and transfers.

In Expenditure budget, the Central government expenditure is classified into six broad categories as below:

A. Centre's Expenditure:

- Establishment Expenditure of the Centre;
- Central sector schemes, and
- Other central expenditures including those on CPSEs and Autonomous Bodies

B. Centrally Sponsored Schemes and other Transfers:

The transfers include

- Centrally sponsored schemes
- Finance Commission transfers and
- Other transfers to states

Establishment expenditure includes establishment-related expenditure of the ministries /departments, and attached and subordinates offices.

Central Sector Schemes (CS) include those schemes which are entirely funded and implemented by the central agencies under union government ministries/departments.

3.5 Public Debt Management

Government debt from internal and external sources contracted in the Consolidated Fund of India is defined as Public Debt.

The government raises funds primarily from the domestic market using market-based and fixed-rate instruments to finance its fiscal deficit.

Public debt, in simple words, means debt incurred by the government in mobilizing savings of the people in the form of loans, which are to be repaid at a future date with interest.

Debt servicing is a continuous exercise as a portion of debt falls due each month, government does not usually cut expenditure or raise taxes to provide funds to retire or repay the maturing bonds.

Rather, the government simply refinances the debt, i.e. it sells new bonds and uses the proceeds to pay off holders of the maturity bonds.

Productive use of public debt contributes to economic growth and welfare of the society.

Sustainability of sovereign debt has always been an important indicator of the overall macroeconomic health of a country.

Debt sustainability is in great part a function of the level of debt and the government's capacity to service the outstanding debt.

Public debt management refers to the task of determining, by the fiscal and monetary authorities, the size and composition of debt, the maturity pattern, interest rates, redemption of debt etc. It is the process of setting up and implementing the strategy for managing public debt in order to raise the required amount of funding at the desired risk and cost levels.

The overall objective of the central government's debt management policy is to "meet the central government's financing needs at the lowest possible long term borrowing costs and also to keep the total debt within sustainable levels. Additionally, it aims at supporting development of a well-functioning and vibrant domestic bond market".

Debt management strategy is based on three broad pillars namely, low cost of borrowing, risk mitigation and market development.

The institutions responsible for public debt management are:

1. Reserve Bank of India – domestic marketable debt i.e., dated securities, treasury bills and cash management bills.
2. Ministry of Finance (MOF); – external debt
3. Ministry of Finance; Budget Division and Reserve Bank of India – Other liabilities such as small savings, deposits, reserve funds etc.

The responsibility of managing the domestic debt of the central government and of 28 state governments and two union territories is entrusted with the Internal Debt Management Department (IDMD) of the Reserve Bank of India.

The RBI acts as the debt manager for marketable internal debt. While treasury bills are issued to meet short term cash requirements of the government, dated securities are issued to mobilise longer term resources to finance the fiscal deficit.

From 1997 onwards, the Reserve Bank also provides short-term credit up to three months to state governments banking with it in the form of Ways and Means Advances (WMA) to bridge temporary mismatches in cash flows.

External debt (bilateral and multilateral loans) is managed by the Department of Economic Affairs in the Ministry of Finance (MoF). Most of the external debt is sourced from multilateral agencies (International Bank for Reconstruction and Development, Asian Development Bank, etc.).

There is no sovereign borrowing from international capital markets.

The entire external debt, in terms of original maturity, is on long -term basis and a major part is at fixed interest rates.

The risk associated with external debt is the depreciation in the value of the domestic currency vis-à-vis the currency of denomination of external loans leading to increase in the government's debt servicing cost.

The Fiscal Responsibility and Budget Management (FRBM) was passed in 2003 to provide a legislative framework for reduction of deficit and thereby debt of the central government to a sustainable level. The objectives of the act are:

- inter-generational equity in fiscal management,
- long run macroeconomic stability,
- better coordination between fiscal and monetary policy, and
- transparency in fiscal operation of the government.

The Public Debt Management Cell (PDMC) was created in 2016 under the Department of Economic Affairs. The Medium Term Debt Management Strategy or MTDS 2021-24 is a framework to determine the appropriate composition of the debt portfolio.

The objective of the debt management strategy is to efficiently raise debt at the lowest possible cost in the medium term while ensuring that financing requirements are met without disruption.

The sheer size of India's public debt can be understood from the following table:

Debt Position of the Government of India **(in Rs. crores)**

	As on 31st March 2023	As on 31st March 2024
Internal debt and other liabilities	147,77,724.43	164,23,983.04
External debt#	4,83,397.69	5,22,683.81
Total	152,61,122.12	169,46,666.85

The Reserve Bank has been proactively engaged in the development of the government securities (G-sec) market including broadening of investor participation. As part of continuing efforts to increase retail participation in G-sec, 'RBI Retail Direct' facility was announced on February 5, 2021:

- for improving the ease of access by retail investors through online access to the primary and secondary government securities market
- to provide the facility to open their government securities account ('Retail Direct') with the Reserve Bank.

Budget concepts Type of budgets

- Balanced budget: - A balanced budget is a budget in which revenues are equal to expenditures. Thus, neither a budget deficit nor a budget surplus exists. Revenue does not fall short of expenditure. i.e., revenue is equal to expenditure (Revenue= Expenditure).
- Unbalanced budget: The budget may either be surplus or deficit.
  - ✓ A surplus budget: when estimated government receipts are more than the estimated government expenditure it is termed as surplus budget. When the government spends less than the receipts the budget becomes surplus. Briefly put, public revenue exceeds public expenditure (R>E.)
  - ✓ A deficit budget: when estimated government receipts are less than the government expenditure, it is termed as a deficit budget. A deficit budget increases the liability of the government or decreases its reserves. In modern economies, most of the countries follow deficit budgeting.

Capital Receipts

Capital receipts are those receipts that lead to a reduction in the assets or an increase in the liabilities of the government. Examples include recoveries of loans, earnings from disinvestment and debt.

Revenue Receipts

Revenue receipts can be defined as those receipts which neither create any liability nor cause any reduction in the assets of the government. There are two sources of revenue receipts for the government — tax revenues and non-tax revenues.

Revenue Expenditure

Revenue expenditure is expenditure incurred for purposes other than creation of physical or financial assets of the central government. It relates to those expenses incurred for the normal functioning of the government departments and various services, interest payments on debt incurred by the government, and grants given to state governments and other parties (even though some of the grants may be meant for creation of assets).

Capital Expenditure

There are expenditures of the government which result in creation of physical or financial assets or reduction in financial liabilities. This includes expenditure on the acquisition of land, building, machinery and equipment, investment in shares, and loans and advances by the central government to state and union territory governments, PSUs and other parties.

When a government spends more than it collects by way of revenue, it incurs a budget deficit. There are various measures that capture government deficit and they have their own implications for the economy.

Budgetary Deficit or Overall Deficit

Budgetary Deficit is defined as the excess of total estimated expenditure over total estimated revenue is the difference between all receipts and expenditure, both revenue and capital.

Revenue Deficit

The revenue deficit refers to the excess of government's revenue expenditure over revenue receipts. It shows the shortfall of government's current receipts over current expenditure. It shows the

government revenue is insufficient to meet the regular expenditures in connection with the normal functioning of the government, or the government is diverting resources from other sectors to finance its current expenditure.

Revenue deficit = Revenue expenditure – Revenue receipts

Fiscal Deficit

When the government's non-borrowed receipts fall short of its entire expenditure, it has to borrow money from the public to meet the shortfall. The excess of total expenditure over total receipts excluding borrowings during a given fiscal year is called the fiscal deficit. In other words, fiscal deficit is the difference between the government's total expenditure and its total receipts excluding borrowing. It is often presented as a percentage of the gross domestic product (GDP).

Total Receipts excluding borrowing = Revenue Receipts + Capital Receipts excluding borrowing or (Non debt creating capital receipts). Non debt creating capital receipts include recoveries of loans advanced by the government and sale proceeds of government assets, including those realized from divestment of government equity in public sector undertakings (PSUs).

Fiscal deficit = Total Expenditure –Total Receipts excluding borrowing
Fiscal Deficit = (Revenue Expenditure + Capital Expenditure) – (Revenue Receipts + Capital Receipts excluding borrowing)
Fiscal Deficit = (Revenue Expenditure- Revenue Receipts) + (Capital Expenditure – Capital Receipts excluding borrowing)
Fiscal Deficit = Revenue Deficit + (Capital Expenditure - Capital Receipts excluding borrowing)

The fiscal deficit will have to be financed by borrowing. Therefore fiscal deficit points to the total borrowing requirements of the government from all sources. In case revenue deficit occupies a substantial share of fiscal deficit, it is an indication that a large part of borrowing is used for consumption purposes rather than for investment.

Primary Deficit

Primary deficit is defined as fiscal deficit of current year minus interest payments on previous borrowings. In other words whereas fiscal deficit indicates borrowing requirement inclusive of interest payment, primary deficit indicates borrowing requirement exclusive of interest payment.
It tells how much of the government's borrowings are going towards meeting expenses other than interest payments. Primary deficit thus gives an estimate of borrowings on account of current expenditure exceeding current revenues. The goal of measuring primary deficit is to focus on present fiscal imbalances.

Primary deficit = Fiscal deficit – Net Interest liabilities

Net interest liabilities interest payments minus interest receipts by the government on domestic lending.

Finance Bill

The Bill produced immediately after the presentation of the union budget detailing the Imposition, abolition, alteration or regulation of taxes proposed in the budget.

Outcome budget

The outcome budget establishes a direct link between budgetary allocations of schemes and its annual performance targets measured through output and outcome indicators.

The outcome budget is a progress card on what various ministries and departments have done with the outlays in the previous annual budget. It measures the development outcomes of all government programs and whether the money has been spent for the purpose it was sanctioned including the outcome of the fund usage.

Guillotine

The parliament has very limited time for examining the expenditure demands of all the ministries. So, once the prescribed period for the discussion on demands for grants is over, the speaker of Lok Sabha puts all the outstanding demands for grants, whether discussed or not, to the vote of the house. This process is popularly known as 'Guillotine'.

Cut Motions

Motions for reduction to various demands for grants are made in the form of cut motions seeking to reduce the sums sought by government on grounds of economy or difference of opinion on matters of policy or just in order to voice a grievance.

Consolidated Fund of India

All revenues received, loans raised and all moneys received by the government in repayment of loans are credited to the Consolidated Fund of India and all expenditures of the government are incurred from this fund. Money can be spent through this fund only if appropriated by the parliament. The consolidated Fund has further been divided into 'revenue' and 'capital' divisions.

Contingency Fund of India

A fund placed at the disposal of the President to enable him/her to make advances to the executive/Government to meet urgent unforeseen expenditure. Contingency fund enables the government to meet unforeseen expenditure and does not require prior legislative approval, unlike with the Consolidated Fund. For meeting such exigencies, advances are made to the executive from the contingency fund which is subsequently reported to the Parliament for recoupment from the Consolidated Fund of India.

Public Account

Under provisions of Article 266(1) of the Constitution of India, public account is used in relation to all the fund flows where government is acting as a banker. Examples include Provident Funds and Small Savings. This money does not belong to government but is to be returned to the depositors. The expenditure from this fund need not be approved by the parliament.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. The difference between the budget deficit of a government and its debt service payments is
   (a) Fiscal deficit (b) Budget deficit (c) Primary deficit (d) None of the above

2. *The following hypothetical figures relate to country A* *₹ Crores*

Revenue receipts	*20,000*
Recovery of loans	*1,500*
Borrowing	*15,000*
Other Receipts	*5,000*
Expenditure on revenue account	*24,500*
Expenditure on capital account	*26,000*
Interest payments	*2,000*

2.1 The revenue deficit for country A is
(a) 5,000 (b) 24,000 (c) 4,500 (d) None of the above

2.2 Fiscal deficit of country A is
(a) 14,000 (b) 24,000 (c) 23,500 (d) None of the above

2.3 Primary deficit of Country A is
(a) 26,000 (b) 26,500 (c) 22,000 (d) 24,500

3. In NITI Aayog, NITI stands for
   (a) National Initiative for Transforming India
   (b) National Institution for Transforming India
   (c) National Institute for Technology and Innovation
   (d) None of the above

4. The Appropriation Bill is intended to
   (a) reduce unnecessary expenditure on the part of the government
   (b) give authority to government to incur expenditure from and out of the Consolidated Fund of India
   (c) give authority to government to incur expenditure from the revenue receipts only
   (d) be passed before the budget is taken for discussion

5. Public debt management aims at
   (a) An efficient budgetary policy to avail of domestic debt facilities
   (b) Raising loans from international agencies at lower rates of interest
   (c) Raising the required amount of funding at the desired risk and cost levels
   (d) Management of public expenditure to reduce public debt

6. The railway budget is
   (a) Part of the general budget, but is presented by the railway minister
   (b) Part of the general budget from the budget for financial year 2017 -18.
   (c) Part of the general budget from the budget for financial year 2021 -22
   (d) Part of the general budget but presented on the next day of the general budget

7. Outcome budgeting
   (a) *shares information about the money allocated for various purposes in a budget*
   (b) *establishes a direct link between budgetary allocations and performance targets measured through output and outcome indicators*
   (c) *establishes a direct link between budgetary performance targets and public account disbursals*
   (d) *shares information about public policies and programmes under the budget*

8. Corporate tax
   (a) *is collected by the union government and can be a capital receipt or revenue receipt*
   (b) *may be collected by the respective states and fall under revenue receipts*
   (c) *may be collected either by the centre or states and fall under revenue receipts*
   (d) *is collected by the union government and is a revenue receipt*

9. Government borrowings from foreign governments and institutions
   (a) *Capital receipt* (b) *Revenue receipt*
   (c) *Accounts for fiscal deficit* (d) *Any of the above depending on the purpose of borrowing*

10. *The following table relates to the revenue and expenditure figures of a hypothetical economy*

(a)	*Recovery of loans*	*5.1*
(b)	*Salaries of govt. servants*	*41.1*
(c)	*Capital Expenditure*	*45.0*
(d)	*Interest payments*	*1.3*
(e)	*Payments towards subsidies*	*3.2*
(f)	*Other receipts (mainly from disinvestment)*	*11.6*
(g)	*Tax revenue (net of states' share)*	*26.3*
(h)	*Non-tax revenue*	*12.3*
(i)	*Borrowings and other liabilities*	*6.8*
(j)	*States' share in tax revenue*	*11.9*

10.1 The capital receipts are
   (a) *23.5* (b) *19.7* (c) *11.3* (d) *None of the above*

10.2 Revenue deficit is
   (a) *23.6* (b) *13.0* (c) *7.0* (d) *2.6*

10.3 The non–debt capital receipts of this country is
   (a) *45.1* (b) *16.7* (c) *15.8* (d) *None of the above*

11 A budget is said to be unbalanced when
   (a) *when government's revenue exceeds government's expenditure*
   (b) *when government's expenditure exceeds government's revenue*
   (c) *either budget surplus of budget deficit occurs*
   (d) *All the above*

12 Fiscal deficit refers to
   (a) *the excess of government's revenue expenditure over revenue receipts*
   (b) *The excess of total expenditure over total receipts excluding borrowings*
   (c) *Primary deficit - interest payments*
   (d) *None of these*

13 Budget of the government generally impacts
(a) *the resource allocation in the economy*
(b) *redistribution of income and enhance equity*
(c) *stability in the economy by measures to control price fluctuations*
(d) *all the above*

14 Which of the following is a statement submitted along with the budget as a requirement of FRBM Act
(a) *Annual Financial Statement*
(b) *Macro -Economic Framework Statement*
(c) *Medium-Term Fiscal Policy cum Fiscal Policy Strategy Statement*
(d) *(b) and (c) above*

15 Government borrowing is treated as capital receipt because
(a) *It is mainly used for creating assets by government*
(b) *It creates a liability for the government*
(c) *Both a) and b) above are correct*
(d) *None of the above is correct*

16 'Retail Direct 'scheme is
(a) *Initiated by the Reserve Bank of India*
(b) *facilitate investment in government securities by individual investors.*
(c) *Direct sale of goods and services by government departments*
(d) *Both (a) and (b) are correct*

17 Non-debt capital receipts
(a) *do not add to the assets of the government and therefore not treated as capital receipts*
(b) *are those that do not create any future repayment burden for the government*
(c) *are those that create future liabilities for the government*
(d) *facilitate capital investments at low cost*

18 Which of the following is a capital receipt?
(a) *Licence fee received*
(b) *Sale proceeds from disinvestment*
(c) *Assistance from Japan for covid vaccine*
(d) *Dividend from a public sector enterprise*

19 Grants given by the central government to state governments is
(a) *A revenue expenditure as it is meant to meet the current expenditure of the states*
(b) *A revenue expenditure as it does neither creates any asset, nor reduces any liability of the government*
(c) *A capital expenditure because it increase the capital base of the states*
(d) *It is a grant and so does not come under revenue expenditure or capital expenditure.*

20 Short-term credit from the Reserve Bank to state governments to bridge temporary mismatches in cash flows is known as
(a) *RBI credit to states*
(b) *Commercial credit of RBI*
(c) *Ways and Means Advances (WMA)*
(d) *Short term facility*

PEQ SEPTEMBER - 2024

21 Under which Supreme Court verdict do the Union and State Legislatures have "equal, simultaneous and unique powers" to make laws on Goods and Services Tax (GST)?

(a) July 2021 verdict *(b) June 2020 verdict*
(c) Feb. 2022 verdict *(d) May 2022 verdict* *(1 mark)*

22 Under institutional responsibility for public debt management, external debt is the responsibility of:

(a) Ministry of Finance *(b) Reserve Bank of India*
(c) Department of Economic Affairs *(d) State Government* *(1 mark)*

23 If the primary deficit is ₹ 20,500 crores and the net interest liabilities of a country are ₹ 3,500 crores then what will be the fiscal deficit?

(a) ₹ 21,500 crores *(b) ₹ 17,000 crores* *(c) ₹ 19,500 crores* *(d) ₹ 24,000 crores*
(1 mark)

24 The following figures relate to country A for a particular financial year:

Particulars	Amount (₹) in crores
Revenue deficit	10,000
Fiscal deficit	24,000
Net interest liability	2,000
Borrowing	6,000
Expenditure on revenue account	3,000

What will be the primary deficit of country 'A'?

(a) ₹ 22,000 crores *(b) ₹ 8,000 crores* *(c) ₹ 18,000 crores* *(d) ₹ 21,000 crores*
(1 mark)

PEQ JANUARY - 2025

25 Non-debt capital receipts of government include:

(a) Market loans for different purposes *(b) State provident fund (Net)*
(c) Securities issued against small savings *(d) Recoveries of loans and advances*
(1 mark)

26 The excess of the government's total expenditure over its total receipts excluding borrowings is termed as:

(a) Revenue deficit *(b) Fiscal deficit* *(c) Primary deficit* *(d) Budgetary deficit*
(1 mark)

27 The receipts which neither create any liability nor cause any reduction in the assets of government are called:

(a) Non-debt capital receipts *(b) Debt capital receipts*
(c) Revenue receipts *(d) Estimated receipts* *(1 mark)*

PEQ MAY - 2025

28. Article 112 of Indian Constitution provides that in respect of every financial year the President shall cause to be laid before both houses of the parliament a statement of estimated receipts and expenditure of the government of India for that year. This statement is referred as:

(a) Budget
(b) Annual Financial Statement
(c) Statement of Income & Expenditure
(d) Interim Budget

29. Which of the following information regarding the receipts and expenditure of the Government is not presented in the budget documents?
(a) *Budget estimates of Current financial year*
(b) *Budget estimates of the Ensuing financial year*
(c) *Revised estimates of the Ensuing financial year*
(d) *Actual expenditure and Income of Current financial year*

30. Which of the following is the nodal department for overseeing the public financial management system in the Central Government?
(a) *NITI Aayog*
(b) *Department of Revenue under Finance Ministry*
(c) *Department of Expenditure under Finance Ministry*
(d) *Reserve Bank of India*

31. During the Budget proceedings, the speaker of Lok Sabha once the prescribed time is over, puts all the outstanding demand for grants, whether discussed or not, to the vote of the house, this process is known as:
(a) *Cut Motion*
(b) *Presenting the appropriation bill*
(c) *Outcome Budget*
(d) *Guillotine*

PEQ – SEPTEMBER - 2025

32. If a government's revenue receipts for a year are ₹ 10,000 crore, capital receipts excluding borrowing are ₹ 2,000 crore, revenue expenditure is ₹ 12,000 crore and capital expenditure is ₹ 5,000 crore. What is the fiscal deficit?
(a) *₹ 5,000 crore*
(b) *₹ 7,000 crore*
(c) *₹ 10,000 crore*
(d) *₹ 9,000 crore*

33. Under Article 266(1) of the Constitution of India, which of the following is used in relation to all fund flows where the government acts as a banker?
(a) *Public Account*
(b) *Consolidated Fund*
(c) *Contingency Fund*
(d) *State Development Fund*

34. _____ is defined as the excess of total estimated expenditure over total estimated revenue and is the difference between all receipts and expenditures, both revenue and capital.
(a) *Fiscal Deficit*
(c) *Budgetary Deficit*
(8) *Revenue Deficit*
(d) *Primary Deficit*

PEQ – JANUARY - 2025

Using below mentioned data, answer Question No. 35 and 36:

Revenue receipts	₹ 25,000 Cr.
Expenses on revenue account	₹ 30,000 Cr.
Expenses on capital account	₹ 30,000 Cr.
Estimated total expenditure	₹ 60,000 Cr.
Estimated total revenue	₹ 54,000 Cr.

35. Calculate the revenue deficit:
   (a) *₹ 5,000 crores*
   (b) *₹ 12,000 crores*
   (c) *₹ 2,000 crores*
   (d) *₹ 8,000 crores*

36. Calculate the budgetary deficit.
   (a) *₹ 25,000 Crores*
   (b) *₹ 5,000 Crores*
   (c) *₹ 6,000 Crores*
   (d) *₹ 12,000 Crores*

ANSWERS / HINTS

I. Multiple Choice Type Questions

1.	(c)	2.1	(c)	2.2	(b)	2.3	(c)	3.	(b)	4.	(b)
5.	(c)	6.	(b)	7.	(b)	8.	(d)	9.	(a)	10.1	(a)
10.2	(c)	10.3	(b)	11.	(d)	12.	(d)	13.	(d)	14.	(d)
15.	(b)	16.	(d)	17	(b)	18.	(b)	19.	(b)	20.	(c)
21.	(a)	22.	(c)	23.	(d)	24.	(a)	25.	(d)	26.	(b)
27.	(c)	28.	(b)	29.	(d)	30.	(c)	31.	(d)	32.	(a)
33.	(a)	34.	(c)	35.	(a)	36.	(c)				

For Your Notes

For Your Notes

For Your Notes

For Your Notes

UNIT – IV
FISCAL POLICY

Topic References:

4.1 Introduction
4.2 Objectives of Fiscal Policy
4.3 Instruments of Fiscal Policy
4.4 Types of Fiscal Policy
4.5 Fiscal Policy for long-run Economic Groth
4.6 Fiscal Policy for Reduction in Inequalities of Income and Wealth
4.7 Limitation to Fiscal Policy

4.1 Introduction

Governments of all countries pursue innumerable policies to accomplish their economic goals such as rapid economic growth, equitable distribution of wealth and income, reduction of poverty, price stability, exchange rate stability, full-employment, balanced regional development etc.

Government budget is one among the most powerful instruments of economic policy. The important tools in the budgetary policy could be broadly classified into public revenue including taxation, public expenditure, public debt and finally deficit financing to bridge the gap between public receipts and payments.

Fiscal policy involves the use of government spending, taxation and borrowing to influence both the pattern of economic activity and level of growth of aggregate demand, output and employment.

Fiscal policy is designed to influence the pattern and level of economic activity in a country. Fiscal policy is in the nature of a demand-side policy.

4.2 Objectives of Fiscal Policy

The most common objectives of fiscal policy are:

- Achievement and maintenance of full employment,
- maintenance of price stability,
- acceleration of the rate of economic development, and
- equitable distribution of income and wealth,

The importance as well as order of priority of these objectives may vary from country to country and from time to time.

For instance, while stability and equality may be the priorities of developed nations, economic growth, employment and equity may get higher priority in developing countries.

Governments may directly as well as indirectly influence the way resources are used in an economy.

We know that GDP is the value of all final economic goods and services produced in an economy during a given period of time.

$$GDP = C + I + G + NX.$$

The right side of the equation shows the different sources of aggregate spending or demand namely, private consumption (C), private investment (I), government expenditure i.e. purchases of goods and services by the government (G), and net exports, (exports minus imports) (NX).

It is evident from the equation that governments can influence economic activity (GDP) by controlling G directly and influencing C, I, and NX indirectly, through changes in taxes, transfer payments and expenditure.

4.3 Instruments of Fiscal Policy

The ability of fiscal policy to influence output by affecting aggregate demand makes it a potential instrument for stabilization of the economy.

The Keynesian school is of the opinion that fiscal policy can have very powerful effects in altering aggregate demand, employment and output in an economy when the economy is operating at less than full employment levels and when there is need to offer stimulus to demand.

The tools of fiscal policy are taxes, government expenditure, public debt and the government budget.

4.3.1 Government Expenditure

Public expenditures are income generating and include all types of government expenditure such as capital expenditure on public works, relief expenditures, subsidy payments of various types, transfer payments and other social security benefits.

Government expenditure is an important instrument of fiscal policy. It includes governments' expenditure towards consumption, investment, and transfer payments. Government expenditures include:

1. current expenditures to meet the day to day running of the government,
2. capital expenditures which are in the form of investments made by the government in capital equipment and infrastructure, and
3. transfer payments i.e. government spending which does not contribute to GDP because income is only transferred from one group of people to another without any direct contribution from the receivers.

Government may spend money on performance of its large and ever-growing functions and also for deliberately bringing in stabilization.

RECESSION:

During a recession, it may initiate a fresh wave of public works, such as construction of roads, irrigation facilities, sanitary works, ports, electrification of new areas etc.

Government expenditure involves employment of labour as well as purchase of multitude of goods and services. These expenditures directly generate incomes to labour and suppliers of materials and services.

Apart from the direct effect, there is also indirect effect in the form of working of multiplier.

The incomes generated are spent on purchase of consumer goods. The extent of spending by people depends on their marginal propensity to consume (MPC).

Primary employment in public works programmes will induce secondary and tertiary employment, and before long the economy is put on an expansion track.

INFLATION:

Public expenditure is also used as a policy instrument to reduce the severity of inflation and to bring down the prices.

This is done by reducing government expenditure when there is a fear of inflationary rise in prices. Reduced incomes on account of decreased public spending helps to eliminate excess aggregate demand.

4.3.2 Taxes

Taxes form the most important source of revenue for governments.

Taxation policies are effectively used for establishing stability in an economy.

Tax as an instrument of fiscal policy consists of changes in government revenues or in rates of taxes aimed at encouraging or restricting private expenditures on consumption and investment.

Taxes determine the size of disposable income in the hands of the general public which in turn determines aggregate demand and possible inflationary and deflationary gaps.

The structure of tax rates is varied in the context of the overall economic conditions prevailing in an economy.

RECESSION:

During recession and depression, the tax policy is framed to encourage private consumption and investment.

A general reduction in income taxes leaves higher disposable incomes with people inducing higher consumption.

Low corporate taxes increase the prospects of profits for business and promote further investment.

The extent of tax reduction and /or increase in government spending required depends on the size of the recessionary gap and the magnitude of the multiplier.

INFLATION:

During inflation, new taxes can be levied and the rates of existing taxes are raised to reduce disposable incomes and to wipe off the surplus purchasing power.

However, excessive taxation usually stifles new investments and therefore the government has to be cautious about a policy of tax increase.

4.3.3 Public Debts

A rational policy of public borrowing and debt repayment is a potent weapon to fight inflation and deflation.

Public debt may be internal or external;

When the government borrows from its own people in the country, it is called internal debt. On the other hand, when the government borrows from outside sources, the debt is called external debt.

Public debt takes two forms namely, market loans and small savings.

In the case of market loans, the government issues treasury bills and government securities of varying denominations and duration which are traded in debt markets. For financing capital projects, long-term capital bonds are floated and for meeting short-term government expenditure, treasury bills are issued.

The small savings represent public borrowings, which are not negotiable and are not bought and sold in the market. In India, various types of schemes are introduced for mobilising small savings e.g., National Savings Certificates, National Development Certificates, etc.

Borrowing from the public through the sale of bonds and securities curtails the aggregate demand in the economy.

Repayments of debt by governments increase the availability of money in the economy and increase aggregate demand.

4.3.4 Budget

Government's budget is widely used as a policy tool to stimulate or contract aggregate demand as required.

The budget is simply a statement of revenues earned from taxes and other sources and expenditures made by a nation's government in a year.

A government's budget can either be balanced, surplus or deficit.

BALANCED BUDGET - A balanced budget results when expenditures in a year equal its revenues for that year. Such a budget will have no net effect on aggregate demand since the leakages from the system in the form of taxes collected are equal to the injections in the form of expenditures made.

SURPLUS BUDGET - A budget surplus that occurs when the government collects more than what it spends, though sounds like a highly attractive one, has in fact a negative net effect on aggregate demand since leakages exceed injections.

DEFICIT BUDGET - A budget deficit wherein the government expenditure in a year is greater than the tax revenue it collects has a positive net effect on aggregate demand since total injections exceed leakages from the government sector.

While a budget surplus reduces national debt, a budget deficit will add to the national debt.

A nation's debt is the difference between its total past deficits and its total past surpluses.

If a government has borrowed money over the years to finance its deficits and has not paid it back through accumulated surpluses, then it is said to be in debt.

4.4 Types of Fiscal Policy

a) Expansionary Fiscal Policy

Expansionary fiscal policy is designed to stimulate the economy during the contractionary phase of a business cycle or when there is an anticipation of a business cycle contraction.

A recession is said to occur when the overall economic activity declines, or in other words, when the economy 'contracts'.

A 'demand-deficient' recession sets in with a period of falling real GDP, low aggregate demand and reduced consumer spending and rising unemployment.

To combat such a slump in overall economic activity, the government can resort to expansionary fiscal policies. We may technically refer to this as a policy measure to close a 'recessionary gap'. How does the government achieve this?

- The government may cut all types of taxes, direct and indirect, leaving the taxpayers with extra money to spend so that there is more purchasing power and more demand for goods and services. Consequently aggregate demand, output and employment increase.
- An increase in government expenditure (discussed in detail below) will pump money into the economy and increase aggregate demand. This in turn will increase output and employment.
- A combination of increase in government spending and decrease in personal income taxes and/or business taxes

While resorting to expansionary fiscal policy, the government may run into budget deficits because tax cuts reduce government income and the government expenditures exceed tax revenues in a given year.

b) *Contractionary fiscal policy*

Contractionary fiscal policy is basically the opposite of expansionary fiscal policy. Contractionary fiscal policy is designed to restrain the levels of economic activity of the economy during an inflationary phase or when there is anticipation of a business-cycle expansion which is likely to induce inflation.

Contractionary fiscal policy refers to the deliberate policy of government applied to curtail aggregate demand and consequently the level of economic activity.

Contractionary fiscal policy works through:

- Decrease in government spending: With decrease in government spending, the total amount of money available in the economy is reduced which in turn has the effect of reducing the aggregate demand.
- Increase in personal income taxes and/or business taxes: An increase in personal income taxes reduces disposable incomes leading to fall in consumption spending and aggregate demand. An increase in taxes on business profits reduces the surpluses available to businesses, and as a result, firms' investments shrink causing aggregate demand to fall. Increased taxes also dampen the prospects of profits of potential entrants who will respond by holding back fresh investments.
- A combination of decrease in government spending and increase in personal income taxes and/or business taxes.

Contractionary fiscal policy should ideally lead to a smaller government budget deficit or a larger budget surplus.

We have understood in general that governments influence the economy through their policies in respect of taxation, expenditure and borrowing. To sum up:

- during inflation or when there is excessive levels of utilization of resources, fiscal policy aims at controlling excessive aggregate spending, and
- during deflation or during a period of sluggish economic activity when the rate of utilization of resources is less, fiscal policy aims to compensate the deficiency in effective demand by boosting aggregate spending.

4.5 Fiscal Policy for Long-run Economic Growth

Fiscal policy influence economic growth through its effects on the incentives faced by individuals and firms. For example;

- Fiscal policies such as those involving infrastructure spending generally have positive supply-side effects. When government supports building a modern infrastructure, the private sector is provided with the requisite overheads it needs.
- Government provision of public goods such as education, healthcare, nutrition, research and development etc. provide momentum for long-run economic growth through human capital formation. Increase in human capital makes physical capital more productive.
- Taxes can have either positive or negative impact on economic growth depending on whether it encourages or discourages saving and investment.
- A well designed tax policy that rewards innovation and entrepreneurship, without discouraging incentives will promote private businesses who wish to invest and thereby help the economy grow. For example, an increase in corporate taxes to raise extra revenue may have adverse consequences on incentives and output.
- Tax and spending policies (e.g. subsidies) can be effectively used to correct market failures resulting from externalities.
- Increase in environment taxes increase the cost of firms and reduce their output
- Subsidies on inputs and support prices to producers (e.g. farmers) generate higher output.

4.6 Fiscal Policy for Reduction in Inequalities of Income and Wealth

Fiscal policy is a chief instrument available for governments to influence income distribution and plays a significant role in reducing inequality and achieving equity and social justice.

The distribution of income in the society is influenced by fiscal policy both directly and indirectly.

While current disposable incomes of individuals and corporates are dependent on direct taxes, the potential for future earnings is indirectly influenced by the nation's fiscal policy choices.

Government revenues and expenditure have traditionally been regarded as important instruments for carrying out desired redistribution of income. A few measures as to how each of these can be manipulated to achieve desired distributional effects are -

- A progressive direct tax system ensures that those who have greater ability to pay contribute more towards defraying the expenses of government and that the tax burden is distributed fairly among the population.
- Indirect taxes can be differential: for example, the commodities which are primarily consumed by the richer income group, such as luxuries, are taxed heavily and the commodities the expenditure on which form a larger proportion of the income of the lower income group, such as necessities, are taxed light.
- A carefully planned policy of public expenditure helps in redistributing income from the rich to the poorer sections of the society. This is done through spending programmes targeted on welfare measures for the disadvantaged , such as

(i) poverty alleviation programmes
(ii) free or subsidized medical care, education, housing, essential commodities etc. to improve the quality of living of poor
(iii) infrastructure provision on a selective basis
(iv) various social security schemes under which people are entitled to old-age pensions, unemployment relief, sickness allowance etc.
(v) subsidized production of products of mass consumption
(vi) public production and/ or grant of subsidies to ensure sufficient supply of essential goods,
(vii) Strengthening of human capital for enhancing employability etc..

Choice of a progressive tax system with high marginal taxes may act as a strong deterrent to work, save and invest.

Therefore, the tax structure has to be carefully framed to mitigate possible adverse impacts on production and efficiency.

4.7 Limitation to Fiscal Policy

There are some significant limitations in respect of choice and implementation of fiscal policy.

1. One of the biggest problems with using discretionary fiscal policy to counteract fluctuations is the different types of lags involved in fiscal-policy action:

- **Recognition lag**: The economy is a complex phenomenon and the state of the macro economic variables is usually not easily comprehensible. Just as in the case of any other policy, the government must first recognize the need for a policy change.
- **Decision lag**: Once the need for intervention is recognized, the government has to evaluate the possible alternative policies. Delays are likely to occur to decide on the most appropriate policy.
- **Implementation lag**: Even when appropriate policy measures are decided on, there are possible delays in bringing in legislation and implementing them.
- **Impact lag**: This lag occurs when the outcome of a policy are not visible for some time.

2. Fiscal policy changes may at times be badly timed due to the various lags so that it is highly possible that an expansionary policy is initiated when the economy is already on a path of recovery and vice versa.
3. There are difficulties in instantaneously changing governments' spending and taxation policies.
4. It is practically difficult to reduce government spending on various items such as defence, social security and huge capital projects which are already midway.
5. Public works cannot be adjusted easily along with movements of the trade cycle because many huge projects such as highways and dams have long gestation period. Besides, some urgent public projects cannot be postponed for reasons of expenditure cut to correct fluctuations caused by business cycles.
6. Due to uncertainties, there are difficulties of forecasting when a period of inflation or deflation may set in and also promptly determining the accurate policy to be undertaken.
7. There are possible conflicts between different objectives of fiscal policy such that a policy designed to achieve one goal may adversely affect another. For example, an expansionary fiscal policy may worsen inflation in an economy
8. Supply-side economists are of the opinion that certain fiscal measures will cause **disincentives**.

For example, increase in profits tax may adversely affect the incentives of firms to invest and an increase in social security benefits may adversely affect incentives to work and save.

9. Deficit financing increases the purchasing power of people. The production of goods and services, especially in under developed countries may not catch up simultaneously to meet the increased demand. This will result in prices spiraling beyond control.

10. Increase is government borrowing creates perpetual burden on even future generations as debts have to be repaid. If the economy lags behind in productive utilization of borrowed money, sufficient surpluses will not be generated for servicing debts. External debt burden has been a constant problem for India and many developing countries.

11. If governments compete with the private sector to borrow money for spending, it is likely that interest rates will go up, and firms' willingness to invest may be reduced. Individuals too may be reluctant to borrow and spend and the desired increase in aggregate demand may not be realized. This phenomenon is described below.

4.7.1 Crowding Out

Some economists are of the opinion that government spending would sometimes substitute private spending and when this happens the impact of government spending on aggregate demand would be smaller than what it should be and therefore fiscal policy may become ineffective.

An increase in the size of government spending during recessions will 'crowd-out' private spending in an economy and lead to reduction in an economy's ability to self-correct from the recession, and possibly also reduce the economy's prospects of long-run economic growth..

Crowding out effect is the negative effect fiscal policy may generate when money from the private sector is 'crowded out' to the public sector.

In other words, when spending by government in an economy replaces private spending, the latter is said to be crowded out.

For example, If government provides free computers to students, the demand from students for computers may not be forthcoming.

When government increases it's spending by borrowing from the loanable funds from market, the demand for loans increases and this pushes the interest rates up. Private investments are sensitive to interest rates and therefore some private investment spending is discouraged.

Similarly, when government increases the budget deficit by selling bonds or treasury bills, the amount of money with the private sector decreases and consequently interest rates will be pushed up.

As a result, private investments, especially the ones which are interest–sensitive, will be reduced.

Fiscal policy becomes ineffective as the decline in private spending partially or completely offset the expansion in demand resulting from an increase in government expenditure.

Nevertheless, during deep recessions, crowding-out is less likely to happen as private sector investment is already minimal and therefore there is only insignificant private spending to crowd out.

Moreover, during a recession phase the government would be able to borrow from the market without increasing interest rates.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. Fiscal policy refers to the
   (a) *use of government spending, taxation and borrowing to influence the level of economic activity*
   (b) *government activities related to use of government spending for supply of essential goods*
   (c) *use of government spending, taxation and borrowing for reducing the fiscal deficits*
   (d) *(b) and (c) above*

2. If real GDP is continuously declining and the rate of unemployment in the economy is increasing, the appropriate policy should be to
   (a) *Increase taxes and decrease government spending*
   (b) *Decrease both taxes and government spending*
   (c) *Decrease taxes and increase government spending*
   (d) *Either (a) or (c)*

3. Which of the following are likely to occur when an economy is in an expansionary phase of a business cycle?
   (A) Rising unemployment rate
   (B) Falling unemployment rate
   (C) Rising inflation rate
   (D) Deflation
   (E) Falling or stagnant wage for workers
   (F) Increasing tax revenue
   (G) Falling tax revenue
   (a) *A, B and F are most likely to occur*
   (b) *B, C and F are most likely to occur*
   (c) *D, E and F are most likely to occur*
   (d) *A, E and G are most likely to occur*

4. During recession the fiscal policy of the government should be directed towards
   (a) *Increasing the taxes and reducing the aggregate demand*
   (b) *Decreasing taxes to ensure higher disposable income*
   (c) *Increasing government expenditure and increasing taxes*
   (d) *None of the above*

5. According to Keynesian economics, when we have inflation an effective fiscal policy should not include
   (a) *increase corporate taxes.*
   (b) *decrease aggregate demand.*
   (c) *Increase government purchases.*
   (d) *None of the above is correct*

6. Keynesian economists believe that
   (a) *fiscal policy can have very powerful effects in altering aggregate demand, employment and output in an economy*
   (b) *when the economy is operating at less than full employment levels and when there is a need to offer stimulus to demand fiscal policy is of great use*
   (c) *Wages are flexible and therefore business fluctuations would be automatically adjusted*
   (d) *(a) and (b) above*

7. Which of the following may ensure a decrease in aggregate demand during inflation?
   (a) *decrease in all types of government spending and/ or an increase in taxes*
   (b) *increase in government spending and/ or a decrease in taxes*
   (c) *decrease in government spending and/ or a decrease in taxes*
   (d) *All the above*

8. A recession is characterized by
   (a) *Declining prices and rising employment*
   (b) *Declining unemployment and rising prices*
   (c) *Declining real income and rising unemployment.*
   (d) *Rising real income and rising prices*

9. Which one of the following is an example of fiscal policy?
   (a) *A tax cut aimed at increasing the disposable income and spending*
   (b) *A reduction in government expenditure to contain inflation*
   (c) *An increase in taxes and decrease in government expenditure to control inflation*
   (d) *All the above*

10. Which of the following would illustrate a recognition lag?
   (a) *The time required to identify the appropriate policy*
   (b) *The time required to identify to pass a legislation*
   (c) *The time required to identify the need for a policy change*
   (d) *The time required to establish the outcomes of fiscal policy*

11. An expansionary fiscal policy, taking everything else constant, would in the short-run have the effect of
   (a) *a relative large increase in GDP and a smaller increase in price*
   (b) *a relative large increase in price, a relatively smaller increase in GDP*
   (c) *both GDP and price will be increasing in the same proportion*
   (d) *both GDP and price will be increasing in a smaller proportion*

12. Which statement (s) is (are) correct about crowding out?

I. A decline in private spending may be partially or completely offset by the expansion of demand resulting from an increase in government expenditure.
II. Crowding out effect is the negative effect fiscal policy may generate when money from the private sector is 'crowded out' to the public sector.
III When spending by government in an economy increases government spending would be crowded out.
IV. Private investments, especially the ones which are interest –sensitive, will be reduced if interest rates rise due to increased spending by government

   (a) *I and III only* (b) *I, II, and III* (c) *I, II, and IV* (d) *III only*

13. Which of the following policies is likely to shift an economy's aggregate demand curve to the right?
   (a) *Increase in government spending* (b) *Decrease in taxes*
   (c) *A tax cut along with increase in public expenditure* (d) *All the above*

14. Identify the incorrect statement
   (a) *A progressive direct tax system ensures economic growth with stability because it distributes the burden of taxes unequally*
   (b) *A carefully planned policy of public expenditure helps in redistributing income from the rich to the poorer sections of the society.*

(c) *There are possible conflicts between different objectives of fiscal policy such that a policy designed to achieve one goal may adversely affect another*

(d) *An increase in the size of government spending during recessions may possibly 'crowd-out' private spending in an economy.*

15. Read the following statements

I. Fiscal policy is said to be contractionary when revenue is higher than spending i.e., the government budget is in surplus

II. Other things constant, a fiscal expansion will raise interest rates and "crowd out" some private investment

III. During inflation new taxes can be levied and the rates of existing taxes are raised to reduce disposable incomes

IV. Classical economists advocated contractionary fiscal policy to solve the problem of inflation

Of the above statements

(a) *I and II are correct* (b) *I, II and III are correct*

(c) *Only III is correct* (d) *All are correct*

16. While resorting to expansionary fiscal policy

(a) *the government may possibly have a budget surplus as increased expenditure will bring more output and more tax revenue*

(b) *the government may run into budget deficits because tax cuts reduce government income and the government expenditures exceed tax revenues in a given year*

(c) *it is important to have a balanced budget to avoid inflation and bring in stability*

(d) *None of the above will happen*

17. Contractionary fiscal policy

(a) *is resorted to when government expenditure is greater than tax revenues of any particular year*

(b) *increase the aggregate demand to sustain the economy*

(c) *to increase the disposable income of people through tax cuts and to enable greater demand*

(d) *is designed to restrain the levels of economic activity of the economy during an inflationary phase*

18. When government spending is deliberately reduced to bring in stability

(a) *the government is resorting to contractionary fiscal policy*

(b) *the government is resorting to expansionary fiscal policy*

(c) *trying to limit aggregate demand to sustainable levels*

(d) *(a) and c) above*

19. An increase in personal income taxes

(a) *reduces disposable incomes leading to fall in consumption spending and aggregate demand*

(b) *is desirable during inflation or when there is excessive levels of aggregate demand*

(c) *is to compensate the deficiency in effective demand by boosting aggregate spending*

(d) *both a) and b) are correct*

20. While the government resorts to deliberate fiscal policy it may not attempt to manipulate

(a) *Government expenditures on public works*

(b) *The rates of personal income taxes and corporate taxes*

(c) *Government expenditures on goods and services purchased by government*

(d) *The rate of interest prevailing in the economy*

21. Which of the following fiscal remedy would you advice when an economy is facing recession
   (a) *the government may cut interest rates to encourage consumption and investment*
   (b) *the government may cut taxes to increase aggregate demand*
   (c) *the government may follow a policy of balanced the budget.*
   (d) *None of the above will work*

22. While if governments compete with the private sector to borrow money for securing resources for expansionary fiscal policy
   (a) *it is likely that interest rates will go up and firms may not be willing to invest*
   (b) *it is likely that interest rates will go up and the individuals too may be reluctant to borrow and spend*
   (c) *it is likely that interest rates will go up and the desired increase in aggregate demand may not be realized*
   (d) *All the above are possible.*

PEQ - JUNE – 2024

23. Fiscal federalism in India is maintained by:
   (a) *Central Government* (b) *State Government*
   (c) *RBI* (d) *Finance Commission* *(1 mark)*

24. The policy designed to restrain the level of economic activities of economy during the inflationary phase is known as:
   (a) *Expansionary fiscal policy* (b) *Contractionary fiscal policy*
   (c) *Taxation policy* (d) *Budgetary policy* *(1 mark)*

25. In the period of sluggish economic activities when rate of utilization of resources is less, fiscal policy aims to compensate the deficiency in _____ by boosting _____.
   (a) *demand, private investment* (b) *supply, export activities*
   (c) *effective demand, aggregate spending* (d) *production, supply* *(1 mark)*

PEQ – SEPTEMBER – 2024

26. Which of the following are considered tools of fiscal policy?
   (a) *Taxes, government expenditure, public debt and the budget*
   (b) *Taxes, subsidies and tariffs*
   (c) *Monetary policy, interest rates and inflation targeting*
   (d) *Exchange rates, capital controls and quantitative easing* *(1 mark)*

PEQ – JANUARY – 2025

27. When the outcomes of a policy are not visible for some time, it is called:
   (a) *Recognition lag* (b) *Decision lag* (c) *Implementation lag* (d) *Impact lag*

PEQ – MAY – 2025

28. Which of the following is not true regarding the fiscal policy of Government?
   (a) *It is deliberate policy of Government.*
   (b) *An economy producing full employment does not require Government action in the form of fiscal policy.*
   (c) *Taxation policy is part of Fiscal policy.*
   (d) *Fiscal Policy is supply side policy.*

PEQ – SEPTEMBER - 2025

29. Contractionary fiscal policy typically involves:
    (a) *Increasing government spending and reducing taxes.*
    (b) *Reducing government spending and increasing taxes.*
    (c) *Increasing government spending and increasing taxes.*
    (d) *Reducing both government spending and taxes.*

PEQ – JANUARY - 2025

30. Which of the following is not listed as a common objective of fiscal policy?
    (a) *Equitable distribution of income and wealth*
    (b) *Maintenance of price stability*
    (c) *Promotion of exports and imports*
    (d) *Achievement* and maintenance of full employment

31. Which of the following is not an example of leakages from income stream?
    (a) *Progressive rates of taxation*
    (b) *High liquidity preference*
    (c) *Distributed profits of corporations*
    (d) *Demand for imported goods*

32. Laissez faire approach advocated:
    (a) *Minimum government interference in or regulation of economic activity*
    (b) *Increase in government spending to combat the recessionary forces*
    (c) *Having a more active fiscal policy*
    (d) *Increasing taxes to curtail aggregate demand*

ANSWERS / HINTS

I. Multiple Choice Type Questions

1.	(a)	2	(c)	3	(b)	4.	(b)	5.	(c)	6.	(d)	7.	(a)	8.	(c)
9.	(d)	10.	(c)	11.	(a)	12.	(c)	13.	(d)	14.	(a)	15.	(b)	16.	(b)
17.	(d)	18.	(d)	19.	(d)	20.	(d)	21.	(b)	22.	(d)	23.	(d)	24.	(b)
25.	(c)	26.	(a)	27.	(d)	28.	(d)	29.	(b)	30.	(c)	31.	(c)	32.	(a)

For Your Notes

For Your Notes

For Your Notes

For Your Notes

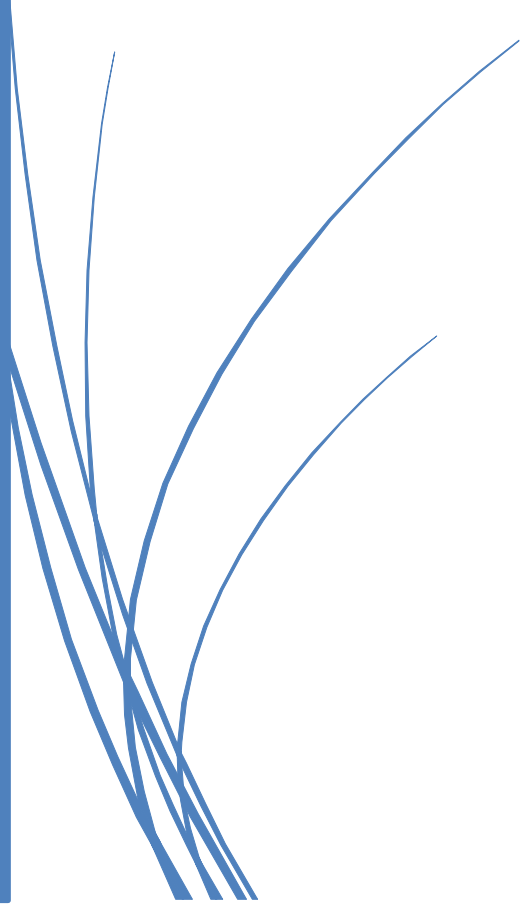
CHAPTER - 08

MONEY MARKET

Unit 1 – The Concept of Money Demand

Unit 2 – The Concept of Money Supply

Unit 3 – Monetary Policy



UNIT - I

THE CONCEPT OF MONEY DEMAND

Topic References:

1.1 Introduction to Money
1.2 The Demand of Money
1.3 Theories of Demand for Money
1.4 Post Keynesian Developments in Theory of Demand for Money

1.1 Introduction to Money

- Money is at the centre of every economic transaction and plays a significant role in all economies.
- Money provides us with a convenient means to access goods and services.
- Money refers to assets which are commonly used and accepted as
  - Means of payment ; or
  - Medium of exchange ; or
  - Transfer of purchasing power.
- Money may be defined as -
  - Set of liquid financial assets,
  - the variation in the stock of whichs
  - will have impact on aggregate economic activity.
- Money has generalized purchasing power and is generally acceptable in settlement of all transactions and in discharge of other kinds of business obligations including future payments.
- Anything that would act as a medium of exchange is not necessarily money.
  For example, a bill of exchange may also be a medium of exchange, but it is not money since it is not generally accepted as a means of payment.
- Money is a totally liquid asset as it can be used directly, instantly, conveniently and without any costs or restrictions to make payments.
- Money represents a certain value, but currency which represents money does not necessarily have intrinsic value. Fiat money has no intrinsic value, but is used as a medium of exchange because the government has, by law, made them "legal tender," which means that they serve by law as means of payment.
- Money is not necessarily a physical item; it may also constitute electronic records.
- Money is, in fact, only one among many kinds of financial assets which households, firms, governments and other economic units hold in their asset portfolios.

Functions of Money

Q. Money has 4 function: "a medium, a measure, a standard, and a store." Elucidate.
[PEQ-May 19 - 3 Marks]

(a) Medium of exchange
(b) Unit of value / unit of account
(c) Unit of Deferred Payment
(d) Store of Value
(e) Perfect Reversibility

Characteristics of Money

Q. Mention the general characteristics of Money? [PEQ-Nov 18 - 2 Marks]

(i) generally acceptable
(ii) durable or long-lasting
(iii) Effortlessly recognizable
(iv) difficult to counterfeit i.e. not easily reproducible by people
(v) relatively scarce, but has elasticity of supply
(vi) portable or easily transported
(vii) possessing uniformity
(viii) divisible into smaller parts in usable quantities or fractions without losing value

1.2 The Demand of Money

- If people desire to hold money, we say there is demand for money. The demand for money is in the nature of derived demand; it is demanded for its purchasing power.
- Basically, people demand money because they wish to have command over real goods and services with the use of money. Demand for money is actually demand for liquidity and demand to store value.

Variables on which demand for money depends -

- Higher the income of individuals, higher the expenditure and richer people hold more money to finance their expenditure.
- The quantity is directly proportional to the prevailing price level; higher the prices, higher should be the holding of money.
- One may hold his wealth in any form other than money, say as an interest yielding asset. It follows that the opportunity cost of holding money is the interest rate a person could earn on other assets. Therefore, higher the interest rate, higher would be opportunity cost of holding cash and lower the demand for money.
- Innovations such as internet banking, application based transfers and automatic teller machines reduce the need for holding liquid money.

1.3 Theories of Demand for Money

1.3.1 The Quantity Theory of Money (QTM) : Classical Approach:

Q. Explain the following modified equation of exchange as given by Irving Fisher :
MV + M'V' = PT [PEQ-May 18 -3 Marks]

The quantity theory of money, one of the oldest theories in Economics, was first propounded by Irving Fisher of Yale University in his book 'The Purchasing Power of Money' published in 1911.

All versions of the QTM demonstrate that there is strong relationship between value of money and price level and the quantity of money is the main determinant of the price level or the value of money.

Changes in the general level of commodity prices or changes in the value or purchasing power of money are determined first and foremost by changes in the quantity of money in circulation. Fisher's version, also termed as 'equation of exchange' or 'transaction approach' is stated as follows:

$$MV = PT$$

Where,

- M = the total amount of money in circulation (on an average) in an economy
- V = transactions velocity of circulation i.e. the average number of times across all transaction a unit of money (say Rupee) is spent in purchasing goods and services
- P = average price level
- T = the total number of transactions

Later, Fisher extended the equation of exchange to include demand (bank) deposits (M') and their velocity (V') in the total supply of money. Thus, the expanded form of the equation of exchange becomes:

$$M V + M' V' = P T$$

Where

- M' = the total quantity of credit money
- V' = velocity of circulation of credit money

Assumptions:

- The total supply of money in the community consists of the quantity of actual money (M) and its velocity of circulation (V).
- Velocity of money in circulation (V) and the velocity of credit money (V') remain constant.
- T is a function of national income.
- Since full employment prevails, the volume of transactions T is fixed in the short run.

The demand for money (PT) is equal to the supply of money (MV + M'V')

In any given period, the total value of transactions made is equal to PT and the value of money flow is equal to MV+ M'V'.

Fisher did not specifically mention anything about the demand for money; but the same is embedded in his theory as dependent on the total value of transactions undertaken in the economy. Thus, there is an aggregate demand for money for transactions purpose and more the number of transactions people want, greater will be the demand for money.

1.3.2 The Cambridge Approach : Neo-classical Approach

Q. Explain the neo-classical approach to demand for money? [PEQ – Nov 19 - 3 Marks]

In the early 1900's, Cambridge Economists Alfred Marshall, A.C. Pigou, D.H. Robertson and John Maynard Keynes put forward a fundamentally different approach to quantity theory, known as neoclassical theory or cash balance approach.

The Cambridge version holds that money increases utility in the following two ways:

1. Enabling the possibility of split-up of sale and purchase to two different points of time rather than being simultaneous, and
2. Being a hedge against uncertainty.

While the first above represents transaction motive, just as Fisher envisaged, the second points to money's role as a store of wealth. Since sale and purchase of commodities by individuals do not take place simultaneously, they need a 'temporary abode' of purchasing power as a hedge against uncertainty.
As such, demand for money also involves a precautionary motive in Cambridge approach.

How much money will be demanded?
It depends partly on income and partly on other factors like wealth and interest rates.
The former determinant of demand i.e. income, points to transactions demand such that higher the income, the greater the quantity of purchases and as a consequence greater will be the need for money as a temporary abode of value to overcome transactions costs.
The Cambridge equation is stated as:

$$M_d = k P Y$$

Where

- M_d = demand for money
- P = average price level of currently produced goods and services
- Y = real national income
- PY = nominal income
- k = proportion of nominal income (PY) that people want to hold as cash balances
  The term 'k' in the above equation is called 'Cambridge k'.

The equation above explains that the demand for money (M_d) equals k proportion of the total money income.
Thus we see that the neoclassical theory changed the focus of the quantity theory of money to money demand and hypothesized that demand for money is a function of only money income.
Both these versions are chiefly concerned with money as a means of transactions or exchange, and therefore, they present models of the transaction demand for money.

1.3.3 The Keynesian Theory of Demand for Money

Keynes' theory of demand for money is known as 'Liquidity Preference Theory'. 'Liquidity preference', a term that was coined by John Maynard Keynes in his masterpiece 'The General Theory of Employment, Interest and Money' (1936), denotes people's desire to hold money rather than securities or long-term interest-bearing investments.

Q. Explain why people hold money according to liquidity preference theory? [PEQ-May.18-3 Marks]

According to Keynes, people hold money (M) in cash for three motives:

(a) Transactions Motive

The transactions motive for holding cash relates to 'the need for cash for current transactions for personal and business esxchange.'

The need for holding money arises because there is lack of synchronization between receipts and expenditures.

The transaction motive is further classified into income motive and business (trade) motive, both of which stressed on the requirement of individuals and businesses respectively to bridge the time gap between receipt of income and planned expenditures.

Keynes did not consider the transaction balances as being affected by interest rates.

The transaction demand for money is directly related to the level of income and is stated as follows:

$$Lr = k\,Y$$

Where

- Lr, = transactions demand for money,
- k = ratio of earnings which is kept for transactions purposes
- Y = earnings.

Keynes considered the aggregate demand for money for transaction purposes as the sum of individual demand and therefore, the aggregate transaction demand for money is a function of national income.

(b) Precautionary Motive

Many unforeseen and unpredictable contingencies involving money payments occur in our day to day life.

Individuals as well as businesses keep a portion of their income to finance such unanticipated expenditures.

The amount of money demanded under the precautionary motive depends on the size of income, prevailing economic as well as political conditions and personal characteristics of the individual such as optimism/ pessimism, farsightedness etc.

Keynes regarded the precautionary balances just as balances under transactions motive as income elastic and by itself not very sensitive to rate of interest.

(c) Speculative Motive

The speculative motive reflects people's desire to hold cash in order to be equipped to exploit any attractive investment opportunity requiring cash expenditure.

According to Keynes, people demand to hold money balances to take advantage of the future changes in the rate of interest, which is the same as future changes in bond prices. It is implicit in Keynes theory, that the 'rate of interest', '**i**', is really the return on bonds.

Keynes assumed that that the expected return on money is zero, while the expected returns on bonds are of two types, namely:

(i) interest payment

(ii) expected rate of capital gain.

The market value of bonds and the market rate of interest are inversely related. A rise in the market rate of interest leads to a decrease in the market value of the bond, and vice versa.

Current rate of interest v/s 'normal' or 'critical' rate of interest

Case : A - Current rate of interest > 'normal' or 'critical' rate of interest

If wealth-holders consider that the current rate of interest is high compared to the 'normal or critical rate of interest', they expect a fall in the interest rate (rise in bond prices). At the high current rate of interest, they will convert their cash balances into bonds because:

(i) they can earn high rate of return on bonds

(ii) they expect capital gains resulting from a rise in bond prices consequent upon an expected fall in the market rate of interest in future.

Case : B - Current rate of interest < 'normal' or 'critical' rate of interest

If the wealth-holders consider the current interest rate as low, compared to the 'normal or critical rate of interest', i.e., if they expect the rate of interest to rise in future (fall in bond prices), they would have an incentive to hold their wealth in the form of liquid cash rather than bonds because:

(i) the loss suffered by way of interest income forgone is small,

(ii) they can avoid the capital losses that would result from anticipated increase in interest rates, and

(iii) the return on money balances will be greater than the return on alternative assets

(iv) If the interest rate does increase in future, the bond prices will fall and the idle cash balances held can be used to buy bonds at lower price and can thereby make a capital-gain.

Summing up, so long as

- the current rate of interest is higher than the critical rate of interest, a typical wealth-holder would hold in his asset portfolio only government bonds
- if the current rate of interest is lower than the critical rate of interest, his asset portfolio would consist wholly of cash.
- When the current rate of interest is equal to the critical rate of interest, a wealth-holder is indifferent to holding either cash or bonds.

The inference from the above is that the speculative demand for money and interest are inversely related.

Conclusion of Keynes Liquidity Preference Theory:

- An increase in income increases the transaction and precautionary demand for money and a rise in the rate of interest decreases the demand for speculative demand of money.
- higher the rate of interest, lower the speculative demand for money
- lower the rate of interest, higher the speculative demand for money
- The sum of the transaction demand, precautionary demand and the speculative demand is the total demand for money.

The speculative demand for money of individuals can be diagrammatically presented as follows:

The concept of Liquidity Trap

Liquidity trap is a situation when expansionary monetary policy (increase in money supply) does not increase the interest rate, income and hence does not stimulate economic growth. Liquidity trap is the extreme effect of monetary policy. It is a situation in which the general public is prepared to hold on to whatever amount of money is supplied, at a given rate of interest. They do so because of the fear of adverse events like deflation, war. In that case, a monetary policy carried out through open market operations has no effect on either the interest rate, or the level of income. In a liquidity trap, the monetary policy is powerless to affect the interest rate.

There is a liquidity trap at short term zero percent interest rate. When interest rate is zero, public would not want to hold any bond, since money, which also pays zero percent interest, has the advantage of being usable in transactions.

In other words, investors would maintain cash savings rather than hold bonds. The speculative demand becomes perfectly elastic with respect to interest rate and the speculative money demand curve becomes parallel to the X-axis. This situation is called a 'Liquidity trap'.

Since the opportunity cost of holding money is zero, even if the monetary authority increases money supply to stimulate the economy, people would prefer to hoard money. Consequently, excess funds may not be converted into new investment. The liquidity trap is synonymous with ineffective monetary policy.

1.4 Post Keynesian Developments in Theory of Demand for Money

1.4.1 Inventory Approach to Transaction Balances

Baumol (1952) and Tobin (1956) developed a deterministic theory of transaction demand for money, known as Inventory Theoretic Approach, in which money or 'real cash balance' was essentially viewed as an inventory held for transaction purposes.

Inventory models assume that there are two media for storing value:

money and an interest-bearing alternative financial asset.

There is a fixed cost of making transfers between money and the alternative assets e.g. broker charges.

Baumol asserts that individuals hold money (inventory of money) for the transaction purposes.

According to him, individuals have to keep optimum inventory of money for their day to day transaction purposes. They also incur cost when they hold inventories of money and the cost for gone is the interest rate which they could have earned if they had kept their wealth in saving

deposits or fixed deposits or invested in bonds or shares. This forgone cost is also called opportunity cost. Money that people hold in the form of currency and demand deposits which are very safe and riskless but pays no interest. While bonds or shares provide returns (interest) but are risky and may also involve capital loss if people invest in them.

Baumol and Tobin proclaim that transactions demand for money depends on the rate of interest.

As interest rates on savings deposits go up people will hold less money in the form of currency or cash or demand deposits and vice versa. So, individuals compare the costs and benefits of funds in the form of money with no interest with the money in the form of savings deposits with some interest. According to Baumol, the cost forgone when people hold money is the opportunity cost of these funds.

Baumol has proved that the average amount of cash withdrawal which minimises cost is given

$$C = \sqrt{2bY"} / r$$

This means that the average amount of cash withdrawal which minimizes cost is the square root of the two times broker's fee multiplied by the size of an individual's income and divided by the interest rate. This is also called Square Root Rule.

The inventory – theoretic approach also suggests that the demand for money and bonds depend on the cost of making a transfer between money and bonds e.g. the brokerage fee. An increase in the brokerage fee raises the marginal cost of bond market transactions and consequently lowers the number of such transactions. An individual combines his asset portfolio of cash and bond in such proportions that his overall cost of holding the assets is minimized.

1.4.2 Friedman's Restatement of the Quantity Theory

Q. Describe the determinants for demand for money as identified by Miltion Friedman in his re-statement of Quantity Theory of demand for money? [PEQ - May 19 - 3 Marks]

Milton Friedman (1956) extended Keynes' speculative money demand within the framework of asset price theory. Friedman treats the demand for money as nothing more than the application of a more general theory of demand for capital assets. Demand for money is affected by the same factors as demand for any other asset, namely

1. Permanent income.
2. Relative returns on assets. (which incorporate risk)

Friedman maintains that it is permanent income and not current income as in the Keynesian theory that determines the demand for money.

Permanent income which is Friedman's measure of wealth is the present expected value of all future income.

Friedman identifies the following four determinants of the demand for money. The nominal demand for money:

- is a function of total wealth, which is represented by permanent income divided by the discount rate, defined as the average return on the five asset classes in the monetarist theory world, namely money, bonds, equity, physical capital and human capital.
- is positively related to the price level, P. If the price level rises the demand for money increases and vice versa.

- rises if the opportunity costs of money holdings (i.e. returns on bonds and stock) decline and vice versa.
- is influenced by inflation, a positive inflation rate reduces the real value of money balances

1.4.3 The Demand for Money as Behaviour toward Risk

James Tobin, an American economist, in his analysis makes a valid assumption that people prefer more wealth to less. According to him, an investor is faced with a problem of what proportion of his portfolio financial assets he should keep in the form of ready money (which earns no interest) and in the form of investment (which earns interest) such as bonds.

According to Tobin, when individuals are faced with various safe and risky assets, they diversify their portfolio by holding a balanced combination of safe and risky assets.

According to Tobin, an individual's behavior shows risk aversion, which means they prefer less risk to more risk at a given rate of return.

If an individual chooses to hold a greater proportion of risky assets such as bonds or shares in his portfolio, then he will be earning a higher average return but will bear a higher degree of risk. Tobin argues that a risk averter will not choose such a portfolio with all risky bonds or a greater proportion of them.

In the other case, an individual who, in his portfolio of wealth, holds only safe and riskless assets such as money in form of cash or demand deposits, he will be taking almost zero risk but will also be getting no return. Therefore, people prefer a mixed or diversified portfolio of money, bonds and shares, with each person opting for a little different balance between risk and return.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. Choose the incorrect statement
   (a) *Anything that would act as a medium of exchange is money*
   (b) *Money has generalized purchasing power and is generally acceptable in settlement of all transactions*
   (c) *Money is a totally liquid asset and provides us with means to access goods and services*
   (d) *Currency which represents money does not necessarily have intrinsic value.*

2. Money performs all of the three functions mentioned below, namely
   (a) *medium of exchange, price control, store of value*
   (b) *unit of account, store of value, provide yields*
   (c) *medium of exchange, unit of account, store of value*
   (d) *medium of exchange, unit of account, income distribution*

3. Demand for money is
   (a) *Derived demand* (b) *Direct demand*
   (c) *Real income demand* (d) *Inverse demand*

4. Higher the ____, higher would be _____ of holding cash and lower will be the _____
   (a) *demand for money, opportunity cost, interest rate*
   (b) *price level , opportunity cost, interest rate*
   (c) *real income , opportunity cost, demand for money*
   (d) *interest rate, opportunity cost, demand for money*

5. The quantity theory of money holds that
   (a) *changes in the general level of commodity prices are caused by changes in the quantity of money*
   (b) *there is strong relationship between money and price level and the quantity of money is the main determinant of the price*
   (c) *changes in the value of money or purchasing power of money are determined first and foremost by changes in the quantity of money in circulation*
   (d) *All the above*

6. The Cambridge approach to quantity theory is also known as
   (a) *Cash balance approach* (b) *Fisher's theory of money*
   (c) *Classical approach* (d) *Keynesian Approach*

7. Fisher's approach and the Cambridge approach to demand for money consider
   (a) *money's role in acting as a store of value and therefore, demand for money is for storing value temporarily.*
   (b) *money as a means of exchange and therefore demand for money is termed as for liquidity preference*
   (c) *money as a means of transactions and therefore demand for money is only transaction demand for money*
   (d) *None of the above*

8. Real Money is
   (a) *Nominal Money / Price Level* (b) *real national income*
   (c) *money demanded at given rate of interest* (d) *nominal GNP / price level*

9. The precautionary money balances people want to hold
   (a) *as income elastic and not very sensitive to rate of interest*
   (b) *as income inelastic and very sensitive to rate of interest*
   (c) *are determined primarily by the level of transactions they expect to make in the future.*
   (d) *are determined primarily by the current level of transactions*

10. Speculative demand for money
   (a) *is not determined by interest rates* (b) *is positively related to interest rates*
   (c) *is negatively related to interest rates* (d) *is determined by general price level*

11. According to Keynes, if the current interest rate is high
   (a) *People will demand more money because the capital gain on bonds would be less than return on money*
   (b) *People will expect the interest rate to rise and bond price to fall in the future.*
   (c) *People will expect the interest rate to fall and bond price to rise in the future.*
   (d) *Either (a) or (b) will happen*

12. The inventory-theoretic approach to the transactions demand for money
   (a) *Explains the negative relationship between money demand and the interest rate.*
   (b) *Explains the positive relationship between money demand and the interest rate.*
   (c) *Explains the positive relationship between money demand and general price level*
   (d) *Explains the nature of expectations of people with respect to interest rates and bond prices*

13. According to Baumol and Tobin approach to demand for money, the optimal average money holding is :
   (a) *a positive function of income Y and the price level P*
   (b) *a positive function of transactions costs c,*
   (c) *a negative function of the nominal interest rate i*
   (d) *all of the above*

14. _____ Considered demand for money as an application of a more general theory of demand for capital assets
   (a) *Baumol* (b) *James Tobin* (c) *J M Keynes* (d) *Milton Friedman*

15. The nominal demand for money rises if
   (a) *the opportunity costs of money holdings - i.e. bonds and stock returns, r_B and r_E, respectively- decline and vice versa*
   (b) *the opportunity costs of money holdings - i.e. bonds and stock returns, r_B and r_E , respectively- rises and vice versa*
   (c) *the opportunity costs of money holdings - i.e. bonds and stock returns, r_B and r_E , respectively remain constant*
   (d) *(b) and (c) above*

PEQ – JUNE - 2024

16. If velocity = 20, average price = 120 and volume of transactions T = 150 crores, then money supply is:
   (a) ₹ 2,500 crores (b) ₹ 25 crores (c) ₹ 900 crores (d) ₹ 1,000 crores (1 mark)

17. Which of the following is fiat money?
   (a) Gold coins (b) Silver coins
   (c) Currency notes (d) Gold coins, silver coins as well as currency notes

18. The transactionany demand for money is directly proportional to and is a positive function of:
(a) Level of price (b) Level of income
(c) Level of demand (d) Level of interest rate (1 mark)

19. People's desire to hold cash in order to be equipped to exploit any attractive investment opportunity requiring cash expenditure reflects .
(a) Transaction motive (b) Speculative motive
(c) Precautionary motive (d) Personal and business exchange (1 mark)

20. What does the concept of liquidity trap mean?
(a) Liquidity trap is a situation where the desire to hold bonds is very low and approaches 0 and demand to hold money in liquid form as an alternative approaches infinity
(b) Even if public fear adverse events then they prefer to hold only bonds at given rate of interest
(c) The speculative money demand curve is parallel to Y axis
(d) None of these (1 mark)

PEQ - SEPTEMBER - 2024

21. Real money refers to:
(a) Money demanded at given rate of interest (b) Real national income
(c) Nominal GNP divided by price level (d) Nominal money adjusted to the price level (1 mark)

22. Which of the following is NOT a function of money?
(a) Acting as a medium of exchange (b) Providing a common measure of value
(c) Serving as a unit of account (d) Easily reproducible by people (1 mark)

23. The concept of "aversion of risk" is propounded by:
(a) Milton Friedman (b) James Tobin
(c) John Maynard Keynes (d) Alfred Marshall (1 mark)

PEQ - JANUARY - 2025

24. Which of the following is not a characteristic of money?
(a) Generally acceptable (b) Effortlessly recognisable
(c) Easily transportable (d) Easily reproducible by people (1 mark)

PEQ - MAY - 2025

25. What is the term Y represent in the expression Md = k PY?
(a) Real National income
(b) Price level
(c) Money supply
(d) Interest rate

26. Liquidity trap occurs when:
(a) Interest rates are high, and people prefer bonds over cash balances.
(b) Interest rates are near zero, and people prefer holding cash over bonds.
(c) Inflation rates are high, reducing purchasing power of money balances.
(d) Central banks increase CRR drastically.

27. Which of the following is NOT one of the four determinants of the Friedman's demand for money?
(a) *Nominal Demand for money is a function of total wealth*
(b) *It is positively related to price level P, if price level rises the demand for money increases*
(c) *Nominal demand for money falls if the opportunity costs of money holding declines*
(d) *Nominal Demand for money is influenced by inflation*

PEQ - SEPTEMBER - 2025

28. According to Friedman, how does the price level, as a determinant, affect the nominal demand for money?
(a) *It has no effect.*
(b) *It is negatively related to the demand for money.*
(c) *A rise in the price level increases the demand for money and vice versa.*
(d) *A rise in the price level decreases the demand for money.*

29. When the speculative demand for money becomes perfectly elastic with respect to the interest rate, making the speculative money demand curve parallel to the X-axis, this situation is called a/an _____
(a) *Inflationary gap*
(b) *Liquidity trap*
(c) *Monetary expansion*
(d) *Deflationary spiral*

PEQ - JANUARY - 2026

30. Which equation accurately represents the expanded version of the equation of exchange?
(a) $P=MV$
(b) $P=M'V'$
(c) $PT = MV + M'V'$
(d) $PT = MV - M'V'$

31. Speculative demand for money can be closely linked to which of the following situation?
(a) *Unforeseen and unpredictable contingencies*
(b) *Current transaction for business exchange*
(c) *Mismatch between receipt of salary and household expenses*
(d) *Attractive investment opportunity*

32. Tobin, in his theory of liquidity preference function, framed the relationship between which of the following?
(a) *Demand for money and supply of money*
(b) *Rate of inflation and demand for money*
(c) *Rate of interest and demand for money*
(d) *Rate of interest and disposable income*

33. The optimal average money holding is a negative function of which of the following?
(a) *Income*
(b) *Nominal interest rate*
(c) *Transaction cost*
(d) *Price level*

PEQ - SEPTEMBER - 2026

33. Money demand is said to be 'derived' because:
   (a) *It is decided by the government*
   (b) *People hold money because it earns interest*
   (c) *People demand money to facilitate expenditure*
   (d) *Money supply controls demand*

ANSWERS / HINTS

I. Multiple Choice Type Questions

1	(a)	2	(c)	3	(a)	4	(d)	5	(d)	6	(a)	7	(c)	8	(a)
9.	(a)	10.	(c)	11.	(c)	12.	(a)	13.	(d)	14.	(d)	15.	(a)	16.	(c)
17.	(c)	18.	(b)	19.	(b)	20.	(a)	21.	(d)	22.	(d)	23.	(b)	24.	(d)
25.	(a)	26.	(b)	27.	(c)	28.	(c)	29.	(b)	30.	(c)	31.	(d)	32.	(c)
33.	(b)														

For Your Notes

For Your Notes

For Your Notes

For Your Notes

UNIT – II

THE CONCEPT OF MONEY SUPPLY

Topic References:

2.1 Introduction
2.2 Rationale of Measuring Money Supply
2.3 The Sources of Money Supply
2.4 Measurement of Money Supply
2.5 Determinates of Money Supply
2.6 The concepts of Money Multiplier
2.7 The Money Multiplier Approach to Supply of Money
2.8 Monetary Policy And Money Supply
2.9 Effect of Government Expenditure on Money Supply
2.10 Credit Multiplier

2.1 Introduction

- Economic stability requires that the supply of money at any time should be maintained at an optimum level. A pre-requisite for achieving this is to accurately estimate the stock of money supply on a regular basis and appropriately regulate it in accordance with the monetary requirements of the country.
- The term money supply denotes the total quantity of money available to the people in an economy.
- The quantity of money at any point of time is a measurable concept. It is important to note two things about any measure of money supply:
  1. The supply of money is a stock variable i.e. it refers to the total amount of money at any particular point of time. It is the change in the stock of money (say, increase or decrease per month or year) which is a flow.
  2. The stock of money always refers to the stock of money available to the 'public' as a means of payments and store of value. This is always smaller than the total stock of money that really exists in an economy.

- ***The term 'public' includes***
  - all economic units i.e. households, firms and institutions
  - local authorities
  - non-banking financial institutions
  - non - departmental public-sector undertakings
  - foreign governments / foreign central banks
  - International Monetary Fund which holds a part of Indian money in India in the form of deposits with the RBI.

- ***The term 'public' excludes***
  - the government (i.e. the central government and all state governments ; local bodies)
  - the banking system (the RBI and all the banks that accept demand deposits i.e. deposits from which money can be withdrawn by cheque mainly CASA deposits)

2.2 Rationale of Measuring Money Supply

1. It facilitates analysis of monetary developments in order to provide a deeper understanding of the causes of money growth.
2. It is essential from a monetary policy perspective as it provides a framework to evaluate whether the stock of money in the economy is consistent with the standards for price stability and to understand the nature of deviations from this standard.

2.3 The Sources of Money Supply

The supply of money in the economy depends on:

(a) the decision of the central bank based on the authority conferred on it; and

(b) the supply responses of the commercial banking system of the country to the changes in policy variables initiated by the central bank to influence the total money supply in the economy.

CENTRAL BANK

- The central banks of all countries are empowered to issue currency and, therefore, the central bank is the primary source of money supply in all countries.
- High powered money issued by monetary authorities is the source of all other forms of money.
- The currency issued by the central bank is **'fiat money'** and is backed by supporting reserves and its value is guaranteed by the government.
- The currency issued by the central bank is, in fact, a liability of the central bank and the government. Therefore, in principle, it must be backed by an equal value of assets mainly consisting of gold and foreign exchange reserves.
- In practice, however, most countries have adopted a 'minimum reserve system' wherein the central bank is empowered to issue currency to any extent by keeping only a certain minimum reserve of gold and foreign securities.

COMMERCIAL BANKS

- The second major source of money supply is the banking system of the country.
- The total supply of money in the economy is also determined by the extent of credit created by the commercial banks in the country.
- Banks create money supply in the process of borrowing and lending transactions with the public. Money so created by the commercial banks is called 'credit money'.

The high powered money and the credit money broadly constitute the most common measure of money supply, or the total money stock of a country.

The concept of money has experienced evolution from Commodity to Metallic Currency to Paper Currency to Digital Currency.

Advancement in technology has made it possible for the development of new form of money viz. Central Bank Digital Currencies (CBDCs).

Reserve Bank broadly defines CBDC as the legal tender issued by a central bank in a digital form. It is akin to sovereign paper currency but takes a different form, exchangeable at par with the existing currency and shall be accepted as a medium of payment, legal tender and a safe store of value. CBDCs would appear as liability on a central bank's balance sheet.

The Crypto currencies face significant legislative uncertainties and are not legally recognized in India as currency. Hence, these are not categorized as money.

2.4 Measurement of Money Suppy

There is virtually a profusion of different types of money, especially credit money, and this makes measurement of money supply a difficult task.

Just as other countries do, a range of monetary and liquidity measures are compiled and published by the RBI.

Since July 1935, the Reserve Bank of India has been compiling and disseminating monetary statistics.

Till 1967-68, the RBI used to publish only a single narrow measure of money supply (M1) defined as the sum of currency and demand deposits held by the public.

From 1967-68, a 'broader' measure of money supply, called 'aggregate monetary resources' (AMR) was additionally published by the RBI.

From April 1977, following the recommendations of the Second Working Group on Money Supply (SWG), the RBI has been publishing data on four alternative measures of money supply denoted by M1, M2, M3 and M4 besides the reserve money.

The RBI regards these 4 measures of money stock as representing different degrees of liquidity.

It has specified them in the descending order of liquidity, M1 being the most liquid and M4 the least liquid of the four measures.

The respective empirical definitions of these measures are given below:

M1 = Currency notes and coins with the people + Net demand deposits of banks (Current and Saving deposit accounts) + other deposits with the RBI.
M2 = M1 + savings deposits with post office savings banks.
M3 = M1 + net time deposits with the banking system.
M4 = M3 + total deposits with the Post Office Savings Organisation (Excluding NSC)

Currency with Public = Notes in circulation + circulation of Rupees coin + circulation of small coin – Cash on Hand with Bank

Components of M1 -

1. ***Currency Notes and Coins With People -***
   - Currency consists of paper currency as well as coins.

2. ***Net Demand Deposit of Banks -***
   - Demand deposits comprise the current-account deposits and the demand deposit portion of savings deposits, all held by the public.
   - These are also called CASA deposits and these are cheapest sources of finance for a commercial bank.
   - It is the net demand deposits of banks, and not their total demand deposits that get included in the measure of money supply.
   - The total deposits include both deposits from the public as well as inter-bank deposits.
   - Money is deemed as something held by the 'public'. Since inter- bank deposits are not held by the public, they are netted out of the total demand deposits to arrive at net demand deposits.

3. *Other Deposits With RBI -*

- Other deposits with the RBI are its deposits other than those held by the government (the Central and state governments),
- It includes demand deposits of quasi-government institutions, other financial institutions, balances in the accounts of foreign central banks and governments, and accounts of international agencies such as IMF and the World Bank.
- Empirically, whatever the measure of money supply, these 'other deposits' with the RBI constitute a very small proportion (< 1%) of the total money supply.

In the monetary literature, money is usually defined in alternative ways ranging from narrow to broad money.

Empirically the M1 (narrow money) is defined as the sum of currency held by the public, demand deposits of the banks and other deposits with RBI.

Reserve money is comprised of the currency held by the public, cash reserves of banks and other deposits with RBI.

On comparison, we find that the difference between M1 and reserve money is that the former includes the demand deposits while the latter includes the cash reserves of banks. Reserves are commercial banks' deposits with the central bank for maintaining cash reserve ratio (CRR) and as working funds for clearing adjustments.

Reserve money, also known as central bank money, base money or high-powered money, determines the level of liquidity and price level in the economy and, therefore, its management is of crucial importance to stabilize liquidity, growth, and price level in an economy.

2.5 Determinates of Money Supply

There are two alternate theories in respect of determination of money supply.
According to the first view, money supply is determined exogenously by the central bank.
The second view holds that the money supply is determined endogenously by changes in the economic activities which affect people's desire to hold currency relative to deposits, rate of interest, etc.

2.6 The concepts of Money Multiplier

The money supply is defined as

$$M = m \times M B$$

Where

- M is the money supply,
- m is money multiplier, and
- MB is the monetary base or high powered money.

From the above equation we can derive the money multiplier (m) as

Money Multiplier = Money Supply / Monetary Base

$$M = \frac{1 + c}{r + e + c}$$

Money multiplier m is defined as a ratio that relates the changes in the money supply to a given change in the monetary base.
It is the ratio of the stock of money to the stock of high- powered money.

For instance, if there is an injection of Rs.100 Cr through an open market operation by the central bank of the country and if it leads to an increment of Rs.500 Cr. of final money supply, then the money multiplier is said to be 5. Hence, the multiplier indicates the change in monetary base which is transformed into money supply.

What determines the size of the money multiplier? The money multiplier is the reciprocal of the reserve ratio. Deposits, unlike currency held by people, keep only a fraction of the high-powered money in reserves and the rest is lent out and culminate in money creation. If R is the reserve ratio in a country for all commercial banks, then each unit of (say Rupee) money reserves generates 1/R money.

Therefore, for any value of R, the Money Multiplier is 1/R

For example, if R =10%, the value of money multiplier will be 10. If the reserve ratio is only 5%, then money multiplier is 20. Thus, the higher the reserve ratio, the less of each deposit banks loan out, and the smaller the money multiplier.

If some portion of the increase in high-powered money finds its way into currency, this portion does not undergo multiple deposit expansion. The size of the money multiplier is reduced when funds are held as cash rather than as demand deposits. In other words, as a rule, an increase in the monetary base that goes into currency is not multiplied, whereas an increase in monetary base that goes into supporting deposits is multiplied.

2.7 The Money Multiplier Approach to Supply of Money

The money multiplier approach to money supply propounded by Milton Friedman and Anna Schwartz, (1963) considers three factors as immediate determinants of money supply, namely:

(a) the stock of high-powered money (H)
(b) the ratio of reserves to deposits, e = {ER / D} and
(c) the ratio of currency to deposits, c = {C / D}

These represent the behaviour of the central bank, behaviour of the commercial banks and the behaviour of the general public respectively.

a) The Behaviour of the Central Bank

The behaviour of the central bank which controls the issue of currency is reflected in the supply of the nominal high-powered money.

Money stock is determined by the money multiplier and the monetary base is controlled by the monetary authority.

If the behaviour of the public and the commercial banks remains unchanged over time, the total supply of nominal money in the economy will vary directly with the supply of the nominal high-powered money issued by the central bank.

b) *The Behaviour of Commercial Banks*

By creating credit, the commercial banks determine the total amount of nominal demand deposits.

The behaviour of the commercial banks in the economy is reflected in the ratio of their cash reserves to deposits known as the 'reserve ratio'.

If the required reserve ratio on demand deposits increases while all the other variables remain the same, more reserves would be needed. This implies that banks must contract their loans, causing a decline in deposits and hence in the money supply.

If the required reserve ratio falls, there will be greater expansions of deposits because the same level of reserves can now support more deposits and the money supply will increase.

However, for the commercial banking system as a whole, the actual reserves ratio is greater than the required reserve ratio since the banks keep with them a higher than the statutorily required percentage of their deposits in the form of cash reserves.

The excess reserves (ER) which are funds that a bank keeps back beyond what is required by regulation form a very important determinant of money supply. 'Excess reserves' are the difference between total reserves (TR) and required reserves (RR).

Therefore, ER=TR-RR.

If total reserves are Rs 800 billion, whereas the required reserves are Rs. 600 billion, then the excess reserves are Rs 200 billion.

Therefore, if the central bank injects money into the banking system and these are held as excess reserves by the banking system, there will be no effect on deposits or currency and hence no effect on money supply.

When the costs of holding excess reserves rise, we should expect the level of excess reserves to fall; when the benefits of holding excess reserves rise, we would expect the level of excess reserves to rise.

Two primary factors namely market interest rates and expected deposit outflows affect these costs and benefits and hence in turn affect the excess reserves ratio.

We know that the cost to a bank while holding excess reserves is in terms of its opportunity cost, i.e. the interest that could have been earned on loans or securities if the bank had chosen to invest in them instead of excess reserves.

If interest rate increases, it means that the opportunity cost of holding excess reserves rises because the banks have to sacrifice possible higher earnings and hence the desired ratio of excess reserves to deposits falls.

Conversely, a decrease in interest rate will reduce the opportunity cost of excess reserves, and excess reserves will rise.

Therefore, we conclude that the banking system's excess reserves ratio e is negatively related to the market interest rate.

If banks fear that deposit outflows are likely to increase (that is, if expected deposit outflows increase), they will want more assurance against this possibility and will increase the excess reserves ratio.

Conversely, a decline in expected deposit outflows will reduce the benefit of holding excess reserves and excess reserves will fall.

c) *The Behaviour of Public*

The public, by their decisions in respect of the amount of nominal currency in hand (how much money they wish to hold as cash) is in a position to influence the amount of the nominal demand deposits of the commercial banks.

The behaviour of the public influences bank credit through the decision on ratio of currency to the money supply designated as the 'currency ratio'.

What would be the behaviour of money supply when depositors decide to increase currency holding, with all other variables unchanged? In other words, you decide to keep more money in your pocket and less money in your bank. That means you are converting some of your demand deposits into currency.

If many people like you do so, technically we say there is an increase in currency ratio. As we know, demand deposits undergo multiple expansions while currency in your hands does not. Hence, when bank deposits are being converted into currency, banks can create only less credit money. The overall level of multiple expansion declines, and therefore, money multiplier also falls. Therefore, we conclude that money multiplier and the money supply are negatively related to the currency ratio c.

The currency-deposit ratio (c) represents the degree of adoption of banking habits by the people. This is related to the level of economic activities or the GDP growth and is influenced by the degree of financial sophistication in terms of ease and access to financial services, availability of a richer array of liquid financial assets, financial innovations, institutional changes etc.

The time deposit-demand deposit ratio i.e. how much money is kept as time deposits compared to demand deposits, also has an important implication for the money multiplier and, hence for the money stock in the economy. An increase in TD/DD ratio means that greater availability of free reserves and consequent enlargement of volume of multiple deposit expansion and monetary expansion.

To summarise the money multiplier approach, the size of the money multiplier is determined by the required reserve ratio (r) at the central bank, the excess reserve ratio (e) of commercial banks and the currency ratio (c) of the public. The lower these ratios are, the larger the money multiplier is.

In other words, the money supply is determined by high powered money (H) and the money multiplier (m) and varies directly with changes in the monetary base, and inversely with the currency and reserve ratios.
Consequently, these variables are designated as the 'proximate determinants' of the nominal money supply in the economy.

We may now rewrite the money multiplier including the above variables.

M = C+D **(1)**
H = C+ reserves **(2)**

Where

C is currency and

D is deposits which are assumed to be demand deposits.

We summarise the behaviour of the public, banks and the central bank by three variables namely,

Currency- Deposit ratio, $c = \frac{C}{D}$,

Reserve-ratio, r= Reserves/D, and

the stock of high-powered money (H)

Rewriting equation (1) and (2) above as

M = (c+1) D,
H = (c+ r) D
M = $\frac{1+c}{r+c}$ x H = m x H
m= $\frac{1+c}{r+c}$

When there are excess reserves, the money multiplier m is expressed as

m = $\frac{1+c}{r+e+c}$.
Money Supply M = $\frac{1+c}{r+e+c}$ x H

The money multiplier is a function of:

(a) the currency ratio set by depositors c which depends on the behaviour of the public

(b) excess reserves ratio set by banks e, and

(c) the required reserve ratio set by the central bank r, which depends on prescribed CRR and the balances necessary to meet settlement obligations.

A simple example will explain the concept

Numerical Illustration

(a) In Gladys land,

r = 10%= 0.10

Currency = 400 billion

Deposits = 800 billion

Excess Reserves = 0.8 billion = 800 million

Money Supply is M = Currency + Deposits = 1200 billion

c = C/D = 400 billion/800 billion = 0.5 or depositors hold 50 % of their money as currency

e = 0.8 billion /800 billion = 0.001 or banks hold 0.1% of their deposits as excess reserves.

Multiplier m= $\frac{1+c}{r+e+c}$.

$= \frac{1+0.5}{0.1+0.001+0.5} = \frac{1.5}{0.601} = 2.5$

Therefore, a 1 unit increase in H leads to a 2.50 units increase in M.

The simple deposit multiplier in this example would be 1/r= 1/0.1=10

(b) If the reserve ratio is increased to 15 %, the value of the money multiplier will be,

$$= \frac{1+0.5}{0.15+0.001+0.50} = \frac{1+0.50}{0.15+0.001+0.50} = \frac{1.5}{0.651} = 2.3$$

Obviously, r and m are negatively related: m falls when r rises, and m rises when r falls. The reason is that less multiple deposit creation can occur when r rises, while more multiple deposit creation can occur when r falls.

2.8 Monetary Policy And Money Supply

If the central bank of a country wants to stimulate economic activity it does so by infusing liquidity into the system. Let us take the example of open market operations (OMO) by central banks. Purchase of government securities injects high powered money (monetary base) into the system.

Assuming that banks do not hold excess reserves and people do not hold more currency than before, and also that there is demand for loans from businesses, the credit creation process by the banking system in the country will create money to the tune of

Δ Money supply = (1/R) X Δ Reserves

The effect of an open market sale is very similar to that of open market purchase, but in the opposite direction. In other words, an open market purchase by central bank will reduce the reserves and thereby reduce the money supply.

Is it possible that the value of money multiplier is zero? It may happen when the interest rates are too low and the banks prefer to hold the newly injected reserves as excess reserves with no risk attached to it.

2.9 Effect of Government Expenditure on Money Supply

Whenever the central and the state governments' cash balances fall short of the minimum requirement, they are eligible to avail of a facility called Ways and Means Advances (WMA)/ overdraft (OD) facility.

When the Reserve Bank of India lends to the governments under WMA /OD, it results in the generation of excess reserves (i.e., excess balances of commercial banks with the Reserve Bank). This happens because when government incurs expenditure, it involves debiting the government balances with the Reserve Bank and crediting the receiver (for e.g., salary account of government employee) account with the commercial bank.

The excess reserves thus created can potentially lead to an increase in money supply through the money multiplier process.

2.10 Credit Multiplier

The Credit Multiplier also referred to as the deposit multiplier or the deposit expansion multiplier, describes the amount of additional money created by commercial bank through the process of lending the available money it has in excess of the central bank's reserve requirements.

This measure tells us how much new money will be created by the banking system for a given increase in the high- powered money.
It reflects a bank's ability to increase the money supply.

Credit Multiplier = 1 / Required Reserve Ratio

The credit multiplier is the reciprocal of the required reserve ratio.

If reserve ratio is 20%, then credit multiplier = 1/0.20 = 5.

The existence of the credit multiplier is the outcome of fractional reserve banking. It explains how increase in money supply is caused by the commercial banks' use of depositors' funds to lend money. When a bank uses the deposited money for lending, the bank generates another claim on a given amount of deposited money.

For example:

If A deposits ₹ 1,000/ in cash at a bank (Bank X), this constitutes the bank's current total cash deposits. If the required reserve is 10%, the bank would lend ₹ 900/ to B. By lending B ₹ 900/, the bank creates a deposit for ₹ 900/ that B can now use. It is as though B owns ₹ 900/. This in turn means that A will continue to have a claim against ₹ 1,000/while B will have a claim against ₹ 900/. The bank has ₹1,000/in cash against claims of ₹1,900/. In short, the bank has created ₹900/ out of "thin air" since these ₹900/ are not supported by any genuine money. At any time, the fractional reserve commercial banks have more cash liabilities than cash in their vaults.

Now suppose B buys goods worth ₹ 900/from C and pays C by cheque. C places the cheque with his bank, Bank Y. After clearing the cheque, Bank Y will have an increase in cash of ₹ 900/, which it may take advantage of and use to lend out ₹ 810/ to D which may again be deposited in another bank, say Bank Z. Again 10 % of ₹ 810 (₹ 81) has to be kept as required reserves and the remaining ₹ 719/ can be lent out, say to E. This sequence keeps on continuing until the initial deposit amount ₹1,000 grows exactly by the multiple of required reserves (in this case, 10%). Ultimately, the expanded credit availability would be 1000 + 900 (90% of 1000) + 810 (90% of 900) + 729 (90% of 810) + (90% of 719) + This summation would end with an amount which is equivalent to 1/10% of 1,000, which is ₹ 10,000.

Thus, in our example, the initial deposit is capable of multiplying itself out 10 times. In short, we find that the fact that banks make use of demand deposits for lending it sets in motion a series of activities leading to expansion of money that is not backed by money proper. It is interesting to know that there is no difference between the type of money created by commercial banks and that which are issued by the central bank.

The deposit multiplier and the money multiplier though closely related are not identical because:

a) Generally banks do not lend out all of their available money but instead maintain reserves at a level above the minimum required reserve.
b) All borrowers do not spend every Rupee they have borrowed. They are likely to convert some portion of it to cash.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. Reserve money is also known as
   (a) *Central bank money* (b) *Base Money*
   (c) *High powered money* (d) *All of the above*

2. Choose the correct statement from the following
   (a) *Money is deemed as something held by the public and therefore only currency held by the public is included in money supply.*
   (b) *Money is deemed as something held by the public and therefore inter- bank deposits are included in money supply.*
   (c) *Since inter-bank deposits are not held by the public, therefore inter-bank deposits are excluded from measures of money supply*
   (d) *Both (a) and (c) above*

3. Reserve Money is composed of
   (a) *Currency in circulation + demand deposits of banks (Current and Saving accounts) + other deposits with the RBI.*
   (b) *Currency in circulation + Bankers' deposit with the RBI + other deposits with the RBI.*
   (c) *Currency in circulation + demand deposits of banks + other deposits with the RBI.*
   (d) *Currency in circulation + demand and time deposits of banks + other deposits with the RBI.*

4. M1 is the sum of
   (a) *Currency and coins with the people + demand deposits of banks (Current and Saving accounts) + other deposits with RBI.*
   (b) *Currency and coins with the people + demand and time deposits of banks (Current and Saving accounts) + other deposits with RBI.*
   (c) *Currency in circulation + Bankers' deposits with the RBI + Other deposits with the RBI*
   (d) *None of the above*

5. Under the 'minimum reserve system' the central bank is
   (a) *empowered to issue currency to any extent by keeping an equivalent reserve of gold and foreign securities.*
   (b) *empowered to issue currency to any extent by keeping only a certain minimum reserve of gold and foreign securities.*
   (c) *empowered to issue currency in proportion to the reserve money by keeping only a minimum reserve of gold and foreign securities.*
   (d) *empowered to issue currency to any extent by keeping a reserve of gold and foreign securities to the extent of ₹350 crores*

6. The primary source of money supply in all countries is
   (a) *The Reserve Bank of India* (b) *The Central bank of the country*
   (c) *The Bank of England* (d) *The Federal Reserve*

7. The supply of money in an economy depends on
   (a) *the decision of the central bank based on the authority conferred on it.*
   (b) *the decision of the central bank and the supply responses of the commercial banking system.*
   (c) *the decision of the central bank in respect of high powered money.*
   (d) *both (a) and (c) above.*

8. Banks in the country are required to maintain deposits with the central bank
   (a) *to provide the necessary reserves for the functioning of the central bank*
   (b) *to meet the demand for money by the banking system*
   (c) *to meet the central bank prescribed reserve requirements and to meet settlement obligations.*
   (d) *to meet the money needs for the day to day working of the commercial banks*

9. If the behaviour of the public and the commercial banks is constant, then
   (a) *the total supply of nominal money in the economy will vary directly with the supply of the nominal high- powered money issued by the central bank*
   (b) *the total supply of nominal money in the economy will vary directly with the rate of interest and inversely with reserve money*
   (c) *the total supply of nominal money in the economy will vary inversely with the supply of high powered money*
   (d) *all the above are possible*

10. Under the fractional reserve system
   (a) *the money supply is an increasing function of reserve money (or high powered money) and the money multiplier.*
   (b) *the money supply is an decreasing function of reserve money (or high powered money) and the money multiplier.*
   (c) *the money supply is an increasing function of reserve money (or high powered money) and a decreasing function of money multiplier.*
   (d) *None of the above as the determinants of money supply are different.*

11. The money multiplier and money supply are
   (a) *positively related to the excess reserve ratio* ***'e'***
   (b) *negatively related to the excess reserve ratio* ***'e'***
   (c) *not related to the excess reserve ratio* ***'e'***
   (d) *proportional to the excess reserve ratio* ***'e'***

12. The currency ratio represents
   (a) *the behavior of central bank in the issue of currency.*
   (b) *the behavior of central bank in respect cash reserve ratio.*
   (c) *the behavior of the public.*
   (d) *the behavior of commercial banks in the country.*

13. The size of the money multiplier is determined by
   (a) *the currency ratio (c) of the public*
   (b) *the required reserve ratio (r) at the central bank*
   (c) *the excess reserve ratio (e) of commercial banks.*
   (d) *all the above*

14. _____ tells us how much new money will be created by the banking system for a given increase in the high-powered money.
   (a) *The currency ratio* (b) *The excess reserve ratio (e)*
   (c) *The money multiplier* (d) *The currency ratio (c)*

15. The money multiplier will be large
   (a) *for higher currency ratio (c), lower required reserve ratio (r) and lower excess reserve ratio (e)*
   (b) *for constant currency ratio (c), higher required reserve ratio (r) and lower excess reserve ratio (e)*
   (c) *for lower currency ratio (c), lower required reserve ratio (r) and lower excess reserve ratio (e)*
   (d) *None of the Above*

16. The ratio that relates the change in the money supply to a given change in the monetary base is called the
   (a) *required reserve ratio.* (b) *Money multiplier*
   (c) *deposit ratio.* (d) *discount rate.*

17. For a given level of the monetary base, an increase in the required reserve ratio will denote
   (a) *a decrease in the money supply.* (b) *an increase in the money supply*
   (c) *an increase in demand deposits.* (d) *Nothing precise can be said*

18. For a given level of the monetary base, an increase in the currency ratio causes the money multiplier to _____ and the money supply to _____
(a) *decrease; increase* (b) *increase; decrease*
(c) *decrease; decrease* (d) *increase; increase*

19. If commercial banks reduce their holdings of excess reserves
(a) *the monetary base increases.* (b) *the monetary base falls.*
(c) *the money supply increases.* (d) *the money supply falls.*

PEQ - JUNE - 2024

20. Calculate broad Money M3?
(a) *Currency with public + demand deposits with banks*
(b) *Currency with public + demand deposits with bank + other deposits with RBI + net time deposits with the bank system*
(c) *Currency with public + savings with gov. banks*
(d) *Currency with public + demand deposits with banks + other deposits with RBI* *(1 mark)*

21. Calculate currency with public:

Particulars	(in crores) (₹)
Notes with public in circulation	23,26,500
Currency of small coins in circulation	500
Currency of rupee coins in circulation	34,500
Cash in hand with the bank	80,615

(a) *₹ 22,80,885 cr.* (b) *₹ 24,42,115 cr.*
(c) *₹ 22,45,885 cr.* (d) *₹ 23,62,000 cr.* *(1 mark)*

22. Money Multiplier means:
(a) *It decreases the total money supply for given monetary base*
(b) *It dictates interest rates set by RBI*
(c) *The ratio of money supply to monetary base*
(d) *It controls amount of gold reserves held by bank* *(1 mark)*

PEQ – SEPTEMBER - 2024

23. Which of the following is the correct formula to calculate Excess Reserves?
(a) *Excess Reserves = Total reserves + Required reserves*
(b) *Excess Reserves = Total reserves - Required reserves*
(c) *Excess Reserves = Total reserves / Required reserves*
(d) *Excess Reserves = Total reserves x Required reserves* *(1 mark)*

24. What type of currency is issued by the central bank??
(a) *Digital money* (b) *Commodity money*
(c) *Fiat money* (d) *Representative money* *(1 mark)*

25. In an economy, the money supply (M) is ₹ 500 crores, the velocity of money (V) is 5 and the total number of transactions (T) is 10,000. Calculate the average price level (P) in the economy.
(a) *₹ 25 lakhs* (b) *₹ 25 thousand*
(c) *₹ 50 lakhs* (d) *₹ 50 thousand* *(1 mark)*

26. If the monetary base in an economy is ₹ 250 crores through an open market operation by the central bank and the money supply is ₹ 1,000 crores, calculate the money multiplier.
(a) *1* (b) *0.25* (c) *4* (d) *0.40* *(1 mark)*

27. If the reserve ratio (R) is 0.10 and there is an increase in the reserves by ₹ 1,000, what will be the change in the money supply?
(a) *₹ 1,000* (b) *₹ 10,000* (c) *₹ 100* (d) *₹ 10* *(1 mark)*

PEQ - JANUARY - 2025

28. The currency issued by the Central Bank is known as _____ and is backed by supporting reserves and its value is a sovereign guarantee,
(a) Real money *(b) Credit money*
(c) Fiat money *(d) Sovereign bonds* *(1 mark)*

29. Considering that with a money multiplier of 1.5 there has been an increment of ₹ 600 cr of money supply. Find out the monetary base,
(a) ₹ 800 cr *(b) ₹ 200cr* *(c) ₹ 400cr* *(d) ₹ 900 cr* *(1 mark)*

30. Calculate Narrow Money M1 from the following data:

Currency with public	₹ 88,000 cr
Demand deposit with the banking system	₹ 2,20,000 cr
Time deposit with the banking system	₹ 2,40,000 cr
Other deposits with RBI	₹ 2,60,000 cr
Saving deposits with Post Office Saving Bank	₹ 50,000 cr

(a) ₹ 5,68,000 cr *(b) ₹ 6,18,000 cr*
(c) ₹ 5,98,000 cr *(d) ₹ 6,38,000 cr* *(1 mark)*

31. Calculate currency with the public from the following data:

Notes in circulation	₹ 45,000 cr
Circulation of rupee coins	₹ 1,500 cr
Circulation of small coins	₹ 750 cr
Cash on hand with banks	₹ 27,500 cr

(a) ₹ 74,750 cr *(b) ₹19,750 cr* *(c) ₹ 73,250 cr* *(d) ₹ 29,750 cr* *(1 mark)*

32. Money created by the commercial banks is called .
(a) Real money *(b) High powered money*
(c) Fiat money *(d) Credit money* *(1 mark)*

33. Under the concept of money supply, the term 'public' do not include .
(a) Households *(b) Institutions*
(c) Government and banking system *(d) Firms* *(1 mark)*

34. Compute the total credit money created by the banking system if the required reserved ratio is 15% for every ₹ 12,00,000 deposited in the banking system?
(a) ₹ 1,00,00,000 *(b) ₹ 80,00,000*
(c) ₹ 1,25,00,000 *(d) ₹ 1,50,00,000* *(1 mark)*

PEQ - MAY - 2025

35. Calculate currency with Public from following Data (' in lakhs)

Notes in Circulation	2,59,121
Coins in Circulation	23,345
Cash on hands with Banks	19,009
Coin on hands with Banks	909

(a) 3,02,384
(b) 2,62,548
(c) 2,53,876
(d) 2,15,858

36. What is the likely impact of an increase in the Time Deposit to Demand Deposit (TD/DD) ratio on the banking system and monetary expansion?
   (a) It leads to higher availability of free reserves and consequent enlargement of volume of multiple deposit expansion and monetary expansion.
   (b) It reduces the availability of free reserves, thereby restricting the process of monetary expansion.
   (c) It has no impact on the banking system as time deposits and demand deposits are interchangeable.
   (d) It directly reduces the money supply as time deposits are not considered part of the money supply.

37. Broad money includes currency deposits with an agreed maturity of up to years, _____deposits redeemable at notice up to _____ months and repurchase agreements, money market fund shares/units and debt security up to _____ years
   (a) 3 years, 6 months, 3 years
   (b) 2 years, 6 months, 2 years
   (c) 3 years, 3 months, 3 years,
   (d) 2 years, 3 months, 2 years

38. Which of the following is NOT included in MI?
   (a) Currency with the public
   (b) Demand deposits with banks
   (c) Time (term) deposits with banks
   (d) Other deposits with RBI

PEQ - SEPTEMBER - 2025

39. For initial deposit of ₹ 6,00,000, the credit creation is calculated at ₹ 50,00,000. What is Required Reserved Ratio (RRR)?
   (a) 12%
   (b) 10%
   (c) 12.5%
   (d) 8%

40. If the money supply is ₹ 1,000 billion and the monetary base is ₹ 200 billion, what is the money multiplier (m)?
   (a) 5
   (b) 2
   (c) 10
   (d) 0.2

41. Calculate Broad Money (M3) from the following data:

Particulars	(₹ in Crores)
Currency with the public	2250
Demand Deposits with the banking system	630
Total post office deposits	478
Time deposits with banks	535
Savings deposits with Post office savings banks	312

   (a) ₹3,415 crores
   (b) ₹ 3,225 crores
   (c) ₹ 3,260 crores
   (d) ₹ 3,420 crores

PEQ - JANUARY - 2026

42. Given the required reserve ratio (RRR) is 8% and 12.5% for every 1,00,000 deposited in the banking system, compute the credit multipliers.
    (a) 12.5; 8
    (b) 8; 12.5
    (c) 12.5; 10
    (d) 8; 10

43. The money multiplier would be larger in which of the following scenarios?
    (a) Smaller expenditure-deposit ratio
    (b) Higher currency-deposit ratio
    (c) Smaller currency-expenditure ratio
    (d) Smaller currency-deposit ratio

II. Numerical illustrations

ILLUSTRATION 1 **[SM]**

Calculate Narrow Money (M1) from the following data

Currency with public	₹ 90000 crore
Demand Deposits with Banking System	₹ 200000 crore
Time Deposits with Banking System	₹ 220000 crore
Other Deposits with RBI	₹ 280000 crore
Saving Deposits of Post office saving banks	₹ 60000 crore

[ANS. Rs. 5,70,000 crore]

ILLUSTRATION 2 **[SM]**

Compute credit multiplier if the required reserved ratio is 10% and 12.5% for every ₹ 1, 00,000 deposited in the banking system. What will be the total credit money created by the banking system in each case?

[ANS. Rs, 10, 00,000, Rs, 8, 00,000]

ILLUSTRATION 3 **[SM]**

Calculate currency with the Public from the following data (₹ Crore)

1.1 Notes in Circulation	2496611
1.2 Circulation of Rupee Coin	25572
1.3 Circulation of Small Coins	743
1.4 Cash on Hand with Banks	98305

[ANS. 2424621]

ILLUSTRATION 4 **[SM]**

Calculate M2 from the following data

	(₹ Crore)
Notes in Circulation	2420964
Circulation of Rupee Coin	25572
Circulation of Small Coins	743
Post Office Saving Bank Deposits	141786
Cash on Hand with Banks	97563

Deposit Money of the Public	1776199
Demand Deposits with Banks	1737692
'Other' Deposits with Reserve Bank	38507
Total Post Office Deposits	14896
Time Deposits with Banks	178694

[ANS. 4267701]

ILLUSTRATION 5

If the required reserve ratio is 10 %, currency in circulation is ₹ 400 billion, demand deposits are 1000 billion, and excess reserves total ₹ 1 billion, find the value of money multiplier.

[ANS. 2.79]

ILLUSTRATION 6

Compute Reserve Money from the following data published by RBI

Components	As on 07.07.2017 (In Billions of ₹)
Currency in circulation	15,428.40
Bankers' deposit with RBI	4,596.18
Other deposits with RBI	183.30

[ANS. ₹ 20,207.88]

ILLUSTRATION 7

Compute M3 from the following data published by RBI

Components	As on 31.03.2017 (In Billions of ₹)
Currency with the public	12,637.10
Demand deposits with bank	14,106.30
Time deposits with bank	1,01,489.50
Other deposits with RBI	210.90

[ANS. ₹ 1,28,443.80]

ILLUSTRATION 8

What will be the total credit created by the commercial banking system for an initial deposit of ₹1,000/- for required reserve ratio 2%, 5% and 10% percent respectively? Compute credit multiplier

[ANS. 50,000; 20,000; 10,000]

IV. Past Exam Questions

1. What would be the impact of each of the following on credit multiplier and money supply?
   - *i.* If commercial bank keep 100% reserves
   - *ii.* If commercial bank do not keep reserves
   - *iii.* If commercial bank keeps excess reserve

   [PEQ – May 18 - 3 Marks]

2. What will be the total credit created by the commercial banking system for an initial deposit of ₹ 3,000 at a Required Reserve Ratio of 0.05 and 0.08 respectively. Also compute credit multiplier?

   [PEQ - May 19 - 2 Marks]

3. Compute credit multiplier if Required Reserve Ratio is 10% and 12.5% for every ₹ 1,00,000 deposited in the banking system. What will be the total credit money created by the banking system in each case?

   [PEQ – Nov 19 - 2 Marks]

4. The RBI published the following data as on 31st March, 2018. You are required to compute M4

Particulars	(₹ in crores)
Currency with the public	1,12,206.6
Demand Deposits with Banks	1,93,300.4
Net Time Deposits with Banks	2,67,310.2
Other Deposits of RBI	614.8
Post Office Savings Deposits	277.5
Post Office National Savings Certificates (NSC's)	110.5

[ANS. ₹ 5,73,709.50 crore] **[PEQ – Nov 18 - 3 Marks]**

5. Compute M1 supply of money from the data given below

Particulars	(₹ in crores)
Currency with the public	2,13,279.80
Time Deposits with Banks	3,45,000.70
Demand Deposits with Banks	1,62,374.50
Post Office Savings Deposits	382.90
Other Deposits of RBI	765.10

[ANS. 3,76,419.40 crore] **[PEQ – May 19 - 3 Marks]**

6. Compute Reserve Money from the following data published by RBI

Particulars	(₹ in crores)
Net RBI credit to the government	8,51,651
RBI Credit to commercial sector	2,62,115
RBI's claim on bank	4,10,315
Government's currency liability to the public	1,85,060
RBI's net foreign asset	72,133
RBI's net non-monetary liabilities	68,032

[ANS. ₹ 17,13,242 crore] **[PEQ – Nov 19 - 3 Marks]**

ANSWERS/HNTS

I. Multiple Choice Type Questions

1	(d)	2	(c)	3	(b)	4	(a)	5	(b)	6	(b)	7	(b)	8	(c)
9	(a)	10	(a)	11	(b)	12	(c)	13	(d)	14	(c)	15	(c)	16	(b)
17	(a)	18	(c)	19	(c)	20	(b)	21	(a)	22	(c)	23	(b)	24	(c)
25	(a)	26	(c)	27	(b)	28	(c)	29	(c)	30	(a)	31	(b)	32	(d)
33	(c)	34	(b)	35	(b)	36	(a)	37	(d)	38	(a)	39	(a)	40	(a)
41	(a)	42	(a)	43	(d)										

II. Numerical illustrations

1. M1 = Currency with public + Demand Deposits with Banking System + Other Deposits with the RBI = 90000 crore + 200000 crore + 280000 crore = 57 0000crore

2. Credit Multiplier is the reciprocal of required reserved ratio.
Credit Multiplier = 1 / Required Reserverd Ratio
For RRR = 0.10 i.e. 10% the credit multiplier =1/0.10 = 10
For RRR = 0.125i.e. 12.5% the credit multiplier = 1/0.125 = 8
Credit creation = Initial deposits * 1/RRR
For RRR 0.10 credit creation will be 1, 00,000× 1/0.10 = Rs, 10, 00,000
For RRR 0.125 credit creation will be 1, 00,000× 1/0.125= Rs, 8, 00,000

3. Currency with the Public (1.1 + 1.2 + 1.3 – 1.4) = (2496611+25572+743) – 98305 =2424621

4. M2 = M1+ Post Office Saving Bank Deposits
where M1 = (Notes in Circulation + Circulation of Rupee Coin + Circulation of Small Coins - Cash on Hand with Banks) + Deposit Money of the Public
= (2420964+25572+743- 97563) +1776199 =4125915
M2 = M1+ Post Office Saving Bank Deposits = 4125915 +141786= 4267701

5. r = 10% = 0.10
Currency = 400 billion
Deposits = 1000 billion
Excess Reserves = 1 billion
Money Supply is M = Currency + Deposits = 1400 billion
c = C/D = 400 billion/1000 billion = 0. 4 or Depositors hold 40 percent of their money as currency
e= 1billion /1000 billion = 0.001 or banks hold 0.1% of their deposits as excess reserves.
Multiplier

$$= \frac{1+0.4}{0.1+0.001+0.40} = \frac{14}{0.501} = 2.79$$

Therefore, a 1 unit increase in MB leads to a 2.79 units increase in M.

6. Reserve Money = ₹ 20,207.88/- [In Billion]
7. Ms = ₹ 1,28,443.80/- [In Billion]
8. Credit Multiplier = 50,000; 20,000; 10,000

IV. Past Exam Questions

1.

(i) If commercial banks keep 100% reserves, the reserve deposit ratio is one and the value of money multiplier is one. Deposits simply substitute for the currency that is held by banks as reserves and therefore, no new money is created by banks.

(ii) If commercial banks do not keep reserves and lends the entire deposits, it is a case of zero required reserve ratio and credit multiplier will be infinite and therefore money creation will also be infinite.

(iii) Excess reserves are reserves over and above what banks are legally required to hold against deposits. The additional units of money that goes into 'excess reserves' of the commercial banks do not lead to any additional loans, and therefore, these excess reserves do not lead to creation of money. The increase in banks' excess reserves reduces the credit multiplier, causing the money supply to decline.

2. Credit Multiplier = 20 ; Total Credit = ₹ 60,000
Credit Multiplier = 12.50 ; Total Credit = ₹ 37,500

3. Credit Multiplier = 10 ; Total Credit = ₹ 10,00,000
Credit Multiplier = 8 ; Total Credit = ₹ 8,00,000

4. M4 = ₹ 5,73,709.50 crore

5. M1 = ₹ 3,76,419.40 crore

6. Mo = ₹ 17,13,242 crore

For Your Notes

For Your Notes

For Your Notes

For Your Notes

UNIT - III
MONETARY POLICY

Topic References:

3.1 Introduction
3.2 Monetary Policy
3.3 The Monetary Policy Framework
3.4 The Organisational Structure for Monetary Policy Decisions

3.1 Introduction

We observe that the Reserve Bank of India is occasionally manipulating policy rates for maneuvering liquidity conditions with reasons thereof explicitly notified.

The discussion that follows is an attempt to throw light on the well-acknowledged monetary measures undertaken by governments to fight economic instability.

3.2 Monetary Policy

Monetary policy refers to the

- use of monetary policy instruments
- which are at the disposal of the central bank
- to regulate the availability, cost and use of money and credit
- to promote economic growth, price stability, optimum levels of output and employment, balance of payments equilibrium, stable currency or any other goal of government's economic policy.

Reserve Bank of India uses monetary policy to manage economic fluctuations and achieve price stability, which means that inflation is low and stable.

Reserve Bank of India conducts monetary policy by adjusting the supply of money, usually through buying or selling securities in the open market.

Open market operations affect short-term interest rates, which in turn influence longer-term rates and economic activity.

When central banks lower interest rates, monetary policy is easing. When it raises interest rates, monetary policy is tightening.

3.3 The Monetary Policy Framework

The central bank, in its execution of monetary policy, functions within an articulated monetary policy framework which has three basic components, viz.

(i) the objectives of monetary policy,
(ii) the analytics of monetary policy which focus on the transmission mechanisms, and
(iii) the operating procedure which focuses on the operating targets and instruments.

3.3.1 The Objectives of Monetary Policy

The objectives set for monetary policy are important because they provide explicit guidance to policy makers.

Monetary policy of a country is in fact a reflection of its economic policy and therefore, the

objectives of monetary policy generally coincide with the overall objectives of economic policy.

The Reserve Bank of India Act, 1934, in its preamble sets out the objectives of the Bank as 'to regulate the issue of bank notes and the keeping of reserves with a view to securing monetary stability in India and generally to operate the currency and credit system of the country to its advantage'.

Given the development needs of developing countries, the monetary policy of such countries also incorporate explicit objectives such as:

(i) maintenance the economic growth,
(ii) ensuring an adequate flow of credit to the productive sectors,
(iii) sustaining - a moderate structure of interest rates to encourage investments, and
(iv) creation of an efficient market for government securities.

3.3.2 Transmission of Monetary Policy

The process or channels through which the change of monetary aggregates affects the level of product and prices is known as 'monetary transmission mechanism'.

It describes how policy-induced changes in the nominal money stock or in the short-term nominal interest rates impact real variables such as aggregate output and employment.
The transmission of the monetary policy describes how changes made by the Reserve Bank to its monetary policy settings flow through to economic activity and inflation.
In simple terms, the transmission can be summarised in two stages.

1. Changes to monetary policy affect interest rates in the economy.
2. Changes to interest rates affect economic activity and inflation.

Q. Explain the different mechanism of monetary policy which influences the price-level and national income? [PEQ - Nov 18 - 3 Marks]

CHANNELS OF MONETARY POLICY TRANSMISSION

Saving and Investment Channel

Monetary policy influences economic activity by changing the incentives for saving and investment. This channel typically affects consumption, housing investment, and business investment.

- Lower interest rates on bank deposits reduce the incentives households must save their money. Instead, there is an increased incentive for households to spend their money on goods and services.
- Lower interest rates for loans can encourage households to borrow more as they face lower repayments. Because of this, lower lending rates support higher demand for assets, such as housing.
- Lower lending rates can increase investment spending by businesses (on capital goods like new equipment or buildings). This is because the cost of borrowing is lower, and because of increased demand for the goods and services they supply. This means that returns on these projects are now more likely to be higher than the cost of borrowing, helping to justify going ahead with the projects. This will have a more direct effect on businesses that borrow to fund their projects with debt rather than those that use the business owners' funds.

Cash-flow Channel

- Monetary policy influences interest rates, which affects the decisions of households and businesses by changing the amount of cash they have available to spend on goods and services. This is an important channel for those that are liquidity constrained (for example, those who have already borrowed up to the maximum that banks will provide).
- A reduction in lending rates reduces interest repayments on debt, increasing the amount of cash available for households and businesses to spend on goods and services. For example, a reduction in interest rates lowers repayments for households with variable-rate mortgages, leaving them with more disposable income.
- At the same time, a reduction in interest rates reduces the amount of income that households and businesses get from deposits, and some may choose to restrict their spending.

These two effects work in opposite directions, but a reduction in interest rates can be expected to increase spending in the Indian economy through this channel (with the first effect larger than the second)

Asset Prices and Wealth Channel

- Asset prices and people's wealth influence how much they can borrow and how much they spend in the economy. The asset prices and wealth channel typically affects consumption and investment.
- Lower interest rates support asset prices (such as housing and equities) by encouraging demand for assets. One reason for this is that the present discounted value of future income is higher when interest rates are lower.
- Higher asset prices also increase the equity (collateral) of an asset that is available for banks to lend against. This can make it easier for households and businesses to borrow.
- An increase in asset prices increases people's wealth. This can lead to higher consumption and housing investment as households generally spend some share of any increase in their wealth.

Exchange Rate Channel

The exchange rate can have an important influence on economic activity and inflation. It is typically more important for sectors that are export-oriented or exposed to competition from imported goods and services.

- If the Reserve Bank lowers the cash rate it means that interest rates in India have fallen compared with interest rates in the rest of the world (all else being equal).
- Lower interest rates reduce the returns investors earn from assets in India (relative to other countries). Lower returns reduce demand for assets in India (as well as for Indian rupees) with investors shifting their funds to foreign assets (and currencies) instead.
- A reduction in interest rates (compared with the rest of the world) results in a lower exchange rate, making foreign goods and services more expensive compared with those produced in India. This leads to an increase in exports and domestic activity. A lower exchange rate also adds to inflation because imports become more expensive in Indian rupees.

3.3.3 Operating Procedures and Instruments

Quantitative tools –

The tools applied by the policy that impact money supply in the entire economy, including sectors such as manufacturing, agriculture, automobile, housing, etc.

Reserve Ratio

Banks are required to keep aside a set percentage of cash reserves or RBI approved assets. Reserve ratio is of two types:

Q. How do changes in Cash Reserve Ratio (CRR) impact the economy? [PEQ-May -18 - 2 Marks]

Cash Reserve Ratio (CRR) – Banks are required to set aside this portion in cash with the RBI. The bank can neither lend it to anyone nor can it earn any interest rate or profit on CRR.

Statutory Liquidity Ratio (SLR) – Banks are required to set aside this portion in liquid assets such as gold or RBI approved securities such as government securities. Banks are allowed to earn interest on these securities, however it is very low.

Open Market Operations (OMO)

In order to control money supply, the RBI buys and sells government securities in the open market. These operations conducted by the Central Bank in the open market are referred to as Open Market Operations.

When the RBI sells government securities, the liquidity is sucked from the market, and the exact opposite happens when RBI buys securities. The latter is done to control inflation. The objective of OMOs are to keep a check on temporary liquidity mismatches in the market, owing to foreign capital flow.

Qualitative tools -

Unlike quantitative tools which have a direct effect on the entire economy's money supply, qualitative tools are selective tools that have an effect in the money supply of a specific sector of the economy.

Margin requirements – The RBI prescribes a certain margin against collateral, which in turn impacts the borrowing habit of customers. When the margin requirements are raised by the RBI, customers will be able to borrow less.

Moral suasion – By way of persuasion, the RBI convinces banks to keep money in government securities, rather than certain sectors.

Selective credit control – Controlling credit by not lending to selective industries or speculative businesses.

Market Stabilisation Scheme (MSS) –

Policy Rates -

Bank rate – The interest rate at which RBI lends long term funds to banks is referred to as the bank rate. However, presently RBI does not entirely control money supply via the bank rate. It uses Liquidity Adjustment Facility (LAF) – repo rate as one of the significant tools to establish control over money supply.

Bank rate is used to prescribe penalty to the bank if it does not maintain the prescribed SLR or CRR.

Liquidity Adjustment Facility (LAF) – RBI uses LAF as an instrument to adjust liquidity and money supply. The following types of LAF are:

Repo rate: Repo rate is the rate at which banks borrow from RBI on a short-term basis against a repurchase agreement. Under this policy, banks are required to provide government securities as collateral and later buy them back after a pre-defined time.

Q. Explain Reverse Repo Rate? [PEQ - Nov 19 - 2 Marks]

Reverse Repo rate: It is the reverse of repo rate, i.e., this is the rate RBI pays to banks in order to keep additional funds in RBI. It is linked to repo rate in the following way:
Reverse Repo Rate = Repo Rate – 0.25

Marginal Standing Facility (MSF) Rate:
MSF Rate is the penal rate at which the Central Bank lends money to banks, over the rate available under the rep policy. Banks availing MSF Rate can use a maximum of 1% of SLR securities.
MSF Rate = Repo Rate + 0.25

Q. Explain the difference between Liquidity Adjustment Facility (LAF) and Marginal Standing Facility (MSF) [PEQ - May 18 - 3 Marks]

3.4 The Organisational Structure for Monetary Policy Decision

Q. Explain the Monetary Policy Framework Agreement? [PEQ - Nov 18 - 2 Marks]

The Reserve Bank of India (RBI) Act, 1934 was amended on June 27, 2016, for giving a statutory backing to the Monetary Policy Framework Agreement and for setting up a Monetary Policy Committee (MPC).

The Monetary Policy Framework Agreement is an agreement reached between the Government of India and the Reserve Bank of India (RBI) on the maximum tolerable inflation rate that the RBI should target to achieve price stability.

The amended RBI Act (2016) provides for a statutory basis for the implementation of the 'flexible inflation targeting framework'.

Announcement of an official target range for inflation is known as inflation targeting.

The inflation target is to be set by the Government of India, in consultation with the Reserve Bank, once in every five years.

Accordingly,

- The Central Government has notified 4% Consumer Price Index (CPI) inflation as the target for the period from August 5, 2016 to March 31, 2021 with the upper tolerance limit of 6 per cent and the lower tolerance limit of 2%.
- The RBI is mandated to publish a Monetary Policy Report every six months, explaining the sources of inflation and the forecasts of inflation for the coming period of six to eighteen months.
- The following factors are notified by the central government as constituting a failure to achieve the inflation target:

(a) the average inflation is more than the upper tolerance level of the inflation target for any three consecutive quarters; or

(b) the average inflation is less than the lower tolerance level for any three consecutive quarter

The choice of CPI was made because it closely reflects cost of living and has larger influence on inflation expectations compared to other anchors. With this step, India is following countries such as the New Zealand, the USA, the UK, European Union, and Brazil. Although in recent times many of the countries are moving away from this approach and the targeting nominal GDP growth.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. Which of the following is the function of monetary policy?
   (a) *regulate the exchange rate and keep it stable*
   (b) *regulate the movement of credit to the corporate sector*
   (c) *regulate the level of production and prices*
   (d) *regulate the availability, cost and use of money and credit*

2. The main objective of monetary policy in India is:
   (a) *reduce food shortages to achieve stability*
   (b) *economic growth with price stability*
   (c) *overall monetary stability in the banking system*
   (d) *reduction of poverty and unemployment*

3. The monetary transmission mechanism refers to
   (a) *how money get circulated in different sectors of the economy post monetary policy*
   (b) *the ratio of nominal interest and real interest rate consequent on a monetary policy*
   (c) *the process or channels through which the evolution of monetary aggregates affects the level of product and prices*
   (d) *None of the above*

4. A contractionary monetary policy-induced increase in interest rates
   (a) *increases the cost of capital and the real cost of borrowing for firms*
   (b) *increases the cost of capital and the real cost of borrowing for firms and households*
   (c) *decreases the cost of capital and the real cost of borrowing for firms*
   (d) *has no interest rate effect on firms and households*

5. During deflation
   (a) *the RBI reduces the CRR in order to enable the banks to expand credit and increase the supply of money available in the economy*
   (b) *the RBI increases the CRR in order to enable the banks to expand credit and increase the supply of money available in the economy*
   (c) *the RBI reduces the CRR in order to enable the banks to contract credit and increase the supply of money available in the economy*
   (d) *the RBI reduces the CRR but increase SLR in order to enable the banks to contract credit and increase the supply of money available in the economy*

6. Which of the following statements is correct?
   (a) *The governor of the RBI in consultation with Ministry of Finance decides the policy rate and implements the same.*
   (b) *While CRR has to be maintained by banks as cash with the RBI, the SLR requires holding of approved assets by the bank itself*
   (c) *When Repo Rate increases, it means that banks can now borrow money through Open Market Operation*
   (d) *None of the above*

7. RBI provides financial accommodation to commercial banks through repos / reverse repos under
   (a) *Market Stabilization Scheme (MSS)*
   (b) *The Marginal Standing Facility (MSF)*
   (c) *Liquidity Adjustment Facility (LAF)*
   (d) *Statutory Liquidity Ratio (SLR)*

8. _____ is a money market instrument, which enables collateralized short term borrowing and lending through sale/purchase operations in debt instruments.
   (a) *OMO* (b) *CRR* (c) *SLR* (d) *Repo*

9. In India, the term 'Policy rate' refers to
   (a) *The bank rate prescribed by the RBI in its half yearly monetary policy statement*
   (b) *The CRR and SLR prescribed by RBI in its monetary policy statement*
   (c) *the fixed repo rate quoted for sovereign securities in the overnight segment of Liquidity Adjustment Facility*
   (d) *the fixed repo rate quoted for sovereign securities in the overnight segment of Marginal Standing Facility*

10. Reverse repo operation takes place when
   (a) *RBI borrows money from banks by giving them securities*
   (b) *Bank borrows money from the RBI by giving them securities*
   (c) *Bank borrows money in the overnight segment of the money market*
   (d) *RBI borrows money from the Central Government*

11. The Monetary Policy Framework Agreement is on
   (a) *The maximum repo rate that RBI can charge from government*
   (b) *The maximum tolerable inflation rate that RBI should target to achieve price stability*
   (c) *The maximum repo rate that RBI can charge from commercial banks*
   (d) *The maximum reverse repo rate that RBI can charge from commercial banks*

12. An open market operation is an instrument of monetary policy which involves buying or selling of _____ from or to the public and banks
   (a) *bonds and bills of exchange* (b) *debentures and shares*
   (c) *government securities* (d) *none of these*

13. Which statement (s) is (are) true about Monetary Policy Committee?
   I. *The Reserve Bank of India (RBI) Act, 1934 was amended on June 27, 2016, for giving a statutory backing to the Monetary Policy Framework Agreement and for setting up a Monetary Policy Committee*
   II. *The Monetary Policy Committee shall determine the policy rate through debate and majority vote by a panel of experts required to achieve the inflation target.*
   III. *The Monetary Policy Committee shall determine the policy rate through consensus from the governor of RBI*
   IV. *The Monetary Policy Committee shall determine the policy rate through debate and majority vote by a panel of bankers chosen for the purpose*
   (a) *I only* (b) *I and II* (c) *III and IV* (d) *III only*

PEQ – JUNE – 2024

14. Which of the following is not a channel of monetary transmission mechanism?
   (a) *Savings and investment channel* (b) *Cash flow channel*
   (c) *Exchange rate channel* (d) *International Trade channel* *(1 mark)*

15. Open Market purchases of government securities by RBI will:
   (a) *Reduce money supply* (b) *Reduce reserves*
   (c) *Increase reserves* (d) *Reduce reserves and thereby reduce money supply* *(1 mark)*

16. The "Bank Rate" is one of the key policy used by the Reserve Bank of India (RBI). An increase in the Bank Rate is likely to impact the economy by:
   (a) *Encouraging banks to increase their lending activities*
   (b) *Reducing interest rates for consumers and business*
   (c) *Discouraging borrowing and spending*
   (d) *Promoting exports and foreign trade* *(1 mark)*

PEQ – SEPTEMBER – 2024

17. What does RBI publish every six months, providing explanations of the sources of inflation and forecasts for the upcoming period of six to eighteen months?
(a) *Financial Stability Report* (b) *Economic Outlook Report*
(c) *Monetary Policy Report* (d) *Inflation Targeting Framework* *(1 mark)*

18. Reverse Repo rate is linked to repo rate in the following way:
(a) *Reverse Repo Rate = Repo Rate -1* (b) *Reverse Repo Rate = Repo Rate + 1*
(c) *Reverse Repo Rate = Repo Rate x 0.1%* (d) *Reverse Repo Rate = Repo Rate x 0.01%*
(1 mark)

PEQ – JANUARY - 2025

19. _____ is a penal rate at which RBI lends money to banks, above the rate available under the repo policy.
(a) *Marginal standing facility rate* (b) *Bankrate*
(c) *Repo rate* (d) *Reverse repo rate* *(1 mark)*

20. Liquidity Adjustment Facility (LAF) was introduced by RBI on the basis of the recommendation of the _____ Committee on the reforms in banking sector.
(a) *Tandon* (b) *Narsimham* (c) *Chore* (d) *Basel*
(1 mark)

PEQ – MAY - 2025

21. Banks availing Marginal Standing Facility Rate can use a maximum of how much percentage of Statutory Liquidity Ratio Securities?
(a) *1%*
(b) *2%*
(c) *3%*
(d) *4%*

22. Which of the following is NOT a quantitative tool of monetary policy?
(a) *Cash Reserve Ratio (CRR)*
(b) *Statutory Liquidity Ratio (SLR)*
(c) *Open Market Operations (OMO)*
(d) *Liquidity Adjustment Facility (LAF)*

23. Which of the following expressions is true?
(a) *Reverse Repo Rate = Repo Rate - 1*
(b) *Reverse Repo Rate = Repo Rate + 1*
(c) *Repo Rate = Reverse Repo Rate - 1*
(d) *Repo Rate = 1- Reverse Repo Rate*

PEQ - SEPTEMBER - 2025

24. Which of the following is not a quantitative tool as used by RBI to control the money supply in the entire economy?
(a) *Open Market Operations*
(b) *Statutory Liquidity Ratio*
(c) *Selective Credit Control*
(d) *Cash Reserve Ratio*

25. Which of the following statements are true in relation to the Crypto currency?
Statement-I: These have achieved significant legislative recognition. Statement-II: These are not legally recognized in India as currency. Statement-III: These are not categorized as money.
Statement-IV: In India, it serves as a legitimate form of payment for goods and services in the same way as traditional money does.
(a) Statement I and IV
(b) Statement I and II
(c) Statement I and III
(d) Statement II and III

26. What is the primary objective of monetary policy in most countries, including India?
(a) To ensure a high level of government spending.
(b) To maintain a balance between price stability and economic growth.
(c) To promote export-led growth.
(d) To control exchange rates at all costs.

27. Marginal Standing Facility (MSF) Rate is the penal rate at which the Central Banks lend money to Banks, over the rate available under the Repo Policy. Which of the following are the correct options to calculate MSF Rate ?
Option (1): MSF Rate Repo Rate + 1 MSF Rate
Option (2): MSF Rate-Repo Rate-1 MSF Rate
Option (3): MSF Rate = Repo Rate + 1
Option (4): MSF Rate Repo Rate - 1
(a) Option (1) and (4)
(b) Option (2) and (3)
(c) Option (1) and (3)
(d) Option (2) and (4)

28. Which monetary tool is used to check temporary liquidity mismatches in the market due to foreign capital flow?
(a) Cash Reserve Ratio (CRR)
(b) Statutory Liquidity Ratio (SLR)
(c) Open Market Operations (OMO)
(d) Marginal Standing Facility (MSF)

PEQ - JANUARY - 2026

29. Which of the following is not a channel of monetary policy transmission?
(a) Saving and Investment channel
(c) Exchange Rate channel
(8) Cash-flow channel
(d) Open Market channel

30. When the RBI uses persuasion to convince banks to hold government securities instead of lending to certain sectors, it is called:
(a) Margin requirements
(b) Statutory Liquidity Ratio
(c) Market Stabilization Ratio
(d) Moral Suasion

31. Who is responsible for determining the inflation target for the RBI under the Monetary Policy Framework?
    (a) *RBI independently*
    (b) *The Central Government in collaboration with the RBI*
    (c) *The Finance Ministry*
    (d) *The Monetary Policy Committee*

32. Which action does the central bank perform during Open Market Operations (OMO) to introduce high-powered money into the economy?
    (a) *Purchase of government securities*
    (b) *Sale of corporate bonds*
    (c) *Printing additional currency*
    (d) *Raising interest rates*

PEQ - SEPTEMBER - 2026

33. Monetary policy primarily aims to regulate:
    (a) *Tax rates*
    (b) *Money supply and credit conditions*
    (c) *Government expenditure*
    (d) *Export levels*

34. Which policy action is most likely to reduce inflation?
    (a) *Expansionary monetary policy*
    (b) *Contractionary monetary policy*
    (c) *Lower reserve ratio*
    (d) *Higher government spending*

ANSWERS/HNTS

I. Multiple Choice Type Questions

1	(d)	2	(b)	3	(c)	4	(b)	5	(a)	6	(b)	7	(c)	8	(d)
9	(c)	10	(a)	11	(b)	12	(c)	13	(b)	14.	(d)	15.	(c)	16.	(c)
17.	(c)	18.	(a)	19.	(a)	20.	(b)	21.	(a)	22.	(d)	23.	(a)	24.	(c)
25.	(d)	26.	(b)	27.	(c)	28.	(c)	29.	(d)	30.	(d)	31.	(b)	32.	(a)
33.	(b)	34.	(b)												

For Your Notes

For Your Notes

For Your Notes

For Your Notes

CHAPTER-09

INTERNATIONAL TRADE

Unit 1 – Theories of International Trade

Unit 2 – The Instruments of Trade Policy

Unit 3 – Trade Negotiations

Unit 4 – Exchange Rate and Its Economic Effects

Unit 5 – International Capital Movements

UNIT – I

THEORIES OF INTERNATIONAL TRADE

Topic References:

1.1 Introduction
1.2 Advantages of International Trade
1.3 Disadvantages of International Trade
1.4 Theories of International Trade

1.1 Introduction

International trade is the exchange of goods and services as well as resources between countries. It involves transactions between residents of different countries.

As distinguished from domestic trade or internal trade which involves exchange of goods and services within the domestic territory of a country using domestic currency, international trade involves transactions in multiple currencies.

Compared to internal trade, international trade has greater complexity as it involves heterogeneity of customers and currencies, differences in legal systems, more elaborate documentation, diverse restrictions in the form of taxes, regulations, duties, tariffs, quotas, trade barriers, standards, restrains to movement of specified goods and services and issues related to shipping and transportation.

At present, liberal international trade is an integral part of international relations and has become an important engine of growth in developed as well as developing countries.

1.2 Advantages of International Trade

(i) Economic Growth –

International trade is a powerful stimulus to economic efficiency and contributes to economic growth and rising incomes.

As a result, they would enlarge their manufacturing capabilities and benefit from economies of large scale production.

The gains from international trade are reinforced by the increased competition that domestic producers are confronted with on account of globalization of production and marketing requiring businesses to compete against global businesses.

Competition from foreign goods compels manufacturers, especially in developing countries, to enhance efficiency and profitability by adoption of cost reducing technology and business practices.

Efficient deployment of productive resources to their best uses is a direct economic advantage of foreign trade.

Greater efficiency in the use of natural, human, industrial and financial resources ensures productivity gains.

Since international trade also tends to decrease the likelihood of domestic monopolies, it is always beneficial to the community.

(ii) Access to New Market –

Trade provides access to new markets and new materials and enables sourcing of inputs and components internationally at competitive prices.

This reflects in innovative products at lower prices and wider choice in products and services for consumers.

Also, international trade enables consumers to have access to wider variety of goods and services that would not otherwise be available.

It also enables nations to acquire foreign exchange reserves necessary for imports which are crucial for sustaining their economies.

(iii) Skilled Labour / Mechanistion –

International trade augments the scope for mechanization and specialization.
Trade necessitates increased use of automation, supports technological change, stimulates innovations, and facilitates greater investment in research and development and productivity improvement in the economy.

(iv) Employment –

Exports stimulate economic growth by creating jobs, which could potentially reduce poverty and augmenting factor incomes and in so doing raising standards of livelihood and overall demand for goods and services.

Trade also provides greater stimulus to innovative services in banking, insurance, logistics, consultancy services etc.

(v) FDI –

Employment generating investments, including foreign direct investment, inevitably follow trade.

For emerging economies, improvement in the quality of output of goods and services, superior products, finer labour and environmental standards etc. enhance the value of their products and enable them to move up the global value chain.

(vi) Price Stability –

Opening up of new markets results in broadening of productive base and facilitates export diversification so that new production possibilities are opened up.

Countries can gainfully dispose-off their surplus output and, thus, prevent undue fall in domestic prices caused by overproduction.

Trade also allows nations to maintain stability in prices and supply of goods during periods of natural calamities like famine, flood, epidemic etc.

(vii) Human Resource Development –

Trade can also contribute to human resource development, by facilitating fundamental and applied research and exchange of know-how and best practices between trade partners.

(viii) Bonding –

Trade strengthens bonds between nations by bringing citizens of different countries together in mutually beneficial exchanges and, thus, promotes harmony and cooperation among nations.

1.3 Disadvantages of International Trade

(i) Unskilled Labour –

Possible negative labour market outcomes in terms of labour - saving technological change that depress demand for unskilled workers, loss of labourers' bargaining power, downward pressure on wages of semi-skilled and unskilled workers and forced work under unfair circumstances and unhealthy occupational environments.

(ii) Unequal Market Access –

International trade is often not equally beneficial to all nations. Potential unequal market access and disregard for the principles of fair trading system may even amplify the differences between trading countries, especially if they differ in their wealth.

Economic exploitation is a likely outcome when underprivileged countries become vulnerable to the growing political power of corporations operating globally.

The domestic entities can be easily outperformed by financially stronger transnational companies.

(iii) Stress on Exports –

International trade is often criticized for its excessive stress on exports and profit-driven exhaustion of natural resources due to unsustainable production and consumption.

Substantial environmental damage and exhaustion of natural resources in a shorter span of time could have serious negative consequences on the society at large.

(iv) Foreign Goods –

Probable shift towards a consumer culture and change in patterns of demand in favour of foreign goods which are likely to occur in less developed countries may have adverse effect on the development of domestic industries and may even threaten the survival of infant industries.

Trade cycles and the associated economic crises occurring in different countries are also likely to get transmitted rapidly to other countries.

(v) Political Soverignity –

Risky dependence of underdeveloped countries on foreign nations impairs economic autonomy and endangers their political sovereignty. Such reliance often leads to widespread exploitation and loss of cultural identity.

Substantial dependence may also have severe adverse consequences in times of wars and other political disturbances.

(vi) Inflation –

Welfare of people may be ignored or jeopardized for the sake of profit. Excessive exports may cause shortages of many commodities in the exporting countries and lead to high inflation (e.g. onion price rise in 2014). Also, import of harmful products may cause health hazards and environmental damage. (E.g. Chinese products).

(vii) Economic Exploitation –

Too much export orientation may distort actual investments away from the genuine investment needs of a country.

(viii) Rivalry –

Instead of cooperation among nations, trade may breed rivalry on account of severe competition

(ix) Lack of Transperency –

Finally, there is often lack of transparency and predictability in respect of many aspects related to trade policies of trading partners. There are also many risks in trade which are associated with changes in governments' policies of participating countries, such as imposition of an import ban or trade embargoes.

1.4 Theories of International Trade

1.4.1 The Mercantilists' View of International Trade

Mercantilism, was the policy of Europe's great powers,
It was based on the premise that national wealth and power are best served by increasing exports and collecting precious metals in return.

Mercantilists also believed that the more gold and silver a country accumulates, the richer it becomes.

Mercantilism advocated maximizing exports in order to bring in more "specie" (precious metals) and minimizing imports through the state imposing very high tariffs on foreign goods.

This view argues that trade is a 'zero-sum game', with winners who win does so only at the expense of losers and one country's gain is equal to another country's loss, so that the net change in wealth or benefits among the participants is zero.

1.4.2 The Theory of Absolute Advantage

Adam Smith was the first to put across the possibility that international trade is not a zero- sum game.
According to him, absolute cost advantage is the determinant of mutually beneficial international trade.

The absolute cost advantage theory points out that a country will specialize in the production and export of a commodity in which it has an absolute cost advantage.

Exchange of goods between two countries will take place only if each of the two countries can produce one commodity at an absolutely lower production cost than the other country.

Smith's thoughts on the principle of division of labour constitute the basis for his theory of international trade and therefore, the value of goods is determined by measuring the labour incorporated in them.

The theory is generally presented with an example of a hypothetical two countries and two commodities model (2x2 model).

Absolute advantage exists between nations when they differ in their ability to produce goods. Each nation can produce one good with less expenditure of human labour or more cheaply than the other. As a result, each nation has an absolute advantage in the production of one good.

Absolute advantage can be explained with a simple numerical example:

Table
Output per Hour of Labour

Commodity	Country A	Country B
Wheat (bushels/hour)	6	1
Cloth (yards/hour)	4	5

As can be seen from the above table-

One hour of labour time produces 6 bushels and 1 bushel of wheat respectively in Country A and Country B.

On the other hand, one hour of labour time produces 4 yards of cloth in Country A and 5 in Country B.

Country A is more efficient than Country B, or has an absolute advantage over Country B in production of wheat. Similarly, Country B is more efficient than country A, or has an absolute advantage over Country A in the production of cloth.

If both nations can engage in trade with each other, each nation will specialize in the production of the good it has an absolute advantage in and obtain the other commodity through international trade.

Therefore, Country A would specialize completely in production of wheat and Country B in cloth.

EXCHANGE GAIN:

If country A exchanges six bushels of wheat (6W) for six yards of country B's cloth (6C), then Country A gains 2C or saves 30 minutes of labour time (since the country A can only exchange 6W for 4C domestically).

Similarly, the 6W that country B receives from country A is equivalent to or would require six hours of labour time to produce in country B. These same six hours can produce 30C in country B (6 hours x 5 yards of cloth per hour). By being able to exchange 6C (requiring one hour and twelve minutes to produce in the country B) for 6W, Country B gains 24C, or saves nearly four hours forty eight minutes of work.

This shows trade is advantageous, although gains may not be distributed equally, because their given resources are utilised more efficiently, and, therefore, both countries can produce larger quantities of commodities which they specialize in.

By specializing and trading freely, global output is, thus, maximized and more of both goods are available to the consumers in both the countries.

If they specialize but do not trade freely, Country A's consumers would have no cloth, and country B's consumers would have no wheat. That is not desirable situation.

What happens if a country had higher productivity in both commodities compared to another country?

Let us now think of a situation where country A makes both wheat and cloth with fewer resources than country B. In other words, country A has absolute advantage in the production of both commodities and country B has absolute disadvantage in the production of both commodities.

This is the question that Ricardo attempted to answer when he formalized the concept of 'comparative advantage' to espouse the argument that even when one country is technologically superior in both goods, it could still be advantageous for them to trade.

1.4.3 The Theory of Comparative Advantage

David Ricardo developed the classical theory of comparative advantage in his book 'Principles of Political Economy and Taxation' published in 1817.

The law of comparative advantage states that even if one nation is less efficient than (has an absolute disadvantage with respect to) the other nation in the production of all commodities, there is still scope for mutually beneficial trade.

The first nation should specialize in the production and export of the commodity in which its absolute disadvantage is smaller (this is the commodity of its comparative advantage) and import the commodity in which it's absolute disadvantage is greater (this is the commodity of its comparative disadvantage)

The theory can be explained with a simple example given in table:

Table
Output per Hour of Labour

Commodity	Country A	Country B
Wheat (bushels/hour)	6	1
Cloth (yards/hour)	4	2

Table 4.1.2 differs from table 4.1.1 only in one respect; i.e., in this example, country B can produce only two yards of cloth per hour of labour.

Country B has now absolute disadvantage in the production of both wheat and cloth.

However, since B's labour is only half as productive in cloth but six times less productive in wheat compared to Country A, Country B has a comparative advantage in cloth.

On the other hand, Country A has an absolute advantage in both wheat and cloth with respect to the country B, but since its absolute advantage is greater in wheat (6:1) than in cloth (4:2), country A has a comparative advantage in production and exporting wheat.

In a two-nation, two-commodity world, once it is established that one nation has a comparative advantage in one commodity, then the other nation must necessarily have a comparative advantage in the other commodity.

Put in other words, country A's absolute advantage is greater in wheat, and so country A has a comparative advantage in producing and exporting wheat.

Country B's absolute disadvantage is smaller in cloth, so its comparative advantage lies in cloth production.

According to the law of comparative advantage, both nations can gain if country A specializes in the production of wheat and exports some of it in exchange for country B's cloth. Simultaneously, Country B should specialize in the production of cloth and export some of it in exchange for country A's wheat.

EXCHANGE GAIN:

Assume that country A could exchange 6W for 6C with country B.

Then, country A would gain 2C (or save half an hour of labour time) since the country A could only exchange 6W for 4C domestically.

Country B would also gain from trade. 6W that the country B receives from the country A would require six hours of labour time to produce in country B. With trade, country B can instead use these six hours to produce 12C and give up only 6C for 6W from the country A. Thus, the country B would gain 6C or save three hours of labour time.

However, the gains of both countries are not likely to be equal.

We need to recognize that this is not the only rate of exchange at which mutually beneficial trade can take place. Country A would gain if it could exchange 6W for more than 4C from country B; because 6W for 4 C is what it can exchange domestically (both require the same one hour labour time). The more C it gets, the greater would be the gain from trade.

Conversely, in country B, 6W = 12C (in the sense that both require 6 hours to produce). Anything less than 12C that country B must give up to obtain 6W from country A represents a gain from trade for country B.

Thus, the range for mutually advantageous trade is 4C < 6W < 12C.

The spread between 12C and 4C (i.e., 8C) represents the total gains from trade available to be shared by the two nations by trading 6W for 6C.

The closer the rate of exchange is to 4C = 6W (the domestic, or internal rate in country A), the smaller is the share of the gain going to country A and the larger is the share of the gain going to country B.

Alternatively, the closer the rate of exchange is to 6W = 12C (the domestic or internal rate in country B), the greater is the gain of country A relative to that of country B.

However, if the absolute disadvantage that one nation has with respect to another nation is the same in both commodities, there will be no comparative advantage and no trade.

However, the Ricardian theory of comparative advantage suffers from many limitations. Its emphasis is on supply conditions and excludes demand patterns. Moreover, the theory does not examine why countries have different costs. The answer to this question is provided by the Heckscher-Ohlin theory.

1.4.4 The Heckscher-Ohlin Theory of Trade

The Heckscher-Ohlin theory of trade, (named after two Swedish economists, Eli Heckscher and his student Bertil Ohlin), also referred to as Factor-Endowment Theory of Trade or Modern Theory of Trade, is considered as a very important theory of international trade.

In view of the contributions made by P. A. Samuelson, this theory is also sometimes referred to as Heckscher-Ohlin-Samuelson theorem.

The Heckscher-Ohlin (H-O) model studies the case that two countries have different factor endowments under identical production function and identical preferences.

The difference in factor endowment results in two countries having different factor prices in the beginning. Consequently, H-O model implies that the two countries will have different cost functions.

In a general sense of the term, 'factor endowment' refers to the overall availability of usable resources including both natural and man-made means of production. Nevertheless, in the exposition of the modern theory, only the two most important factors—labour and capital—are taken into account.

According to this theory, international trade is but a special case of inter-regional trade.

Different regions have different factor endowments i.e. some regions have abundance of labour, but scarcity of capital; whereas other regions have abundance of capital, but scarcity of labour.

Different goods have different production functions, that is, factors of production are combined in different proportions to produce different commodities.

While some goods are produced by employing a relatively larger proportion of labour and relatively small proportion of capital, other goods are produced by employing a relatively small proportion of labour and relatively large proportion of capital.

Thus, each region is suitable for the production of those goods for whose production it has relatively plentiful supply of the requisite factors.

A region is not suitable for production of those goods for whose production it has relatively scarce or zero supply of essential factors.

Therefore, difference in factor endowments is the main cause of international trade as well as inter-regional trade.

According to Ohlin, the immediate cause of inter-regional trade is that goods can be bought cheaper in terms of money than they can be produced at home and this is the case of international trade as well. The cause of difference in the relative prices of goods is the difference the amount of factor endowments, like capital and labour, between two countries.

The theory states that a country's exports depend on its resources endowment i.e. whether the country is capital-abundant or labour-abundant.

If a country is a capital abundant one, it will produce and export capital-intensive goods relatively more cheaply than another country.

Likewise, a labour-abundant country will produce and export labour-intensive goods relatively more cheaply than another country.

Table
Comparison of Theory of Comparative Costs and Modern Theory

Theory of Comparative Costs	Modern Theory
The basis is the difference between countries is comparative costs	Explains the causes of differences in comparative costs as differences in factor endowments
Based on labour theory of value	Based on money cost which is more realistic.
Considered labour as the sole factor of production and presents a one-factor (labour) model	Widened the scope to include labour and capital as important factors of production. This is 2-factor model and cand be extended to more factors.
Treats international trade as quite distinct from domestic trade	International trade is only a special case of inter-regional trade.
Studies only comparative costs of the goods concerned	Considers the relative prices of the factors which influence the comparative costs of the goods
Attributes the differences in comparative advantage to differences in productive efficiency of workers	Attributes the differences in comparative advantage to the differences in factor endowments.
Does not take into account the factor price differences	Considers factor price differences as the main cause of commodity price differences
Does not provide the cause of differences in comparative advantage.	Explains the differences in comparative advantage in terms of differences in factor endowments.
Normative; tries to demonstrate the gains from international trade	Positive; concentrates on the basis of trade

1.4.5 Globalization and New International Trade Theory

New Trade Theory (NTT) is an economic theory that was developed in the 1970s as a way to understand international trade patterns.

NTT came about to help us understand why developed and big countries are trade partners when they are trading similar goods and services. These countries constitute more than 50% of world trade.

This is particularly true in key economic sectors such as electronics, IT, food, and automotive. We have cars made in the India, yet we purchase many cars made in other countries.

These are usually products that come from large, global industries that directly impact international economies. The mobile phones we use are a good example. India produces them and also imports them.

NTT argues that, because of substantial economies of scale and network effects, it pays to export phones to sell in another country. Those countries with the advantages will dominate the market, and the market takes the form of **monopolistic competition**.

According to NTT, two key concepts give advantages to countries that import goods to compete with products from the home country:

- ***Economies of Scale:*** As a firm produces more of a product its cost per unit keeps going down. So if the firm serves domestic as well as foreign market instead of just one, it can reap the benefit of large scale of production consequently the profits are likely to be higher.

- ***Network effects:*** These are the way one person's value of a good or service is affected by the value of that good or service to others. The value of the product or service is enhanced as the number of individuals using it increases. This is also referred to as the 'bandwagon effect'. Consumers like more choices, but they also want products and services with high utility, and the network effect offers increased utility from these products over others.
  Example: Mobile App such as WhatsApp and software like Microsoft Windows

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. Which of the following does not represent a difference between internal trade and international trade?
   (a) *transactions in multiple currencies* (b)*homogeneity of customers and currencies*
   (c) *differences in legal systems* (d) *none of the above*

2. The theory of absolute advantage states that
   (a) *national wealth and power are best served by increasing exports and decreasing imports*
   (b) *nations can increase their economic well-being by specializing in the production of goods they produce more efficiently than anyone else*
   (c) *that the value or price of a commodity depends exclusively on the amount of labour going into its production and therefore factor prices will be the same*
   (d) *differences in absolute advantage explains differences in factor endowments in different countries*

3. Which of the following theories advocates that countries should produce those goods for which it has the greatest relative advantage
   (a) *Modern theory of international trade* (b) *The factor endowment theory*
   (c) *The Heckscher-Ohlin Theory* (d) *None of the above*

4. Which of the following holds that a country can increase its wealth by encouraging exports and discouraging imports
   (a) *Capitalism* (b) *Socialism*
   (c) *Mercantilism* (d) *Laissez faire*

5. Given the number of labour hours to produce cloth and grain in two countries, which country should produce grain?

Labour cost (hours) for production of one unit

	Country A	*Country B*
Cloth	*40*	*80*
Grain	*80*	*40*

   (a) *Country A* (b) *Country B*
   (c) *Neither A nor B* (d) *Both A and B*

6. According to theory of comparative advantage
   (a) *Trade is a zero sum game so that the net change in wealth or benefits among the participants is zero*
   (b) *trade is not a zero-sum game so that the net change in wealth or benefits among the participants is positive*
   (c) *nothing definite can be said about the gains from trade*
   (d) *gains from trade depends upon factor endowment and utilization*

7. Given the number of labour hours to produce wheat and rice in two countries and that these countries specialise and engage in trade at a relative price of 1:1 what will be the gain of country X?

Labour cost (hours) for production of one unit

	Wheat	*Rice*
Country X	*10*	*20*
Country Y	*20*	*10*

   (a) *20 labour hours* (b) *10 labour hours*
   (c) *30 labour hours* (d) *Does not gain anything*

8. Assume India and Bangladesh have the unit labour requirements for producing tables and mats shown in the table below. It follows that:

Labour cost (hours) for production of one unit

	India	*Bangladesh*
Tables	*3*	*8*
Mats	*2*	*1*

(a) Bangladesh has a comparative advantage in mats
(b) India has a comparative advantage in tables
(c) Bangladesh has an absolute advantage in mats
(d) All the above are true

9. Comparitive advantage refers to
(a) A country's ability to produce some goods or service at the lowest possible cost as compared to other countries
(b) A country's ability to produce some goods or service at the lower opportunity cost than other countries
(c) Choosing a productive method which uses minimum of the abundant factor
(d) (a) and (b) above

10. Ricardo explained the law of comparative advantage on the basis of
(a) opportunity costs *(b) the law of diminishing returns*
(c) economies of scale *(d) the labour theory of value*

PEQ - JUNE - 2024

11. Theory of comparative cost advantage is _____ in nature and it _____ takes into account of factor price difference.
(a) Positive, does *(b) Normative, does not* *(c) Positive, does not* *(d) Normative, does*

12. Mercantilism means:
(a) Aggressive exports over imports to accumulate wealth *(b) Comparative advantage*
(c) Absolute cost advantage *(d) Factor endowment*
(1 mark)

PEQ - SEPTEMBER - 2024

13. What does the principle of Absolute Advantage refer to?
(a) The ability to produce a greater quantity of a goods using the same amount of resources as competitors
(b) The ability to produce a greater quantity of a goods using fewer resources
(c) The ability to produce a lesser quantity of a goods using the same amount of resources as competitors
(d) The ability to produce a greater variety of goods than competitors *(1 mark)*

PEQ – JANUARY - 2025

14. Factor Endowment Theory of trade is also known as _____.
(a) Baumol and Tobin theory *(b) Adam Smith, Absolute Cost Advantage theory*
(c) Heckscher-Ohlin theory *(d) Factor Price Equalisation theory* *(1 mark)*

15. The theory of Comparative Advantage in International Trade was presented by:
(a) Adam Smith *(b) David Ricardo*
(c) John Maynard Keynes *(d) Milton Friedman* *(1 mark)*

PEQ – MAY - 2025

16. In the Theory of Comparative Costs, which of the following statements are true?
    I. It is based on money cost which is more realistic
    II. It is Positive in Nature
    III. It is Normative in nature
    IV. Do not take into account the factor price differences
    (a) I and II are correct
    (b) III and IV are correct
    (c) II and IV are correct
    (d) I and IV are correct

17. The value of the product or service is enhanced as the number of individuals using it increases. What effect is this?
    (a) Veblen Effect
    (b) Bandwagon Effect
    (c) Income Effect
    (d) Snob Effect

PEQ – JANUARY - 2026

18. Which of the following is not a benefit of International Trade?
    (a) Provides greater stimulus to innovative services
    (b) Trade provides access to new markets and new materials
    (c) Underprivileged countries will be less vulnerable to exploitation by corporations operating globally
    (d) Improvement in the quality of output of goods and services for emerging economies

PEQ – SEPTEMBER - 2026

19. The Heckscher-Ohlin model predicts that a country will export goods that:
    (a) Use its scarce factors intensely
    (b) Use its abundant factors intensely
    (c) It cannot produce efficiently
    (d) Are cheap in the world market

ANSWERS / HINTS

I. Multiple Choice Type Questions

1	(b)	2	(b)	3	(d)	4	(c)	5	(b)
6	(b)	7	(b)	8	(d)	9	(b)	10	(d)
11	(b)	12	(a)	13	(b)	14	(c)	15	(b)
16	(b)	17	(b)	18	(c)	19.	(b)		

For Your Notes

For Your Notes

For Your Notes

For Your Notes

UNIT – II

THE INSTRUMENTS OF TRADE POLICY

Topic References:

2.1 Introduction
2.2 Tariff Based Measures
2.3 Non – Tariff Based Measures
2.4 Export Related Measures

2.1 Introduction

Before we go into the subject matter of this unit, we shall take a quick look into a few recent developments in the international trade arena.

- On January 27, 2017 The European Commission, after a detailed investigation which confirmed dumping ,decided to impose anti-dumping measures on two steel products (stainless steel tube and pipe butt-welding fittings) originating in China and Taiwan.
- In April, 2017 India accuses that by deciding to test up to 50 per cent of India's shrimp consignments for antibiotic residues, European Union is using SPS (Sanitary and Phytosanitary) restrictions in the case of seafood, and also on fruits and vegetables) in an exaggerated way and that its specifications sometimes exceed the norms prescribed in the Codex Alimentarius standards of the FAO.
- In April, 2017, India decided to have a domestic purchase preference policy applicable for five years for PSU oil companies. Targets of local contents (LC) also are stipulated for certain oil and gas business activities.
- In view of the avian influenza threat, India has in recent years curbed the import of frozen chicken legs from the US. The WTO upheld the objections raised by the US to India's move that the curbs are beyond international norms and therefore amounted to non-tariff barrier.

The above vignettes are just a few of the multitudes of episodes that arise almost on a daily basis when countries engage in trade. A glance at similar newspaper reports makes it obvious that governments do not conform to free trade despite the potential efficiency and welfare outcomes it will promote; rather, they employ different devices for restricting the free flow of goods and services across their borders.

In unit 1, we have seen that there are clear efficiency benefits from trade in terms of economic growth, job-creation and welfare.

The persuasive academic arguments for open trade presuppose that fair competition, without distortions, is maintained between domestic and foreign producers. However, it is a fact that fair competition does not always exist and unobstructed international trade also brings in severe dislocation to many domestic firms and industries on account of difficult adjustment problems.

Therefore, individuals and organisations continue to pressurize policy makers and regulatory authorities to restrict imports or to artificially boost up the size of exports.

In this unit, we shall describe some of the most frequently used forms of interference with trade. Understanding the uses and implications of the common trade policy instruments will enable formulation of appropriate policy responses and more balanced dialogues on trade policy issues and international trade agreements.

The instruments of trade policy that countries typically use to restrict imports and/ or to encourage exports can be broadly classified into

- Price - related measures such as tariffs; and
- Non - Price measures or non-tariff measures (NTMs).

2.2 Tariff Based Measures

Tariffs, also known as customs duties, are basically taxes or duties imposed on goods and services which are imported or exported.

It is defined as a financial charge in the form of a tax, imposed at the border on goods going from one customs territory to another.

They are the most visible and universally used trade measures that determine market access for goods.

Import duties being pervasive than export duties, tariffs are often identified with import duties and in this unit, the term 'tariff' would refer to import duties.

Tariffs are aimed at altering the relative prices of goods and services imported, so as to contract the domestic demand and thus regulate the volume of their imports.

Tariffs leave the world market price of the goods unaffected; while raising their prices in the domestic market.

The main goals of tariffs are to raise revenue for the government, and more importantly to protect the domestic import-competing industries.

2.2.1 Forms of Import Tariffs

(i) Specific Tariff:

- A specific tariff is an import duty that assigns a fixed monetary tax per physical unit of the good imported.
- It is calculated on the basis of a unit of measure, such as weight, volume, etc., of the imported good.
  Thus, a specific tariff of ₹ 1000/- may be charged on each imported bicycle.
- The disadvantage of specific tariff as an instrument for protection of domestic producers is that its protective value varies inversely with the price of the import.
  For example: if the price of the imported cycle is ₹ 5,000/- then the rate of tariff is 20%; if due to inflation, the price of bicycle rises to ₹ 10,000/-, the specific tariff is only 10% of the value of the import.
- Since the calculation of these duties does not involve the value of merchandise, customs valuation is not applicable in this case.

(ii) Ad valorem tariff:

- An ad valorem tariff is levied as a constant percentage of the monetary value of one unit of the imported good.
  A 20% ad valorem tariff on any bicycle generates a ₹ 1,000/- payment on each imported bicycle priced at ₹ 5,000/- in the world market; and if the price rises to ₹ 10,000/-, it generates a payment of ₹ ₹ 2,000/-
- While ad valorem tariff preserves the protective value of tariff on home producer, it gives incentives to deliberately undervalue the good's price on invoices and bills of lading to reduce the tax burden.
  Ad valorem tariffs are widely used the world over.

VARIATIONS OF ABOVE TARIFF :

(a) ***Mixed Tariffs :*** Mixed tariffs are expressed either on the basis of the value of the imported goods (an ad valorem rate) or on the basis of a unit of measure of the imported goods (a specific duty) depending on which generates the most income (or least income at times) for the nation.
For example, duty on cotton at 5 % ad valorem or ₹ 3,000 per tonne, whichever is higher.

(b) ***Compound Tariff or Compound Duty:*** It is a combination of an ad valorem and a specific tariff. That is, the tariff is calculated on the basis of both the value of the imported goods (an ad valorem duty) and a unit of measure of the imported goods (a specific duty). It is generally calculated by adding up a specific duty to an ad valorem duty.
For example, duty on cheese at 5 % ad valorem plus ₹ 100 per kilogram.

(c) ***Technical or Other Tariff:*** These are calculated on the basis of the specific contents of the imported goods i.e. duties are payable by its components or related items.
For example, ₹ 3000/- on each solar panel plus ₹ 50 per kg on the battery.

(d) ***Tariff Rate Quotas:*** Tariff rate quotas (TRQs) combine two policy instruments: ***quotas and tariffs***. Imports entering under the specified quota portion are usually subject to a lower (sometimes zero) tariff rate. Imports above the quantitative threshold of the quota face a much higher tariff.

(e) ***Most-Favored Nation Tariffs:*** MFN tariffs are what countries promise to impose on imports ***from other members of the WTO***, unless the country is part of a preferential trade agreement (such as a free trade area or customs union).
This means that, in practice, MFN rates are the highest (most restrictive) that WTO members charge one another. Some countries impose higher tariffs on countries that are not part of the WTO.

(f) ***Variable Tariff:*** A duty typically fixed to bring the price of an imported commodity up to the domestic support price for the commodity.

(g) ***Preferential Tariff:*** Nearly all countries are part of at least one preferential trade agreement, under which they promise to give another country's products ***lower tariffs than their MFN rate***. These agreements are reciprocal. A lower tariff is charged from goods imported from a country which is given preferential treatment.
For Example, preferential duties in the EU region under which a good coming into one EU country to another is charged zero tariffs.
North American Free Trade Agreement (NAFTA) among Canada, Mexico and the USA where the preferential tariff rate is zero on essentially all products.

Countries, especially the affluent ones also grant 'unilateral preferential treatment' to select list of products from specified developing countries. The Generalized System of Preferences (GSP) is one such system which is currently prevailing.

(h) ***Bound Tariff:*** A bound tariff is a tariff which a ***WTO member binds itself*** with a legal commitment not to raise it above a certain level. By binding a tariff, often during negotiations, the members agree to limit their right to set tariff levels beyond a certain level. ***The bound rates are specific to individual products*** and represent the maximum level of import duty that can be levied on a product imported by that member. A member is always free to impose a tariff that is lower than the bound level. Once bound, a tariff rate becomes permanent and a member can only increase its level after negotiating with its trading partners and compensating them for possible losses of trade. A bound tariff ensures transparency and predictability.

(i) ***Applied Tariffs:*** An 'applied tariff' is the duty that is actually charged on imports on a most-favoured nation (MFN) basis. A WTO member can have an applied tariff for a product that differs from the bound tariff for that product as long as the applied level is not higher than the bound level.

(j) ***Escalated Tariff:*** This refers to the system wherein the nominal tariff rates on imports of manufactured goods are higher than the nominal tariff rates on intermediate inputs and raw materials, i.e. the tariff on a product increases as that product moves through the value- added chain.
For example, a 4% tariff on iron ore or iron ingots and 12% tariff on steel pipes.

This type of tariff is discriminatory as it protects manufacturing industries in importing countries and dampens the attempts of developing manufacturing industries of exporting countries. This has special relevance to trade between developed countries and developing countries. Developing countries are thus forced to continue to be suppliers of raw materials without much value addition.

(k) ***Prohibitive tariff:*** It is one that is set so high that ***no imports will enter***.

(l) ***Import subsidies:*** In some countries, import subsidies also exist. An import subsidy is simply a payment per unit or as a percent of value for the importation of a good ***(i.e., a negative import tariff).***

(m) ***Tariffs as Response to Trade Distortions:*** Sometimes countries engage in 'unfair' foreign- trade practices which are trade distorting in nature and adverse to the interests of the domestic firms. The affected importing countries, upon confirmation of the distortion, respond quickly by measures in the form of tariff responses to offset the distortion. These policies are often referred to as "trigger-price" mechanisms. The following sections relate to such tariff responses to distortions related to foreign dumping and export subsidies

(i) ***Anti-dumping Duties:***
Dumping occurs when manufacturers sell goods in a foreign country below the sales prices in their domestic market or below their full average cost of the product.

Dumping may be persistent, seasonal, or cyclical.

Dumping may also be resorted to as a predatory pricing practice to drive out established domestic producers from the market and to establish monopoly position.

Dumping is an international price discrimination favouring buyers of exports, but in fact, the exporters deliberately forego money in order to harm the domestic producers of the importing country.

This is unfair and constitutes a threat to domestic producers and therefore when dumping is found, anti-dumping measures which are tariffs to offset the effects of dumping may be initiated as a safeguard instrument by imposition of additional import duties so as to offset the foreign firm's unfair price advantage.

This is justified only if the domestic industry is seriously injured by import competition, and protection is in the national interest (that is, the associated costs to consumers would be less than the benefits that would accrue to producers).

For example, In January 2017, India imposed anti- dumping duties on colour-coated or pre-painted flat steel products imported into the country from China and European nations for a period not exceeding six months and for jute and jute products from Bangladesh and Nepal.

(ii) Countervailing Duties:

Countervailing duties are tariffs that aim to offset the artificially low prices charged by exporters who enjoy export subsidies and tax concessions offered by the governments in their home country.

If a foreign country does not have a comparative advantage in a particular good and a government subsidy allows the foreign firm to be an exporter of the product, then the subsidy generates a distortion from the free-trade allocation of resources.

In such cases, CVD is charged in an importing country to negate the advantage that exporters get from subsidies to ensure fair and market oriented pricing of imported products and thereby protecting domestic industries and firms.

For example, in 2016, in order to protect its domestic industry, India imposed 12.5% countervailing duty on Gold jewellery imports from ASEAN.

2.2.2 Effects of Tariffs

A tariff levied on an imported product affects both the country exporting a product and the country importing that product.

(i) Tariff barriers create obstacles to trade, decrease the volume of imports and exports and therefore of international trade. The prospect of market access of the exporting country is worsened when an importing country imposes a tariff.

(ii) By making imported goods more expensive, tariffs discourage domestic consumers from consuming imported foreign goods. Domestic consumers suffer a loss in consumer surplus because they must now pay a higher price for the good and also because compared to free trade quantity, they now consume lesser quantity of the good.

(iii) Tariffs encourage consumption and production of the domestically produced import substitutes and thus protect domestic industries.

(iv) Producers in the importing country experience an increase in well-being as a result of imposition of tariff. The price increase of their product in the domestic market increases producer surplus in the industry. They can also charge higher prices than would be possible in the case of free trade because foreign competition has reduced.

(v) The price increase also induces an increase in the output of the existing firms and possibly addition of new firms due to entry into the industry to take advantage of the new high profits and consequently an increase in employment in the industry.

(vi) Tariffs create trade distortions by disregarding comparative advantage and prevent countries from enjoying gains from trade arising from comparative advantage. Thus, tariffs discourage efficient production in the rest of the world and encourage inefficient production in the home country.

(vii) Tariffs increase government revenues of the importing country by the value of the total tariff it charges.

Currently, trade policy is focusing increasingly on not so easily observable forms of trade barriers usually called nontariff measures (NTMs). NTMs are thought to have important restrictive and distortionary effects on international trade.

2.3 Non-Tariff Measures (NTM'S)

While tariffs constitute the visible barriers to trade and have the effect of increasing the prices of imported merchandise, the non-tariff measures which have come into greater prominence than the conventional tariff barriers, constitute the hidden or 'invisible' measures that interfere with free trade.

Non-tariff measures (NTMs) are policy measures, other than ordinary customs tariffs, that can potentially have an economic effect on international trade in goods, changing quantities traded, or prices or both (UNCTAD, 2010).

Non-tariff measures comprise all types of measures which alter the conditions of international trade, including policies and regulations that restrict trade and those that facilitate it.

NTMs are not the same as non-tariff barriers (NTBs).

Compared to non-tariff barriers which are simply discriminatory non-tariff measures imposed by governments to favour domestic over foreign suppliers, non-tariff measures encompass a broader set of measures.

According to WTO agreements, the use of NTMs is allowed under certain circumstances. Examples of this include the Technical Barriers to Trade (TBT) Agreement and the Sanitary and Phytosanitary Measures (SPS) Agreement, both negotiated during the Uruguay Round.

However, NTMs are sometimes used as a means to circumvent free-trade rules and favour domestic industries at the expense of foreign competition. In this case they are called non- tariff measures (NTMs).

It is very difficult, and sometimes impossible, to distinguish legitimate NTMs from protectionist NTMs, especially as the same measure may be used for several reasons.

2.3.1 Forms of Non- Tariff Measures:

(i) Technical Measures:

(a) Sanitary and Phytosanitary (SPS) Measures: SPS measures are applied to protect human, animal or plant life from risks arising from additives, pests, contaminants, toxins or disease-causing organisms and to protect biodiversity.

These include ban or prohibition of import of certain goods, all measures governing quality and hygienic requirements, production processes, and associated compliance assessments.

For example, prohibition of import of poultry from countries affected by avian flu, meat and poultry processing standards to reduce pathogens, residue limits for pesticides in foods etc.

(b) Technical Barriers To Trade (TBT): Technical Barriers to Trade (TBT) which cover both food and non-food traded products refer to mandatory 'Standards and Technical Regulations' that define the specific characteristics that a product should have, such as its size, shape, design, labelling, marking, packaging, functionality or performance and production methods, excluding measures covered by the SPS Agreement.

The specific procedures used to check whether a product is really conforming to these requirements (conformity assessment procedures e.g. testing, inspection and certification) are also covered in TBT.

This involves compulsory quality, quantity and price control of goods before shipment from the exporting country.

Just as SPS, TBT measures are standards-based measures that countries use to protect their consumers and preserve natural resources, but these can also be used effectively as obstacles to imports or to discriminate against imports and protect domestic products.

Altering products and production processes to comply with the diverse requirements in export markets may be either impossible for the exporting country or would obviously raise costs hurting the competitiveness of the exporting country.

For example, food laws, quality standards, industrial standards, organic certification, eco-labeling, marketing and label requirements etc.

(ii) Non-Technical Measures:

(a) *Import Quotas:* An import quota is a direct restriction which specifies that only a certain physical amount of the good will be allowed into the country during a given time period, usually one year.

Import quotas are typically set below the free trade level of imports and are usually enforced by issuing licenses. This is referred to as a binding quota; a non-binding quota is a quota that is set at or above the free trade level of imports, thus having little effect on trade.

Import quotas are mainly of two types: *absolute quotas and tariff-rate quotas.*

Absolute quotas or quotas of a permanent nature limit the quantity of imports to a specified level during a specified period of time and the imports can take place any time of the year. No condition is attached to the country of origin of the product.

For Example, 1000 tonnes of fish import, which can take place any time of the year from any country.
When country allocation is specified, a fixed volume or value of the product must originate in one or more countries.

For Example, A quota of 1000 tonnes of fish that can be imported any time of the year, but where 750 tonnes must originate in country A and 250 tonnes in country B.

In addition, there are seasonal quotas and temporary quotas.

With a quota, the government, of course, receives no revenue. The profits received by the holders of such import licenses are known as 'quota rents'.

While tariffs directly interfere with prices that can be charged for an imported good in the domestic market, import quota interferes with the market prices indirectly.

Obviously, an import quota at all times raises the domestic price of the imported good. The license holders are able to buy imports and resell them at a higher price in the domestic market and they will be able to earn a 'rent' on their operations over and above the profit they would have made in a free market.

The welfare effects of quotas are similar to that of tariffs. If a quota is set below free trade level, the amount of imports will be reduced. A reduction in imports will lower the supply of the good in the domestic market and raise the domestic price.

Consumers of the product in the importing country will be worse-off because the increase in the domestic price of both imported goods and the domestic substitutes reduces consumer surplus in the market.

Producers in the importing country are better-off as a result of the quota. The increase in the price of their product increases producer surplus in the industry. The price increase also induces an increase in output of existing firms (and perhaps the addition of new firms), an increase in employment, and an increase in profit.

(b) ***Price Control Measures:*** Such measures (including additional taxes and charges) are steps taken to control or influence the prices of imported goods in order to support the domestic price of certain products when the import prices of these goods are lower.

These are also known as 'para-tariff' measures and include measures, other than tariff measures, that increase the cost of imports in a similar manner, i.e. by a fixed percentage or by a fixed amount.

For example, a minimum import price established for sulphur.

(c) ***Non-automatic Licensing and Prohibitions :*** These measures are normally aimed at limiting the quantity of goods that can be imported, regardless of whether they originate from different sources or from one particular supplier.

These measures may take the form of non-automatic licensing, or through complete prohibitions.

For example, Textiles may be allowed only on a discretionary license by the importing country. India prohibits import/export of arms and related material from/to Iraq.
Also, India prohibits many items (mostly of animal origin) falling under 60 EXIM codes.

(d) ***Financial Measures:*** The objective of financial measures is to increase import costs by regulating the access to and cost of foreign exchange for imports and to define the terms of payment.

It includes measures such as advance payment requirements and foreign exchange controls denying the use of foreign exchange for certain types of imports or for goods imported from certain countries.

For example, an importer may be required to pay a certain percentage of the value of goods imported three months before the arrival of goods or foreign exchange may not be permitted for import of newsprint.

(e) ***Measures Affecting Competition:*** These measures are aimed at granting exclusive or special preferences or privileges to one or a few limited group of economic operators. It may include government imposed special import channels or enterprises, and compulsory use of national services.

For example, a statutory marketing board may be granted exclusive rights to import wheat: or a canalizing agency (like State Trading Corporation) may be given monopoly right to distribute palm oil.

(f) ***Government Procurement Policies:*** Government procurement policies may interfere with trade if they involve mandates that the whole of a specified percentage of government purchases should be from domestic firms rather than foreign firms, despite higher prices than similar foreign suppliers.

For example, in accepting public tenders, a government may give preference to the local tenders rather than foreign tenders.

(g) ***Trade-Related Investment Measures:*** These measures include rules on local content requirements that mandate a specified fraction of a final good should be produced domestically.

For example, requirement to use certain minimum levels of locally made components, (25% of components of automobiles to be sourced domestically) restricting the level of imported components, or limiting the purchase or use of imported products to an amount related to the quantity or value of local products that it exports. (A firm may import only up to 75% of its export earnings of the previous year)

(h) ***Distribution Restrictions:*** Distribution restrictions are limitations imposed on the distribution of goods in the importing country involving additional license or certification requirements. These may relate to geographical restrictions or restrictions as to the type of agents who may resell.

For example: a restriction that imported fruits may be sold only through outlets having refrigeration facilities

(i) ***Restriction on Post-sales Services:*** Producers may be restricted from providing after-sales services for exported goods in the importing country. Such services may be reserved to local service companies of the importing country.

(j) ***Administrative Procedures:*** Another potential obstruction to free trade is the costly and time consuming administrative procedures which are mandatory for import of foreign goods. These will increase transaction costs and discourage imports. The domestic import- competing industries gain by such non-tariff measures.

For example, specifying particular procedures and formalities, requiring licenses, administrative delay, red-tape and corruption in customs clearing frustrating the potential importers, procedural obstacles linked to prove compliance etc.

(k) ***Rules of origin:*** Rules of origin are the criteria needed by governments of importing countries to determine the national source of a product. Their importance is derived from the fact that duties and restrictions in several cases depend upon the source of imports. Important procedural obstacles occur in the home countries for making available certifications regarding origin of goods, especially when different components of the product originate in different countries.

(l) ***Safeguard Measures:*** These are initiated by countries to restrict imports of a product temporarily if its domestic industry is injured or threatened with serious injury caused by a surge in imports.

(m) ***Embargos:*** An embargo is a total ban imposed by government on import or export of some or all commodities to particular country or regions for a specified or indefinite period. This may be done due to political reasons or for other reasons such as health, religious sentiments. This is the most extreme form of trade barrier

2.4 Export Related Measures

(i) ***Ban on exports:*** Export-related measures refer to all measures applied by the government of the exporting country including both technical and non-technical measures. By reducing international supply, export restrictions have been shown to increase international prices.

For example, during periods of shortages, export of agricultural products such as onion, wheat etc. may be prohibited to make them available for domestic consumption.

(ii) ***Export Taxes:*** An export tax is a tax collected on exported goods and may be either specific or ad-valorem. The effect of an export tax is to raise the price of the good and to decrease exports. Since an export tax reduces exports and increases domestic supply, it also reduces domestic prices and leads to higher domestic consumption.

(iii) ***Export Subsidies and Incentives:*** Tariffs on imports hurt exports and therefore countries have developed compensatory measures of different types for exporters like export subsidies, duty drawback, duty-free access to imported intermediates etc.

Governments or government bodies also usually provide financial contribution to domestic producers in the form of grants, loans, equity infusions etc. or give some form of income or price support. If such policies on the part of governments are directed at encouraging domestic industries to sell specified products or services abroad, they can be considered as trade policy tools.

(iv) ***Voluntary Export Restraints:*** VER's refer to a type of informal quota administered by an exporting country voluntarily restraining the quantity of goods that can be exported out of that country during a specified period of time.

Such restraints originate primarily from political considerations and are imposed based on negotiations of the importer with the exporter. The inducement for the exporter to agree to a VER is mostly to appease the importing country and to avoid the effects of possible retaliatory trade restraints that may be imposed by the importer.

VERs cause, as do tariffs and quotas, domestic prices to rise and cause loss of domestic consumer surplus.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. A specific tariff is
   (a) *a tax on a set of specified imported good*
   (b) *an import tax that is common to all goods imported during a given period*
   (c) *a specified fraction of the economic value of an imported good*
   (d) *a tax on imports defined as an amount of currency per unit of the good*

2. A tariff on imports is beneficial to domestic producers of the imported good because
   (a) *they get a part of the tariff revenue*
   (b) *it raises the price for which they can sell their product on the domestic market*
   (c) *it determines the quantity that can be imported to the country*
   (d) *it reduces their producer surplus, making them more efficient*

3. A tax applied as a percentage of the value of an imported good is known as
   (a) *preferential tariff* (b) *ad valorem tariff*
   (c) *specific tariff* (d) *mixed or compound tariff*

4. Escalated tariff refers to
   (a) *nominal tariff rates on raw materials which are greater than tariffs on manufactured products*
   (b) *nominal tariff rates on manufactured products which are greater than tariffs on raw materials*
   (c) *a tariff which is escalated to prohibit imports of a particular good to protect domestic industries*
   (d) *none of the above*

5. Voluntary export restraints involve:
   (a) *an importing country voluntarily restraining the quantity of goods that can be exported into the country during a specified period of time*
   (b) *domestic firms agreeing to limit the quantity foreign products sold in their domestic markets*
   (c) *an exporting country voluntarily restraining the quantity of goods that can be exported out of a country during a specified period of time*
   (d) *quantitative restrictions imposed by the importing country's government.*

6. Anti-dumping duties are
   (a) *additional import duties so as to offset the effects of exporting firm's unfair charging of prices in the foreign market which are lower than production costs*
   (b) *additional import duties so as to offset the effects of exporting firm's increased competitiveness due to subsidies by government*
   (c) *additional import duties so as to offset the effects of exporting firm's unfair charging of lower prices in the foreign market*
   (d) *(b) and (c) above*

7. A countervailing duty is
   (a) *a tariff that aim to offset artificially low prices charged by exporters who enjoy export subsidies and tax concessions in their home country*
   (b) *charged by importing countries to ensure fair and market oriented pricing of imported products*
   (c) *charged by importing countries to protect domestic industries and firms from unfair price advantage arising from subsidies*
   (d) *All the above*

8. Which of the following is an outcome of tariff?
   (a) *create obstacles to trade and increase the volume of imports and exports*
   (b) *domestic consumers enjoy consumer surplus because consumers must now pay only a lower price for the good*

(c) *discourage domestic consumers from consuming imported foreign goods and encourage consumption of domestically produced import substitutes*
(d) *increase government revenues of the importing country by more than value of the total tariff it charges*

9. SPS measures and TBTs are
(a) *permissible under WTO to protect the interests of countries*
(b) *may result in loss of competitive advantage of developing countries*
(c) *increases the costs of compliance to the exporting countries*
(d) *All the above*

10. Which of the following is not a non-tariff barrier.
(a) *Complex documentation requirements*
(b) *Import quotas on specific goods*
(c) *Countervailing duties charged by importing country*
(d) *Pre shipment product inspection and certification requirements*

11. Under tariff rate quota
(a) *countries promise to impose tariffs on imports from members other than those who are part of a preferential trade agreement*
(b) *a country permits an import of limited quantities at low rates of duty but subjects an excess amount to a much higher rate*
(c) *lower tariff is charged from goods imported from a country which is given preferential treatment*
(d) *none of the above*

12. Non -tariff barriers (NTBs) include all of the following except:
(a) *import quotas* (b) *tariffs*
(c) *export subsidies* (d) *technical standards of products*

13. Which of the following is correct?
(a) *49% FDI is allowed in defence production*
(b) *49% FDI is allowed in private sector banking*
(c) *74% FDI is now allowed in multibrand retails*
(d) *100% FDI is allowed in insurance*

PEQ - JUNE - 2024

14. A total ban imposed by government on imports and exports of some or all commodities to particular country or regions for a specified or indefinite period is termed as:
(a) *Embargo* (b) *Safeguard measure*
(c) *Distribution restrictions* (d) *Restrictive measures* *(1 mark)*

15. Which of the following is levied as a fixed/constant percentage on money value of an imported or exported good?
(a) *Ad valorem tariff* (b) *Compound tariff*
(c) *Specific tariff* (d) *Bound tariff* *(1 mark)*

16. Which of the following is not negative outcome of tariffs?
(a) *Decreases revenue to government*
(b) *Producers of importing country increase well-being*
(c) *Domestic consumers suffer a loss in consumer surplus*
(d) *By ignoring comparative advantage, tariffs discourage effective production in rest of world*
(1 mark)

PEQ - SEPTEMBER - 2024

17. A tariff which a WTO member binds itself with a legal commitment not to raise tariff rate above a certain level is known as:
(a) Bound tariff *(b) Applied tariff*
(c) Specific tariff *(d) Most-favoured nation tariffs* *(1 mark)*

18. Which of the following is covered under Technical Barriers to Trade (TBT)?
(a) Only non-food traded products. *(b) Only food products.*
(c) Only technical standard products. *(d) Both food and non-food traded products.*
(1 mark)

PEQ - JANUARY - 2025

19. Which tariff is calculated on the basis of specific contents of the imported goods (duties are payable by its components or related items)?
(a) Compound tariff *(b) Mixed tariff*
(c) Ad valorem tariff *(d) Technical tariff* *(1 mark)*

20. Which tariff is expressed either on the basis of the value of the imported goods or on the basis of a unit of measure of the imported goods depending on which generates the most income (or least income at times) for the country?
(a) Ad valorem tariff *(b) Specific tariff*
(c) Mixed tariff *(d) Compound tariff* *(1 mark)*

21. The system wherein the nominal tariff rates on imports of manufactured goods are higher than the nominal tariff rates on intermediate inputs and raw materials is known as .
(a) Applied tariff *(b) Escalated tariff*
(c) Bound tariff *(d) Preferential tariff* *(1 mark)*

22. Which of the following is a measure to protect human, animal or plant life from risks arising out of additives, pests, toxins, etc. and to protect the biodiversity?
(a) Prohibited tariff *(b) Sanitary and phytosanitary measures*
(c) Technical barriers to trade *(d) Anti-dumping duties* *(1 mark)*

23. A total ban imposed by the Government on imports or exports of some or all commodities to a particular country or regions for a specified or indefinite period is known as .
(a) Prohibitive tariff *(b) Anti-dumping duties*
(c) Embargo *(d) Rules of origin* *(1 mark)*

PEQ - MAY - 2025

24. What is the Ad valorem tariff?
(a) The fixed amount of money per physical unit or weight of commodity imported or exported.
(b) A fixed time period for tariff application per calendar year.
(c) The duty levied as a fixed percentage of the value of the traded commodity.
(d) A flat rate imposed regardless of the product's value.

25. A tariff that is set so high that no imports can enter is known as
(a) Prohibitive Tariff
(b) Bound Tariff
(c) Escalated Tariff
(d) Variable Tariff

PEQ – JANUARY - 2026

26. Which of the following is a combination of two policy instruments quotas and tariffs?
    (a) *Tariff rate quotas*
    (b) Compound tariffs
    (c) Mixed tariffs
    (d) Applied tariffs

27. Which of the following represents a technical measure within Non-Tariff Measures (NTMs)?
    (a) Custom formalities
    (b) Shipping requirements
    (c) Production Processes
    (d) Trade-related finance and Investment measures

28. What is the primary purpose of countervailing duties charged in an importing country?
    (a) To counteract low quality of goods in importing countries
    (b) To neutralize artificially low prices charged by exporter resulting from export subsidies in their home country
    (c) To prevent tariff evasion by importers
    (d) To address variations in the quality of goods in importing country

29. State policy aimed to protect domestic producers against foreign competition through the use of tariff, quotes and non-tariff policy instruments is called:
    (a) Protectionism
    (b) Distortions
    (c) Safeguard
    (d) Blanket cover

PEQ – SEPTEMBER - 2026

30. A specific tariff is different from an ad valorem tariff because it is:
    (a) A percentage of trade value
    (b) A fixed amount per unit of imported goods
    (c) Applied only to exports
    (d) Used only under GATT

31. Non-tariff measures (NTMs) include all except:
    (a) Import quotas
    (b) Export subsidies
    (c) Sanitary and phytosanitary (SPS) measures
    (d) Floating exchange rates

ANSWERS / HINTS

I. Multiple Choice Type Questions

1	(d)	2	(b)	3	(b)	4	(b)	5	(c)	6	(d)
7	(d)	8	(c)	9	(d)	10	(c)	11	(b)	12	(b)
13	(b)	14	(a)	15	(a)	16	(b)	17	(a)	18	(d)
19	(d)	20	(c)	21	(b)	22	(b)	23	(c)	24	(c)
25	(a)	26	(a)	27	(c)	28	(b)	29	(a)	30	(b)
31	(d)										

II. Application Oriented Questions

For Your Notes

For Your Notes

For Your Notes

For Your Notes

UNIT – III
TRADE NEGOTIATION

Topic References:

3.1 Introduction
3.2 The Taxonomy of Regional Trade Agreements
3.3 The General Agreement on Tariff and Trade
3.4 The Uruguay Round and the Establishment of WTO
3.5 The World Trade Organisation
3.6 The Doha Round
3.7 G 20 Economies: Facilitating Trade

3.1 Introduction

The recent years have seen intense bilateral and multilateral negotiations among different nations in the international arena. India, for example, has already become part of 19 such concluded agreements and is currently negotiating more than two dozens of such proposals.

Events such as Britain planning an exit from the European Union, the US deliberations on the impact of NAFTA, and many other unpredictable developments in the trade front make trade negotiations a highly relevant topic in Economics.

International trade negotiations, especially the ones aimed at formulation of international trade rules, are complex interactive processes engaged in by countries having competing objectives.

Trade negotiations are not just face to face discussions; rather they are multilevel or network games and involve intricate and time consuming processes.

They usually involve many parties who have conflicting interests and objectives.

It is not national governments alone who are the stakeholders in a trade negotiation. Many interest groups, lobbying groups, pressure groups and Non-Governmental Organizations (NGO) exert their influence on the process.

For example, in trade negotiations, when one of the parties seems to be bargaining for market access through reduction in tariffs, the other (s) may be clamouring on the issue of possible grant of protection to domestic industries.

3.2 Taxonomy of Regional Trade Agreements (RTA'S)

Regional Trade Agreements (RTAs) are defined as groupings of countries, (not necessarily belonging to the same geographical region) which are formed with the objective of reducing barriers to trade between member countries.
Trade negotiations result in different types of agreements namely;

1. ***Unilateral Trade Agreement:*** These are agreements under which an importing country offers trade incentives in order to encourage the exporting country to engage in international economic activities that will improve the exporting country's economy.
E.g. Generalized System of Preferences.

2. ***Bilateral Trade Agreement:*** These are agreements which set rules of trade between two countries, two blocs or a bloc and a country. These may be limited to certain goods and services or certain types of market entry barriers.
E.g. South Africa Free Trade Agreement, ASEAN – India Free Trade Area

3. ***Reginal Preferential Trade Agreement:*** RPTA among a group of countries reduce trade barriers on a reciprocal and preferential basis for only the members of the group. *E.g. Global System of Trade Preferences among Developing Countries (GSTP)*

4. ***Trading BLOC:*** Trading Bloc has a group of countries that have a free trade agreement between themselves and may apply a common external tariff to other countries. *E.g. Arab League (AL), European Free Trade Association (EFTA)*

5. ***Free Trade Area:*** Free-trade area is a group of countries that eliminate all tariff barriers on trade with each other and retains independence in determining their tariffs with nonmembers. *E.g. NAFTA*

6. ***Customs Union:*** A customs union is a group of countries that eliminate all tariffs on trade among themselves but maintain a common external tariff on trade with countries outside the union (thus technically violating MFN). *E.g. EC, MERCOSUR.*

7. ***Common Market:*** A Common Market deepens a customs union by providing for the free flow of factors of production (labor and capital) in addition to the free flow of outputs. The member countries attempt to harmonize some institutional arrangements and commercial and financial laws and regulations among themselves. There are also common barriers against non-members. *E.g. EU, ASEAN*

8. ***Economic and Monetary Union:*** In this, members share a common currency and macroeconomic policies. *E.g. EU countries adopts a single currency.*
The political institutions that facilitate trade negotiations, and support international trade cooperation by providing the rules of the game have been the former General Agreements on Tariffs and Trade (GATT) and the World Trade Organization (WTO).

3.3 The General Agreement on Tariffs and Trade (GATT)

The General Agreement on Tariffs and Trade (GATT) covers international trade in goods. The workings of the GATT agreement are the responsibility of the Council for Trade in Goods (Goods Council) which is made up of representatives from all WTO member countries.

The Goods Council has 10 committees dealing with specific subjects (such as agriculture, market access, subsidies, anti-dumping measures, and so on). Again, these committees consist of all member countries.

Also reporting to the Goods Council are a working party on state trading enterprises, and the Information Technology Agreement (ITA) Committee.

The GATT lost its relevance by 1980's because

- it was obsolete to the fast evolving contemporary complex world trade scenario characterised by emerging globalization
- international investments had expanded substantially
- intellectual property rights and trade in services were not covered by GATT
- world merchandise trade increased by leaps and bounds and was beyond its scope
- the ambiguities in the multilateral system could be heavily exploited
- efforts at liberalizing agricultural trade were not successful
- there were inadequacies in institutional structure and dispute settlement system
- it was not a treaty and therefore terms of GATT were binding only insofar as they are not incoherent with a nation's domestic rules.

3.4 The Uruguay Round and Establishment of WTO

The need for a formal international organization which is more powerful and comprehensive was felt by many countries by late 1980's.

The Uruguay Round brought about the biggest reform of the world's trading system.

Members established 15 groups to work on limiting restrictions in the areas of tariffs, non- tariff barriers, tropical products, natural resource products, textiles and clothing, agriculture, safeguards against sudden 'surges' in imports, subsidies and countervailing duties, trade related intellectual property restrictions, trade related investment restrictions, services and four other areas dealing with GATT itself, such as, the GATT system, dispute settlement procedures and implementation of the NTB Codes of the Tokyo Round, especially on anti-dumping.

The Round started in Punta del Este in Uruguay in September 1986 and was scheduled to be completed by December 1990. However, due to many differences and especially due to heated controversies over agriculture, no consensus was arrived at.

Finally, in December 1993, the Uruguay Round, the eighth and the most ambitious and largest ever round of multilateral trade negotiations in which 123 countries participated, was completed after seven years of elaborate negotiations.

The agreement was signed by most countries on April 15, 1994, and took effect on July 1, 1995.

It also marked the birth of the World Trade Organization (WTO) which is a single institutional framework encompassing the GATT, as modified by the Uruguay Round.

3.5 The World Trade Organization (WTO)

The most important outcome of the Uruguay Round agreement was the replacement of the General Agreement on Tariffs and Trade (GATT) secretariat with the World Trade Organization (WTO) in Geneva with authority not only in trade in industrial products but also in agricultural products and services.

The bulk of the WTO's present operations come from the 1986-94 negotiations called the Uruguay Round and earlier negotiations under the General Agreement on Tariffs and Trade (GATT).

Despite the fact that the WTO replaced GATT as an international organization, the General Agreement still exists as the WTO's umbrella treaty for trade in goods, updated as a result of the Uruguay Round negotiations.

The objectives of the WTO Agreements as acknowledged in the preamble of the Agreement creating the World Trade Organization, include

- Raising standards of living,
- Ensuring full employment,
- A large and steadily growing volume of real income and effective demand, and
- Expanding the production of and trade in goods and services

The principal objective of the WTO is to facilitate the flow of international trade smoothly, freely, fairly and predictably.

The WTO does its functions by acting as a forum for trade negotiations among member governments, administering trade agreements, reviewing national trade policies, assisting developing countries in trade policy issues, through technical assistance and training programmes and cooperating with other international organizations.

3.5.1 The Structure of the WTO

The WTO activities are supported by a Secretariat located in Geneva, headed by a Director General.

It has a three-tier system of decision making.

1. The WTO's top level decision-making body is the Ministerial Conference which can take decisions on all matters under any of the multilateral trade agreements. The Ministerial Conference meets at least once every two years.
2. The next level is the General Council which meets several times a year at the Geneva headquarters. The General Council also meets as the Trade Policy Review Body and the Dispute Settlement Body.
3. At the next level, the Goods Council, Services Council and Intellectual Property (TRIPS) Council report to the General Council. These councils are responsible for overseeing the implementation of the WTO agreements in their respective areas of specialisation.

   Numerous specialized committees, working groups and working parties deal with the individual agreements and other areas such as the environment, development, membership applications and regional trade agreements.

   The WTO accounting for about 95% of world trade currently has 164 members, of which 117 are developing countries or separate customs territories accounting for about 95% of world trade. Around 25 others are negotiating membership. The WTO's agreements have been ratified in all members' parliaments.

3.5.2 The Guiding Principles of World Trade Organization (WTO)

Following are the major guiding principles:

1. ***Trade without discrimination:*** Most-favoured-nation (MFN): Originally formulated as Article 1 of GATT, this principle states that any advantage, favour, privilege or immunity granted by any contracting party to any product originating in or destined for any other country shall be extended immediately and unconditionally to the like product originating or destined for the territories of all other contracting parties.

   Under the WTO agreements, countries cannot normally discriminate between their trading partners.
   If a country lowers a trade barrier or opens up a market, it has to do so for the same goods or services from all other WTO members.

   Under strict conditions, various permitted exceptions are allowed.

   For example, countries may enter into free trade agreements and trading may be done within the group discriminating against goods from outside; a country can raise barriers against products that are considered to be traded unfairly from specific countries; or they may give special market access to developing countries.

2. ***The National Treatment Principle (NTP):*** The NTP is complementary to the MFN principle. GATT Article III requires that with respect to internal taxes, internal laws, etc. applied to imports, treatment not less favourable than that which is accorded to like domestic products must be accorded to all other members.
   A country should not discriminate between its own and foreign products, services or nationals.
   For Example, once imported apples reach Indian market, they cannot be discriminated against and should be treated at par in respect of marketing opportunities, product visibility or any other aspect with locally produced apples.

3. ***Free trade:*** Lowering trade barriers for opening up markets is one of the most obvious means of encouraging trade. But by the 1980's, the negotiations had expanded to cover non- tariff barriers on goods, and to the new areas such as services and intellectual property.
Since these require adjustments, the WTO agreements permit countries to bring in changes gradually, through "progressive liberalization".

4. ***Predictability:*** Investments will be encouraged only if the business environment is stable and predictable. The foreign companies, investors and governments should be confident that the trade barriers will not be raised arbitrarily. This is achieved through 'binding' tariff rates, discouraging the use of quotas and other measures used to set limits on quantities of imports, establishing market-opening commitments and other measures to ensure transparency. A country can change its bindings, but only after negotiating with its trading partners, which could mean compensating them for loss of trade.

5. ***Promoting fair competition***

The WTO is sometimes described as a "free trade" institution, but that is not entirely accurate. The system does allow tariffs and, in limited circumstances, other forms of protection. More accurately, it is a system of rules dedicated to open, fair, and undistorted competition.

The rules on non-discrimination — MFN and national treatment — are designed to secure fair conditions of trade. So too are those on dumping (exporting at below cost to gain market share) and subsidies. The issues are complex, and the rules try to establish what is fair or unfair, and how governments can respond, in particular by charging additional import duties calculated to compensate for damage caused by unfair trade.

Many of the other WTO agreements aim to support fair competition: in agriculture, intellectual property, services, for example. The agreement on government procurement (a "plurilateral" agreement because it is signed by only a few WTO members) extends competition rules to purchases by thousands of government entities in many countries. And so on.

6. ***Encouraging development and economic reform***

The WTO system contributes to development. On the other hand, developing countries need flexibility in the time they take to implement the system's agreements. And the agreements themselves inherit the earlier provisions of GATT that allow for special assistance and trade concessions for developing countries.

Over three-quarters of WTO members are developing countries and countries in transition to market economies. During the seven and a half years of the Uruguay Round, over 60 of these countries implemented trade liberalization programmes autonomously. At the same time, developing countries and transition economies were much more active and influential in the Uruguay Round negotiations than in any previous round, and they are even more so in the current Doha Development Agenda.

At the end of the Uruguay Round, developing countries were prepared to take on most of the obligations that are required of developed countries. But the agreements did give them transition periods to adjust to the more unfamiliar and, perhaps, difficult WTO provisions — particularly so for the poorest, "least-developed" countries. A ministerial decision adopted at the end of the round says better-off countries should accelerate implementing market access commitments on goods exported by the least-developed countries, and it seeks increased technical assistance for them. More recently, developed countries have started to allow duty-free and quota-free imports for almost all products from least-developed countries.

On all of this, the WTO and its members are still going through a learning process. The current Doha Development Agenda includes developing countries' concerns about the difficulties they face in implementing the Uruguay Round agreements.

3.5.3 Overview of WTO Agreements

The WTO agreements cover goods, services and intellectual property and the permitted exceptions. These agreements are often called the WTO's trade rules, and the WTO is often described as "rules-based", a system based on rules. (The rules are actually agreements that the governments negotiated).

The WTO agreements are voluminous and multifaceted. The 'Legal Texts' consist of a list of about 60 agreements, annexes, decisions and understandings covering a wide range of activities. (The list of WTO agreements is given at the end of this unit).

Following are the important agreements under WTO.

1. Agreement on Agriculture aims at strengthening GATT disciplines and improving agricultural trade. It includes specific and binding commitments made by WTO Member governments in the three areas of market access, domestic support and export subsidies.
2. Agreement on the Application of Sanitary and Phytosanitary (SPS) Measures establishes multilateral frameworks for the planning, adoption and implementation of sanitary and phytosanitary measures to prevent such measures from being used for arbitrary or unjustifiable discrimination or for camouflaged restraint on international trade and to minimize their adverse effects on trade.
3. Agreement on Textiles and Clothing replaced the Multi-Fiber Arrangement (MFA) which was prevalent since 1974 which entailed import protection policies. ATC provides that textile trade should be deregulated by gradually integrating it into GATT disciplines over a 10-year transition period.
4. Agreement on Technical Barriers to Trade (TBT) aims to prevent standards and conformity assessment systems from becoming unnecessary trade barriers by securing their transparency and harmonization with international standards. Often excessive standards or misuse of standards in respect of manufactured goods, and safety/environment regulations act as trade barriers.
5. Agreement on Trade-Related Investment Measures (TRIMs) expands disciplines governing investment measures in relation to cross-border investments by stipulating that countries receiving foreign investments shall not impose investment measures such as requirements, conditions and restrictions inconsistent with the provisions of the principle of national treatment and general elimination of quantitative restrictions. For example: measures such as local content requirements and trade balancing requirements should not be applied on investing corporations.
6. Anti-Dumping Agreement seeks to tighten and codify disciplines for calculating dumping margins and conducting dumping investigations, etc. in order to prevent anti-dumping measures from being abused or misused to protect domestic industries
7. Customs Valuation Agreement specifies rules for more consistent and reliable customs valuation and aims to harmonize customs valuation systems on an international basis by eliminating arbitrary valuation systems.
8. Agreement on Pre-shipment Inspection (PSI) intends to secure transparency of pre-shipment inspection wherein a company designated by the importing country conducts inspection of the quality, volume, price, tariff classification, customs valuation, etc. of merchandise in the territory of the exporting country on behalf of the importing country's custom office and issues certificates. The agreement also provides for a mechanism for the solution of disputes between PSI agencies and exporters.
9. Agreement on Rules of Origin provides for the harmonization of rules of origin for application to all non-preferential commercial policy instruments. It also provides for dispute settlement procedures and creates the rules of origin committee.

10. Agreement on Import Licensing Procedures relates to simplification of administrative procedures and to ensure their fair operation so that import licensing procedures of different countries may not act as trade barriers.

11. Agreement on Subsidies and Countervailing Measures aims to clarify definitions of subsidies, strengthen disciplines by subsidy type and to strengthen and clarify procedures for adopting countervailing tariffs

12. Agreement on Safeguards clarify disciplines for requirements and procedures for imposing safeguards and related measures which are emergency measures to restrict imports in the event of a sudden surge in imports.

13. General Agreement on Trade in Services (GATS): This agreement provides the general obligations regarding trade in services, such as most- favored-nation treatment and transparency. In addition, it enumerates service sectors and stipulates that in the service sectors for which it has made commitments, a member country cannot maintain or introduce market access restriction measures and discriminatory measures that are severer than those that were committed during the negotiations.

14. Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS): This agreement stipulates most-favored-nation treatment and national treatment for intellectual properties, such as copyright, trademarks, geographical indications, industrial designs, patents, IC layout designs and undisclosed information. In addition, it requires member countries to maintain high levels of intellectual property protection and to administer a system of enforcement of such rights. It also stipulates procedures for the settlement of disputes related to the agreement.

15. Trade Policy Review Mechanism (TPRM) provides the procedures for the trade policy review mechanism to conduct periodical reviews of members' trade policies and practices conducted by the Trade Policy Review Body (TPRB).

16. Plurilateral Trade Agreements:

- Agreement on Trade in Civil Aircraft: Negotiations were ongoing alongside the Uruguay Round to revise the Civil Aircraft Agreement (an agreement from the Tokyo Round) and to strengthen disciplines on subsidies. However, no agreement has yet been reached and the agreement reached under the Tokyo Round continues.

- Agreement on Government Procurement: This agreement requires national treatment and non-discriminatory treatment in the area of government procurement and calls for fair and transparent procurement procedures. The agreement covers the procurement of services (in addition to goods) and the procurement by sub-central government entities and government-related agencies (in addition to central government).

All the above mentioned agreements entered into by the members are not static; they are renegotiated from time to time and new agreements evolve from negotiations. Many are now being negotiated under the Doha Development Agenda, launched by WTO trade ministers in Doha, Qatar, in November 2001.

3.6 The Doha Round

The Doha Round, formally the Doha Development Agenda, which is the ninth round since the Second World War was officially launched at the WTO's Fourth Ministerial Conference in Doha, Qatar, in November 2001.

The round seeks to accomplish major modifications of the international trading system through lower trade barriers and revised trade rules. The negotiations include 20 areas of trade, including agriculture, services trade, market access for non-agricultural products, and certain intellectual property issues. The most controversial topic in the yet to conclude Doha Agenda has been agriculture trade.

3.7 G 20 Economies: Facilitating Trade

While some trade-restrictive measures have been lifted by G20 countries, the report indicates that the trend has been going in the wrong direction.

Export restrictions contribute to shortages, price volatility, and uncertainty.

G20 economies must build on their collective pledges from the 12th Ministerial Conference and demonstrate leadership to keep markets open and predictable, so that food and fertilizer in particular can flow to where they are needed," said WTO Director-General Ngozi Okonjo-Iweala, who will be attending the G20 Leaders' Summit in Bali, Indonesia, on 15-16 November.

The report indicates that supply chains on the whole have thus far proved to be resilient, despite the war in Ukraine, the continuing impacts of the COVID-19 pandemic, the highest inflation many countries have experienced in decades, and the impacts of monetary tightening by central banks seeking to limit price increases.

That said, specific industries and regions have been differently impacted.

Overall, the pace of implementation of new export restrictions by WTO members has increased since 2020, first in the context of the pandemic and subsequently with the war in Ukraine and the food crisis. Some of these export restrictions have been gradually lifted, but several still remain in place.

As of mid-October 2022, WTO members still had in place 52 export restrictions on food, feed and fertilizers, in addition to 27 export restrictions on products essential to combat COVID-19. Of these, 44% of the export restrictions on food, feed and fertilizers, and 63% of the pandemic-related export restrictions, were maintained by G20 economies.

During the review period, G20 economies introduced 66 new trade-facilitating measures (covering trade worth USD 451.8 billion) and 47 trade-restrictive measures on goods (with a trade coverage of USD 160.1 billion). These measures were not related to the pandemic.

At the same time, the accumulated stockpile of G20 import restrictions continued to grow. By mid-October, 11.6% of G20 imports were affected by trade-restricting measures implemented since 2009 and still in force.

Initiations of trade remedy investigations by G20 economies declined sharply during the review period (17 initiations), after a peak in 2020 that was the highest since the beginning of the trade monitoring exercise in 2009.

Anti-dumping measures continued to be the most frequent trade remedy action in terms of initiations and terminations.

Similarly, the implementation of new COVID-19-related trade measures by G20 economies decelerated over the past five months, with four new such measures recorded on goods and one on services.

The number of new COVID-19-related support measures to mitigate the social and economic impacts of the pandemic also fell sharply over the past five months.

Since the beginning of the pandemic, 201 COVID-19 trade and trade-related measures in goods were implemented by G20 economies. Most (61%) were trade facilitating, while the rest (39%) could be considered trade restrictive.

G20 economies also continued to phase out pandemic-related import and export measures. By mid-October 2022, 77% of export restrictions had been repealed, leaving 17 restrictions in place. Although the number of the pandemic-related trade restrictions in place decreased, their trade coverage remained significant, at USD 122.0 billion.

The WTO trade monitoring reports have been prepared by the WTO Secretariat since 2009. G20 members are: Argentina; Australia; Brazil; Canada; China; the European Union; France; Germany; India; Indonesia; Italy; Japan; the Republic of Korea; Mexico; the Russian Federation; Saudi Arabia; South Africa; Türkiye; the United Kingdom; and the United States.

YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. Which of the following culminated in the establishment of the World Trade Organization?
   (a) *The Doha Round* (b) *The Tokyo Round*
   (c) *The Uruguay Round* (d) *The Kennedy Round*

2. Choose the correct statement
   (a) *The GATT was meant to prevent exploitation of poor countries by richer countries*
   (b) *The GATT dealt with trade in goods only, while, the WTO covers services as well as intellectual property.*
   (c) *All members of the World Trade Organization are required to avoid tariffs of all types*
   (d) *All the above*

3. The 'National treatment' principle stands for
   (a) *the procedures within the WTO for resolving disagreements about trade policy among countries*
   (b) *the principle that imported products are to be treated no worse in the domestic market than the local ones*
   (c) *exported products are to be treated no worse in the domestic market than the local ones*
   (d) *imported products should have the same tariff , no matter where they are imported from*

4. 'Bound tariff' refers to
   (a) *clubbing of tariffs of different commodities into one common measure*
   (b) *the lower limit of the tariff below which a nation cannot be taxing is imports*
   (c) *the upper limit on the tariff that a country can levy on a particular good, according to its commitments under the GATT and WTO.*
   (d) *the limit within which the country's export duty should fall so that there is cheaper exports*

5. The essence of 'MFN principle' is
   (a) *equality of treatment of all member countries of WTO in respect of matters related to trade*
   (b) *favour one, country , you need to favour all in the same manner*
   (c) *every WTO member will treat all its trading partners equally without any prejudice and discrimination*
   (d) *all the above*

6. The World Trade Organization (WTO)
   (a) *has now been replaced by the GATT*
   (b) *has an inbuilt mechanism to settle disputes among members*
   (c) *was established to ensure free and fair trade internationally.*
   (d) *(b) and (c) above*

7. The Agreement on Agriculture includes explicit and binding commitments made by WTO Member governments
   (a) *on increasing agricultural productivity and rural development*
   (b) *market access and agricultural credit support*
   (c) *market access, domestic support and export subsidies*
   (d) *market access, import subsidies and export subsidies*

8. The Agreement on Textiles and Clothing
   (a) *provides that textile trade should be deregulated by gradually and the tariffs should be increased*
   (b) *replaced the Multi-Fiber Arrangement (MFA) which was prevalent since 1974*
   (c) *granted rights of textile exporting countries to increase tariffs to protect their domestic textile industries*
   (d) *stipulated that tariffs in all countries should be the same*

9. The Agreement on Trade-Related Aspects of Intellectual Property Rights
   (a) *stipulates to administer a system of enforcement of intellectual property rights.*
   (b) *provides for most-favored-nation treatment and national treatment for intellectual properties*
   (c) *mandates to maintain high levels of intellectual property protection by all members*
   (d) *all the above*

10 The most controversial topic in the yet to conclude Doha Agenda is
   (a) *trade in manufactured goods*
   (b) *trade in intellectual property rights based goods*
   (c) *trade in agricultural goods*
   (d) *market access to goods from developed countries*

11. The WTO commitments
   (a) *affect developed countries adversely because they have comparatively less agricultural goods*
   (b) *affect developing countries more because they need to make radical adjustments*
   (c) *affect both developed and developing countries equally*
   (d) *affect none as they increase world trade and ensure prosperity to all*

PEQ -JUNE - 2024

12. GATT was established in year:
   (a) 1945 (b) 1948 (c) 1995 (d) 2014 (1 mark)

13. Under WTO agreements, the countries cannot normally discriminate between their trading partners. This is referred as:
   (a) National treatment (NT) *(b) Most Favoured Nation (MFN)*
   (c) Promoting fair competition (PFC) *(d) Free trade through negotiation (FTN)*
   (1 mark)

14. Group of countries that have a free trade agreement between themselves and may apply a common external tariff to other countries is referred as:
   (a) Trading block *(b) Free trade area*
   (c) Customs union *(d) Economic and monetary union (1 mark)*

PEQ - SEPTEMBER - 2024

15. As of 1st February, 2021, how many Regional Trade Agreements (RTAs) were in force worldwide?
   (a) 339 RTAs (b) 239 RTAs (c) 439 RTAs (d) 539 RTAs (1 mark)

16. What does TRIPS stand for?
   (a) Trade-Related Aspects of Intellectual Property Rights
   (b) Trade-Related Agreements on Investment Policies
   (c) Transnational Regulations for International Product Standards
   (d) Trade Regulations for International Patent Systems (1 mark)

17. The North American Free Trade Agreement (NAFTA), is a trade agreement signed between:
   (a) United States of America, Canada and Mexico
   (b) United States of America and Canada
   (c) United States of America and Mexico
   (d) Canada and Mexico (1 mark)

PEQ JANUARY - 2025

18. With regards to international trade the European Union can be categorised as a .
   (a) Trading bloc *(b) Free trade area*
   (c) Bilateral agreements *(d) Customs union (1 mark)*

19. Which of the following country is not a member of the G20 economies?
(a) *Argentina* (b) *India*
(c) *Tunisia* (d) *Mexico* *(1 mark)*

PEQ - MAY - 2025

20. Which of these countries is part of the USMCA Agreement?
(a) *Mexico*
(b) *Malta*
(c) *Malaysia*
(d) *Mongolia*

21. Which one of these is the main reason why GATT lost its relevance by 1980?
(a) *Efforts at liberalising agricultural trade were successful.*
(b) *International investments did not expand substantially.*
(c) *GATT was a treaty.*
(d) *There were inadequacies in institutional structure and dispute settlement system.*

22. The most controversial topic for Doha Development Agenda was
(a) *The Labour Welfare*
(b) *World Peace*
(c) *Globalization*
(d) *Agriculture Trade*

PEQ - JANUARY - 2026

23. Which of the following is not an objective of WTO?
(a) *To set and enforce rules for international trade.*
(b) *To maintain secrecy of decision making.*
(c) *To resolve trade disputes.*
(d) *To provide a forum for negotiating and monitoring further trade liberalization*

PEQ - SEPTEMBER - 2026

24. An important principle of GATT and WTO is the Most-Favoured-Nation (MFN) treatment, which means:
(a) *Preferential tariffs for select partners only*
(b) *Equal trade advantages given to all members*
(c) *National treatment for domestic goods only*
(d) *Exclusive bilateral treaties outside WTO*

ANSWERS / HINTS

I. Multiple Choice Type Questions

1	(c)	2	(b)	3	(b)	4	(c)	5	(d)	6	(d)	7	(c)
8	(b)	9	(d)	10	(c)	11	(b)	12	(b)	13	(b)	14	(a)
15	(a)	16	(a)	17	(a)	18	(d)	19	(c)	20	(a)	21	(d)
22	(d)	23	(b)	24	(b)								

For Your Notes

For Your Notes

For Your Notes

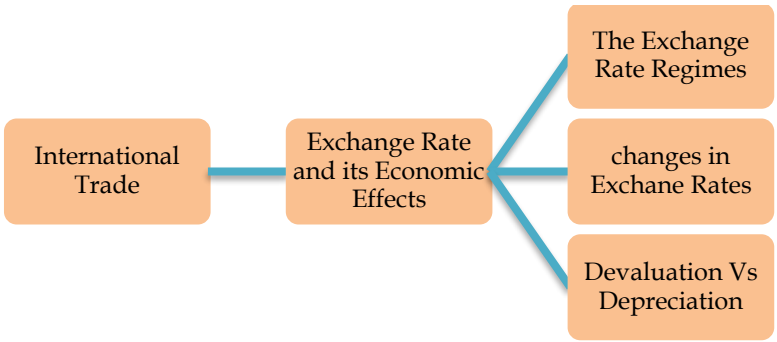
For Your Notes

UNIT – IV

EXCHANGE RATE AND ITS ECONOMIC EFFECTS

Topic References:

4.1 Introduction
4.2 What is Exchange Rate
4.3 Concept of Exchange Rate Regimes
4.4 Free- Floating Exchange rate System
4.5 Fixed Exchange Rates
4.6 Free float VS Fixed Exchange Rate
4.7 Nominal Exchange Rate Vs Real Exchange rate
4.8 Computation of Real Exchange rate
4.9 What is the Foreign Exchange Market
4.10 Determination of Nominal Exchange Rate
4.11 Changes in Exchange Rates
4.12 Devaluation (Revaluation) Vs Depreciation (Appreciation)
4.13 Impacts of Exchange rate Fluctuations on domestic Economy



4.1 Introduction:

- We are Basically going to Study About Foreign currency market and Its in open Economy.
- International Market involves the determination of Price of one Currency in terms of Another.

4.2 What is Exchange Rate:

- It is a transaction that is denominated in or requires settlement in a foreign currency,
- including transactions arising when an enterprise either:
  a) buys or sells goods or services whose price is denominated in a foreign currency.
  b) borrows or lends funds when the amounts payable or receivable are denominated in a foreign currency.
  c) becomes a party to an unperformed forward exchange contract; or
  d) otherwise acquires or disposes of assets, or incurs or settles liabilities, denominated in a foreign currency.

4.3 Concept of Exchange Rate Regimes:

- Exchange rates are determined by demand and supply.
- But governments can influence those exchange rates in various ways.
- The extent and nature of government involvement in currency markets define alternative systems of exchange rates
- *"An exchange rate regime is the system by which a country manages its currency with respect to foreign currencies. It refers to the method by which the value of the domestic currency in terms of foreign currencies is determined".*

- Three Broad Categories of Exchange rate System:
  - A) ***Private Market:*** Exchange rates are set purely by private market forces with no government involvement.
  - B) ***Gov-Private Market:*** Currency values are allowed to change, but governments participate in currency markets in an effort to influence those values.
  - C) ***Government Market:*** Governments may seek to fix the values of their currencies, either through participation in the market or through regulatory policy.
- There are two major types of exchange rate regimes at the extreme ends; namely:
  - A) Floating exchange rate regime (also called a flexible exchange rate), And
  - B) Fixed exchange rate regime

4.4 Free- Floating Exchange rate System:

- In this Governments and central banks do not participate in the market for foreign exchange.
- The relationship between governments and central banks on the one hand and currency markets on the other is much the same as the typical relationship between these institutions and stock markets.
- Governments may regulate stock markets to prevent fraud, but stock values themselves are left to float in the market.
- The primary difficulty with free-floating exchange rates lies in their unpredictability.
- Fluctuating exchange rates make international transactions riskier and thus increase the cost of doing business with other countries.

Managed Float System:

- Governments and central banks often seek to increase or decrease their exchange rates by buying or selling their own currencies.
- Exchange rates are still free to float, but governments try to influence their values.
- Government or central bank participation in a floating exchange rate system is called a managed float.

4.5 Fixed Exchange Rates:

- The exchange rate between two currencies is set by government policy.
- There are several mechanisms through which fixed exchange rates may be maintained. Whatever the system for maintaining these rates, however, all fixed exchange rate systems share some important features.

- In an open economy, the main advantages of a fixed rate regime are:
  1) A fixed exchange rate avoids currency fluctuations and eliminates exchange rate risks and transaction costs.
  2) Greatly enhance international trade and investment.
  3) A reduction in speculation.
  4) Imposes discipline on a country's monetary authority and therefore is more likely to generate lower levels of inflation.
  5) the central bank is required to maintain an adequate amount of foreign exchange reserves for this purpose.

4.6 Free float VS Fixed Exchange Rate

- A floating exchange rate has many advantages:
  1) A floating exchange rate has the greatest advantage of allowing a Central bank and/or government to pursue its own independent monetary policy.
  2) Floating exchange rate regime allows exchange rate to be used as a policy tool: for example, policy-makers can adjust the nominal exchange rate to influence the competitiveness of the tradable goods sector.
  3) As there is no obligation or necessity to intervene in the currency markets, the central bank is not required to maintain a huge foreign exchange reserves.

4.7 Nominal Exchange Rate Vs Real Exchange rate

A) Nominal Exchange rate:

- Nominal exchange rate which refers to the rate at which a person can trade the currency of one country for the currency of another country.
- For any country, there are many nominal exchange rates because its currency can be used to purchase many foreign currencies.
- While studying exchange rate changes, economists make use of indexes that average these many exchange rates.
- An exchange rate index turns these many exchange rates into a single measure of the international value of currency.
- Nominal Exchange Rates can be used to find the domestic price of foreign goods.

B) Real Exchange rate:

- The real exchange rate is the rate at which a person can trade the goods and services of one country for the goods and services of another.
- It describes 'how many' of a good or service in one country can be traded for 'one' of that good or service in a foreign country.
- A country's real exchange rate is a key determinant of its net exports of goods and services.

4.8 Computation of Real Exchange rate:

- For calculating real exchange rate, in the case of trade in a single good, we must first use the nominal exchange rate to convert the prices into a common currency.
- The real exchange rate (RER) between two currencies is the product of the nominal exchange rate and the ratio of prices between the two countries. It is calculated as

Real exchange Rate = (Nominal exchange Rate) x Domestic price
Foreign price

Real exchange rate = Nominal exchange rate x Domestic Price
Foreign price

- Thus, real exchange rate depends on the nominal exchange rate and the prices of the good in two countries measured in the local currencies.
- When studying the economy as a whole, we use price indices which measure the price of a basket of goods and services. Real exchange rate will then be:

Real exchange rate = Nominal exchange rate x Domestic Price Index
Foreign price Index

Conclusion:

- The Real Effective Exchange Rate (REER) is the nominal effective exchange rate (a measure of the value of a domestic currency against a weighted average of various foreign currencies) divided by a price deflator or index of costs.
- An increase in REER implies that exports become more expensive and imports become cheaper; therefore, an increase in REER indicates a loss in trade competitiveness.

4.9 What is the Foreign Exchange Market:

- FEM is a Place Where Participants Come to buy and Sell Foreign Currency.
- Forex market participants mainly are commercial banks executing orders from exporters, importers, investment institutions, insurance and retirement funds, hedgers, and private investors.
- *Market Makers:* Commercial Banks and Brokerage Houses do not only execute currency exchange operations at prices set by other active players but come out with their own prices as well, actively influencing the price formation process and the market life. That is why they are called market makers.

- *Two types of Transactions:*

A) Current Transaction: Which are carried out in the spot market and the exchange involves immediate delivery.

B) Future Transaction: Contracts are agreed upon to buy or sell currencies for future delivery which are carried out in forward and/or futures markets.

Some Definitions:

A) **Spot Exchange rate:** Exchange rates prevailing for spot trading (for which settlement by and large takes two days) are called spot exchange rates.

B) **Forward exchange rates:** The exchange rates quoted in foreign exchange transactions that specify a future date are called forward exchange rates.
"A forward premium is said to occur when the forward exchange rate is more than a spot exchange rates. On the contrary, if the forward trade is quoted at a lower rate than the spot rate, then there is a forward discount."

C) **Vehicle Currency:** While a foreign exchange transaction can involve any two currencies, most transactions involve exchanges of foreign currencies for the U.S. dollars even when it is not the national currency of either the importer or the exporter. On account of its critical role in the forex markets, the dollar is often called a 'vehicle currency'.

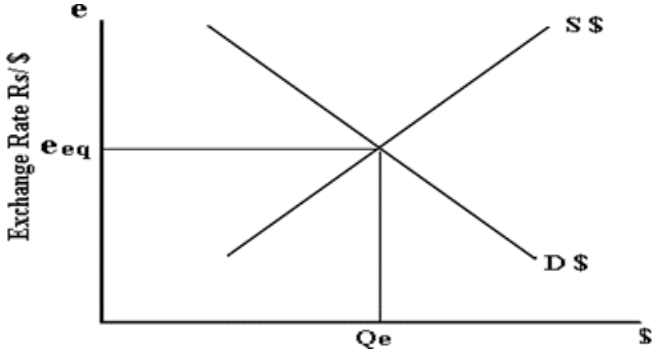
4.10 Determination of Nominal Exchange Rate:

A) *Concept:*

- Usually, the supply of and demand for foreign exchange in the domestic foreign exchange market determine the external value of the domestic currency, or in other words, a country's exchange rate.
- There are two Sides of Determination, one is ***Demand*** side and another is Supply Side.
- On Demand Side people desire foreign currency to:
  1) ***Purchase goods*** and services from another country
  2) for ***unilateral transfers*** such as gifts, awards, grants, donations or endowments
  3) to make ***investment*** income payments abroad
  4) to purchase ***financial assets***, stocks or bonds abroad
  5) to open a foreign ***bank account*** to acquire direct ownership of real capital, and
  6) for ***speculation and hedging*** activities related to risk-taking or risk-avoidance activity.
- On Supply Side It Operates for Similar reasons.
- Thus, the supply of foreign currency to the home country results from purchases of home exports, unilateral transfers to home country, investment income payments, foreign direct investments and portfolio investments, placement of bank deposits and speculation.

B) *How Foreign Exchange Market works:*

- Similar to any standard market, the exchange market also faces a downward-sloping demand curve and an upward- sloping supply curve.



Determination of Nominal Exchange Rate

- The equilibrium rate of exchange is determined by the interaction of the supply and demand for a particular foreign currency. In figure , the demand curve (D$) and supply curve (S$) of dollars intersect to determine equilibrium exchange rate e_{eq} with Q_e as the equilibrium quantity of dollars exchanged.

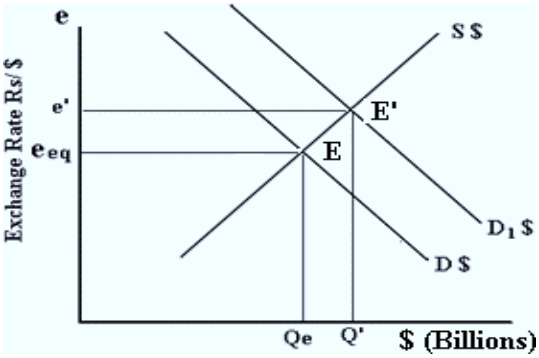
4.11 Changes in Exchange Rates:

- Changes in exchange rates portray depreciation or appreciation of one currency.
- The terms, 'currency appreciation' and 'currency depreciation' describe the movements of the exchange rate.
- Currency appreciates when its value increases with respect to the value of another currency or a basket of other currencies.
- On the contrary, currency depreciates when its value falls with respect to the value of another currency or a basket of other currencies.

For example, the Rupee dollar exchange rate in the month of January is $1 = Rs 70. and, we find that in the month of April it is $1 = Rs 75. What does this indicate?
It Means Rs value Depreciated as value of rupees has become less valuable with Respect to the US Dollar.

- **Home- Currency Depreciation or foreign-currency Appreciation:** It takes place when there is an increase in the home currency price of the foreign currency (or, alternatively, a decrease in the foreign currency price of the home currency). The home currency thus becomes relatively less valuable.
- **Home-currency appreciation or foreign-currency depreciation):** It takes place when there is a decrease in the home currency price of foreign currency (or alternatively, an increase in the foreign currency price of home currency). The home currency thus becomes relatively more valuable.

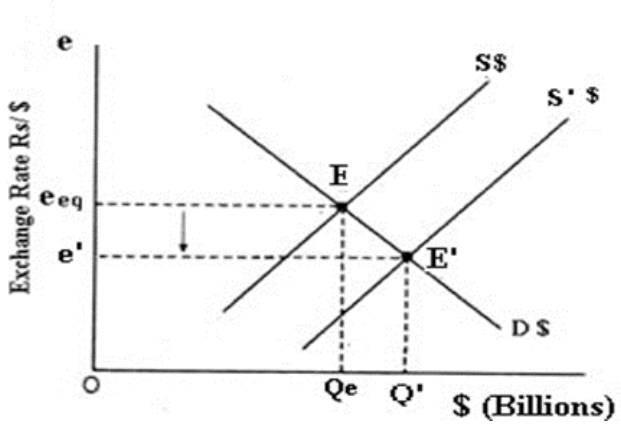
Under a floating rate system, if for any reason, the demand curve for foreign currency shifts to the right representing increased demand for foreign currency, and supply curve remains unchanged, then the exchange value of foreign currency rises and the domestic currency depreciates in value.



Home-Currency Depreciation under Floating Exchange Rates

1. The market initially is in equilibrium at point E with equilibrium exchange rate e eq. An increase in domestic demand for the foreign currency, with supply of dollars remaining constant, is represented by a rightward shift of the demand curve to D1$.
2. The equilibrium exchange rate rises to e1. This indicates that more units of domestic currency (here Indian Rupees) are required to buy one unit of foreign currency (here dollar) and that the domestic currency (the Rupee) has depreciated.

We shall now examine what happens when there is an increase in the supply of dollars in the Indian market. This is illustrated in figure



Home-Currency Appreciation under Floating Exchange Rates

1. An increase in the supply of foreign exchange shifts the supply curve to the right to S1 $ and as a consequence, the exchange rate declines to e1.
2. It means, that lesser units of domestic currency (here Indian Rupees) are required to buy one unit of foreign currency (dollar), and that the domestic currency (the Rupee) has appreciated.

4.12 Devaluation (Revaluation) Vs Depreciation (Appreciation)

- ***Devaluation:***
  a) It is a deliberate downward adjustment in the value of a country's currency relative to another country's currency or group of currencies or standard.
  b) It is a monetary policy tool used by countries that have a fixed exchange rate or nearly fixed exchange rate regime and involves a discrete official reduction in the otherwise fixed par value of a currency.
  c) The monetary authority formally sets a new fixed rate with respect to a foreign reference currency or currency basket.
- ***Depreciation:***
  a) Depreciation is a decrease in a currency's value (relative to other major currency benchmarks) due to market forces of demand and supply under a floating exchange rate and b) not due to any government or central bank policy actions.
- ***Revaluation:***
  a) It is the opposite of devaluation and the term refers to a discrete official increase of the otherwise fixed par value of a nation's currency.

- *Appreciation:*
  a) It is an increase in a currency's value (relative to other major currencies) due to market forces of demand and supply under a floating exchange rate and
  b) not due to any government or central bank policy interventions.

4.13 Impacts of Exchange rate Fluctuations on domestic Economy:

The developments in the foreign exchange markets affect the domestic economy both directly and indirectly. The direct impact of fluctuations in rates is initially felt by economic agents who are directly involved in international trade or international finance.

A) *Impacts of Fluctuations:*

a) Fluctuations in the exchange rate have a significant role in determining the nature and extent of a country's trade.

b) Fluctuations in the exchange rate affect the economy by changing the relative prices of domestically-produced and foreign-produced goods and services.

c) Exchange rate changes affect economic activity in the domestic economy. A depreciation of domestic currency primarily increases the price of foreign goods relative to goods produced in the home country and diverts spending from foreign goods to domestic goods.

d) For an economy where exports are significantly high, a depreciated currency would mean a lot of gain. In addition, if exports originate from labour-intensive industries, increased export prices will have positive effect on employment and potentially on wages.

e) Depreciation is also likely to add to consumer price inflation in the short run, directly through its effect on prices of imported consumer goods and also due to increased demand for domestic goods. The impact will be greater if the composition of domestic consumption baskets consists more of imported goods.

f) The fiscal health of a country whose currency depreciates is likely to be affected with rising export earnings and import payments and consequent impact on current account balance.

g) Companies that have borrowed in foreign exchange through external commercial borrowings (ECBs) but have been careless and did not sufficiently hedge these loans against foreign exchange risks, would also be negatively impacted as they would require more domestic currency to repay their loans.

h) Countries with foreign currency denominated government debts, currency depreciation will increase the interest burden and cause strain to the exchequer for repaying and servicing foreign debt. Fortunately, India's has small proportion of public debt in foreign currency.

i) Exchange rate fluctuations make financial forecasting more difficult for firms and larger amounts will have to be earmarked for insuring against exchange rate risks through hedging.

B) *Impact of Appreciation on Real Economy:*

An appreciation will have the following consequences on real economy:

a) An appreciation of currency raises the price of exports and, therefore, the quantity of exports would fall. Since imports become cheaper, we may expect an increase in the

quantity of imports. Combining these two effects together, the domestic aggregate demand falls and, therefore, economic growth is likely to be negatively impacted.

b) An appreciation may cause reduction in the levels of inflation because imports are cheaper.

c) With increasing export prices, the competitiveness of domestic industry is adversely affected and therefore, firms have greater incentives to introduce technological innovations and capital-intensive production to cut costs to remain competitive.

d) Increasing imports and declining exports are liable to cause larger deficits and worsen the current account.

e) Loss of competitiveness will be insignificant if currency appreciation is because of strong fundamentals of the economy.

4.14 Summary

- Exchange rate is the rate at which the currency of one country exchanges for the currency of another country.
- A direct quote (European Currency Quotation) is the number of units of a local currency exchangeable for one unit of a foreign currency. For example, ` 65/US$.
- An indirect quote (American Currency Quotation) is the number of units of a foreign currency exchangeable for one unit of local currency; for example: $ 0.0151 per rupee.
- In a direct quotation, the foreign currency is the base currency and the domestic currency is the counter currency. In an indirect quotation, the domestic currency is the base currency and the foreign currency is the counter currency.
- The rate between Y and Z which is derived from the given rates of another set of two pairs of currency (say, X and Y, and, X and Z) is called cross rate.
- An exchange rate regime is the system by which a country manages its currency with respect to foreign currencies.
- There are two major types of exchange rate regimes at the extreme ends; namely floating exchange rate regime, (also called a flexible exchange rate) and fixed exchange rate regime.
- Under floating exchange rate regime, the equilibrium value of the exchange rate of a country's currency is market determined i.e. the demand for and supply of currency relative to other currencies determines the exchange rate.
- A fixed exchange rate, also referred to as pegged exchange rate, is an exchange rate regime under which a country's government announces, or decrees, what its currency will be worth in terms of either another country's currency or a basket of currencies or another measure of value, such as gold.
- A central bank may implement soft peg policy under which the exchange rate is generally determined by the market or a hard peg where the central bank sets a fixed and unchanging value for the exchange rate.
- A fixed exchange rate avoids currency fluctuations and eliminates exchange rate risks and transaction costs, enhances international trade and investment and lowers the levels of inflation. But the central bank has to maintain an adequate amount of reserves and be always ready to intervene in the foreign exchange market.

- A floating exchange rate allows a government to pursue its own independent monetary policy and there is no need for market intervention or maintenance of reserves. However, volatile exchange rates generate a lot of uncertainties with regard to international transactions.
- The 'real exchange rate' incorporates changes in prices and describes 'how many' of a good or service in one country can be traded for 'one' of that good or service in a foreign country.
- Real exchange rate = Nominal exchange rate X $\frac{\text{Domestic price Index}}{\text{Foreign price Index}}$
- Real Effective Exchange Rate (REER) is the nominal effective exchange rate (a measure of the value of a currency against a weighted average of various foreign currencies) divided by a price deflator or index of costs.
- The wide-reaching collection of markets and institutions that handle the exchange of foreign currencies is known as the foreign exchange market. Being an over-the-counter market, it is not a physical place; rather, it is an electronically linked network bringing buyers and sellers together and has only very narrow spreads.
- On account of arbitrage, regardless of physical location, at any given moment, all markets tend to have the same exchange rate for a given currency. Arbitrage refers to the practice of making risk-less profits by intelligently exploiting price differences of an asset at different dealing places.
- There are two types of transactions in a forex market: current transactions which are carried out in the spot market and future transactions involving contracts to buy or sell currencies for future delivery which are carried out in forward and futures markets.
- Generally, the supply of and demand for foreign exchange in the domestic foreign exchange market determine the external value of the domestic currency, or in other words, a country's exchange rate.
- Changes in exchange rates portray depreciation or appreciation of one currency. The terms, 'currency appreciation' and 'currency depreciation' describe the movements of the exchange rate.
- Currency appreciates when its value increases with respect to the value of another currency or a basket of other currencies. On the contrary, currency depreciates when its value falls with respect to the value of another currency or a basket of other currencies.
- Devaluation is a deliberate downward adjustment by central bank in the value of a country's currency relative to another currency, group of currencies or standard.
- An appreciation of a country's currency cause changes in import and export prices will lead to changes in import and export volumes, causing resulting in import spending and export earnings.
- Exchange rate depreciation lowers the relative price of a country's exports, raises the relative price of its imports, increases demand both for domestic import-competing goods and for exports, leads to output expansion, encourages economic activity, increases the international competitiveness of domestic industries, increases the volume of exports and improves trade balance.
- Currency appreciation raises the price of exports, decrease exports; increase imports, adversely affect the competitiveness of domestic industry, cause larger deficits and worsens the trade balance.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. Based on the supply and demand model of determination of exchange rate, which of the following ought to cause the domestic currency of Country X to appreciate against dollar?
   (a) *The US decides not to import from Country X*
   (b) *An increase in remittances from the employees who are employed abroad to their families in the home country*
   (c) *Increased imports by consumers of Country X*
   (d) *Repayment of foreign debts by Country X*

2. All else equal, which of the following is true if consumers of India develop taste for imported commodities and decide to buy more from the US?
   (a) *The demand curve for dollars shifts to the right and Indian Rupee appreciates*
   (b) *The supply of US dollars shrinks and, therefore, import prices decrease*
   (c) *The demand curve for dollars shifts to the right and Indian Rupee depreciates*
   (d) *The demand curve for dollars shifts to the left and leads to an increase in exchange rate*

3. 'The nominal exchange rate is expressed in units of one currency per unit of the other currency. A real exchange rate adjusts this for changes in price levels'. The statements are
   (a) *wholly correct* (b) *partially correct*
   (c) *wholly incorrect* (d) *None of the above*

4. Match the following by choosing the term which has the same meaning

i)	floating exchange rate	a.	pegged exchange rate
ii)	fixed exchange rate	b.	depreciation
iii)	devaluation	c.	revaluation
iv)	appreciation	d.	flexible exchange rate

   (a) *(i c); (ii d); (iii b); (iv a))* (b) *(i b); (ii a); (iii d); (iv c)*
   (c) *(i a); (ii d) ; (iii b); (iv c)* (d) *(i d); (ii a); (iii b); (iv c)*

5. Choose the correct statement
   (a) *An indirect quote is the number of units of a local currency exchangeable for one unit of a foreign currency*
   (b) *the fixed exchange rate regime is said to be efficient and highly transparent.*
   (c) *A direct quote is the number of units of a local currency exchangeable for one unit of a foreign currency*
   (d) *Exchange rates are generally fixed by the central bank of the country*

6. Which of the following statements is true?
   (a) *Home-currency appreciation or foreign-currency depreciation takes place when there is a decrease in the home currency price of foreign currency*
   (b) *Home-currency depreciation takes place when there is an increase in the home currency price of the foreign currency*
   (c) *Home-currency depreciation is the same as foreign-currency appreciation and implies that the home currency has become relatively less valuable.*
   (d) *All the above*

7. An increase in the supply of foreign exchange
   (a) *shifts the supply curve to the right and as a consequence, the exchange rate declines*
   (b) *shifts the supply curve to the right and as a consequence, the exchange rate increases*
   (c) *more units of domestic currency are required to buy a unit of foreign exchange*
   (d) *the domestic currency depreciates and the foreign currency appreciates*

8. Currency devaluation
   (a) *may increase the price of imported commodities and, therefore, reduce the international competitiveness of domestic industries*
   (b) *may reduce export prices and increase the international competitiveness of domestic industries*
   (c) *may cause a fall in the volume of exports and promote consumer welfare through increased availability of goods and services*
   (d) *(a) and (c) above*

9. At any point of time, all markets tend to have the same exchange rate for a given currency due to
   (a) *Hedging* (c) *Arbitrage* (b) *Speculation* (d) *Currency futures*

10. 'Vehicle Currency' refers to
   (a) *a currency that is widely used to denominate international contracts made by parties because it is the national currency of either of the parties*
   (b) *a currency that is traded internationally and, therefore, is in high demand*
   (c) *a type of currency used in euro area for synchronization of exchange rates*
   (d) *a currency that is widely used to denominate international contracts made by parties even when it is not the national currency of either of the parties*

PEQ JUNE - 2024

11. An increase in Real effective exchange rate indicates:
   (a) *A loss in trade competitiveness* (b) *An increase in trade competitiveness*
   (c) *Exports become more cheaper* (d) *Imports become more costlier* *(1 mark)*

12. No. of units of foreign currency can be exchanged for one unit of local currency is referred to as:
   (a) *Direct Quote* (b) *Indirect Quote*
   (c) *European currency* (d) *Cross rates* *(1 mark)*

13. What is the impact of a depreciation of a country's currency on its exports?
   (a) *Exports decrease because foreign goods become cheaper for domestic consumers*
   (b) *Exports increase because domestic goods become cheaper for foreign consumers*
   (c) *Exports remain unchanged as the depreciation has no effect on trade*
   (d) *Exports increase because foreign goods become more expensive for domestic consumers*
   (1 mark)

PEQ SEPTEMBER - 2024

14. Suppose the exchange rate between INR and EUR changes from 1 EUR = 80 INR to 1 EUR = 85 INR. What can be said about the change in the value of the INR relative terms to the EUR?
   (a) *INR has depreciated against the EUR.* (b) *INR has appreciated against EUR.*
   (c) *EUR has no effect against INR.* (d) *EUR has depreciated against INR.* *(1 mark)*

15. What term is used for the rate between currencies Y and Z, which is derived from the given rates of another set of two pairs of currency (say, X and Y, and X and Z)?

(a) Ask rate *(b) Bid rate*

(c) Spot rate *(d) Cross rate* *(1 mark)*

PEQ - MAY - 2025

16. An appreciation of currency or strong currency makes the domestic currency more _____ therefore it can be exchanged for a _____amount of foreign currency.

(a) Weaker, smaller
(b) Valuable, larger
(c) Weaker, larger
(d) Valuable, smaller

PEQ - JANUARY - 2026

17. Calculate the Nominal exchange rate considering the following data:

Real exchange rate	25
Domestic price index	14070
Foreign price index	70

Choose the correct answer from the following options:

(a) 50.0
(b) 12.5
(c) 40.0
(d) 35.0

PEQ - SEPTEMBER - 2026

18. Real exchange rate increases if:

(a) Domestic prices rise faster than foreign prices
(b) Foreign prices rise relative to domestic prices
(c) Nominal exchange rate is fixed
(d) Trade barriers increase only

ANSWERS / HINTS

I. Multiple Choice Type Questions

1.	(b)	2.	(c)	3.	(a)	4.	(d)	5.	(c)	6	(d)
7.	(a)	8.	(b)	9.	(c)	10.	(d)	11.	(a)	12.	(b)
13.	(b)	14.	(a)	15.	(d)	16	(b)	17	(b)	18	(b)

For Your Notes

For Your Notes

For Your Notes

For Your Notes

UNIT – V

INTERNATIONAL CAPITAL MOVEMENTS

Topic References:

5.1 Introduction
5.2 Types of Foreign Capital
5.3 Foreign Direct Investment
5.4 Foreign Portfolio Investment
5.5 Reasons for FDI
5.6 Modes of FDI
5.7 Benefits of FDI
5.8 Problems with FDI
5.9 FDI in India
5.10 ODI by Indian Companies

5.1 Introduction

In this unit, we shall look into international capital movements; more precisely into why do capital move across national boundaries and what are the consequences of such capital movements. We shall also briefly touch upon the FDI situation in India.

5.2 Types of Foreign Capital

The term 'foreign capital' is a comprehensive one and includes any inflow of capital into the home country from abroad and therefore, we need to be clear about the distinction between movement of capital and foreign investment. Foreign capital may flow into an economy in different ways. Some of the important components of foreign capital flows are:

1. ***Foreign aid or assistance*** which may be:
(a) Bilateral or direct inter government grants
(b) Multilateral aid from many governments who pool funds to international organizations like the World Bank
(c) Tied aid with strict mandates regarding the use of money or untied aid where there are no such stipulations
(d) Foreign grants which are voluntary transfer of resources by governments, institutions, agencies or organizations

2. ***Borrowings*** which may take different forms such as:
(a) Direct inter government loans
(b) Loans from international institutions (e.g. world bank, IMF, ADB)
(c) Soft loans for e.g. from affiliates of World Bank such as IDA
(d) External commercial borrowing, and
(e) Trade credit facilities

3. ***Deposits*** from non-resident Indians (NRI)

4. ***Investments*** in the form of :
(a) Foreign portfolio investment (FPI) in bonds, stocks and securities, and
(b) Foreign direct investment(FDI) in industrial, commercial and similar other enterprises

5.3 Foreign Direct Investment (FDI)

Foreign direct investment is defined as a process whereby -

- the resident of one country (i.e. home country)
- acquires ownership of an asset in another country (i.e. the host country)
- and such movement of capital involves ownership, control as well as management of the asset in the host country.

This occurs through acquisition of more than 10% of the shares of the target asset.

Direct investment comprises not only the initial transaction establishing the relationship between the investor and the enterprise, but also all subsequent transactions between them and among affiliated enterprises, both incorporated and unincorporated.

According to the IMF and OECD definitions, the acquisition of at least 10% of the ordinary shares or voting power in a public or private enterprise by non-resident investors makes it eligible to be categorized as foreign direct investment (FDI).

Components of FDI - FDI has three components,

1. equity capital ;
2. reinvested earnings ; and
3. other direct capital in the form of intra-company loans between direct investors (parent enterprises) and affiliate enterprises.

Foreign direct investors may be individuals, incorporated or unincorporated private or public enterprises, associated groups of individuals or enterprises, governments or government agencies, estates, trusts, or other organizations or any combination of the above mentioned entities.

Direct investments are real investments in factories, assets, land, inventories etc. and involve foreign ownership of production facilities.

The investor retains control over the use of the invested capital and also seeks the power to exercise control over decision making to the extent of its equity participation.

The lasting interest implies the existence of a long-term relationship between the direct investor and the enterprise and a significant degree of influence by the investor on the management of the enterprise.

Based on the nature of foreign investments, FDI may be categorized as:

i) ***Horizontal FDI:*** A horizontal direct investment is said to take place when the investor establishes the same type of business operation in a foreign country as it operates in its home country,
For example, a cell phone service provider based in the United States moving to India to provide the same service.

ii) ***Vertical FDI:*** A vertical investment is one under which the investor establishes or acquires a business activity in a foreign country which is different from the investor's main business activity yet in some way supplements its major activity.
For example, an automobile manufacturing company may acquire an interest in a foreign company that supplies parts or raw materials required for the company.

iii) ***Conglomerate FDI:*** A conglomerate type of foreign direct investment is one where an investor makes a foreign investment in a business that is unrelated to its existing business in its home country. This is often in the form of a joint venture with a foreign firm already operating in the industry as the investor has no previous experience.

Yet another category of investments is 'two-way direct foreign investments' which are reciprocal investments between countries that occur when some industries are more advanced in one nation (for example, the computer industry in the United States), while other industries are more efficient in other nations (such as the automobile industry in Japan).

5.4 Foreign Portfolio Investment (FPI)

Foreign portfolio investment is the flow of what economists call 'financial capital' rather than 'real capital' and does not involve ownership or control on the part of the investor.

Examples of FPI are the deposit of funds in an Indian or a British bank by an Italian company or the purchase of a bond (a certificate of indebtedness) of a Swiss company or of the Swiss government by a citizen or company based in France.

Unlike FDI, portfolio capital, in general, moves to investment in financial stocks, bonds and other financial instruments and is effected largely by individuals and institutions through the mechanism of capital market. These flows of financial capital have their immediate effects on balance of payments or exchange rates rather than on production or income generation.

FPI is not concerned with either manufacture of goods or with provision of services.

Such investors also do not have any intention of exercising voting power or controlling or managing the affairs of the company in whose securities they invest.

The singular intention of a foreign portfolio investor is to earn a remunerative return through investment in foreign securities and is primarily concerned about the safety of their capital, the likelihood of appreciation in its value, and the return generated.

Logically, portfolio capital moves to a recipient country which has revealed its potential for higher returns and profitability.

Portfolio investments are characterised by lower stake in companies with their total stake in a firm at below 10%.

It is also noteworthy that unlike the FDIs, these investments are typically of short term nature, and therefore, are not intended to enhance the productive capacity of an economy by the creation of capital assets.

Portfolio investors will evaluate, on a separate basis, the prospects of each independent unit in which they might invest and may often shift their capital with changes in these prospects.

Therefore, portfolio investments are largely expected to be speculative. Once investor confidence is shaken, such capital has a tendency to speedily shift from one country to another, occasionally creating financial crisis for the host country.

Table
Foreign direct investment (FDI) VS Foreign portfolio investment (FPI)

Foreign direct investment (FDI)	Foreign portfolio investment (FPI)
Investment involves creation of physical assets	Investment is only in financial assets
Has a long term interest and therefore remain invested for long	Only short term interest and generally remain invested for short periods
Relatively difficult to withdraw	Relatively easy to withdraw
Not inclined to be speculative	Speculative in nature
Often accompanied by technology transfer	Not accompanied by technology transfer
Direct impact on employment of labour and wages	No direct impact on employment of labour and wages

Enduring interest in management and control	No abiding interest in management and control
Securities are held with significant degree of influence by the investor on the management of the enterprise	Securities are held purely as a financial investment and no significant degree of influence on the management of the enterprise

5.5 Reasons for Foreign Direct Investment

The chief motive for shifting of capital between different regions or between different industries is the expectation of higher rate of return than what is possible in the home country.

Investment in a host country may be found profitable by foreign firms because of some firm-specific knowledge or assets (such as superior management skills or an important patent) that enable the foreign firm to gainfully outperform the host country's domestic firms.

There are many other reasons (as listed below) for international capital movements which have found adequate empirical support. Investments move across borders on account of:

(i) the increasing interdependence of national economies and the consequent trade relations and international industrial cooperation established among them

(ii) internationalization of production and investment of transnational corporations in their subsidiaries and affiliates.

(iii) desire to reap economies of large-scale operation arising from technological growth

(iv) lack of feasibility of licensing agreements with foreign producers in view of the rapid rate of technological innovations

(v) necessity to retain direct control of production knowledge or managerial skill (usually found in monopolistic or oligopolistic markets) that could easily and profitably be utilized by corporations

(vi) desire to procure a promising foreign firm to avoid future competition and the possible loss of export markets

(vii) risk diversification so that recessions or downturns may be experienced with reduced severity

(viii) shared common language or common boundaries and possible saving in time and transport costs because of geographical proximity

(ix) necessity to retain complete control over its trade patents and to ensure consistent quality and service or for creating monopolies in a global context

(x) promoting optimal utilization of physical, human, financial and other resources

(xi) desire to capture large and rapidly growing high potential emerging markets with substantially high and growing population

(xii) ease of penetration into the markets of those countries that have established import restrictions such as blanket bans, high customs duties or non-tariff barriers which make it difficult for the foreign firm to sell in the host-country market by 'getting behind the tariff wall'.

(xiii) lower environmental standards in the host country and the consequent relative savings in costs

(xiv) stable political environment and overall favourable investment climate in the host country.

(xv) higher degree of openness to foreign capital exhibited by the recipient country and the prevalence of preferential investment systems such as special economic zones to encourage direct foreign investments

(xvi) the strategy to obtain control of strategic raw material or resource so as to ensure their uninterrupted supply at the lowest possible price; usually a form of vertical integration

(xvii) desire to secure access to minerals or raw material deposits located elsewhere and earn profits through processing them to finished form (Eg. FDI in petroleum)

(xviii) the existence of low relative wages in the host country because of relative labour abundance coupled with shortage and high cost of labour in capital exporting countries, especially when the production process is labour intensive.

(xix) lower level of economic efficiency in host countries and identifiable gaps in development

(xx) tax differentials and tax policies of the host country which support direct investment. However, a low tax burden cannot compensate for a generally fragile and unattractive FDI environment

(xxi) inevitability of defensive investments in order to preserve a firm's competitive position

(xxii) high gross domestic product and high per capita income coupled with their high rate of growth. There are also other philanthropic objectives such as strengthening of socio- economic infrastructure, alleviation of poverty and maintenance of ecological balance of the host country and

(xxiii) prevalence of high standards of social amenities and possibility of good quality of life in the host country

Factors in the host country discouraging inflow of foreign investments are infrastructure lags, high rates of inflation, balance of payment deficits, poor literacy and low labour skills, rigidity in the labour market, bureaucracy and corruption, unfavourable tax regime, cumbersome legal formalities and delays, small size of market and lack of potential for its growth, political instability, absence of well- defined property rights, exchange rate volatility, poor track-record of investments, prevalence of non-tariff barriers, stringent regulations, lack of openness, language barriers, high rates of industrial disputes, lack of security to life and property, lack of facilities for immigration and employment of foreign technical and administrative personnel, double taxation and lack of a general spirit of friendliness towards foreign investors.

5.6 Modes of Foreign Direct Investment

Foreign direct investments can be made in a variety of ways, such as:

(i) Opening of a subsidiary or associate company in a foreign country,
(ii) Equity injection into an overseas company,
(iii) Acquiring a controlling interest in an existing foreign company,
(iv) Mergers and acquisitions (M&A)
(v) Joint venture with a foreign company.
(vi) Green field investment (establishment of a new overseas affiliate for freshly starting production by a parent company).
(vii) Brownfield investments (a form of FDI which makes use of the existing infrastructure by merging, acquiring or leasing, instead of developing a completely new one. For e.g. in India 100% FDI under automatic route is allowed in Brownfield Airport projects.

5.7 Benefits of Foreign Direct Investment

The benefits from and concerns about FDI are widely discussed and well documented. While recognizing the fact that there are also benefits and costs to the home country from capital outflow, in this unit we focus only on host-country effects of FDI with particular attention to the developing countries. Following are the benefits ascribed to foreign investments:

1. Entry of foreign enterprises usually fosters competition and generates a competitive environment in the host country. The domestic enterprises are compelled to compete with the foreign enterprises operating in the domestic market. This results in positive outcomes in the form of cost-reducing and quality-improving innovations, higher efficiency and increasing variety of better products and services at lower prices ensuring wider choice and welfare for consumers
2. International capital allows countries to finance more investment than can be supported by domestic savings. The provision of increased capital to work with labour and other resources available in the host country can enhance the total output (as well as output per unit of input) flowing from the factors of production.
3. From the perspective of emerging and developing countries, FDI can accelerate growth and foster economic development by providing the much needed capital, technological know- how, management skills and marketing methods and critical human capital skills in the form of managers and technicians. The spill-over effects of the new technologies usually spread beyond the foreign corporations. In addition, the new technology can clearly enhance the recipient country's production possibilities.
4. Competition for FDI among national governments also has helped to promote political reforms important to attract foreign investors, including legal systems and macroeconomic policies.
5. Since FDI involves setting up of production base (in terms of factories, power plants, etc.) it generates direct employment in the recipient country. Subsequent FDI as well as domestic investments propelled in the downstream and upstream projects that come up in multitude of other services generate multiplier effects on employment and income.
6. FDI not only creates direct employment opportunities but also, through backward and forward linkages, generate indirect employment opportunities.This impact is particularly important if the recipient country is a developing country with an excess supply of labour caused by population pressure.
7. Foreign direct investments also promote relatively higher wages for skilled jobs. More indirect employment will be generated to persons in the lower-end services sector occupations thereby catering to an extent even to the less educated and unskilled persons engaged in those units.
8. Foreign corporations provide better access to foreign markets. Unlike portfolio investments, FDI generally entails people-to-people relations and is usually considered as a promoter of bilateral and international relations. Greater openness to foreign capital leads to higher national dependence on international investors, making the cost of discords higher.
9. There is also greater possibility for the promotion of ancillary units resulting in job creation and skill development for workers.
10. Foreign enterprises possessing marketing information with their global network of marketing are in a unique position to utilize these strengths to promote the exports of developing countries. If the foreign capital produces goods with export potential, the host country is in a position to secure scarce foreign exchange which can be used to import needed capital equipments or materials to assist the country's development plans or to ease its external debt servicing.
11. If the host country is in a position to implement effective tax measures, the foreign investment projects also would act as a source of new tax revenue which can be used for development projects.
12. It is likely that foreign investments enter into industries in which scale economies can be realized so that consumer prices might be lowered. Domestic firms might not always be able to generate the necessary capital to achieve the cost reductions associated with large-scale production.

13. Increased competition resulting from the inflow of foreign direct investments facilitates weakening of the market power of domestic monopolies resulting in a possible increase in output and fall in prices.
14. Since FDI has a distinct advantage over the external borrowings, it is considered to have a favourable impact on the host country's balance of payment position, and
15. Better work culture and higher productivity standards brought in by foreign firms may possibly induce productivity related awareness and may also contribute to overall human resources development.

5.8 Problems with Foreign Direct Investment

Following are the general arguments put forth against the entry of foreign capital.

1. FDIs are likely to concentrate on capital-intensive methods of production and service so that they need to hire only relatively few workers. Such technology is inappropriate for a labour-abundant country as it does not support generation of jobs which is a crucial requirement to address poverty and unemployment which are the two fundamental areas of concern for the less developed countries.
2. The inherent tendency of FDI flows to move towards regions or states which are well endowed in terms of natural resources and availability of infrastructure has the potential to accentuate regional disparity. Foreign capital is also criticized for accentuating the already existing income inequalities in the host country.
3. In the context of developing countries, it is usually alleged that the inflow of foreign capital may cause the domestic governments to slow down its efforts to generate more domestic savings, especially when tax mechanisms are difficult to implement. If the foreign corporations are able to secure incentives in the form of tax holidays or similar provisions, the host country loses tax revenues.
4. Often, the foreign firms may partly finance their domestic investments by borrowing funds in the host country's capital market. This action can raise interest rates in the host country and lead to a decline in domestic investments through 'crowding-out' effect. Moreover, suppliers of funds in developing economies would prefer foreign firms due to perceived lower risks and such shifts of funds may divert capital away from investments which are crucial for the development needs of the country.
5. The expected benefits from easing of the balance of payments situation might remain unrealised or narrowed down due to the likely instability in the balance of payments and the exchange rate. Obviously, FDI brings in more foreign exchange, improves the balance of payments and raises the value of the host country's currency in the exchange markets. However, when imported inputs need to be obtained or when profits are repatriated, a strain is placed on the host country's balance of payments and the home currency leading to its depreciation. Such instabilities jeopardize long-term economic planning. Foreign corporations also have a tendency to use their usual input suppliers which can lead to increased imports. Also, large scale repatriation of profits can be stressful on the balance of payments.
6. Jobs that require expertise and entrepreneurial skills for creative decision making may generally be retained in the home country and therefore the host country is left with routine management jobs that demand only lower levels of skills and ability. The argument of possible human resource development and acquisition of new innovative skills through FDI may not be realized in reality.
7. High profit orientation of foreign direct investors tend to promote a distorted pattern of production and investment such that production could get concentrated on items of elite and popular consumption and on non-essential items.

8. Foreign entities are usually accused of being anti-ethical as they frequently resort to methods like aggressive advertising and anticompetitive practices which would induce market distortions.
9. A large foreign firm with deep pockets may undercut a competitive local industry because of various advantages (such as in technology) possessed by it and may even drive out domestic firms from the industry resulting in serious problems of displacement of labour. The foreign firms may also exercise a high degree of market power and exist as monopolists with all the accompanying disadvantages of monopoly. The high growth of wages in foreign corporations can influence a similar escalation in the domestic corporations which are not able to cover this increase with growth of productivity. The result is decreasing competitiveness of domestic companies which might prove detrimental to the long term interests of industrial development of the host country.
10. FDI usually involves domestic companies 'off–shoring', or shifting jobs and operations abroad in pursuit of lower operating costs and consequent higher profits. This has deleterious effects on employment potential of home country.
11. The continuance of lower labour or environmental standards in host countries is highly appreciated by the profit seeking foreign enterprises. This is of great concern because efforts to converge such standards often fail to receive support from interested parties.
12. At times, there is potential national security considerations involved when foreign firms function in the territory of the host country, especially when acute hostilities prevail.
13. FDI may have adverse impact on the host country's commodity terms of trade (defined as the price of a country's exports divided by the price of its imports). This could occur if the investments go into production of export goods and the country is a large country in the sale of its exports. Thus, increased exports drive down the price of exports relative to the price of imports.
14. FDI is also held responsible by many for ruthless exploitation of natural resources and the possible environmental damage.
15. With substantial FDI in developing countries there is a strong possibility of emergence of a dual economy with a developed foreign sector and an underdeveloped domestic sector.
16. Perhaps the most disturbing of the various charges levied against foreign direct investment is that a large foreign investment sector can exert excessive amount of power in a variety of ways so that there is potential loss of control by host country over domestic policies and therefore the less developed host country's sovereignty is put at risk. Mighty multinational firms are often criticized of corruption issues, unduly influencing policy making and evasion of corporate social responsibility.

5.9 Foreign Direct Investment in India

Foreign Direct Investment (FDI), in addition to being a key driver of economic growth, has been a significant non-debt financial resource for India's economic development. Foreign corporations invest in India to benefit from the country's particular investment privileges such as tax breaks and comparatively lower salaries.

This helps India develop technological know- how and create jobs as well as other benefits. These investments have been coming into India because of the government's supportive policy framework, vibrant business climate, rising global competitiveness and economic influence.

The government has recently made numerous efforts, including easing FDI regulations in various industries, PSUs, oil refineries, telecom and defence.

India's FDI inflows reached record levels during 2020-21. The total FDI inflows stood at US$ 81,973 million, a 10% increase over the previous financial year. According to the World

Investment Report 2022, India was ranked eighth among the world's major FDI recipients in 2020, up from ninth in 2019. Information and technology, telecommunication and automobile were the major receivers of FDI in FY22.

With the help of significant transactions in the technology and health sectors, multinational companies (MNCs) have pursued strategic collaborations with top domestic business groupings, fuelling an increase in cross-border M&A of 83% to US$ 27 billion.

5.10 Overseas Direct Investment by Indian Companies

India is primarily a domestic demand-driven economy, with consumption and investments contributing to 70% of the economic activity. With an improvement in the economic scenario and the Indian economy recovering from the Covid-19 pandemic shock, India is relatively well placed than the rest of the world.

Despite major headwinds that continue to pose risks in the short term, the Indian economy has remained strong owing to robust policy measures in place. This gives Indian businesses an advantage to make investments abroad and broaden their operational footprint in such nations. New innovations from abroad would be brought to India with the help of knowledge spillover, and India itself would contribute to the growth of other nations.

In this manner, a mutual benefit is achieved. In this regard, there have been several overseas investments made by Indian companies. Some of the key overseas investments and developments that have taken place in the recent past are mentioned as follows:

According to data released by the Reserve Bank of India (RBI), overseas direct investment stood at US$ 1,922.51 million in September 2022.

The critical investments are as follows:

- In June 2022, Tata Steel announced plans to invest 7 million pounds (US$ 837.95 billion) for its Hartlepool Tube Mill in North-East England.
- Tata Communications invested US$ 690 million in its wholly-owned subsidiary in Singapore.
- Jindal Steel and Power invested US$ 366 million in its wholly owned subsidiary in Mauritius
- Wipro invested US$ 204.96 million in its wholly-owned subsidiary in Cyprus.
- Jindal Saw invested US$ 64.5 million in its wholly-owned subsidiary in the United Arab Emirates.
- Restaurant Brand Aisa and Lupin Ltd invested US$ 141.34 million and US$ 131.25 million in their JVs in Indonesia and the US, respectively.
- Reliance New Energy invested US$ 87.73 million in its wholly owned subsidiary in Norway.
- Mohalla Internet Pvt. Ltd. invested US$ 86 million in its fully owned unit in Mauritius.
- ONGC Videsh invested US$ 83.31 million in a joint-venture in Russia.
- ICICI Bank ties up with Santander in Britain in a pact aimed at facilitating the banking requirements of corporates operating across both countries.
- ANI Technologies, the promoter of OLA, invested US$ 675 million in its wholly-owned subsidiary in Singapore.
- Dr Reddy invested US$ 149.99 million in a joint- venture (JV) in the US.
- A total of US$ 168.9 million was invested by Reliance New Energy in a JV and wholly- owned subsidy in Germany and Norway.
- Gail India, energy PSU invested US$ 70.17 million in a JV and wholly-owned unit in Myanmar and the US.
- ONGC invested US$ 74.15 million during the month in various countries in 5 different ventures.

- In July 2022, Reliance Brands Ltd. signed a distribution agreement with Maison Valentino, an Italian luxury fashion house, to open its first boutique in Delhi, followed by a flagship store in Mumbai.
- In July 2022, Reliance Retail Limited entered into a long-term partnership with Gap Inc. to bring the iconic American fashion brand, Gap, to India.
- In July 2022, Tata Steel signed a Memorandum of Understanding (MoU) with BHP, a leading global resources company, with the intention to jointly study and explore low- carbon iron and steelmaking technology.
- In January 2022, Ola Electric, the ride-hailing company's electric vehicle (EV) subsidiary, announced its plans to establish Ola Future foundry, a global hub for advanced engineering and vehicle design in the UK, investing US$ 100 million over the next 5 years.
- In January, Essar Group of India announced that it had created a joint venture with Progressive Energy of the UK to invest US$ 1.34 billion in a hydrogen manufacturing plant at its Essar Stanlow refinery complex.
- In January, Hindalco Ltd's US subsidiary, Novelis, announced its plans to invest US$ 365 million in a state-of-the-art vehicle recycling facility in North America.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. Which of the following statements is incorrect?
   (a) *Direct investments are real investments in factories, assets, land, inventories etc. and involve foreign ownership of production facilities.*
   (b) *Foreign portfolio investments involve flow of 'financial capital'*
   (c) *Foreign direct investment (FDI) is not concerned with either manufacture of goods or with provision of services.*
   (d) *Portfolio capital moves to a recipient country which has revealed its potential for higher returns and profitability.*

2. Which of the following is a component of foreign capital?
   (a) *Direct inter government loans*
   (b) *Loans from international institutions (e.g. World Bank, IMF, ADB)*
   (c) *Soft loans for e.g. from affiliates of World Bank such as IDA*
   (d) *All the above*

3. Which of the following would be an example of foreign direct investment from Country X?
   (a) *A firm in Country X buys bonds issued by a Chinese computer manufacturer.*
   (b) *A computer firm in Country X enters into a contract with a Malaysian firm for the latter to make and sell to it processors*
   (c) *Mr. Z a citizen of Country X buys a controlling share in an Italian electronics firm*
   (d) *None of the above*

4. Which of the following types of FDI includes creation of fresh assets and production facilities in the host country?
   (a) *Brownfield investment* (b) *Merger and acquisition*
   (c) *Greenfield investment* (d) *Strategic alliances*

5. Which is the leading country in respect of inflow of FDI to India?
   (a) *Mauritius* (b) *USA* (c) *Japan* (d) *USA*

6. An argument in favour of direct foreign investment is that it tends to
   (a) *promote rural development* (b) *increase access to modern technology*
   (c) *protect domestic industries* (d) *keep inflation under control*

7. Which of the following is a reason for foreign direct investment?
   (a) *secure access to minerals or raw materials*
   (b) *desire to capture of large and rapidly growing emerging markets*
   (c) *desire to influence home country industries*
   (d) *(a) and (b) above*

8. A foreign direct investor
   (a) *May enter India only through automatic route*
   (b) *May enter India only through government route*
   (c) *May enter India only through equity in domestic enterprises*
   (d) *Any of the above*

9. Foreign investments are prohibited in
   (a) *Power generation and distribution* (b) *Highways and waterways*
   (c) *Chit funds and Nidhi company* (d) *Airports and air transport*

10. Which of the following statement is false in respect of FPI?
(a) *portfolio capital in general, moves to investment in financial stocks, bonds and other financial instruments about foreign portfolio investment?*
(b) *is effected largely by individuals and institutions through the mechanism of capital market*
(c) *is difficult to recover as it involves purely long term investments and the investors have controlling interest*
(d) *investors also do not have any intention of exercising voting power or controlling or managing the affairs of the company*

PEQ – JUNE – 2024

11. Not a component of FDI according to IMF:
(a) *Equity capital* (b) *Re-investment earnings*
(c) *Portfolio investments* (d) *Intra company loans* *(1 mark)*

PEQ – SEPTEMBER – 2024

12. What is the main difference between Foreign Direct Investment (FDI) and Foreign Portfolio Investment (FPI)?
(a) *FDI involves financial assets, while FPI creates physical assets.*
(b) *FDI creates physical assets, while FPI involves only financial assets.*
(c) *Both FDI and FPI create physical assets.*
(d) *Both FDI and FPI involve only financial assets.* *(1 mark)*

13. Foreign corporations invest in India to benefit from the country's particular investment privileges such as tax breaks and comparatively lower salaries. This type of investment is an example of:
(a) *Joint Venture* (b) *Foreign Portfolio Investment*
(c) *Foreign Direct Investment* (d) *Strategic Alliance* *(1 mark)*

PEQ – JANUARY – 2025

14. Investments which are reciprocal investments between countries are referred to as _____.
(a) *Horizontal direct investment* (b) *Vertical direct investments*
(c) *Two-way direct foreign investment* (d) *Conglomerate foreign investment* *(1 mark)*

PEQ – MAY – 2025

15. Match the following:

Table A	Table B
(a) Has a long term interest and is invested for long	(i) Foreign Portfolio Investment
(b) Speculative in Nature	(ii) Foreign Direct Investment
(c) Help developing countries benefit fully from global trading system	(iii) Free Floating exchange rate system
(d) Government and central banks do not participate in the market for foreign exchange	(iv) World Trade Organisation

(a) *(a)(ii), (b) - (i), (c) - (iv), (d) - (iii)*
(b) *(a)(i), (b) (ii), (c) - (iv), (d) - (iii)*
(c) *(a)(ii), (b) - (i), (c) - (iii), (d) - (iv)*
(d) *(a)(i), (b) (ii), (c) - (iii), (d) - (iv)*

16. An investment in which one investor establishes a business activity in a foreign country which is different from investor's main business activity but in some way supplements its major activity is called _____
(a) *Horizontal direct investment*
(b) *Vertical investment*
(c) *Conglomerate type of FDI*
(d) *Two way direct foreign investment*

PEQ - JANUARY - 2026

17. What does a Brownfield investment in Foreign Direct Investment (FDI) imply:
(a) *Developing entirely new infrastructure*
(b) *Utilizing existing infrastructure by means of mergers, acquisitions or leasing*
(c) *Forming partnerships with foreign companies*
(d) *Establishing a new subsidiary in another country*

18. Which sector accounts for the largest recipient of FDI inflows in India?
(a) *Service*
(b) *Manufacturing*
(c) *Primary*
(d) *Secondary*

19. Which of the following currency is often referred to as the vehicle currency, in the foreign exchange market?
(a) *US Dollar*
(b) *Euro*
(c) *Russian Ruble*
(d) *INR*

20. When a foreign investor having no previous experience initiates a FDI in a business which is not related to its existing business in its home country and is usually in the form of a joint venture with a foreign firm operating in the industry, it is known as a _____.
(a) *Two way FDI*
(b) *Vertical FDI*
(c) *Horizontal FDI*
(d) *Conglomerate FDI*

PEQ - SEPTEMBER - 2026

21. Foreign Direct Investment (FDI) is best distinguished from Foreign Portfolio Investment (FPI) by its:
(a) *Ownership of foreign currency*
(b) *Intent to gain management control in foreign enterprises*
(c) *Duration of investment only*
(d) *Government backing*

ANSWERS / HINTS

I. Multiple Choice Type Questions

1.	(c)	2.	(d)	3.	(c)	4.	(c)	5.	(a)
6.	(b)	7.	(d)	8.	(d)	9.	(c)	10.	(c)
11.	(c)	12.	(b)	13.	(c)	14.	(c)	15	(a)
16	(b)	17	(b)	18	(a)	19	(a)	20	(d)
21	(b)								

For Your Notes

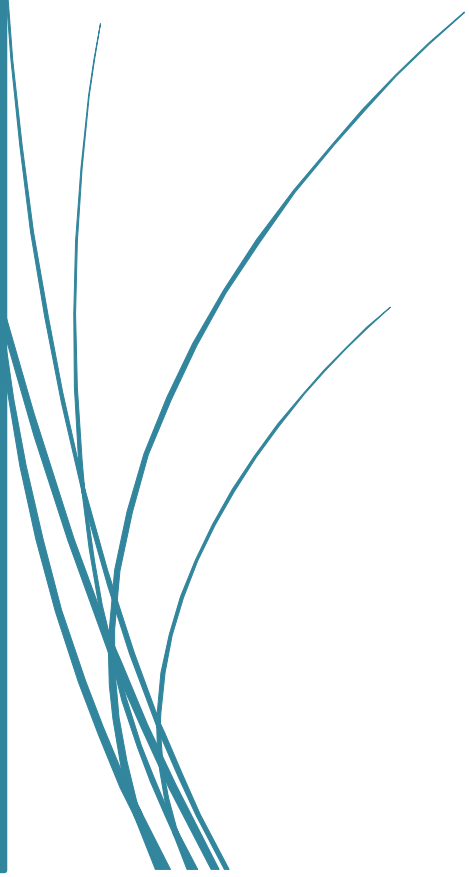
For Your Notes

For Your Notes

For Your Notes

CHAPTER-10

INDIAN ECONOMY



INDIAN ECONOMY

Topic References:

10.1 Status of Indian Economy: Pre Independence-Period (1850 -1947)
10.2 Indian Economy: Post-Independence (1947- 1991)
10.3 The Era of Reforms
10.4 The Economic Reforms of 1991
10.5 GDP Growth Rates Post 1991 Reforms
10.6 NITI AAYOG: A Bold Step for Transforming India
10.7 The Current State of The Indian Economy: A Brief Overview
10.8 Conclusion

10.1 Status of Indian Economy: Pre Independence-Period (1850 -1947)

Between the first and the seventeenth century AD, India is believed to have had the largest economy of the ancient and the medieval world.

It was prosperous and self-reliant and is believed to have controlled between one third and one fourth of the world's wealth.

The economy consisted of self-sufficient villages as well as cities which were centres of commerce, pilgrimage and administration.

Compared to villages, cities presented more opportunities for diverse occupations, trades and gainful economic activities.

Simple division of labour intertwined with attributes such as race, class, and gender was the basis of the structure of the villages and acted as a built-in mechanism of economic and social differentiation.

Though agriculture was the dominant occupation and the main source of livelihood for majority of people, the country had a highly skilled set of artisans and craftsmen who produced manufactures, handicrafts and textiles of superior quality and fineness for the worldwide market.

Ancient Economic Philosophy of India

The earliest known treatise on ancient Indian economic philosophy is 'Arthashastra' the pioneering work attributed to Kautilya (Chanakya) (321–296 BCE) Arthashastra is recognized as one of the most important works on statecraft in the genre of political philosophy.

It is believed to be a kind of handbook for King Chandragupta Maurya, the founder of Mauryan empire, containing directives as to how to reign over the kingdom and encouraging direct action in addressing political concerns without regard for ethical considerations.

Artha is not wealth alone; rather it encompasses all aspects of the material well-being of individuals. Arthashastra is the science of 'artha' or material prosperity, or "the means of subsistence of humanity," which is, primarily, 'wealth' and, secondarily, 'the land'.

The major focus of the work is on the means of fruitfully maintaining and using land. Kautilya emphasizes the importance of robust agricultural initiatives for an abundant harvest which will go toward filling the state's treasury.

Taxes, which were charged equal for private and state-owned businesses, must be fair to all and should be easily understood by the king's subjects.

Being a multidisciplinary discourse on areas such as politics, economics, military strategy, diplomacy, function of the state, and the social organization, Kautilya's writings relate to statecraft, political science, economic policy and military strategy.
True kingship is defined as a ruler's subordination of his own desires and ambitions to the good of his people; i.e. a king's policies should reflect a concern for the greatest good of the greatest number of his subjects. The preservation and advancement of this good was comprised of seven vital elements, namely the King, Ministers, Farmlands, Fortresses, Treasury, Military and the Allies.

The advent of the Europeans and the British marked a shift in the economic history of India. The period of British rule can be divided into two sub periods:

1. The rule of East India Company from 1757 to 1858
2. British government in India from 1858 to 1947

The historical legacy of British colonialism is an important starting point to illustrate the development path of India.

With the onset of Industrial revolution in the latter half of the 18th century, the manufacturing capabilities of Britain increased manifold, and consequently there arose the need to augment raw material supply as well as the need for finding markets for finished goods. This led to a virtual reversal of the nature of India's foreign trade from an exporter of manufactures to an exporter of raw materials.

The Indian exports of finished goods were subjected to heavy tariffs and the imports were charged lower tariffs under the policy of discriminatory tariffs followed by the British. This made the exports of finished goods relatively costlier and the imports cheaper. In this backdrop, the Indian goods lost their competitiveness.

Consequently, the external as well as the domestic demand for indigenous products fell sharply culminating in the destruction of Indian handicrafts and manufactures. The destruction of Indian manufactures, mainly due to the hostile imperial policies to serve the British interests and the competition from machine- made goods, had far reaching adverse consequences on the Indian manufacturing sector.

The problem was aggravated by the shift in patterns of demand by domestic consumers favoring foreign goods as many Indians wanted to affiliate themselves with western culture and ways of life.

The damage done to the long established production structure had far reaching economic and social consequences as it destroyed the internal balance of the traditional village economy which was characterized by the harmonious blending of agriculture and handicrafts. These were manifest as:

1. Large scale unemployment and absence of alternate sources of employment which forced many to depend on agriculture for livelihood
2. The increased pressure on land caused sub division and fragmentation of land holdings, subsistence farming, reduced agricultural productivity and poverty.
3. The imports of cheap machine made goods from Britain and an overt shift of tastes and fashion of Indians in favour of imported goods made the survival of domestic industries all the more difficult.
4. The systems of land tenure, especially the zamindari system created a class of people whose interests were focused on perpetuating the British rule.
5. Excessive pressure on land increased the demand for land under tenancy, and the zamindars got the opportunity to extract excessive rents and other payments

6. Absentee landlordism, high indebtedness of agriculturists, growth of a class of exploitative money lenders and low attention to productivity enhancing measures led to a virtual collapse of Indian agriculture.

We shall now have a look into the stagnated nature of industrialisation during the colonial era. Factory-based production did not exist in India before 1850.The 'Modern' industrial enterprises in colonial India started to grow in the mid-19th century.

The cotton milling business grew steadily throughout the second half of the 19th century, and achieved high international competitiveness. The cotton mill industry in India had 9 million spindles in the 1930s, which placed India in the fifth position globally in terms of number of spindles.

Jute mills also expanded rapidly in and around Calcutta in response to a mounting global demand for ropes and other products, and Indian jute occupied a large share of the international market by the late 19th century. At the end of the 19th century, the Indian jute mill industry was the largest in the world in terms of the amount of raw jute consumed in production.

In addition, brewing, paper-milling, leather-making, matches, and rice-milling industries also developed during the century. Heavy industries such as the iron industry were also established as early as 1814 by British capital. India's iron industry was ranked eighth in the world in terms of output in 1930.

Due to progress in modern industrial enterprises, some industries even reached global standards by the beginning of the 20th century. Just before the Great Depression, India was ranked as the twelfth largest industrialised country measured by the value of manufactured products.

The producer goods industries, however, did not show high levels of expansion. Perhaps, the most important of the factors that led to this state of affairs was the pressure exerted by the English producers in matters of policy formulation to positively discourage the development of industries which were likely to compete with those of the English producers.

India's industrial growth was insufficient to bring in a general transformation in its economic structure. The share in the net domestic product (NDP) of the manufacturing sector (excluding small scale and cottage industries) had barely reached 7% even in 1946.Considering its slow progress, the share of factory employment in India was also small (i.e. 0.4% of the total population in 1900 and 1.4% in 1941).

10.2 Indian Economy: Post-Independence (1947- 1991)

At the time of independence, India was overwhelmingly rural inhabited by mostly illiterate people who were exceedingly poor. We had a deeply stratified society characterized by extreme heterogeneity on many counts. With the literacy rate just above 18 percent and barely 32 years of life expectancy in 1951, India's poverty was not just in terms of income alone, but also in terms of human capital,

For historical reasons, the Nehruvian model which supported social and economic redistribution and industrialization directed by the state came to dominate the post- Independence Indian economic policy.

Centralized economic planning and direction was at the core of India's development strategy and the economic policies were crafted to accomplish rapid economic growth accompanied by equity and distributive justice.

The Planning Commission of India was established to meticulously plan for the economic development of the nation in line with the socialistic strategy. This was carried through the five-year plans which were developed, implemented and monitored by the Planning Commission.

It is pertinent here to have a look at the ideology of industrialization prevailed in the early days of independence. India's political leadership was keen on establishing an economic system in which the central government would have authority to design the economic strategy and to carry out the necessary investments in coordination with the private sector. Rapid industrialization of the economy was the cornerstone of Nehru's development strategy.

The concept of 'planned modernization' meant a systematic planning to support industrialization. The bureaucrats and the technocrats envisioned a substantially significant role for the state in industrialisation.
The Industrial Policy Resolution (1948) envisaged an expanded role for the public sector and licensing to the private sector. It granted state monopoly for strategic areas such as atomic energy, arms and ammunition and railways. Also, the rights to new investments in basic Industries were exclusively given to the state.

The policies in 1950's were guided by two economic philosophies:

1. The then prime minister Nehru's visualization to build a socialistic society with emphasis on heavy industry, and
2. The Gandhian philosophy of small scale and cottage industry and village republics

The Industrial Policy Resolution of 1956 though provided a comprehensive framework for industrial development, was lopsided as its guiding principle supported enormous expansion of the scope of the public sector. A natural outcome of the undue priority for public sector was the dampening of private initiative and enterprise. For obvious reasons, private investments were discouraged and this had long-lasting negative consequences for industrial growth.

India followed an open foreign investment policy and a relatively open trade policy until the late 1950s.A balance of payments crisis emerged in 1958 causing concerns regarding foreign exchange depletion. Consequently, there emerged a gradual tightening of trade and reduction in investment-licensing of new investments requiring imports of capital goods. The comprehensive import controls were maintained until 1966.

In the first three decades after independence (1950–80), India's average annual rate of growth of GDP- often referred to as the 'Hindu growth rate'- was a modest 3.5 percent. While agriculture was not neglected, the thrust of the first decade and a half was on capital goods— capital-intensive projects such as dams, power plants, and heavy industrialization—rather than consumer goods.

The first major shift in Indian economic strategy was in the mid-1960s. Agriculture was not given adequate priority during the second plan and the outlays were reduced. The strategy for agricultural development till then was reliance on institutional model i.e. land reforms, farm cooperatives etc. and not much importance was given to technocratic areas such as research and development, irrigation etc.

These institutional reforms were only modestly successful and the productivity increase in agriculture was meagre.
With continuous failures of monsoon, two severe and consecutive droughts struck India in 1966 and 1967. The agricultural sector recorded substantial negative growth and India faced a serious food problem. India had to depend on the United States for food aid under PL 480. A quantum jump in the

food grain production was the need of the hour. Increasing productivity in agriculture was given the highest priority.

This, in fact, kick-started a strategic change in the government's agricultural policies. The new wave of change relied less on the earlier efforts at institutional change and relied more on enhancing productivity of agriculture, especially of wheat.

A thorough restructuring of agricultural policy referred to as the 'green revolution' was initiated. The green revolution was materialised by innovative farm technologies, including high yielding seed varieties and intensive use of water, fertilizer and pesticides. The green revolution was successful in increasing agricultural productivity through technical progress and significantly increased food grain production enabling India to tide over the food problem.

While India drastically changed its agricultural policies, the government introduced extra stringent administrative controls on both trade and industrial licensing and launched a wave of nationalization. The government nationalized 14 banks in 1969 and then followed it up with nationalizing another 6 in 1980. The wide sweep of the interventionist policies that had come to exist in the 1960s had irreparable consequences in the next decade.

The economic performance during the period of 1965-81 is the worst in independent India's history. The decline in growth during this period is attributed mainly to decline in productivity. The license-raj, the autarchic policies that dominated the 1960s and 1970s, the external shocks such as three wars (in 1962, 1965, and 1971), major droughts (especially 1966 and 1967), and the oil shocks of 1973 and 1979contributed to the decelerated growth that lasted two decades. India being practically a closed economy missed out on the opportunities created by a rapidly growing world economy.

Many government policies aimed at equitable distribution of income and wealth effectively killed the incentive for creating wealth. Equity driven policies were also largely anti-growth. The Monopolies and Restrictive Trade Practices (MRTP) Act, 1969 was aimed at regulation of large firms which had relatively large market power. Several restrictions were placed on them in terms of licensing, capacity addition, mergers and acquisitions.
Thus, policies restricting the possibility of expansion of big business houses kept their entry away from nearly all but a few highly capital intensive sectors.

In 1967, the policy of reservation of many products for exclusive manufacture by the small scale sector was initiated with the objective of promotion of small scale industries. It was argued that this policy will encourage labour-intensive economic growth and allow redistribution of income by shifting incomes towards lower wage earners.

However, this policy excluded all big firms from labour intensive industries and India was not able to compete in the world market for these products. Stringent labour laws which were in place also discouraged starting of labour intensive industries in the organized sector.

There was a growing realisation among policymakers and industrialists that the prevailing strict regime is invariably counterproductive and that most of the controls and regulations had not delivered in the absence of adequate incentives and openness which are necessary conditions for sustained rapid growth.

10.3 The Era of Reforms

The seeds of early liberalisation and reforms were sown during the 1980s, especially after 1985. In early 1980s considerable efforts were initiated in different directions to restore reasonable price stability through a combination of tight monetary policy, fiscal moderation and a few structural reforms.

These initiatives, spanning 1981 to 1989, practically referred to as 'early liberalization' were specifically aimed at changing the prevailing thrust on 'inward- oriented' trade and investment practices. In fact, this liberalization is often referred to as 'reforms by stealth' to denote its ad hoc and not widely publicized nature.

Despite the fact that these efforts were not in the form of a comprehensive package (as the one in 1991) to reverse the centralised controls and the protectionist bias in policies, they started bearing fruits in the form of higher growth rate during the 1980s as compared with the previous three decades. The average annual growth rate of GDP during the sixth plan period (1980 –1985) and the seventh plan period (1985–1990) were 5.7 and 5.8 percent respectively.

The early reforms of 1980's broadly covered three areas, namely industry, trade and taxation. Simultaneously, the government also embarked on a policy of skillful exchange rate management. The prominent industrial policy initiatives during this period directed towards removing constraints on growth and creating a more dynamic industrial environment were:

- In 1985 delicensing of 25 broad categories of industries was done. This was later extended to many others
- The facility of 'broad-banding' was accorded for industry groups to allow flexibility and rapid changes in their product mix without going in for fresh licensing. In other words, the firms in the engineering industry were allowed to change their product mix within their existing capacity. For example, firms may switch production between different production lines such as trucks and car without a new licence
- To relax the hold of the licensing and capacity constraints on larger MRTP firms, in 1985–86, the asset limit above which firms were subject to MRTP regulations was raised from 20 crore to 100 crore.
- The multipoint excise duties were converted into a modified value-added (MODVAT) tax which significantly reduced the taxation on inputs and the associated distortions.
- Establishment of the Securities and Exchange Board of India (SEBI) as a non-statutory body on April 12, 1988 through a resolution of the Government of India
- The open general licence (OGL) list was steadily expanded. The number of capital goods items included in the OGL list expanded steadily reaching1,329 in April 1990.
- Several export incentives were introduced and expanded
- The exchange rate was set at a realistic level which helped expand exports and in turn reduced pressure on foreign exchange needed for imports
- Price and distribution controls on cement and aluminum were entirely abolished.
- Based on the real effective exchange rate (REER), the rupee was depreciated by about 30.0 per cent from 1985–86 to 1989–90. This reflects a considerable change in the official attitude towards exchange rate depreciation
- The budget for 1986 introduced policies of cutting taxes further, liberalising imports and reducing tariffs.

However, the growth performance of the economy was thwarted due to structural inadequacies and distortions. The private sector investments were inhibited due to reasons such as convoluted licensing policies, public sector reservations and excessive government controls.

Due to reservation of goods to small scale sector as well as excessive price and distribution controls, the private sector was virtually discouraged from making investments. The public sector which led the manufacturing and service sectors was plagued by inefficiency, government controls and bureaucratic procedures.

Despite the fact that they were of massive in size and enjoyed monopoly in their respective areas, their performance was far from satisfactory and yielded very low returns on investment.

The MRTP act had many restrictive conditions creating barriers for entry, diversification and expansion for large industrial houses. Import controls in the form of tariffs, quotas and quantitative restrictions ensured that foreign manufactures and components did not cross the borders and compete with the domestic industries.

Foreign investments and foreign competition were not allowed on grounds of affording protection to domestic industries. Briefly put, the rules and regulations which were aimed at promoting and regulating the economic activities became major hindrances to growth and development.

Though the reforms in 1980's were limited in scope and were without a clearly observable road map as compared to the New Economic Policy in 1990, they were instrumental in bringing confidence in the minds of politicians and policy makers regarding the efficacy of policy changes to produce sustained economic growth.

The belief that well-regulated competitive markets can ensure economic growth and also increase total welfare got fostered in the minds of policy makers. In other words, the idea that government intervention in markets need not always be accepted as 'the standard' and that markets should be given priority over government in the conduct of a good number of economic activities gained a broad acceptance. Thus, the liberalization in the 1980s served as the necessary foundation for the more universal and organized reforms of the 1990s.

10.4 The Economic Reforms of 1991

India embarked on a bold set of economic reforms in 1991 under the Narsimha Rao government.
The causes attributed to the immediate need for such a drastic change are:

1. The fiscal initiatives for enhanced economic growth in 1980s saw the government revenue expenditure consistently exceeding revenue receipts. The fiscal deficit was financed by huge amounts of domestic as well as external debt. The high level current expenditure proved clearly unsustainable and got manifested on extremely large fiscal deficits and adverse balance of payments.
2. Persistent huge deficits led to swelling public debt and a large proportion of government revenues had to be earmarked for interest payments.
3. The surge in oil prices triggered by the gulf war in 1990 and the consequent severe strain on a balance of payments.
4. The foreign exchange reserves touched the lowest point with a reserve of only $1.2 billion which was barely sufficient for two weeks of imports. This was the major context that triggered economic reforms.

5. Tightening of import restrictions to muster forex for essential imports resulted in reduction in industrial output.
6. India had to depend on external borrowing from the International Monetary Fund which in turn put forth stringent conditions in terms of corrective policy measures before additional drawings could be made.
7. The fragile political situation along with the crises in the economic front ballooned into what may be called a 'crisis of confidence'.

The year 1991 marked a paradigm shift in the Indian policy reforms. The nation which had embraced the 'socialist model', with the state playing an overriding role in the economy had the history of the government persistently intervening in the markets.

Collapse of the Soviet Union and the spectacular success of China, based on outward oriented policies were lessons for the Indian policy makers. The reforms instituted in 1991 aimed to move the economy toward greater market orientation and external openness.

The reforms, popularly known as liberalization, privatization and globalisation, spelt a major shift in economic philosophy and fundamental change in approach and had two major objectives:

1. reorientation of the economy from a centrally directed and highly controlled one to a 'market friendly' or market oriented economy.
2. macroeconomic stabilization by substantial reduction in fiscal deficit.

A detailed description of reform measures is beyond the scope of this unit. We shall now have a brief account of the major measures taken in 1991.

As we know, the momentum for reforms originated in the critical economic, fiscal and balance of payments crises. Therefore, the reform package was structured as a core package of mutually supportive reforms to address the balance of payment crisis and the structural rigidities. The policy paradigm focused on shifting from central direction to market orientation.

The policies can be broadly classified as :

1. stabilisation measures which were short term measures to address the problems of inflation and adverse balance of payment and
2. the structural reform measures which are long term and of continuing nature aimed at bringing in productivity and competitiveness by removing the structural rigidities in different sectors of the economy.

10.4.1 The Fiscal Reforms

The escalating deficit levels rendered the stabilisation efforts rather complicated. Bringing in fiscal discipline by reducing the fiscal deficit was vital because the crisis was caused by excess domestic demand, surge in imports and the widening of the current account deficit (CAD) which was to be financed by drawing down on reserves.

This was attempted by radical measures to augment revenues and to curtail government expenditure. Measures to this effect included:

1. Introduction of a stable and transparent tax structure,
2. Ensuring better tax compliance,
3. Thrust on curbing government expenditure

4. Reduction in subsidies and abolition of unnecessary subsidies
5. Disinvestment of part of government's equity holdings in select public sector undertakings and
6. Encouraging private sector participation.

In order to bring in fiscal discipline, it was essential to do away with the temptation to finance deficit thorough the easy path of money creation. Therefore, the government entered into a historic agreement with the Reserve Bank in September 1994 to bring down the fiscal deficit in a phased manner to nil by 1997–98.

10.4.2 Monetary and Financial Sector Reforms

Drastic monetary and financial sector reforms were introduced with the objective of making the financial system more efficient and transparent. The focus was mostly on reducing the burden of nonperforming assets on government banks, introducing and sustaining competition, and deregulating interest rates. These included many measures, important among them are:

1. Interest rate liberalization and reduction in controls on banks by the Reserve Bank of India in respect of interest rates chargeable on loans and payable on deposits.
2. Opening of new private sector banks and facilitating greater competition among public sector, private sector and foreign banks and simultaneously removal of administrative constraints that reduced efficiency
3. Reduction in reserve requirements namely, statutory liquidity ratio (SLR) and cash reserve ratio (CRR) in line with the recommendations of the Narasimham Committee Report, 1991.
4. Liberalisation of bank branch licensing policy and granting of freedom to banks in respect of opening, relocating or closure of branches
5. Prudential norms of accounting in respect of classification of assets, disclosure of income and provisions for bad debt were introduced in tune with the Narasimham Committee recommendations to ensure that the books of commercial banks reflect the accurate and truthful picture of their financial position.

10.4.3 Reforms in Capital Markets

The Securities and Exchange Board of India (SEBI) which was set up in 1988 was given statutory recognition in 1992. SEBI has been mandated as an independent regulator of the capital market so as to create a transparent environment which would facilitate mobilization of adequate resources and their efficient allocation.

10.4.4 The 'New Industrial Policy'

The 'New Industrial Policy' announced by the government on 24 July 1991 sought to substantially deregulate industry so as to promote growth of a more efficient and competitive industrial economy. In order to provide greater competitive stimulus to the domestic industry, a series of reforms were introduced

1. The New Economic Policy put an end to the 'License Raj' by removing licensing restrictions for all industries except for 18 that 'related to security and strategic concerns, social reasons, problems related to safety and overriding environmental issues'.

   Consequently, 80 percent of the industry was taken out of the licensing framework.

This is subsequently reduced to 5, namely, arms and ammunition, atomic substances, narcotic drugs and hazardous chemicals, distillation and brewing of alcoholic drinks and cigarettes and cigars as these have severe implications on health, safety, and environment.

2. Public sector was limited to eight sectors based on security and strategic grounds. Subsequently only two items remained – railway transport and atomic energy
3. The Monopolies and Restrictive Trade Practices (MRTP) Act was restructured and the provisions relating to merger, amalgamation, and takeover were repealed. This has eliminated the need for pre-entry scrutiny of investment decisions and prior approval for large companies for capacity expansion or diversification.
4. Many goods produced by small-scale industries have been de reserved enabling entry of large scale industries.
5. The policy ended the public sector monopoly in many sectors The number of areas reserved for public sector was narrowed down to ensure liberal participation by the private sector. Only eight industries which are of importance due to strategic and security concerns were reserved for the public sector. The changes continued and we find that now the industries reserved for the public sector are only a part of atomic energy generation and some core activities in railway transport.
6. Foreign investment was also liberalised. The concept of automatic approval was introduced for foreign direct investments up to 51 percent which was later extended to nearly all industries except the reserved ones. FDI is prohibited only in four sectors viz. retail trade, atomic energy, lottery business and betting and gambling.
7. External trade was further liberalised by substituting 'the positive list approach' of listing license-free items on the OGL list with the negative list approach. The policy did away with import licensing on all but a handful of intermediate and capital goods. The consumer goods which remained under licensing was made free 10 years later. Today, except for a handful of goods disallowed on health, environmental and safety grounds, and few others such as edible oil, fertilizer and petroleum products all goods can be imported
8. In 1990-91, the highest tariff rate was 355%, The top tariff rate was brought down to 85% in 1993-94 and to 50% in 1995-96 and by 2007-08, it has come down to 10% with some exceptions such as automobile at 100%
9. Rupee was devalued by 18% against the dollar. From 1994 onwards, all current account transactions including business, education, medical and foreign travel were permitted at market exchange rate and rupee became officially convertible on current account
10. The disinvestment of government holdings of equity share capital of public sector enterprises was a very bold step. The hitherto constrained public sector units were provided with greater autonomy in decision making and opportunity for professional management for ensuring reasonable returns. The budgetary support to public sector was progressively reduced.

10.4.5 Trade Policy Reforms

The trade policy reforms aimed at:

- dismantling of quantitative restrictions on imports and exports
- focusing on a more outward oriented regime with phased reduction and simplification of tariffs, and
- removal of licensing procedures for imports.

A number of export incentives were continued and new ones were initiated for boosting exports. Export duties were removed to increase the competitive position of Indian goods in the international markets.

In 1991, India still had a fixed exchange rate system, under which the rupee was pegged to the value of a basket of currencies of major trading partners. In July 1991 the Indian government devalued the rupee by between 18 and 19 percent.

In March 1992 the government decided to establish a dual exchange rate regime. The government allowed importers to pay for some imports with foreign exchange valued at free-market rates and other imports could be purchased with foreign exchange purchased at a government- mandated rate

In March 1993 the government unified the exchange rate and allowed, for the first time, the rupee to float. From 1993 onwards, India has followed a managed floating exchange rate system.

India has witnessed vast changes over the last 31 years of economic reforms. Changes enumerated below are only broad observations and are in no way comprehensive.

- India has increasingly integrated its economy with the global economy.
- India has progressively moved towards a market oriented economy, with a sizeable reduction in government's market intervention and controls
- There is an unprecedented growth of private sector investment and initiatives
- A number of sectors such as auto components, telecommunications, software, pharmaceuticals, biotechnology, and professional services have achieved very high levels of international competitiveness
- Easing of trade controls has enabled easier access to foreign technology, inputs, know- how and finance
- Stable foreign direct investment inflows and substantial foreign portfolio investments
- India enjoys a solid cushion of foreign exchange reserves close to eight months of import cover. India has one of the largest holdings of international reserves in the world.
- Robust demand for information technology and financial services has kept the services trade surplus high at around 3.7 percent of GDP
- Pressure on the Indian rupee is lower compared to other emerging market economies (EMEs)
- Increased incomes, large domestic market and high levels of aggregate demand sustains the economy.
- India is better placed than most of the emerging market economies to deal with global headwinds
- Poverty has reduced substantially
- Reforms led to increased competition in sectors like banking, insurance and other financial services leading to greater customer choice and increased efficiency. It has also led to increased investment and growth of private players in these sectors.
- Infrastructure sectors have achieved phenomenal growth
- Value-added share of agriculture and allied activities has declined steadily over the past four decades.
- India's financial sector has also deepened considerably due to increased financial sector liberalisation.

However, the country is constrained by high levels of fiscal deficit, inflation and a high level of debt as a share of GDP at 86 percent of GDP in FY21/22. Among the emerging market and developing economies (EMDEs), India's debt is higher than their average of 64.5% for 2022(IMF).

10.5 GDP growth rates post 1991 reforms

As we are aware, GDP growth rate is regarded as the most reliable indicator of economic growth. The following table and graphical presentation present data on GDP growth rate post 1991 reforms.

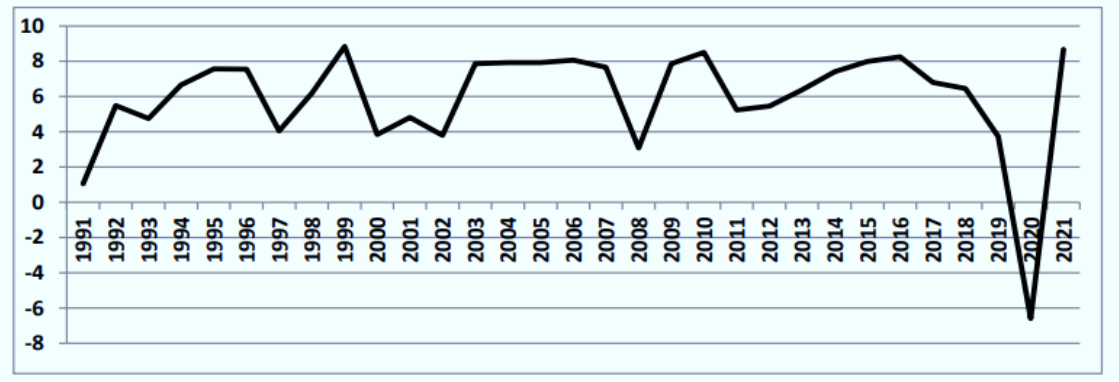
Table

GDP Growth (Annual %) – India from 1991 to 2021

Year	,GDP Growth (Annual %)	Year	GDP Growth (Annual %)
1991	1.056831	2006	8.060733
1992	5.482396	2007	7.660815
1993	4.750776	2008	3.086698
1994	6.658924	2009	7.861889
1995	7.574492	2010	8.497585
1996	7.549522	2011	5.241315
1997	4.049821	2012	5.456389
1998	6.184416	2013	6.386106
1999	8.845756	2014	7.410228
2000	3.840991	2015	7.996254
2001	4.823966	2016	8.256306
2002	3.803975	2017	6.795383
2003	7.860381	2018	6.453851
2004	7.922937	2019	3.737919
2005	7.923431	2020	-6.59608
		2021	8.681229

Figure

GDP Growth (Annual %) – India from 1991 to 2021



10.6 NITI AAYOG: A Bold Step for Transforming India

For nearly sixty four years, the Planning Commission of India - a powerful advocate of public investment-led development - was one of the most important institutions within India's central government.

The new ideologies of the neoliberal era with their centre of attention on market orientation and shrinking roles of the government and the collapse of the planning system called for a change in the nature, composition and scope of institutions of governance.

On 1st January 2015, the apex policy-making body namely Planning Commission, was replaced by the National Institution for Transforming India (NITI) Aayog. The major objective of such a move was to 'spur innovative thinking by objective 'experts' and promote 'co-operative federalism' by enhancing the voice and influence of the states'. NITI Aayog is expected to serve as a 'Think Tank' of the government. [and] a 'directional and policy dynamo'.

NITI Ayog will work towards the following objectives*:

1. To evolve a shared vision of national development priorities, sectors and strategies with the active involvement of states.
2. To foster cooperative federalism through structured support initiatives and mechanisms with the states on a continuous basis, recognizing that strong states make a strong nation.
3. To develop mechanisms to formulate credible plans at the village level and aggregate these progressively at higher levels of government.
4. To ensure, on areas that are specifically referred to it, that the interests of national security are incorporated in economic strategy and policy.
5. To pay special attention to the sections of our society that may be at risk of not benefiting adequately from economic progress.
6. To design strategic and long-term policy and programme frameworks and initiatives, and monitor their progress and their efficacy
7. To provide advice and encourage partnerships between key stakeholders and national and international like-minded think tanks, as well as educational and policy research institutions.
8. To create a knowledge, innovation and entrepreneurial support system through a collaborative community of national and international experts, practitioners and other partners.
9. To offer a platform for the resolution of inter-sectoral and inter departmental issues in order to accelerate the implementation of the development agenda.
10. To maintain a state-of-the-art resource centre, be a repository of research on good governance and best practices in sustainable and equitable development as well as help their dissemination to stake-holders.
11. To actively monitor and evaluate the implementation of programmes and initiatives, including the identification of the needed resources so as to strengthen the probability of success and scope of delivery.
12. To focus on technology up gradation and capacity building for implementation of programmes and initiatives.
13. To undertake other activities as may be necessary in order to further the execution of the national development agenda, and the objectives mentioned above. *NITI Aaayog

    https://niti.gov.in/objectives-and-features

The key initiatives of NITI Aayog are:

1. 'Life' which envisions replacing the prevalent 'use-and-dispose' economy
2. The National Data and Analytics Platform (NDAP) facilitates and improves access to Indian government data
3. Shoonya campaign aims to improve air quality in India by accelerating the deployment of electric vehicles

4. E-Amrit is a one-stop destination for all information on electric vehicles
5. India Policy Insights (IPI)
6. 'Methanol Economy' programme is aimed at reducing India's oil import bill, greenhouse gas (GHG) emissions, and converting coal reserves and municipal solid waste into methanol, and
7. 'Transforming India's Gold Market' constituted by NITI Aayog to recommend measures for tapping into the potential of the sector and provide a stimulus to exports and economic growth

There are arguments put forth by experts about the weaknesses of the system. They argue that NITI has a limited role; it does not produce national plans, control expenditures, or review state plans. The major shortcoming of NITI is its exclusion from the budgeting process.

It also lacks autonomy and balance of power within the policy making apparatus of the central government. The termination of the Planning Commission has strengthened the hand of the Ministry of Finance, with its 'fixation on near-term macroeconomic stability and the natural instinct to limit expenditure'.

But NITI lacks the independence and power to perform as a 'counterweight' to act as a "voice of development" concerned with inequities.

10.7 The Current State of The Indian Economy: A Brief Overview

On account of the enormity of the economic phenomena and the dynamic nature of economic variables, it is not possible to have an up-to-date and comprehensive documentation on the current state of the economy.
Given the constraints of the unit, an attempt is made in the following sections to present the broad nature of the present day Indian economy based on the three sectors namely, primary, secondary and tertiary.

10.7.1 The Primary Sector

Agriculture, with its allied sectors, is indisputably the largest source of livelihood in India. Till the end of 1960's, India was a food deficient nation and depended on imports. India has emerged as the world's largest producer of milk, pulses, jute and spices.

India has the largest area planted under wheat, rice and cotton. It is the second-largest producer of fruits, vegetables, tea, farmed fish, cotton, sugarcane, wheat, rice, cotton, and sugar. Indian food and grocery market is the world's sixth largest, with retail contributing 70% of the sales. India has the world's largest cattle herd (buffaloes).

The Indian livestock sector attained a record growth of 6.6 per cent during the last decade (2010-19) emerging as a major producer of milk, egg and meat in the world. India grows large varieties of cash crops of which cotton, jute and sugarcane are prominent. Although the share of agriculture has been declining in overall gross value added (GVA) of India, it continues to grow in absolute terms.

According to the latest estimates, 47 per cent of India's population is directly dependent on agriculture for living. It also contributes a significant figure to the Gross Domestic Product (GDP). Gross Value Added by the agriculture and allied sector was 18.8% in 2021 -22 (until 31 January, 2022).

The index numbers of agricultural production in 2021-22 (base: triennium ending 2007- 08=100) for categories namely, all crops, food-grains, cereals, wheat and coarse cereals was above 140; and that of rice and pulses was 138.7 and 196.2 respectively.

For non- food grains, it was 142.9. These figures show sustained increase in agricultural output. Food grains production has reached 315.7 million tonnes in 2021-22. Private investment in agriculture has increased to 9.3% in 2020-21.

As per the economic survey, 2022-23, agriculture remained robust, recording a growth of 3.5 per cent in 2022-23, driven by buoyant rabi sowing and allied activities. The performance of the agriculture and allied sectors has been buoyant over the past several years, much of which is on account of the measures taken by the government to:

- augment crop and livestock productivity,
- ensure certainty of returns to the farmers through price support (The Minimum Support Price (MSP) of all 23 mandated crops is fixed at 1.5 times of all India weighted average cost of production)
- promote crop diversification,
- improve market infrastructure through the impetus provided for the setting up of farmer-producer organisations and
- promotion of investment in infrastructure facilities through the Agriculture Infrastructure Fund.

India has achieved a remarkable shift from a food deficient and import dependent nation during the early nineteen sixties to a food exporting nation. India is among the top ten exporters of agricultural products in the world.

Export of agricultural and allied products has witnessed significant increase during the last few years and touched an all-time peak of 374611 crore during the last one year. Exports of agricultural and processed food products rose by 25 percent within six months of the current financial year 2022-23 (April-September) in comparison to the corresponding period in 2021-22.

Agricultural and Processed Food Export Development Authority (APEDA) is entrusted with the responsibility of export promotion of agri-products.

A number of liberalization measures are adopted by the government. The Government of India has allowed 100% FDI in marketing of food products and in food product E-commerce under the automatic route. Considering the diverse needs of the agricultural sector and the larger farming community, a large number of interventions are undertaken by different governments. A few such recent measures are:

- Income support to farmers through PM KISAN
- Fixing of Minimum Support Price (MSP) at one-and-a half times the cost of production
- Institutional credit for agriculture sector at concessional rates
- Launch of the National Mission for Edible Oils
- Pradhan Mantri Fasal BimaYojana (PMFBY) – a novel insurance scheme for financial support to farmers suffering crop loss/damage
- Mission for Integrated Development of Horticulture (MIDH) for the holistic growth of the horticulture sector
- Provision of Soil Health Cards
- Paramparagat Krishi Vikas Yojana (PKVY) supporting and promoting organic farming, and improvement of soil health.

- Agri Infrastructure Fund, a medium / long term debt financing facility for investment in viable projects for post-harvest management Infrastructure and community farming assets
- Promotion of Farmer Producer Organisations (FPOs) to ensure better income for the producers through an organization of their own.
- Per Drop More Crop (PDMC) scheme to increase water use efficiency at the farm level
- Setting up of Micro Irrigation Fund
- Initiatives towards agricultural mechanization
- Setting up of E-NAM -a pan-India electronic trading portal which networks the existing APMC mandis to create a unified national market for agricultural commodities.
- Introduction of Kisan Rail for improvement in farm produce logistics, and
- Creation of a Start-up Eco system in agriculture and allied sectors
- Despite phenomenal increase in output of both food crops and commercial crops, Indian agriculture faces many issues such as:
- Indian agriculture is dominated by small and medium farmers. Small and fragmented landholdings, low farm productivity and subsistence farming result in very little marketable surplus and the consequent lower income levels of the agriculturists. These also reduce their ability to participate in the domestic as well as export market.
- Indian agriculture is resource intensive, cereal centric and regionally biased. There is Increasing stress on water resources and soil fertility. Unscientific and wasteful agricultural practices lead to desertification and land degradation in many parts of the country.
- Inadequate agro-processing infrastructure and failure to build competitive value chains from producers to urban centers and export markets
- Sluggish agricultural diversification to higher-value commodities
- Inadequate adoption of environmentally sustainable and climate resistant new farm technology
- Poor adoption of new agricultural technologies
- Lopsided marketing practices and ineffective credit delivery
- Complexities associated with adaptation to climate change disturbances
- High food price volatility
- Heavy dependence on monsoons and loss of crops and livelihood due to vagaries of nature
- Issues related to marketing and warehousing of agricultural products
- Inability to tap the full export potential of primary as well as value added products
- Inability to effectively channelize huge surpluses in some commodities to alternative profitable destinations
- Inadequate post-harvest infrastructure and management practices
- Incidence of poverty and malnutrition

10.7.2 The Secondary Sector

The Indian industry holds a significant position in the Indian economy contributing about 30 percent of total gross value added in the country and employing over 12.1 crores of people. The industrial sector in India broadly comprises of manufacturing, heavy industries, fertilizers, pharmaceuticals, chemicals and petrochemicals, oil and natural gas, food processing, mining, defence products, textiles, retail, micro, small & medium enterprises, cottage industries and tourism.

The share of informal sector in the economy is more than 50% of GVA. Rapid industrial growth of domestic industries and diversification of industrial structure are essential elements for sustainable economic growth. The development of a robust manufacturing sector is a key priority of the Indian Government.

A detailed discussion on industrial development is beyond the scope of this unit. Starting with the industrial growth figures, we shall briefly touch upon the general aspects related to industries. In India, industrial production measures the output of businesses integrated in industrial sector of the economy.

Manufacturing is the most important sector and accounts for 78 percent of total production. The manufacturing GVA at current prices was estimated at US$ 77.47 billion in the third quarter of financial year 2021-22 and has contributed around 16.3% to the nominal GVA during the past ten years.

In 2022- 23 (until September 2022), the combined index of eight core industries* stood at 142.8 driven by the production of coal, refinery products, fertilizers, steel, electricity and cement industries. In Jan 31, 2023 the Manufacturing Purchasing Managers' Index (PMI) in India stood at 55.4. India's rank in the Global Innovation Index (GII) improved to 40th in 2022 from 81st in 2015.

The Department for Promotion of Industry and Internal Trade (DPIIT) has a role in the formulation and implementation of industrial policy and strategies for industrial development in conformity with the development needs and national objectives. Ever since independence, many innovative schemes are undertaken by different governments from time to time to boost industrial performance.
Some of the policies are presented below:

- Introduction of goods and services tax (GST) on 1 July 2017 as a single domestic indirect tax law for the entire country replacing many indirect taxes in India such as the excise duty, VAT, services tax, etc.
- Reduction of corporate tax to domestic companies giving an option to pay income-tax at the rate of 22% subject to condition that they will not avail any exemption/incentive.
- 'Make in India' is a 'Vocal for Local' initiative launched in 2014 to facilitate investment, foster innovation, build excellent infrastructure and make India a hub for manufacturing, design and innovation. Make in India 2.0' is now focusing on 27 sectors, which include 15 manufacturing sectors and 12 service sectors.
- 'Ease of Doing Business' with key focus areas as simplification of procedures, rationalization of legal provisions, digitization of government processes, and decriminalization of minor, technical or procedural defaults. India ranks 63rdin the World Bank's annual Doing Business Report (DBR), 2020 as against 77thrank in 2019 registering a jump of 14 ranks.
- The National Single Window System is a one-stop-shop for investor related approvals and services in the country and aims to provide continuous facilitation and support to investors.

- PM Gati Shakti National Master Plan to facilitate data-based decisions related to integrated planning of multimodal infrastructure, thereby reducing logistics cost.
- National Logistics Policy (NLP) launched in September 2022, aims to lower the cost of logistics and make it at par with other developed countries.
- Keeping in view India's vision of becoming 'Atmanirbhar', the Production Linked Incentive (PLI) Scheme was initiated in March 2020 for 14 key sectors to enhance India's manufacturing capabilities and export competitiveness. PLI Scheme is now extended for white goods (air conditioners and led lights).
- Industrial Corridor Development Programme: Greenfield Industrial regions/areas/nodes with sustainable infrastructure and to make available 'plug and play' infrastructure at the plot level.
- FAME-India Scheme (Faster Adoption and Manufacturing of Hybrid and Electric Vehicles) to promote manufacturing of electric and hybrid vehicle technology and to ensure sustainable growth of the same.
- 'Udyami Bharat' aims at the empowerment of Micro Small and Medium Enterprises (MSMEs).
- PM Mega Integrated Textile Region and Apparel (PM MITRA): to ensure world-class industrial infrastructure which would attract cutting age technology and boost FDI and local investment in the textiles sector.
- Opening up for global investments: To make India a more attractive investment destination, the government has implemented several radical and transformative FDI reforms across sectors such as defence, pension, e-commerce activities etc.
- 100 per cent FDI under automatic route is permitted for the sale of coal, and coal mining activities, including associated processing infrastructure and for insurance intermediaries.
- Foreign Investment Promotion Board (FIPB) was abolished in May 2017, and a new regime namely Foreign Investment Facilitation Portal (FIF) has been put in place. Under the new regime, the process for granting FDI approvals has been simplified. 853 FDI proposals were disposed off in the last 5 years. FDI has increased jumped by 39% since FIF came into being.
- Remission of Duties and Taxes on Export Products (RoDTEP) 2021 formed to replace the existing MEIS (Merchandise Exports from India Scheme) to boost exports. It provides for rebate of all hidden central, state, and local duties/taxes/levies on the goods exported which have not been refunded under any other existing scheme.
- Initiatives towards fostering innovation include incubation, handholding, funding, industry-academia partnership and mentorship and strengthening of IPR regime.
- National Logistics Policy (NLP) is comprehensive policy framework for the Logistics Sector.
- Start-up India Programme acts as the facilitator for ideas and innovation in the country. India's rank in the Global Innovation Index (GII) has improved from 81st in 2015 to 40th in 2022.
- Public Procurement (Preference to Make in India) Order, 2017gives preference to locally manufactured goods, works and services in public procurement thereby giving boost to industrial growth.
- The Emergency Credit Line Guarantee Scheme (ECLGS)is a fully guaranteed emergency credit line to monitor lending institutions.

India is gearing up for the fourth industrial revolution or Industry 4.0 in which manufacturing transformation needs to integrate new technologies such as cloud computing, IoT, machine learning, and artificial intelligence (AI).

The National Manufacturing Policy which aims to increase the share of manufacturing in GDP to 25 percent by 2025 is a step in this direction.

India is an attractive hub for foreign investments in the manufacturing sector. Over the last few years, FDI equity inflows in the manufacturing sector have been progressively rising. India continues to open up its sectors to global investors by raising FDI limits and removing regulatory barriers in addition to developing infrastructure and improving the business environment.

According to the Department for Promotion of Industry and Internal Trade (DPIIT), India received a total foreign direct investment (FDI) inflow of US$ 58.77 billion in 2021-22.

There are many challenges to the industrial sector; a few of these are enumerated below:

- Shortage of efficient infrastructure and manpower and consequent reduced factor productivity.
- Reliance on imports, exchange rate volatility and associated time and cost overruns
- The MSME sector is relatively less favorably placed in terms of credit availability.
- Industrial locations established without reference to cost-effective points tend to experience unsustainable cost structure.
- Heavy losses, inefficiencies, lower productivity and unsustainable returns plaguing public sector industries.
- Strained labor-management relations and loss of man hours.
- Lower export competitiveness, slowing external demand and imposition of non tariff barriers by other countries.
- Global supply chain disruptions and uncertainties.
- Inflation and associated macro economic developments leading to input cost escalations and lower demand.
- Global slowdown and related negative sentiments affecting investment.
- Aggressive tightening of monetary policy and increases in cost of credit.
- High and increasing fuel prices, and
- Mounting presence of informal sector.

10.7.3 The Tertiary Sector

A remarkable feature of the post reform Indian economy is the overarching role of the services sector in generating growth of income and employment. Unlike the usual economic development process of nations where economic growth has led to a shift from agriculture to industries, or from the primary sector to the secondary sector, India has the unique experience of bypassing the secondary sector in the growth trajectory by a shift from agriculture to the services sector.
India's services sector covers a wide variety of activities. (Refer Box 2 Below)

The broad classification of services as per the National Industrial Classification, 2008	
1.	Wholesale and retail trade and repair of vehicles
2.	Transportation and storage
3.	Accommodation and food service activities

4.	Information and communication
5.	Financial and insurance activities
6.	Real estate activities
7.	Professional, scientific and technical activities
8.	Administrative and support services
9.	Public administration, defence and compulsory social security
10.	Education
11.	Human health and social work activities
12.	Arts, entertainments and recreation
13.	Other service activities
14.	Activities of households as employers, undifferentiated goods and services producing activities of households for own use
15.	Activities of extra territorial organizations and bodies

The service sector refers to the industry producing intangible goods viz. services as output. The services sector is the largest sector of India and accounts for 53.89% of total India's GVA. The Gross Value Added (GVA) at current prices for the services sector is estimated at ₹ 96.54 lakh crore in 2020-21.

The service sector is the fastest growing sector in India and has the highest labour productivity. Both domestic and global factors influence the growth of the services sector. The exceptionally rapid expansion of knowledge-based services such as professional and technical services has been responsible for the faster growth of the services sector.

The production and consumption of information-intensive service activities such as computing, accounting, inventory management, quality control, personnel administration, marketing, advertising and legal services has increased manifold due to application of state- of the- art information technology. Services sector growth can also complement growth in the manufacturing sector. The start-ups which have grown remarkably over the last few years mostly belong to the services sector.

India is among the top 10 World Trade Organization (WTO) members in service exports and imports. India's services exports at US$ 27.0 billion recorded robust growth in November 2022 due to software, business, and travel services.

While exports from all other sectors were adversely affected, India's services exports have remained resilient during the Covid-19 pandemic. The reasons are the higher demand for digital support and need for digital infrastructure modernization.

The Indian services sector is the largest recipient of FDI inflows. FDI equity inflows into the services sector accounted for more than 60 per cent of the total FDI equity inflows into India.

The World Investment Report 2022 of UNCTAD places India as the seventh largest recipient of FDI in the top 20 host countries in 2021. In 2021-22, India received the highest-ever FDI inflows of US$ 84.8 billion including US$ 7.1 billion FDI equity inflows in the services sector.

To ensure the liberalisation of investment in various industries, the government has permitted 100 per cent foreign participation in telecommunication services through the Automatic Route including all services and infrastructure providers.

The FDI ceiling in insurance companies was also raised from 49 to 74 per cent. Measures undertaken by the Government, such as the launch of the National Single-Window system and enhancement in the FDI ceiling through the automatic route, have played a significant role in facilitating investment.

10.8 Conclusion

The India Development Update (IDU) of the World Bank published in November 2022, observes that India had to face an unusually challenging external environment following the Russia-Ukraine war, increased crude oil and commodity prices, persistent global supply disruptions, tighter financial conditions and high domestic inflationary pressures.

Despite all these, the real GDP of India grew by 6.3 percent in July-September of 2022-23 driven by strong private consumption and investment. The report observes that India's economy is relatively more insulated from global spillovers than other emerging markets and is less exposed to international trade flows on account of reliance on its large domestic market. As such, compared to other emerging economies, India is much more resilient to withstand adversities in the global arena.

TEST YOUR KNOWLEDGE

I. Multiple Choice Type Questions

1. The Indian industry stagnated under the colonial rule because
   (a) *Indians were keen on building huge structures and monuments only*
   (b) *Deterioration was caused by high prices of inputs due to draught*
   (c) *The Indian manufactures could not compete with the imports of cheap machine made goods*
   (d) *None of the above*

2. The first wave of liberalization starts in India
   (a) *In 1951* (b) *In 1980's* (c) *In 1990* (d) *In 1966*

3. The sequence of growth and structural change in Indian economy is characterized by
   (a) *The historical pattern of prominence of sectors as agriculture, industry, services*
   (b) *The historical pattern of prominence of sectors as industry, services, agriculture*
   (c) *Unique experience of the sequence as agriculture, services, industry*
   (d) *All the above are correct*

4. Merchandise Exports from India Scheme was replaced by -
   (a) *Remission of Duties and Taxes on Export Products (RoDTEP) in 2021*
   (b) *National Logistics Policy (NLP) in 2020*
   (c) *Remission of Duties and Taxes on Export Products (RoDTEP) in 2019*
   (d) *None of the above*

5. The Foreign Investment Promotion Board (FIPB)
   (a) *a government entity through which inward investment proposals were routed to obtain required government approvals*
   (b) *no more exists as the same is replaced by a new regime namely Foreign Investment Facilitation Portal*
   (c) *no more exists as all inward investments are through automatic route and need no approval*
   (d) *is the body which connects different ministries in respect of foreign portfolio investments*

6. FAME-India Scheme aims to
   (a) *Enhance faster industrialization through private participation*
   (b) *to promote manufacturing of electric and hybrid vehicle technology*
   (c) *to spread India's fame among its trading partners*
   (d) *None of the above*

7. In terms of Ease of Doing Business in 2020 India ranks
   (a) *63* (b) *77* (c) *45* (d) *None of the above*

8. E-NAM is -
   (a) *An electronic name card given to citizens of India*
   (b) *National Agriculture Market with the objective of creating a unified national market for agricultural commodities.*
   (c) *a pan-India electronic trading portal which networks the existing APMC mandis*
   (d) *(b) and c) above*

9. Which of the following is not a policy reform included in the new economic policy of 1991 -
   (a) *removing licensing requirements for all industries*
   (b) *Foreign investment was liberalized*
   (c) *Liberalisation of international trade*
   (d) *The disinvestment of government holdings of equity share capital of public sector enterprises*

10. Imports of foreign goods and entry of foreign investments were restricted in India because -
   (a) *The government wanted people to follow the policy of' Be Indian; Buy Indian'*
   (b) *Because foreign goods were costly and meant loss of precious foreign exchange*
   (c) *Government policy was directed towards protection of domestic industries from foreign competition*
   (d) *Government wanted to preserve Indian culture and to avoid influence of foreign culture*

11. The 'Hindu growth rate' is a term used to refer to -
   (a) *the high rate of growth achieved after the new economic policy of 1991*
   (b) *the low rate of economic growth of India from the 1950s to the 1980s, which averaged around 3.5 per cent per year*
   (c) *the low growth of the economy during British period marked by an average of 3.5 percent*
   (d) *the growth rate of the country because India is referred to as 'Hindustan'*

12. In the context of the new economic policy of 1991, the term 'disinvestment' stands for -
   (a) *A policy whereby government investments are reduced to correct fiscal deficit*
   (b) *The policy of sale of portion of the government shareholding of a public sector enterprise*
   (c) *The policy of public partnership in private enterprise*
   (d) *A policy of opening up government monopoly to the privates sector*

13. The objective of introducing Monopolies and Restrictive Trade Practices Act 1969 was -
   (a) *to ensure that the operation of the economic system does not result in the concentration of economic power in hands of a few*
   (b) *to provide for the control of monopolies*
   (c) *to prohibit monopolistic and restrictive trade practice*
   (d) *all the above*

14. Which one of the following is a feature of green revolution -
   (a) *use of soil friendly green manure to preserve fertility of soil*
   (b) *grow more crops by redistributing land to landless people*
   (c) *High yielding varieties of seeds and scientific cultivation*
   (d) *Diversification to horticulture*

15. The strategy of agricultural development in India before green revolution was -
   (a) *High yielding varieties of seeds and chemical fertilizers to boost productivity*
   (b) *Institutional reforms such as land reforms*
   (c) *Technological up gradation of agriculture*
   (d) *All the above*

16. The Industrial Policy Resolution (1948) aimed at -
   (a) *Market oriented economic reforms and opening up of economy*
   (b) *A shift from state led industrialization to private sector led industrialization*
   (c) *an expanded role for the public sector and licensing to the private sector*
   (d) *an expanded role of private sector a limited role of public sector*

17. The new economic policy of 1991 manifest in -
    (a) *State led industrialization and import substitution*
    (b) *Rethinking the role of markets versus the state*
    (c) *Emphasized the role of good governance*
    (d) *Bringing about reduction in poverty and redistributive justice*

18. The post independence economic policy was rooted in -
    (a) *A capitalist mode of production with heavy industrialization*
    (b) *social and economic redistribution and industrialization directed by the state*
    (c) *social and economic redistribution through private sector initiatives*
    (d) *Industrialization led by private entrepreneurs and redistribution by state*

PEQ – JUNE – 2024

19. E Amrit is related to:
    (a) *One stop destination for all information on electric vehicles*
    (b) *Facilitates and improves access to Indian government data*
    (c) *Improves air quality in India by accelerating deployment of electronic vehicles*
    (d) *Provides a stimulus to exports and economic growth* *(1 mark)*

20. Which revolution was materialised by innovative farm technologies including high yielding seed varieties and intensive use of water, fertilizers and pesticides?
    (a) *Green Revolution* (b) *Kisan Revolution*
    (c) *Agriculture Revolution* (d) *Fasal Revolution* *(1 mark)*

21. PM Gati Shakti is related to:
    (a) *Empowerment of MSMEs*
    (b) *Data based decisions related to integrated planning of multimodal infrastructure thereby reducing logistic cost*
    (c) *Manufacturing of electric and hybrid vehicle technology*
    (d) *Automatic route for FDI* *(1 mark)*

22. Which of the following sectors maximum contribution to GVA of India?
    (a) *Primary sector* (b) *Secondary sector*
    (c) *Tertiary sector* (d) *All of above* *(1 mark)*

23. Full form of APEDA.
    (a) *Agricultural product export development agency*
    (b) *Agricultural produce economic development association*
    (c) *Animal product export development authority*
    (d) *Agricultural & processed food export development authority* *(1 mark)*

24. What does E NAM's aim to achieve?
    (a) *Increase farm productivity*
    (b) *Create unified national market for agricultural commodities*
    (c) *Provide financial support to farmers*
    (d) *Promote organic farming* *(1 mark)*

25. Which of the following causes is reducing ability of Indian agriculturalists to participate in domestic as well as export market?
(a) *Low farming productivity and subsistence farming*
(b) *Large amount of marketable surplus with farmers*
(c) *Dominated by big farmers and farm productivity*
(d) *Big landholdings by small farmers* *(1 mark)*

26. What is main cause attributed to immediate need for economic reforms in 1991?
(a) *Fiscal deficit was financed by huge amounts of domestic and external debts*
(b) *Forex reserves touched the highest point*
(c) *Persistent huge deficit led to declining public debt*
(d) *Government's revenue expenditure consistently lowering revenue receipts* *(1 mark)*

27. The Government of India has allowed 100% FDI in marketing of food products and in food product E-commerce under route.
(a) *Automatic* (b) *Green channel*
(c) *Priority route* (d) *Trade route (1 mark)*

28. To develop mechanism to formulate credible plans at village level and aggregate these progressively at higher level of government is an objective of:
(a) *Ministry of rural development* (b) *NITI Aayog*
(c) *Commission of policy and planning* (d) *Planning commission* *(1 mark)*

29. MRTP Act 1969, is aimed at regulation of which had relatively market power.
(a) *Large firms, large* (b) *Large firms, small*
(c) *Small firms, small* (d) *Small firms, large* *(1 mark)*

30. In the post-reform period, if the GDP growth rate in a particular year was 8.5%, and the average growth rate of the previous two years was 7.9% and 8.2%, what was the GDP growth rate of the year before the two previous years?
(a) *7.5%* (b) *7.8%* (c) *8.0%* (d) *8.3%* *(1 mark)*

PEQ – SEPTEMBER – 2024

31. On which date was the apex policy-making body, the Planning Commission, replaced by the National Institution for Transforming India (NITI) Aayog?
(a) *1st July, 2015* (b) *1st April, 2015*
(c) *1st January, 2015* (d) *1st July, 2016* *(1 mark)*

32. Which government scheme supports and promotes organic farming, as well as the improvement of soil health ?
(a) *Rashtriya Krishi Vikas Yojana (RKVY)*
(b) *National Rural Employment Guarantee Act (NREGA)*
(c) *Pradhan Mantri Fasal Bima Yojana (PMFBY)*
(d) *Paramparagat Krishi Vikas Yojana (PKVY)* *(1 mark)*

33. Which initiative aims at the empowerment of Micro Small and Medium Enterprises (MSMEs)?
(a) *Start-up India* (b) *Udyami Bharat*
(c) *Make in India* (d) *Digital India* *(1 mark)*

34. Transportation and storage, real estate service, public administration and education are the activities in India which are covered by:
(a) Secondary sector *(b) Primary sector*
(c) Tertiary sector *(d) Private sector* *(1 mark)*

35. Which of the following is a feature of green revolution?
(a) Use of soil friendly green manure to preserve fertility of soil
(b) Grow more crops by redistributing land to landless people
(c) Diversification to horticulture
(d) Use of high yielding varieties of seeds and scientific cultivation *(1 mark)*

36. On which date was the 'New Industrial Policy' announced by Government of India?
(a) 24th July, 1991 *(b) 15th August, 1947*
(c) 26th January, 1950 *(d) 5th September, 1992* *(1 mark)*

37. Which of the following statement is true in relation to the object of The National Data and Analytics Platform (NDAP)?
(a) To facilitate and improve access to Indian government data
(b) To ensure data security
(c) To convert all government data in electronic form
(c) To formulate a national cyber security law *(1 mark)*

38. The India Development Update (IDU) report published in November 2022 is published by:
(a) International Monetary Fund (IMF)
(b) United Nations Development Programme (UNDP)
(c) World Bank
(d) Asian Development Bank (ADB) *(1 mark)*

39. What is the full form of DPIIT?
(a) Department for Promotion of Industrial and Internal Trade
(b) Department for Promotion of Industry and International Trade
(c) Directorate for Promotion of Industry and Internal Trade
(d) Department for Promotion of Industry and Internal Trade *(1 mark)*

40. Which of the following is not the aim of 'Methanol Economy' programme?
(a) Reducing Greenhouse gas (GHG) emissions
(b) Converting coal reserves and municipal solid waste into methanol
(c) Reducing India's oil import bill
(d) Converting industrial waste into methanol *(1 mark)*

PEQ – JANUARY – 2025

41. _____ facilitates and improves access to Indian Government data.
(a) E-Amrit *(b) E-NAM* *(c) NDAP* *(d) MIDH* *(1 mark)*

42. Which scheme is aimed at promoting manufacture of electric and hybrid vehicle technology and to ensure sustainable growth for the same?
(a) FAME India *(b) E-Amrit* *(c) FIPB* *(d) PDMC* *(1 mark)*

43. Which Act was initially aimed for regulation of large firms which had relatively large market power?
(a) RBI Act *(b) FEMA* *(c) RERA* *(d) MRTP Act, 1969*
(1 mark)

44. From which year onwards, India followed the managed floating exchange rate system?
(a) 1990 (b) 1991 (c) 1995 (d) 1993 (1 mark)

45. Minimum Support Price (MSP) is fixed by the Government of India at _____ of the cost of production.
(a) Two and a half times (b) Half
(c) One and a half time (d) Twice (1 mark)

46. Which policy was adopted to ensure world class industrial infrastructure which would attract cutting edge technology and boost FDI and local investment in the textile sector?
(a) PM-MITRA (b) PM Gati Shakti National Master Plan
(c) National Logistic Policy (d) Production Linked Incentive (PLI) Scheme (1 mark)

47. During the British period modern industrial sector saw lopsided growth with the dominance of _____ industries.
(a) Wool and cotton (b) Nylon and silk
(c) Cotton and jute (d) Silk and cotton (1 mark)

48. Statutory recognition was granted to _____ to facilitate mobilization of adequate resources and their efficient allocation in the capital markets.
(a) RBI (b) BSE (c) SEBI (d) NSE
(1 mark)

49. In which of the following sector FDI is not permissible?
(a) Telecom (b) Aviation (c) Atomic energy (d) Defence (1 mark)

50. Production of milk is included in which sector?
(a) Tertiary sector (b) Service sector (c) Primary sector (d) Secondary sector
(1 mark)

PEQ – MAY – 2025

51. Which of the following are fiscal reforms?
i. Ensuring better tax compliance
ii. Reinvestment of funds in more profitable options
iii. Encouraging private sector participation
iv. Thrust on curbing government expenditure
(a) Only i and iii
(b) Only i, ii, iii
(c) Only i, iii, iv
(d) Only i, ii, iv

52. Which of the following best describes the pre-British Indian economy ?
(a) Dependent on imports for goods
(b) Self-sufficient villages and cities which were centers of commerce
(c) Focus on industrial production
(d) Dominated by foreign trade

53. The trade policy reforms include _____.
(a) Inclusion of licensing restrictions for imports
(b) Removal of licensing procedure for imports
(c) Complication of tariffs.
(d) Inclusion of quantitative restrictions on imports and exports

54. Which initiative by NITI Aayog aims to promote electric vehicles?
(a) *E-Amrit*
(b) *Shoonya Campaign*
(c) *Methanol Economy Program*
(d) *India Policy Insights (IPI)*

55. The early liberalization and reforms started in India in _____.
(a) *1980s*
(b) *1970s*
(c) *1960s*
(d) *1990s*

56. What percentage of India's population depends on agriculture for livelihood as per latest estimates?
(a) *18%*
(b) *25%*
(c) *47%*
(d) *60%*

57. Which government scheme focuses on providing financial support to farmers suffering crop loss or damage?
(a) *PM KISAN*
(b) *Pradhan Mantri Fasal Bima Yojana (PMFBY)*
(c) *Soil Health Card Scheme*
(d) *Paramparagat Krishi Vikas Yojana (PKVY)*

58. Which regime replaced the Foreign Investment Promotion Board (FIPB)?
(a) *SEBI*
(b) *FIF Portal*
(c) *NITI Aayog*
(d) *DPIHT*

59. What was India's rank in 2022 as per Global Innovation Index?
(a) *81st*
(b) *40th*
(c) *63rd*
(d) *25th*

60. Which policy replaced the Merchandise Exports from India Scheme (MEIS) in 2021?
(a) *NDAP*
(b) *GST*
(c) *PLI Scheme*
(d) *RoDTEP*

PEQ – JANUARY – 2026

61. Which initiative of the NITI Aayog is a one-stop destination for all information on electric vehicles?
(a) *Shoonya Campaign*
(b) *E-Amrit*
(c) *Methanol economy*
(d) *Life*

62. Make in India 2.0 focuses on
(a) *15 manufacturing and 12 service sectors*
(b) *12 manufacturing and 15 service sectors*
(c) *22 manufacturing and 18 service sectors*
(d) *18 manufacturing and 22 service sectors*

63. Which of the following industries is included in the eight core sectors assessed by the Index of Industrial Production (IIP)?
(a) *Pharmaceuticals*
(b) *Textiles*
(c) *Coal*
(d) *Tourism*

64. Which of the following was not a major reform introduced during the 1980s in India?
(a) *Industrial delicensing*
(b) *Formation of SEBI*
(c) *Currency demonetization*
(d) *Implementation of MODVAT*

65. How many crops are mandated to receive price support through "Minimum Support Price" (MSP)?
(a) *22*
(b) *21*
(c) *23*
(d) *24*

66. In India, public sector is limited to which of the following sectors?
(a) *Railway transport and arms and ammunition*
(b) *Railway transport and atomic energy*
(c) *Atomic energy and arms and ammunition*
(d) *Hazardous chemicals and narcotic drugs*

67. Introduction of prudential norms of accounting in respect of classification of assets, disclosure of income and provisions for bad debts in case of commercial banks is related to which of the following reform measures?
(a) *Fiscal reforms*
(b) *Monetary and financial sector reforms*
(c) *Capital market reforms*
(d) *Trade policy reforms*

68. Which of the following is a unique feature in the growth trajectory of the Indian economy?
(a) *Shift from services to the agricultural sector*
(b) *Shift from manufacturing to the services sector*
(c) *Shift from agricultural to the manufacturing sector*
(d) *Shift from agriculture to the services sector*

PEQ – SEPTEMBER – 2026

69. One of the early steps in 1991 reforms was:
(a) *Rupee devaluation*
(b) *Nationalisation of banks*
(b) *Abolition of tax reforms*
(d) *Trade protection*

70. Post-reform export growth led to:
   (a) *Decline in forex reserves*
   (b) *Increase in foreign exchange reserves*
   (c) *Reduced trade volume*
   (d) *Collapse in services exports*

71. Reforms in the capital market after 1991 strengthened through:
(a) *RBI takeover of markets*
(b) *Establishment of SEBI autonomy*
(c) *Complete ban on foreign investment*
(d) *Closing stock exchanges*

72. Structural reforms differ from stabilization policies because they:
(a) *Address long-term efficiency issues*
(b) *Focus only on inflation*
(c) *Control exchange rates*
(d) *Increase tariffs*

ANSWERS

I. Multiple Choice Type Questions

1.	(c)	2.	(b)	3.	(c)	4.	(a)	5.	(b)	6	(b)	7.	(a)	8.	(d)	9.	(a)
10.	(c)	11.	(b)	12.	(b)	13.	(d)	14.	(c)	15.	(b)	16.	(c)	17.	(b)	18.	(b)
19.	(a)	20.	(a)	21.	(b)	22.	(c)	23.	(d)	24.	(b)	25.	(a)	26.	(a)	27.	(a)
28.	(b)	29.	(a)	30.	(b)	31.	(b)	32.	(d)	33.	(b)	34.	(c)	35.	(d)	36.	(a)
37.	(a)	38.	(c)	39.	(a)	40.	(d)	41.	(c)	42.	(a)	43.	(d)	44.	(b)	45.	(d)
46.	(a)	47.	(c)	48.	(c)	49.	(c)	50.	(c)	51	(c)	52	(b)	53	(b)	54	(a&b)
55	(a)	56	(c)	57	(b)	58	(b)	59	(b)	60	(d)	61	(b)	62	(a)	63	(c)
64	(c)	65	(c)	66	(b)	67	(b)	68	(d)	69	(a)	70	(b)	71	(b)	72	(a)

For Your Notes

For Your Notes

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